Investor Tax Heterogeneity and the Term Structure: Endogenous Segmentation and the Municipal Bond Puzzle*

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October 11, 2026

Abstract

I develop an equilibrium term-structure model in which heterogeneous investor taxes endogenously and partially segment bond markets. Investors choose bonds jointly across markets and maturities without short sales, so their taxes and balance sheets determine which bonds they hold. I apply the model to taxable and tax-exempt bonds. High-tax investors, the top-bracket households, anchor short tax-exempt yields at their after-tax taxable yields. Lower-tax investors, such as insurers, banks and lower-bracket households, hold taxable bonds, and also long tax-exempt bonds once their lower cost of bearing rate risk outweighs their smaller exemption benefit. The tax rate priced into long tax-exempt yields relative to taxable yields thus also reflects these investors' lower taxes, so it falls with maturity. This explains the municipal bond puzzle: short municipal yields match top-bracket households' after-tax taxable yields, but long ones are far higher. Without targeting municipal yields, the calibrated model explains 83 percent of the 117-basis-point gap at thirty years. Empirically, as the mechanism predicts, household state tax rates matter less and bank- and insurer-specific tax benefits more at longer maturities, contradicting the representative-tax benchmark that applies the top-bracket household's tax rate at every maturity.

Keywords: Term Structure, Investor Heterogeneity, Taxes, Endogenous Segmentation, Asset Ownership, Municipal Bonds
JEL Codes: G12, G14, G22, H24, H63

*I am grateful to Moritz Lenel, Wei Xiong, and Motohiro Yogo for their guidance. I also thank Yacine Ait-Sahalia, Caio Almeida, Markus Brunnermeier, Fousseni Chabi-Yo, Aditya Chaudhry, Daniel Chen, Zhiyu (Julie) Fu, Nicolae Garleanu, Sergei Glebkin, Zhouzhou Gu, Nicolas Hommel, Paymon Khorrami, Jane Li, Shiqi Li, Ziang Li, Jonathan Payne, Yinan Qiu, Julia Selgrad, Amy Huber Wang, Lingxuan (Sean) Wu, Yinjie Yu, Yao Zeng, William (Yinhong) Zhao, Haonan Zhou, and Haoxing Zhu, and participants at the MPSS Conference and Princeton University Seminars for constructive feedback.

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1 Introduction

Under the U.S. tax code, investors face different tax treatment of the same asset. How does this tax heterogeneity shape bond yield curves? I develop an equilibrium term-structure model in which investors with heterogeneous tax treatment choose portfolios jointly across bond markets and maturities, subject to limits on short selling. Their taxes and balance sheets determine which bonds they hold and the returns they require, creating partial segmentation.

I apply the model to the joint pricing of taxable bonds and tax-exempt municipal bonds issued by U.S. state and local governments. Households pay their marginal income tax rate on taxable bond interest but no federal income tax on exempt municipal interest. The municipal exemption is most valuable to top-bracket households because they face higher tax rates on taxable interest than lower-tax households and institutional investors. Thus, top-bracket households can accept the lowest municipal yield. Together with limits on short selling, this tax-sorting logic motivates the representative-tax benchmark used in the literature: at every maturity, municipal yields should equal the household benchmark, taxable yields multiplied by one minus the top-bracket household's tax rate (Longstaff, 2011; Schwert, 2017; Fleckenstein and Longstaff, 2025).

However, the ownership data suggest that top-bracket households hold only about half the municipal market. Lower-tax households, banks and insurers hold substantial portfolios despite smaller net exemption benefits. Yields also depart from the literature's benchmark. Short municipal yields closely match the household benchmark, but long municipal yields exceed it. This is the municipal bond puzzle, documented in the literature for over thirty years (Feenberg and Poterba, 1991). The gap between the observed municipal yield and the household benchmark increases with maturity, reaching 117 basis points at thirty years. Equivalently, the implied tax, the tax rate that equates the after-tax taxable yield with the municipal yield, falls with maturity.

The heterogeneity in municipal ownership is central to the model's explanation of the municipal bond puzzle. Intuitively, when investors with different tax rates participate in the market, equilibrium yields reflect the tax treatment of participating investors, including those with lower tax rates than top-bracket households. Their participation explains why the implied tax rate falls below the household rate and municipal yields exceed the household's after-tax taxable benchmark. The exact equilibrium pattern across maturities depends on investors' endogenous participation and maturity sorting discussed below. The detailed mechanism proceeds in two steps. First, why do lower-tax investors hold municipal bonds despite their smaller exemption benefits? Second, how does this heterogeneous ownership affect equilibrium bond yields? In the model, investors differ in tax treatment and exogenous liability duration exposure. They choose portfolios jointly across maturities in taxable and tax-exempt bond markets, subject to short-sale constraints.

Some lower-tax investors hold municipal bonds despite smaller exemption benefits because they bear interest-rate risk at a lower cost than high-tax households. Balance sheets provide a microfoundation for this advantage. When interest rates rise, the value of long-term liabilities falls, offsetting losses on long bonds. Household mortgage debt, insurers' obligations and banks' assets and deposit liabilities give investors different net liability exposures to rate movements. The

resulting duration benefit grows with a bond's rate exposure, while the difference in exemption benefits is constant across maturities. Lower-tax investors participate where their duration benefit outweighs the difference in exemption benefits, beginning at longer maturities. In the two-investor illustration, the lower-tax investor has greater liability duration exposure. The high-tax household holds shorter municipal bonds, the lower-tax investor holds risky taxable bonds, and both hold municipal bonds beyond the participation threshold.

For the second question, three channels explain how these holdings affect the relative equilibrium returns of taxable and tax-exempt bonds. First, a lower-tax investor keeps more taxable interest than the high-tax household, making exemption less valuable. It therefore accepts a smaller reduction in return for the exemption. At long maturities where it holds both municipal and taxable bonds, this difference contributes to municipal returns above the household benchmark. Second, the cost of bearing rate risk differs: the high-tax household requires more return per unit of rate exposure than the lower-tax investor. At risky maturities below the participation threshold, the household alone holds municipal bonds, while the lower-tax investor holds comparable taxable bonds. Municipal returns therefore exceed the household's after-tax taxable benchmark even where the lower-tax investor holds no municipal bonds. Third, taxes scale bonds' rate loadings. The household benchmark scales taxable loadings by the household's after-tax share. Long municipal bonds also reflect lower-tax holders' larger after-tax shares, making them more sensitive to rates than this benchmark predicts. Household portfolio substitution carries this effect to shorter bonds held only by households: adjustments in long-bond holdings change the household's total rate exposure and the returns it requires on shorter bonds. Holders require additional return for this extra rate risk, widening the gap from the household benchmark. All three channels vanish at the riskless shortest maturity. A bond's yield reflects the equilibrium one-period returns over its remaining life, so these return gaps accumulate into yield differences across maturities.

Liability hedging provides a natural foundation for the quantitative application. Insurers' long-term obligations create demand for duration, while households' outside assets and debt give them different exposures to rate movements. Observed balance sheets and maturity holdings therefore provide a concrete basis for differences in rate-risk costs and data with which to discipline them. The mechanism also accommodates other motives for participation, such as lower trading costs relative to households or a desire for default diversification. When the resulting advantages grow with maturity and apply equally to taxable and tax-exempt bonds, the relative-pricing results hold (Section 4.7). Thus, the pricing mechanism does not rely on a particular microfoundation for lower-tax investors' participation.

I calibrate the model to U.S. bond markets, with multiple investor types, municipal markets across U.S. states and a taxable bond market. Investors include high- and lower-tax households residing in each state, banks, property-and-casualty insurers, life insurers and tax-neutral investors. They differ in tax treatment, state of residence (for households), resources and outside rate exposure. Observed tax rules, balance sheets, bond ownership and the process for taxable interest rates discipline the quantitative model. Without targeting municipal yields, the model explains 83% of

the thirty-year puzzle. Measured differences in default risk, liquidity and capital-gains taxation explain almost all of the remainder.

Three empirical tests, untargeted in the calibration, distinguish the model from the representative-tax benchmark. The model predicts that variation in top-bracket households' exemption benefits is fully reflected in implied taxes at the shortest maturity and less strongly at longer maturities. Changes in lower-tax investors' exemption benefits have no effect at the shortest maturity and matter more at longer maturities. First, states exempt residents' interest on their own municipal bonds from state income tax. Cross-state variation in state tax rates is reflected almost one-for-one in implied state taxes at short maturities and progressively less at longer maturities. Second, bank-qualified designation gives banks an additional tax benefit on a subset of municipal bonds issued by small borrowers. This bank-only benefit is barely reflected in short municipal yields and increasingly reflected at longer maturities. Third, I compare municipal yield changes with changes in the treasury reference yield at the same maturity. This relative response rises with maturity, consistent with the model's rate-loading channel.

Two counterfactual tax reforms measure the fiscal and financing effects of changing investor tax benefits. The first removes a state's home-state exemption for resident households, taxing own-state interest as Illinois, Iowa and Wisconsin generally do. Portfolio reallocation partly offsets the gain in bond-interest receipts calculated at unchanged prices and holdings. Financing costs also rise for issuers in other states and in the taxable market. The total increase in annual financing costs exceeds added state tax receipts. The second is to make muni bonds taxable, equalizing investors' tax treatment of municipal and taxable interest. Taxable municipal issuance has a precedent in the 2009–2010 Build America Bonds program, which allowed state and local governments to issue taxable bonds with federal interest subsidies. In the model's reform, issuers' replacement-financing costs rise by \$1.14 per dollar of added federal and state receipts, including \$0.16 borne by taxable issuers. The excess is an increase in net investor bond income, distributed unevenly: high-tax households and banks lose, while lower-tax households, insurers and tax-neutral investors gain. Evaluating either reform at the old allocation misses both the changing tax base and financing effects outside the targeted market.

I apply the framework to taxable and tax-exempt bonds because their tax rules provide clean variation for testing the mechanism and their relative yields present a longstanding puzzle. The framework applies more broadly wherever investors face different tax treatment across assets, including state-tax differences between treasury and corporate bonds, tax-favored accounts and cross-border withholding. Its central contribution is to show how investor tax heterogeneity shapes the entire yield curve through endogenous participation and joint portfolio choice. Whose tax treatment is reflected in yields varies with maturity, and portfolio substitution transmits tax effects across markets, even to bonds the affected investor never holds. Accounting for this heterogeneity is therefore essential both to understanding relative bond yields and to evaluating the financing and revenue effects of tax policy.

Sections 2–3 review related work and present the tax, ownership and yield evidence. Section 4 develops the model, Sections 5–6 evaluate its quantitative and empirical implications, and Section 7 studies tax policy. Formal results, proofs and supporting evidence are in the appendix.

2 Related Literature

The paper builds on the literature on investor taxes and asset pricing. Miller (1977) shows how personal taxes can offset the corporate tax advantage of debt in equilibrium. Auerbach and King (1983) jointly determine investor portfolios and aggregate debt-equity ratios when taxes and risk aversion differ. Dybvig and Ross (1986) establish when tax clienteles affect holdings alone and when they also affect prices. Dammon and Green (1987) characterize the restrictions on payoffs and tax schedules that permit equilibrium with unrestricted short sales, and Sialm (2009) provides evidence that dividend and capital-gains taxes are capitalized into equity prices. I extend this literature to the term structure: heterogeneous taxes and rate-risk costs determine participation across maturities and jointly shape taxable and tax-exempt yield curves.

Before the Tax Reform Act of 1986, banks could deduct the cost of carrying municipals and held them in large amounts. Early segmentation accounts placed banks at short maturities and households at long maturities (Mussa and Kormendi, 1979; Hendershott and Koch, 1980; Kidwell and Koch, 1983; Skelton, 1983). Fama (1977) and Livingston (1982) develop municipal pricing benchmarks, and Poterba (1986) studies expected tax policy. Clientele explanations were later dismissed because the puzzle persisted after 1986, when banks lost the deduction and largely left the market (Green, 1993). Changing tax rules also complicate tests of fixed clienteles (Mankiw and Poterba, 1996). Brown and Sirmans (2011) relate maturity clienteles to term premiums and the puzzle. Endogenous participation frees the mechanism here from a particular investor group's habitat: it can sustain a declining implied tax both before 1986 and after banks lost the deduction, whichever investor has the largest exemption benefit.

Since the 1986 Act, bond-side explanations of the municipal puzzle have dominated, and most of this literature adopts the representative-tax benchmark of the top-bracket household. Chalmers (1998) finds that the puzzle persists in effectively default-free, treasury-secured municipal bonds. Longstaff (2011) and Schwert (2017) examine taxes, credit risk and liquidity, while Fleckenstein and Longstaff (2025) study the convenience premium associated with avoiding taxes. Lehner et al. (2026) emphasize component-specific taxation in measuring the treasury funding advantage. Longstaff (2011) and Fleckenstein and Longstaff (2025) also document the puzzle's maturity pattern. The municipal swap rates in Longstaff (2011), which avoid bond-specific credit and liquidity risk, imply lower tax rates at longer maturities. The convenience premium in Fleckenstein and Longstaff (2025), measured against the top-bracket after-tax taxable yield after credit and liquidity adjustments, is largest at two to five years and declines at longer maturities. Schwert's finding that default risk dominates comes from bonds that mostly mature within ten years. Evidence on trading costs and dealer intermediation (Harris and Piwowar, 2006; Green et al., 2007; Bagley et al.,

2023) motivates the liquidity adjustment. Green (1993) derives a tax-timing explanation of the relative yield curves, Ang et al. (2010) document the pricing of taxes on market-discount municipal bonds, and Brown (2011) examines the call features of municipal bonds. I measure callability, the issuance premium, capital gains, liquidity and default following the literature and show that investor heterogeneity explains most of the municipal puzzle.[1]

State exemptions connect the paper to municipal market segmentation (Hendershott and Kidwell, 1978; Pirinsky and Wang, 2011). Babina et al. (2021) show how tax-induced home-state ownership limits risk sharing and affects financing conditions. Moussawi (2022) shows that state-tax segmentation shapes how asset purchases affect municipal bond prices, and Moussawi (2024) studies the demand, pricing and interstate effects of state tax shields. I relate declining household state-tax pass-through to participation by investors with smaller exemption benefits and trace financing effects across state municipal markets and taxable debt.

The analysis builds on term-structure models with segmented markets. Vayanos and Vila (2021) link maturity-specific demand to yields through risk-averse arbitrageurs, and Greenwood and Vayanos (2014) study how supply changes compensation for duration risk. Guibaud et al. (2013) derive maturity clienteles from life-cycle horizons. Related models study transmission across partially segmented markets, quantity-driven term premia and exchange rates, and monetary policy with investor balance sheets (Greenwood et al., 2018, 2023; Kekre et al., 2024). I derive participation from heterogeneous taxes and outside exposures, making partial segmentation an equilibrium outcome that responds to prices and tax policy across two jointly priced yield curves.

Institutional balance sheets provide a source of differences in the cost of bearing risk. He and Krishnamurthy (2013) link risk premia to financial intermediaries' capital, and Haddad and Sraer (2020) relate treasury risk premia to banks' rate exposure. Domanski et al. (2017) argue that insurers' duration matching can amplify declines in long-term yields, while Li (2026) links life insurers to long rates and credit spreads. Within municipals, Pati (2023) studies how banks' rate exposure and funds' liquidity needs affect offering yields. I show how a balance-sheet advantage in bearing rate risk can offset a smaller exemption benefit and sustain lower-tax investors' participation in tax-exempt bonds.

A related literature studies how heterogeneous asset demand and limits to arbitrage affect prices. Koijen and Yogo (2019) develop a demand system for asset pricing, Coppola (2025) studies how investor composition affects corporate bond prices and firms' financing over the credit cycle, and Gabaix et al. (2026) study heterogeneous household asset demand. Municipal fund fragility and flows affect pricing and financing (Li et al., 2024; Adelino et al., 2026), while restrictions on tax arbitrage and cross-market risk taking sustain differences across investors (Hines and Horton, 2024; Gromb and Vayanos, 2018; Siriwardane et al., 2025). I identify heterogeneous tax treatment as a determinant of bond demand and trace its effects across maturities and markets through joint portfolio choice.

[1] Cestau et al. (2019) review municipal market structure and pricing, and Spreen and Gerrish (2022) survey the tax-exemption literature. Appendix D.5 compares the model with Green's tax-timing benchmark.

Evidence on bank tax incentives connects investor taxes to municipal financing. Dagostino (2025) studies how bank financing incentives affect municipal issuance and employment, and Benson and Marks (2024) examine how the corporate tax cut changed the bank-qualified borrowing advantage. I contribute a maturity-based test: the yield reduction from bank-qualified designation rises with maturity while household state-tax pass-through falls, the opposite gradients implied by which investors hold each maturity in the model.

Finally, the policy analysis studies the financing effects and revenue cost of tax exemption. Poterba and Ramírez Verdugo (2011) show how portfolio substitution changes revenue estimates, while Garrett et al. (2023) study how tax advantages and auction competition determine municipal borrowing costs. I solve for portfolios, both yield curves and tax receipts after tax reform, accounting for financing effects on taxable issuers as well as municipal issuers.

3 Taxes, Ownership and the Municipal Puzzle

This section establishes the empirical setting for the model. I describe the data and the tax benefits available to different investors, then show that municipal ownership extends well beyond high-tax households. I measure the municipal puzzle and examine whether bond characteristics or the choice of taxable benchmark explain its maturity pattern.

3.1 Data

I combine data on bond issuance, investor holdings and taxes. Mergent provides issuance records, ratings, bond characteristics and call schedules. Insurers' filings in Schedule D of the National Association of Insurance Commissioners (NAIC) and a pension holdings panel collected through Freedom of Information Act requests allow me to examine institutional portfolios at the level of individual bond holdings. I use the Financial Accounts to measure aggregate holdings, the Survey of Consumer Finances to measure household balance sheets, and IRS Statistics of Income to document household tax-exempt interest income across income brackets.

The main yield comparison covers 2000–2020 and uses FRED's A-or-above corporate zero curve as the taxable benchmark. The treasury curves (Gürkaynak et al., 2007) obtained from the Federal Reserve provide the benchmark for the interest-rate response test. To examine trading and liquidity, I use municipal transactions from the MSRB and corporate transactions from TRACE Enhanced. Markit CDS quotes provide credit default swap data for municipal and taxable bond issuers, and Tax Foundation schedules identify state income-tax rates. Appendix E.7 documents the sources and their coverage.

3.2 Municipal issuance

Table 1 describes municipal bonds issued during 2000–2020, with one observation per bond identifier (CUSIP). The full sample contains more than three million issues, with an average maturity of about ten years. Most pay federally exempt interest: taxable municipal bonds account for 5.8% of issues,

Table 1: Municipal issuance: descriptive statistics (sample means).

Variable	Full sample	Tax-exempt, non-AMT	
		AA and above (score ≤ 3)	A− and above (score ≤ 7)
Offering yield (%)	3.11	2.82	2.96
Corporate benchmark yield (%)	4.47	4.33	4.41
Maturity at issuance (years)	10.15	10.76	10.86
Offering price (per $100 par)	102.88	104.68	103.85
Issue rating (1 = AAA, 3 = AA, 7 = A−)	3.80	2.18	3.55
Implied tax rate $1 - y^M/y^C$	0.29	0.35	0.33
Share callable	0.46	0.51	0.51
Share tax-exempt, non-AMT	0.92	1.00	1.00
Sample size (CUSIPs)	3,014,420	984,210	1,990,791

Notes. The table reports bond-level means before the call and issuance-premium adjustments, with one observation per municipal issue (CUSIP). The full sample contains all issues. The AA-and-above and A-or-above columns restrict to federally tax-exempt, non-AMT issues with average agency score ≤ 3 and ≤ 7, the latter matching the A-or-above corporate credit band. The implied-tax row averages the bond-level ratio $1 - y^M/y^C$, winsorized at the first and ninety-ninth percentiles, so it need not equal one minus the ratio of the displayed yield means. Each row uses non-missing observations: the full-sample rating mean uses the 2,292,092 rated issues, and the restricted columns exclude unrated issues.

bonds subject to the alternative minimum tax (AMT) account for 1.9%, and the remaining issues are tax-exempt, non-AMT bonds. Call provisions allow issuers to redeem bonds before maturity and are common in the data. Callable bonds account for 46% of all issues and 51% of the high-grade, tax-exempt samples, making callability important for comparing yields.

The table also reports two high-grade samples of tax-exempt, non-AMT bonds. The broader sample includes A− and above issues (average agency score ≤ 7 on the twenty-two-notch scale) and matches the credit band of the corporate benchmark used for the main yield curves. The main empirical specifications use the AA-and-above sample (≤ 3) to minimize the impact of default risk.

3.3 Bond taxation and the representative-tax benchmark

For a taxable household, interest on taxable bonds is ordinary income, while interest on tax-exempt municipal bonds is federally exempt. Interest includes both coupon payments and annual original-issue-discount accrual. Institutions generally do not receive the full exemption benefit from holding municipal bonds. For example, property-and-casualty (P&C) insurers pay an effective federal tax of 5.25% on municipal interest. Let τ_i^C and τ_i^M denote investor i's effective tax rates on taxable and municipal interest. I define the relative after-tax share as the fraction of taxable interest retained divided by the fraction of municipal interest retained:

$$\xi_i \equiv \frac{1 - \tau_i^C}{1 - \tau_i^M}. \tag{1}$$

Investor i's relative municipal tax benefit, the exemption benefit, is $1 - \xi_i = (\tau_i^C - \tau_i^M)/(1 - \tau_i^M)$. A smaller ξ_i means a larger benefit. When municipal interest is fully exempt, $\xi_i = 1 - \tau_i^C$, so the benefit equals the tax rate on taxable interest. Appendix E explains the underlying tax rules.

The representative-tax benchmark uses the top-bracket household's tax treatment at every maturity. Write y_T^M and y_T^C for municipal and taxable yields at maturity T, and H for this household. Under the benchmark, the two bonds offer the household the same after-tax yield, $y_T^M = \xi_H y_T^C$. I call $\xi_H y_T^C$ the household benchmark and $1 - \xi_H$ the household wedge. The benchmark's yield-implied tax is therefore $1 - y_T^M / y_T^C = 1 - \xi_H$. With fully exempt municipal interest, it equals τ_H^C. Investors with different ξ_i cannot all be indifferent at this yield ratio. Their participation in the municipal bond market is the central question of the model.

3.4 Investor taxes and ownership

The tax benefit of municipal interest varies substantially across investors in the roughly $4 trillion municipal market. Table 2 summarizes these differences before the 2017 Act. For households and insurers, the net exemption benefit is $1 - \xi_i$ in (1). Households in higher tax brackets receive a larger benefit because they pay more tax on taxable interest. Exemption from home-state income tax adds to the benefit on in-state municipal bonds. Appendix E gives the detailed tax treatment of each investor group and the changes after the 2017 Act.

Insurers receive smaller benefits than high-tax households because exempt interest reduces their deductions. The reduction is less severe for P&C insurers than for life insurers, giving P&C insurers the larger benefit. Banks generally lose the interest deduction allocated to municipal holdings, with partial relief for bank-qualified bonds, so the table reports their benefit net of this financing-tax cost.[2] Tax-exempt institutions such as qualified pension funds and endowment funds already receive taxable interest free of current income tax and gain no additional benefit from the municipal exemption. They hold almost no municipal bonds (Appendix D.9).

Figure 1 shows that municipal ownership spans these tax groups. Panel (a) reports aggregate holdings: households and municipal funds hold the largest amounts, while banks and P&C insurers also hold substantial portfolios despite smaller exemption benefits. P&C insurers allocate about 40% of their bond portfolios to municipals and maintain a larger municipal portfolio share than life insurers throughout the sample (Appendix D.9). Panel (b) documents household tax-exempt interest income across income brackets. Assigning municipal-fund holdings to households and allocating household holdings using tax-exempt interest shares puts roughly half the market with high-tax households.[3] The remaining holdings by investors with smaller exemption benefits motivate a model in which investors with different exemption benefits choose municipal and taxable bonds jointly. The analytical benchmark has two investors. The quantitative model includes household groups by state of residence, insurers, banks and a residual group (Section 5.1).

[2] The quantitative model charges the bank financing-tax cost separately. Table 3 therefore reports the bank's exemption benefit $1 - \xi_i$ before that cost.

[3] I use IRS Statistics of Income shares of tax-exempt interest on taxable returns, including fund distributions, to approximate household holdings by income group in 2010–2017. Adjusted gross income of at least $200,000 proxies for the high-tax group.

Table 2: Investor heterogeneity in the municipal market before the 2017 Tax Cuts and Jobs Act

Investor type	Federal treatment	State treatment	Net exemption benefit
High-tax household	Interest exempt	Exempt on in-state bonds only	0.35–0.43
Middle-tax household	Interest exempt	Exempt on in-state bonds only	0.25–0.33
Low-tax household	Interest exempt	Exempt on in-state bonds only	0.10–0.15
P&C insurer	Exempt, less proration	Premium taxes, no distinction	0.31
Life insurer	Exempt, less proration	Premium taxes, no distinction	0.21
Bank	Exempt, funding cost disallowed	Varies by state	0.15–0.33
Other (pensions, etc.)	Varies	Varies	< 0.05

Notes. For households and insurers, the net exemption benefit is $1 - \xi_i$, from statutory rates and proration over 2000–2017. For banks it also deducts the financing-tax cost described below. Appendix Table 50 gives the values after the 2017 Act. Household values are federal marginal rates in three groups (35% and above, 25% to 33%, below 25%), and the top value includes the 3.8% net investment income tax from 2013. The home-state exemption adds to the benefit on in-state bonds. The quantitative model pools middle- and low-tax households into one lower-tax group, whose benefit with home-state taxes is 0.21 to 0.27 across states in 2015–2017, against 0.43 to 0.50 for the high-tax group (Table 3). Proration cuts a P&C insurer's losses-incurred deduction by 15% of its exempt interest and a life insurer's reserve deduction by the policyholders' share. Section 265(b) disallows the share of a bank's interest expense allocable to exempt holdings, of which 80% stays deductible for bank-qualified bonds. The bank value, net of that disallowance, is the break-even implied tax, equal to the bank's tax rate minus its annual financing-tax cost per dollar of municipal holdings divided by the taxable yield. That cost is 1.02% in 2000–2002 and 0.11% in 2015–2017, and the calculation covers the six three-year periods of 2000–2017 and maturities of two to thirty years. States generally tax insurers through premium taxes, which do not distinguish municipal from taxable interest, and banks through franchise or income taxes.

3.5 Limits to arbitrage

Tax rules and trading frictions limit the trades that could exploit yield differences or equalize investors' costs of bearing rate risk. Interest and short-sale payments incurred to carry exempt bonds generally lose their deduction. Consider a household that shorts a long-maturity corporate bond to hedge its municipal holdings. With the deduction denied, an equal-principal position earns coupon carry equal to the pretax municipal-minus-corporate yield difference. This carry is negative along the sample-average par curves and about -95 basis points a year at thirty years. Municipal shorts are scarce and costly because substitute payments to the lender do not retain the interest exemption. Off-exchange swaps face eligibility restrictions, while futures losses can face deduction limits and deferral. The model summarizes these tax asymmetries and trading frictions through nonnegative bond positions and cash balances, ruling out short sales and borrowing. Appendix E.6 examines the tax treatment, carry and residual risk of these trades.

3.6 The municipal puzzle

The municipal puzzle is a maturity pattern: short municipal yields are close to the household benchmark, while long municipal yields substantially exceed it. To measure this pattern, I use the seasoned, option-free A-or-above corporate zero-coupon curve as the taxable comparison. High-grade corporate bonds are closer to municipals in credit and liquidity than treasuries. Treasury yields also contain a premium for safety and liquidity (Krishnamurthy and Vissing-Jorgensen, 2012),

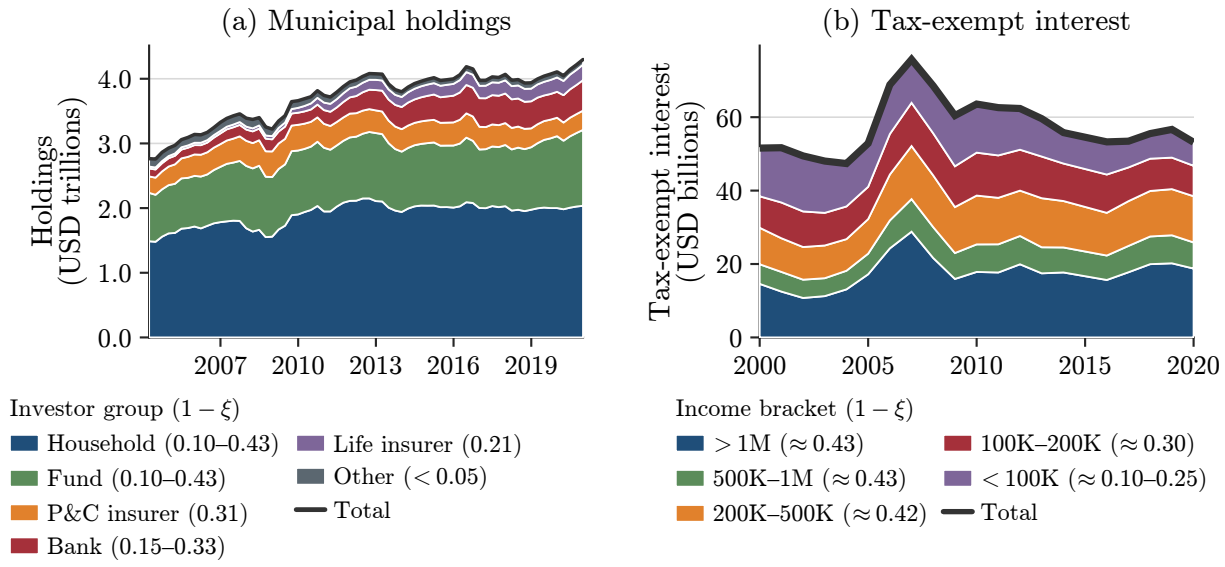


Figure 1: Municipal holdings and tax-exempt interest by investor group. Panel (a): direct holdings of taxable and tax-exempt municipal securities, 2005–2021 (USD trillions, Financial Accounts, with households including nonprofits). Panel (b): tax-exempt interest, an income flow, on taxable individual returns by adjusted-gross-income bracket, 2000–2020 (USD billions, IRS Statistics of Income). 2015 lacks comparable bracket detail and is omitted. Legend values in parentheses are net exemption benefits: in panel (a), the values before the 2017 Act from Table 2, with funds assigned their household shareholders' range, and in panel (b), approximate values by bracket.

which would enter the measured gap if treasuries supplied the benchmark. Previous studies using treasuries as the taxable benchmark also find that the municipal puzzle persists after adjusting for these differences (Chalmers, 1998; Longstaff, 2011).

I construct municipal zero-coupon yields in the same credit band from offering prices and contractual cash flows. Issuance data provide broad coverage and a common measurement date. Offering yields also avoid the direct tax effects of capital gains and market discounts arising from prior secondary-market trades. I remove callability and the issuance premium to make municipal yields comparable with the seasoned, option-free corporate curve.[4] Let $y_t^C(T)$ denote the taxable yield and $y_{s,t}^M(T)$ the municipal yield after these two adjustments, where T is maturity, s is the issuer's state and t is the issue date. The municipal puzzle is the gap between $y_{s,t}^M(T)$ and the household benchmark $\xi_H y_t^C(T)$. The yield-implied tax expresses the same comparison in tax-rate units:

$$ImpliedTax_{s,t}(T) \;\equiv\; 1 - \frac{y_{s,t}^M(T)}{y_t^C(T)}. \tag{2}$$

For a given household's after-tax share ξ_H, the two measures satisfy

$$\underbrace{y_{s,t}^M(T) - \xi_H y_t^C(T)}_{\text{municipal yield gap}} = \underbrace{\left[(1-\xi_H) - ImpliedTax_{s,t}(T)\right]}_{\text{shortfall from the household wedge}} y_t^C(T). \tag{3}$$

[4]The market-discount tax applies to later secondary purchasers. Its expected effect on resale values can also affect offering prices. The capital-gains adjustment includes this effect as well as taxes on realized gains.

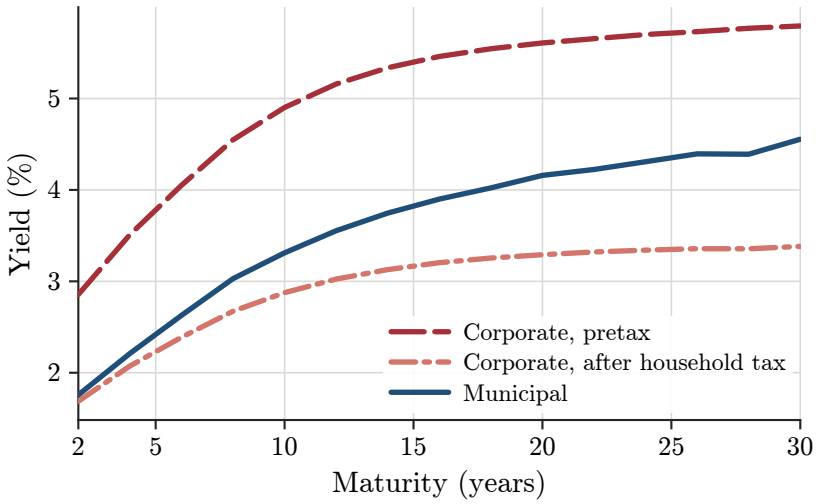


Figure 2: The municipal puzzle: A-or-above zero-coupon yields, 2000–2020. Municipal yields remove callability and the issuance premium. Averaging follows Appendix D.1.1. Appendix D.13.1 gives the AA-and-above counterpart.

For positive taxable yields, a positive gap means an implied tax below the household wedge $1-\xi_H$. In Figure 2, the municipal yield is near the household benchmark at the short end but about 117 basis points above it at thirty years. The implied tax falls from about 38% to 21% while the household wedge remains near 41% (Appendix Figure 14).

The declining implied-tax pattern also appears against treasuries (Chalmers, 1998), within issuer-matched tax-exempt and taxable municipals (Section 3.8), and in the AA-and-above and unlimited-tax general-obligation samples (Appendices D.13.1 and D.4).

3.7 Bond-side adjustments

Studies such as Fleckenstein and Longstaff (2025) use secondary-market prices of option-free bonds to examine the municipal puzzle. Consistent with that comparison, I remove callability and the issuance premium before defining the municipal puzzle as the gap over the household benchmark. I then assess default risk, liquidity and capital-gains taxation as explanations of this gap. Each adjustment measures the difference in compensation between municipal and corporate bonds, expressed in the household's after-tax yield units. Compensation common to both markets therefore cancels. The household wedge, one minus the ratio of averaged after-tax to pretax corporate yields, averages 41.3% and is nearly flat across maturities. Appendix D explains the adjustments, and Appendix Table 26 reports their contributions by maturity. Appendix D.5 discusses the tax timing of taxable bonds in Green (1993).

At thirty years, the call and issuance-premium adjustments reduce the 159.8-basis-point spread over the household benchmark to the 117.2-basis-point municipal puzzle. The default, liquidity and capital-gains differentials explain 51% of the puzzle at ten years but only 16% at thirty. Subtracting them leaves 98.4 basis points at thirty years for comparison with the model in Section 5.4. At two years, their sum exceeds the small puzzle, leaving a residual of −9.1 basis points, consistent with

the small convenience premium discussed in Fleckenstein and Longstaff (2025). The adjustments leave the central pattern intact: an unexplained municipal yield gap that widens with maturity.

3.8 Issuer-matched evidence

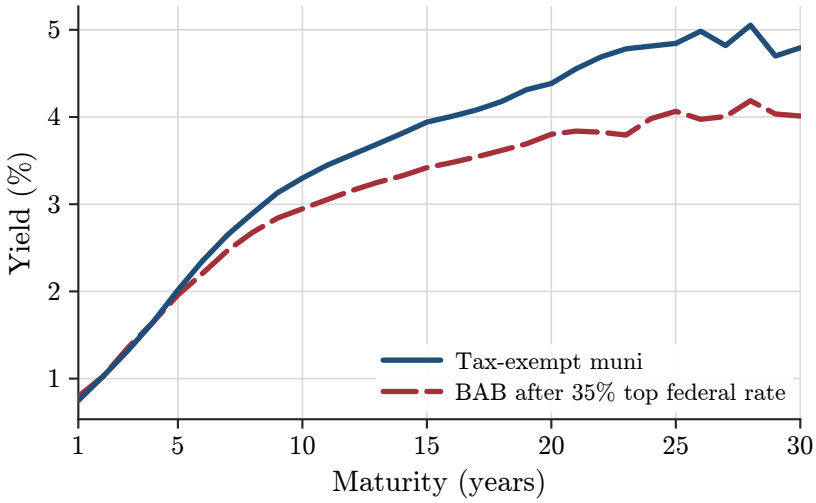


Figure 3: Tax-exempt and Build America Bond offering yields by maturity, AA-and-above issues, April 2009–December 2010. BAB yields are multiplied by 0.65, one minus the 35% top federal rate in 2009–2010. Appendix Table 29 reports regressions with and without call-status matching. Appendix D.3 describes the sample.

Taxable and tax-exempt bonds issued by the same municipal borrower provide a further comparison that does not rely on corporate bonds. During 2009–2010, the Build America Bonds (BAB) program allowed municipal issuers to sell taxable bonds with a federal subsidy equal to 35% of their interest payments. The top federal household income-tax rate was also 35%. I leave tax-exempt offering yields unchanged and multiply BAB yields by 0.65 to put both on the household's after-tax scale. I then compare bonds from the same issuer and estimate how their yield difference varies with maturity:

$$y_b = \alpha_1 \, EXMP_b + \beta \, (EXMP_b \times m_b) + \gamma_C \, y^C_{m_b,t_b} + \lambda_{j(b)\times m_b \times q_b} + \varepsilon_b, \tag{4}$$

where y_b is the offering yield for a tax-exempt bond and 0.65 times the offering yield for a BAB. The indicator $EXMP_b$ equals one for a tax-exempt bond and zero for a BAB, and m_b is maturity in years. The fixed effect $\lambda_{j(b)\times m_b \times q_b}$ compares bonds from issuer $j(b)$ at the same maturity and in the same issue quarter q_b. The call-matched specification also requires the same call status. The corporate yield $y^C_{m_b,t_b}$, measured at the bond's maturity and issue month t_b, controls for market yield movements within that quarter. The coefficient β therefore measures how the tax-exempt yield minus the household after-tax BAB yield changes with maturity. Standard errors are clustered by state. Comparing bonds from the same issuer at the same maturity and with matched call status

holds borrower identity and these bond characteristics fixed. The widening gap persists in this comparison.

The estimated gradient β is 3.4 basis points per year in the AA-and-above sample and 3.3 in the A-or-above sample. Matching call status gives gradients of 3.1 and 3.0 basis points. The binned estimates also rise monotonically: in the AA-and-above sample with call matching, the $[25, 30]$-year coefficient is 63.9 basis points above the one-to-five-year bin. Figure 3 shows mean yields by maturity before issuer matching. Appendix Table 29 reports the issuer-matched estimates. Both show a widening gap, and the regressions establish that it persists within the same issuer, using offering yields for both bond types and matching call status.

4 Model

I develop a term-structure model in which investors differ in tax treatment and balance-sheet exposure to interest rates. They choose bonds jointly across markets and maturities, subject to short-sale constraints. The model answers two questions: why investors with smaller exemption benefits hold municipal bonds, and how their participation changes yields relative to the household benchmark.

The argument begins with portfolio choice. A bond's required return reflects its holder's after-tax cash alternative and the risk it adds to that holder's wealth. I use this comparison to derive participation and the three channels in relative returns. I then show how those returns determine bond prices, rate loadings and yields. The general economy below is also the model used for calibration. The two-investor specialization isolates the mechanism, with formal results and proofs in Appendices A and B.

4.1 Bonds, investors and portfolio choice

Time is discrete. In the quantitative application a period is two years.

Interest rates. A single rate state z_t drives the short rate r, the pretax return on cash, which cannot fall below zero:

$$z_{t+1} = \rho z_t + \sigma_r \varepsilon_{t+1}, \qquad r(z_t) = \max\{\bar{r} + z_t,\, 0\}, \qquad 0 < \rho < 1. \tag{5}$$

Here ε_{t+1} is standard normal, ρ measures persistence, σ_r the size of rate shocks and $\bar{r}$ the mean rate before the floor. Cash earns the known log gross return $r(z_t)$ over the next period.

Bonds and returns. Bonds are zero-coupon and grouped into markets a. The bonds in one market receive the same tax treatment, which can differ across investors. The application has a taxable market C and municipal markets that differ by the issuer's state. A bond with T periods to maturity has price $P_T^a(z)$, pays one unit at maturity and becomes a $(T-1)$-period bond next period. Cash is the one-period taxable asset. Supplies Q_T^a are fixed.

An investor who buys a T-period bond today and sells it next period earns

$$\text{gross holding return} = \frac{P^a_{T-1}(z_{t+1})}{P^a_T(z_t)} \exp\!\big(\sqrt{v^a_T}\,\eta^a_{T,t+1}\big), \qquad P^a_0 = 1.$$

The price ratio is the resale value per dollar invested. It carries rate risk because next period's price depends on next period's rate state. The standard normal shock $\eta^a_{T,t+1}$ is independent across bonds and of the rate shock, adding bond-specific log-return variance v^a_T. This small component represents residual return variation, such as an unexpected change in demand for a particular issue. Without it, different bond portfolios can offer the same after-tax return and rate exposure, leaving their separate holdings indeterminate. Bond-specific risk makes these positions imperfect substitutes by adding a cost that increases with each position. Rate risk drives the mechanism, while the bond-specific component has a small effect on the yield gap (Appendix C.8).

At one period, the payment is known and $v^a_1 = 0$, so the bond is fully riskless. The yield, $y^a_T = -\log P^a_T/T$, summarizes today's price relative to the maturity payment. The holding return covers the next period, including any change in the resale price.

Following Greenwood et al. (2024), I use a log-return approximation for portfolio choice. Write that return as

$$\mathcal{R}^a_{T,t+1} = \mu^a_T(z_t) + u^a_{T,t+1}, \tag{6}$$

where $\mu^a_T(z_t)$ is its conditional mean and $u^a_{T,t+1}$ is the unexpected part, with $\mathbb{E}_t[u^a_{T,t+1}] = 0$. Investors care about both: μ adds to expected wealth, while u changes its risk. In equilibrium, the expected return μ must equal the return a holder requires. The approximation includes the Jensen adjustment and becomes exact in the continuous-time limit.[5]

Investors. Type i has funding W_i, risk aversion γ_i and outside rate exposure ℓ_i from assets and liabilities beyond its bond portfolio. A positive ℓ_i raises net wealth when rates rise, as does a long-term liability whose value falls. A negative ℓ_i adds to bond losses. Investor i keeps the share $\xi^a_i = 1 - \tau^a_i$ of returns in market a, where τ^a_i is its tax rate. Cash is taxed like taxable bonds.

The investor chooses dollar positions $x^a_{i,T} \geq 0$, keeps the rest of its funding in cash, $x^0_i = W_i - \sum_{a,T} x^a_{i,T} \geq 0$, and maximizes expected next-period wealth less a penalty for its variance. With returns measured by $\mathcal{R}$, next-period wealth is $\widehat{W}'_i$, and the problem is

$$\begin{aligned} \max_{x_i \geq 0,\, x^0_i \geq 0} \quad & \mathbb{E}_t[\widehat{W}'_i] - \frac{\gamma_i}{2}\operatorname{Var}_t(\widehat{W}'_i), \\ \text{subject to} \quad & \widehat{W}'_i = W_i + \xi^C_i r(z_t)\, x^0_i + \sum_{a,T} \xi^a_i x^a_{i,T} \mathcal{R}^a_{T,t+1} + \ell_i \sigma_r \varepsilon_{t+1}. \end{aligned} \tag{7}$$

[5] Let R denote the gross holding return above minus one. Precisely, $\mathcal{R} = \mu + u$ approximates R, with $\mu^a_T = \log \mathbb{E}_t[1 + R^a_{T,t+1}]$ and $u^a_{T,t+1} = \log(1 + R^a_{T,t+1}) - \mathbb{E}_t[\log(1 + R^a_{T,t+1})]$. Thus μ includes the Jensen adjustment, and u is the centered log return. Conditional means and variances scale with the time step, as in equations (3)–(8) of Greenwood et al. (2024). Cash uses r in place of $e^r - 1$. Appendix B.1 gives the exact wealth equation and the simple-return comparison.

Next-period wealth adds after-tax cash income, after-tax bond returns and the change in outside wealth to current funding. Investors choose portfolios anew each period, taking price functions as given and conditioning on today's rate state. Nonnegative positions rule out short sales and borrowing (Section 3.5).

Required returns. Moving a dollar from cash into a bond earns its after-tax return over cash and changes portfolio risk. The Kuhn–Tucker conditions are

$$\underbrace{\xi_i^a \mu_T^a - \xi_i^C r}_{\text{after-tax return over cash}} \quad \leq \quad \underbrace{\gamma_i \xi_i^a \operatorname{Cov}_z\big(u_{T,t+1}^a, \widehat{W_i'}\big)}_{\text{cost of the risk the bond adds}} + \underbrace{\beta_i}_{\text{value of funding}}, \tag{8}$$

with equality whenever $x_{i,T}^a > 0$. The covariance measures the risk the bond adds to total wealth, conditional on today's state. The funding multiplier $\beta_i \geq 0$ is zero when the investor holds cash and can invest another dollar without borrowing. A held bond pays for both its added risk and any scarcity of funding. If it offers less, the investor holds none. Differences in taxes and balance sheets therefore allow the same bond to suit one investor but not another.

Equilibrium. Prices clear every market at every rate state:

$$\textstyle\sum_i m_i x_{i,T}^a(z) = Q_T^a, \tag{9}$$

where m_i is the mass of type i, converting its positions and funding into aggregate holdings $m_i x_{i,T}^a$ and resources $m_i W_i$. The rate process, supplies, taxes, funding, risk aversion and outside exposures are inputs. Equilibrium prices imply holding returns for which investors satisfy (8) and all markets clear. Who holds each bond is determined by these conditions.

4.2 Taxes and the cost of bearing rate risk

To isolate the mechanism, consider a high-tax household H and a lower-tax investor I choosing municipal bonds M, taxable bonds C and cash. Municipal maturities begin at $T = 1$, while risky taxable bonds begin at $T = 2$. The two determinants of required returns are the investor's after-tax cash alternative and the cost of adding risk to its portfolio.

The relative after-tax share of Section 3.3 summarizes the tax comparison:

$$\xi_i = \frac{\xi_i^C}{\xi_i^M} = \frac{1 - \tau_i^C}{1 - \tau_i^M}, \qquad \xi_H < \xi_I,$$

The household's smaller ξ_H means a larger exemption benefit, $1 - \xi_H$. I measure each investor's payoffs in units of its after-tax municipal income. Municipal returns are then kept in full, and taxable returns and cash income are multiplied by ξ_i. This change of units preserves portfolio choices.[6]

[6] Divide payoffs by ξ_i^M, and replace risk aversion by $\gamma_i \xi_i^M$ and outside exposure by ℓ_i / ξ_i^M. The objective becomes $1/\xi_i^M$ times the original. Below, γ_i and ℓ_i denote these normalized values.

The analytical benchmark retains bond-specific risk and has slack funding, $\beta_i = 0$, and a floorless short rate $r = \bar{r} + z$. Prices are solved with the same investors holding each bond across rate states, although their positions adjust. This gives log prices linear in z, the affine solution. The resulting holdings are checked for positive positions, slack funding and no desired purchases of unheld bonds at the positive-rate states studied.

For risk, distinguish the exposure of a bond from the exposure of its holder's whole portfolio. A bond's *rate loading*, $A^a_T = -\partial \log P^a_T / \partial z$, measures its proportional price decline when the rate state rises. It is determined by equilibrium prices, as Section 4.5 shows. A bond bought with T periods left becomes a $(T-1)$-period bond next period, so its unexpected holding return uses the shorter bond's loading:

$$u^a_{T,t+1} = -A^a_{T-1}\sigma_r \varepsilon_{t+1} + \sqrt{v^a_T}\,\eta^a_{T,t+1}, \qquad A^a_0 = 0. \tag{10}$$

A one-period bond pays its fixed face value next period and carries no rate risk. Summing the after-tax exposures of all positions gives the investor's *net rate exposure*, $\widehat{D}_i$:

$$\widehat{D}_i = \sum_T x^M_{i,T} A^M_{T-1} + \xi_i \sum_T x^C_{i,T} A^C_{T-1} - \ell_i, \tag{11}$$

The rate-risk term in wealth is $-\widehat{D}_i \sigma_r \varepsilon_{t+1}$. Thus A measures one bond's price exposure, while $\widehat{D}_i$ measures how much the investor's wealth falls when rates rise. Buying a risky bond increases $\widehat{D}_i$.

The investor's cost per unit of additional rate exposure is $\gamma_i \sigma_r^2 \widehat{D}_i$. A larger liability hedge ℓ_i reduces $\widehat{D}_i$ at given bond holdings, lowering the return the investor requires per unit of rate exposure. Multiplying by the bond's loading gives the rate-risk compensation in the portfolio conditions:

$$\begin{aligned} \mu^M_T - \xi_i r &\le \gamma_i \sigma_r^2 \widehat{D}_i\, A^M_{T-1} + \gamma_i v^M_T x^M_{i,T}, \\ \mu^C_T - r &\le \gamma_i \sigma_r^2 \widehat{D}_i\, A^C_{T-1} + \gamma_i \xi_i v^C_T x^C_{i,T}, \end{aligned} \tag{12}$$

Equality holds for a positive position. The municipal condition compares μ^M_T with after-tax cash, $\xi_i r$. Dividing the taxable condition by ξ_i puts it on a pretax scale, so it compares μ^C_T with r. In both lines the right side is rate-risk compensation plus bond-specific compensation, which grows with the position.

These conditions separate the two forces behind participation. The household's larger exemption benefit lowers its required municipal return. A lower unit cost of rate risk also lowers the required return, increasingly so for a bond with a larger loading. The same $\widehat{D}_i$ enters every maturity and market. A purchase of one risky bond therefore changes the compensation the investor requires on its other risky bonds, the link behind portfolio substitution.

4.3 Participation at longer maturities

At the riskless maturity, only the cash alternative matters. With $A^M_0 = v^M_1 = 0$, a municipal holder requires $\mu^M_1 = \xi_i r$. The household accepts the lowest return because it keeps the smallest share

of taxable interest. With a positive short rate and slack household funding, it holds the municipal supply and

$$y_1^M = \mu_1^M = \xi_H r, \qquad y_1^C = r. \tag{13}$$

At the riskless short end the municipal yield equals the household benchmark exactly.

At a risky maturity, subtract the two investors' municipal conditions in (12). The common return μ_T^M cancels, leaving a comparison between their exemption benefits and their costs of bearing the same bond's rate exposure. The lower-tax investor participates whenever[7]

$$\underbrace{\sigma_r^2 A_{T-1}^M (\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I)}_{\text{duration benefit}} > \underbrace{(\xi_I - \xi_H) r}_{\text{difference in exemption benefits}}. \tag{14}$$

All quantities are evaluated at equilibrium. A positive $\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$ means that the lower-tax investor requires less return per unit of rate exposure. When the duration benefit exceeds the difference in exemption benefits, it is willing to accept a lower municipal return than the household, so the household cannot hold the supply alone.

The loading A_{T-1}^M is endogenous and increases with maturity under the affine benchmark's ordering conditions (Appendix A.6). Section 4.5 derives it from equilibrium prices. At a given equilibrium, the difference in unit risk costs is common across maturities. The duration benefit therefore grows with maturity, while $(\xi_I - \xi_H) r$ stays fixed. Lower-tax participation can begin at the long end. Appendix Theorem 1 establishes the conditions for a maturity threshold T^*, retaining bond-specific risk.

Figure 4 shows the resulting allocation in the illustration. The household alone holds shorter municipal bonds, both investors hold longer municipal bonds, and the lower-tax investor holds risky taxable bonds. Bond-specific risk allows the investors to share a range of municipal maturities. This allocation underlies the return comparisons below.

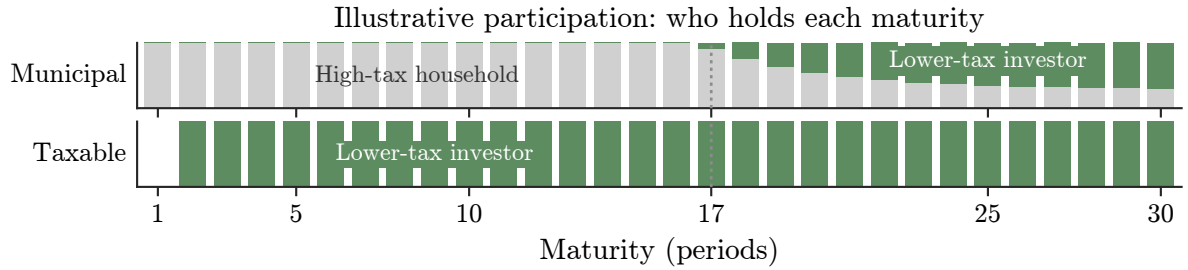


Figure 4: Participation in the two-investor illustration. The bars show each investor's share of the supply at the mean rate state. The lower-tax investor holds municipal bonds from $T^* = 17$ (dotted line). The one-period taxable claim is cash. Appendix B.10 gives the illustrative parameters. Section 5 reports the calibrated allocation.

[7] With $v_T^M > 0$, subtraction and market clearing give the exact criterion $x_{I,T}^M > 0 \iff \sigma_r^2 A_{T-1}^M (\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I) + \gamma_H v_T^M Q_T^M / m_H > (\xi_I - \xi_H) r$. The additional term is the household's bond-specific risk cost if it holds the supply alone. It makes participation easier. Appendix Theorem 1 proves the equivalence.

4.4 Three channels in required returns

Given this allocation, compare municipal and taxable bonds of the same maturity. The municipal return above the household's after-tax taxable return is

$$g_T = \mu_T^M - \xi_H \mu_T^C, \tag{15}$$

The representative-tax benchmark sets $g_T = 0$ at every maturity. The portfolio conditions explain why this equality fails when holders have different taxes and risk costs.

The comparison depends on who holds the bonds. At long maturities, the lower-tax investor holds both municipal and taxable bonds. Below the participation threshold, the household alone holds municipal bonds, while the lower-tax investor holds taxable bonds. These two comparisons reveal the first two channels. A third concerns the bonds' different exposures to rate risk.

After-tax cash return. The lower-tax investor keeps more taxable interest than the household. At jointly held long maturities, it therefore accepts a smaller reduction in return in exchange for the municipal exemption. Relative to the household benchmark, the difference in its cash alternative contributes $(\xi_I - \xi_H)r$ to the municipal return gap. Comparing this investor's conditions for the two bonds holds its unit cost of rate risk fixed. The household also holds the municipal bond because its greater risk cost offsets its larger exemption benefit at the equilibrium allocation.

Duration cost. At shorter risky maturities, the two bonds have different holders. The household requires more compensation per unit of rate exposure than the lower-tax investor. Even with the same household-scaled exposure, the municipal bond therefore pays more than the household benchmark. This contribution is $\sigma_r^2 A_{T-1}^M(\gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I)$. Further household purchases increase its exposure and required compensation, so the household cannot eliminate the gap through unlimited purchases.

Rate loading. The household benchmark scales taxable rate exposure by ξ_H, just as it scales taxable returns. Municipal bonds carry more exposure than this benchmark when $A_{T-1}^M > \xi_H A_{T-1}^C$. Even at the same unit risk cost, their holders then require more compensation. This third contribution is the extra exposure times the lower-tax investor's unit risk cost, $\gamma_I \sigma_r^2 \widehat{D}_I$. It is positive when that investor's net exposure is positive, as in the illustration. Section 4.5 derives the extra loading and explains how portfolio substitution carries it to shorter municipal bonds.

The three contributions follow from subtracting ξ_H times the taxable holder's condition from the municipal holder's condition in (12). To write both maturity ranges in one equation, use $i = H$ below T^* and $i = I$ from T^* onward:

$$\begin{aligned} g_T = \; & \underbrace{(\xi_i - \xi_H)r}_{\text{after-tax cash return}} + \underbrace{\sigma_r^2 A_{T-1}^M(\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I)}_{\text{duration cost}} \\ & + \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I(A_{T-1}^M - \xi_H A_{T-1}^C)}_{\text{rate loading}} + \underbrace{\gamma_i v_T^M x_{i,T}^M - \xi_H \gamma_I \xi_I v_T^C x_{I,T}^C}_{\text{bond-specific compensation}}. \end{aligned} \tag{16}$$

The cash-return term is zero below the threshold, and the duration-cost term is zero from the threshold onward. The loading term applies to both ranges. The final term accounts for independent bond-specific risk and is small in the illustration. Appendix Proposition 2 gives the full derivation.

All three channels vanish at the riskless maturity, where $g_1 = 0$. The loading term also vanishes at $T = 2$, since $A_1^M = \xi_H$ and $A_1^C = 1$, and contributes from $T = 3$ onward. Figure 5 shows how the channels combine across maturities.

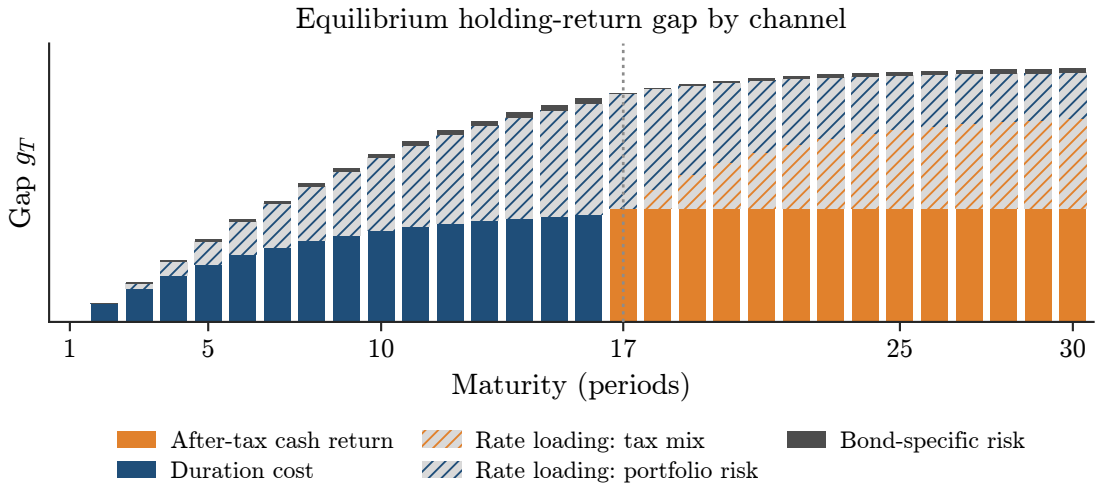


Figure 5: Required-return gap by channel. The allocation is that of Figure 4. The bars decompose $g_T = \mu_T^M - \xi_H \mu_T^C$ using (16). The rate-loading channel is split into its tax-mix and portfolio-risk contributions, with the small bond-specific term on top. The vertical scale is omitted because levels depend on the illustrative parameters.

4.5 From required returns to prices and yields

The portfolio conditions determine required returns given each bond's exposure. Prices must make those exposures consistent with the same returns. Taking expectations of the gross holding return gives the link:

$$P_T^a(z) = e^{-\mu_T^a(z) + v_T^a/2}\, \mathbb{E}_z\big[P_{T-1}^a(z')\big], \qquad P_0^a = 1. \tag{17}$$

A higher required return lowers today's price for a given expected resale price. Starting from the maturity payment, this relation determines prices backward through maturities. The factor $e^{v_T^a/2}$ accounts for the mean of the bond-specific shock (Appendix B.2), whose yield effect is small (Appendix C.8).

In the affine benchmark, write the price and yield as

$$\log P_T^a(z) = -A_T^a z - C_T^a, \qquad y_T^a(z) = \frac{A_T^a z + C_T^a}{T}. \tag{18}$$

The loading A_T^a introduced above determines yield sensitivity, A_T^a/T. The constant C_T^a determines the yield at the mean state, C_T^a/T. To find these coefficients, let $s_T^a = \partial \mu_T^a / \partial z$ denote the response

of the expected holding return to the rate state. Substitution into (17) gives

$$A_T^a = \rho A_{T-1}^a + s_T^a, \qquad C_T^a = C_{T-1}^a + \mu_T^a(0) - \tfrac{1}{2}\left[\sigma_r^2 (A_{T-1}^a)^2 + v_T^a\right], \tag{19}$$

starting from $A_0^a = C_0^a = 0$. The first relation adds today's return response to the loading inherited through the expected resale price. The second accumulates mean-state returns, less the variance correction known as convexity. Thus μ determines yield levels through C, while its rate response s determines yield sensitivity through A.

Why municipal loadings exceed the household benchmark. When rates rise, the difference in exemption benefits, $(\xi_I - \xi_H)r$, grows. The household's incentive to hold jointly held municipal bonds strengthens, and portfolio adjustment shifts net rate exposure toward it. Its unit cost of rate risk rises. Because that cost enters every position's condition, the household requires more compensation even on shorter municipal bonds it holds alone. The lower-tax investor sheds exposure and requires less risk compensation on taxable bonds. Trading shared long bonds therefore affects returns elsewhere in both portfolios.

At a jointly held maturity, the investors' offsetting trades cancel when their portfolio conditions are weighted by risk tolerance. The return response reflects the tax mix $\xi^* = \omega_H \xi_H + \omega_I \xi_I$, where ω_i is each type's share of total risk tolerance.[8] Bond-specific risk remains in these conditions. Its rate derivative is zero for a sole holder, whose position is fixed by supply, and cancels across joint municipal holders by market clearing. Differentiating (12) therefore gives

$$\begin{aligned} s_T^M &= \begin{cases} \xi_H + \gamma_H \sigma_r^2 A_{T-1}^M \, \partial_z \widehat{D}_H, & \text{household alone,} \\ \xi^*, & \text{both investors,} \end{cases} \\ s_T^C &= 1 + \gamma_I \sigma_r^2 A_{T-1}^C \, \partial_z \widehat{D}_I. \end{aligned} \tag{20}$$

Here $\partial_z \widehat{D}_i$ is the investor's exposure change as rates rise. On this allocation, $\partial_z \widehat{D}_H > 0 > \partial_z \widehat{D}_I$ (Appendix Proposition 10). Household-only municipal returns therefore respond by more than ξ_H. Jointly held municipal returns respond by $\xi^* > \xi_H$, and taxable returns by less than one. The recursion carries these responses into $A_T^M > \xi_H A_T^C$ for $T \geq 2$, establishing the extra exposure behind the third gap channel.

Figure 6 separates the two parts of the rate-loading channel. The tax mix raises the municipal return response at jointly held maturities. Portfolio adjustment raises it at shorter risky maturities and lowers the taxable response. Panel (a) shows these responses. Panel (b) accumulates the two municipal contributions into price sensitivity relative to the treasury reference, whose holding return responds one-for-one to the rate state.

From holding returns to yield levels. A long bond becomes a shorter bond each period. Iterating (19) makes its yield the average expected holding return along this path, less convexity

[8] The weights are $\omega_i = (m_i/\gamma_i)/(m_H/\gamma_H + m_I/\gamma_I)$ and sum to one.

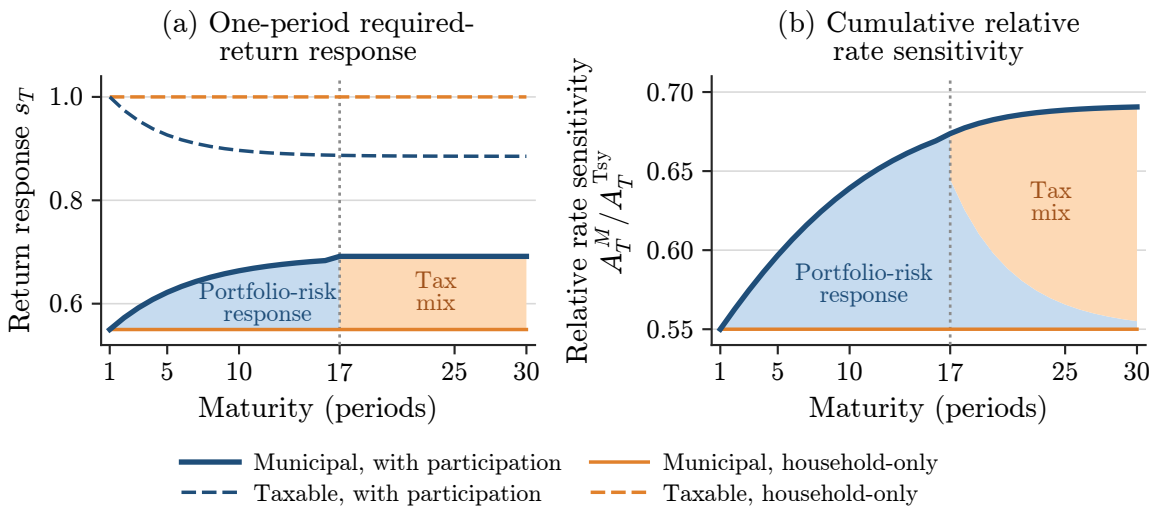


Figure 6: Rate responses in the illustration of Figure 4. Blue lines show equilibrium with lower-tax municipal participation from $T^* = 17$ (dotted line). Orange lines show the household-only benchmark. Panel (a): municipal (solid) and taxable (dashed) holding-return responses. Panel (b): municipal price sensitivity relative to the treasury reference used in P3, with $A_T^{\text{Tsy}} = (1-\rho^T)/(1-\rho)$. Blue and orange shading separate the portfolio-risk and tax-mix contributions to the municipal response and loading.

(Appendix equation (40)). Hence

$$y_T^M - \xi_H y_T^C = \frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t[g_{T-j}(z_{t+j})] - \text{convexity difference.} \tag{21}$$

The yield gap accumulates the three return-gap channels over the bond's remaining life. A positive gap means that the implied tax lies below the household's exemption benefit, by (3). Appendix equation (55) gives the variance correction, and Proposition 4 gives the conditions for implied taxes to decline with maturity.

A tax change also reallocates portfolio exposure. Through the same shared risk costs, an investor's tax benefit can affect yields on bonds it does not hold. Section 4.6 uses this implication. In the quantitative model, participation, funding constraints and the rate floor can vary across states. Expected resale values incorporate those future changes, so prices need not be affine. The same return-to-price relation (17) still applies.

4.6 Empirical predictions

The empirical tests distinguish the household's exemption benefit from benefits received by lower-tax institutions, then test the predicted change in relative rate sensitivity. The full quantitative model supplies the numerical comparisons in Section 6.

For tax changes, the outcome is the yield-implied tax $\tau_T^{Y,\text{impl}} = 1 - y_T^M/y_T^C$ from (2). Investor i's pass-through measures how much this implied tax changes when its exemption benefit changes:

$$\mathcal{P}_i^Y(T) = \frac{d\tau_T^{Y,\text{impl}}}{d(1-\xi_i)}, \qquad i \in \{H, I\}, \tag{22}$$

The derivative allows holdings and both yield curves to adjust. One means full transmission and zero means no effect. At the riskless short end, the household alone holds the municipal bond and $\tau_1^{Y,\text{impl}} = 1 - \xi_H$. Therefore

$$\mathcal{P}_H^Y(1) = 1, \qquad \mathcal{P}_I^Y(1) = 0. \tag{23}$$

At long maturities, both investors' benefits matter through the tax mix ξ^*. Portfolio adjustment further reduces the household's influence and reinforces the lower-tax investor's influence. Under a moderate tax difference and a mean rate high enough for interest income to dominate the tax responses of risk compensation and convexity, Appendix Proposition 6 gives

$$\mathcal{P}_H^Y(\infty) < \omega_H < 1, \qquad \mathcal{P}_I^Y(\infty) > \omega_I > 0. \tag{24}$$

The weights ω_i measure each investor's contribution to the tax mix. A larger household benefit widens the difference in exemption benefits and raises relative municipal exposure, offsetting its own effect on the implied tax. A larger lower-tax-investor benefit does the reverse. Appendix B.8 derives the full yield-ratio derivatives and the conditions above. In particular, $\xi_I < \frac{3}{2}\xi_H$ suffices for the tax-difference condition.

Longer bonds spend more of their remaining life in the jointly held maturity range, giving greater weight to the lower-tax investor's tax treatment (Appendix Proposition 13). In the solved illustration, household pass-through falls and institutional pass-through rises with maturity after portfolio adjustment as well. The endpoint results and this maturity pattern motivate P1 and P2.

Household taxes (P1). A home-state exemption gives resident households an additional benefit on their state's bonds. That benefit is fully reflected in implied state taxes at the riskless short end and less strongly at longer maturities in the solved model. The representative-tax benchmark predicts full pass-through at every maturity.

Institutional taxes (P2). A benefit specific to a lower-tax institution has no effect at the riskless short end and matters more at longer maturities. It affects returns where the institution holds bonds and spreads to other risky maturities through portfolio substitution. Bank qualification and the 2017 corporate tax change for insurers test this prediction. For banks, the measure is the share of the bank-only benefit reflected in yields (Section 6.2). The representative-tax benchmark gives these benefits no role.

Interest-rate responses (P3). Compare municipal and treasury yield changes at the same maturity. The treasury reference has a one-period return that moves one-for-one with the rate state and price loading A_T^{Tsy}. Since yield sensitivity is A_T/T, the relative response is A_T^M/A_T^{Tsy}. The

model gives

$$\frac{A_1^M}{A_1^{\text{Tsy}}} = \xi_H, \qquad \lim_{T\to\infty} \frac{A_T^M}{A_T^{\text{Tsy}}} = \xi^* > \xi_H.$$

Municipal sensitivity rises relative to treasury sensitivity as maturity increases. The test concerns this ratio, so the common increase in both bonds' exposure with maturity does not explain it. The representative-tax benchmark predicts a constant ratio. The treasury reference is distinct from the model's taxable market C, whose return response also reflects portfolio risk.

4.7 Liability hedging and other motives

Liability hedging is one source of the participation advantage. A lower-tax investor may instead trade less often and at lower cost than households, or diversify default risk better than a household concentrated in its home state's bonds. An advantage that grows with maturity can outweigh the difference in exemption benefits and induce participation in long municipal bonds.

The pricing results carry over when the advantage applies equally to taxable and tax-exempt bonds and creates no gap when the household holds both. Under this neutrality condition, participation brings the lower-tax investor's smaller exemption benefit into the yield comparison. Appendix A.7 states the result. Appendix B.13.4 solves trading-cost and default-diversification examples, the latter net of default compensation. A preference for municipal bonds as such can instead lower municipal yields below the household benchmark, so it cannot explain the puzzle.

The quantitative application uses liability hedging because rate risk is common to both markets and observed balance sheets and maturity holdings discipline who bears it. Section 5 solves the general economy of Section 4.1 with many investor groups, home-state exemptions, binding funding constraints and nonlinear prices.

5 Quantitative Results

5.1 The quantitative model

The quantitative model asks how much of the municipal puzzle measured taxes, balance sheets and ownership can explain. It solves the general model of Section 4.1 with municipal markets across the 50 U.S. states and the District of Columbia, a taxable market and six investor groups. Tax rules, quantities and household balance sheets are measured directly. Seventeen common coefficients governing risk aversion, bond-specific risk, outside exposures and within-group resource shares are estimated from twenty-two moment conditions on ownership, maturities, portfolios and the taxable yield curve. Municipal yields are untargeted.

States with a personal income tax have separate municipal markets. The seven states with no state income tax are combined into one market because they are symmetric in the model. Residents value their home-state exemption, while institutions do not receive this state-level benefit under the tax code. To reduce the computational burden, the resulting 45 municipal markets have maturities from two to thirty years in two-year steps. The taxable market pools treasury and corporate

Table 3: Investor groups in the quantitative model, 2015–2017

Group	Types	Types differ by	Exemption benefit $1-\xi_i$	Resources m_i
High-tax households	$7 \times 45 = 315$	Rate exposure, home state	0.43–0.50	6.97
Lower-tax households	$7 \times 45 = 315$	Rate exposure, home state	0.21–0.27	8.09
P&C insurers	2	Rate exposure	0.31	1.11
Life insurers	3	Rate exposure	0.21	3.05
Banks	1	None	0.35	4.83
Residual group	7	Rate exposure	0	26.47
Total	643			50.52

Notes. Each household group has seven types, differing in rate exposure, in each of the 45 markets of the computation, and also seven types in the one-market economy used to estimate the coefficients (Section 5.2). Household exposures are set by each type's measured outside assets and debt; institutional exposures are estimated coefficients (Appendix Table 10). The exemption benefit $1-\xi_i$, with $\xi_i = (1-\tau_i^C)/(1-\tau_i^M)$, follows from tax law. Households multiply the federal and home-state after-tax shares of each SCF tax unit, with the range running across states, and institutions face the 35% corporate rate, with proration for insurers. Banks also pay a financing-tax cost of 0.11% a year on municipal holdings, which lowers their net benefit (Table 2). Resources m_i, in trillions of dollars, are all of a group's debt securities and cash in the Financial Accounts. Holdings are chosen per dollar of resources and clear as $\sum_i m_i x_i = Q$, so m_i scales the group's risk-bearing capacity.

supply. Cash and the two-year municipal claim are riskless. Prices in both markets are determined in equilibrium.

Investors solve (7) with market-specific after-tax shares, and prices satisfy the portfolio conditions (8) and clearing (9) in every market and rate state. Banks' measured financing-tax cost is subtracted from municipal returns according to their tax treatment (Appendix C.7). The model retains the rate floor in (5) and allows funding constraints and participation to change across rate states. I solve $P_T^a(z)$ numerically through the resale relation (17), including bond-specific risk and integrating over next-period states. Returns, portfolio risks, holdings and both yield curves are determined jointly. Prices can be nonlinear in the rate state because future participation, binding funding constraints and the rate floor affect resale values.

The measured inputs cover six three-year blocks, from 2000–2002 through 2015–2017, centered on Survey of Consumer Finances (SCF) waves. Financial Accounts stocks give sector resources and bond quantities. Mergent gives maturity profiles and municipal state shares. The SCF supplies household balance sheets and the income and filing status needed to apply statutory taxes. Historical short rates determine the rate process. Appendix C.7 details the construction, and Table 15 lists the sources.

Table 3 lists the groups. The setup is implemented per dollar of resources: m_i is a type's total resources, W_i the share available for bonds and cash, and $x_{i,T}^a$ its position per dollar. Total investment is $m_i W_i$ and aggregate bond holdings are $m_i x_{i,T}^a$. Households differ by federal tax bracket and outside assets and debt. Outside assets add rate risk and liabilities offset it. Their mapping into exposures is estimated. Insurers differ in exposures and resource shares, banks face municipal financing-tax costs, and the residual group is tax-neutral. Section 3.4 and Appendix E.1 distinguish gross exemption benefits from these financing costs.

5.2 Estimation and fit

I measure directly the inputs available in the data: investor tax treatment, sector resources, bond quantities and maturity profiles, household assets and debt, and the short-rate process. These inputs vary across the six blocks of 2000–2017. The remaining seventeen coefficients govern risk aversion, bond-specific risk and the mapping from balance sheets into outside rate exposures. I estimate one common set of coefficients across all six time blocks.

Observed ownership provides the main discipline on outside rate exposure, which balance-sheet amounts alone cannot measure. Given each group's tax treatment and resources, its municipal and taxable holdings reveal how much rate risk it is willing to bear. For households, three coefficients translate measured outside assets and debt into rate exposure. For institutions, the estimated exposures and within-sector resource shares account for both the amounts held and their maturity composition. The taxable yield curve and long-bond allocations discipline risk aversion and bond-specific risk. The coefficients are estimated jointly because portfolio choices and yields adjust together in equilibrium.

The estimation uses twenty-two targets. Nine are ownership stocks: municipal holdings of the two household groups, P&C insurers, life insurers and banks, and taxable holdings of households and the same three institutional sectors. Four are the mean maturities of insurers' municipal and taxable portfolios. Six describe the distribution of insurer holdings. Two measure whether municipal holdings are spread across insurers or concentrated in portfolios with high municipal shares, giving larger portfolios more weight. Four measure the fractions of each sector's municipal and taxable portfolios maturing beyond twenty years. The remaining three are taxable yields at ten, twenty and thirty years. I evaluate these conditions at observed rate states, average them across the six blocks and minimize their weighted squared deviations from the data.[9]

For computational tractability, I estimate the coefficients in an economy with one national municipal market, using municipal supply weights to average each household type's home-state after-tax shares. I then restore the 45 municipal markets and their measured tax differences, holding every coefficient fixed. No coefficient jointly calibrated to moments is state-specific. [10]

The one-market and full economies give similar relative municipal yields. Their thirty-year gaps average 102.6 and 102.4 basis points across the six blocks, differing by at most 2.5 basis points in any block. Thirty-year taxable yields differ by at most 0.5 basis points. Changing each coefficient group in turn moves the one-market gap by at most about 4 basis points, except for the common risk aversion.[11] Section 5.4 uses the 2000–2020 bond sample's weights and expresses the model's relative municipal yields on the observed corporate curve for the headline comparison.

[9] Each averaged deviation is scaled by 10% of its observed normalizer. Maturity and long-portfolio conditions include model-portfolio weights. Appendix equation (111) gives the criterion, and Tables 10 and 11 report the coefficients and target definitions.

[10] Fourteen of the twenty-two absolute scaled deviations are below one. Appendix C.6 reports the estimation procedure and fit.

[11] Changing risk aversion alone scales both curves and the spread, moving the taxable yield level away from its calibration targets.

Municipal yields, the municipal puzzle and the empirical tax-pass-through coefficients do not enter the estimation criterion. The municipal yield curve and cross-state tax pass-through are untargeted outcomes, evaluated against data in Sections 5.3–5.4 and 6.1. The calibration uses ownership to discipline outside rate exposure. Together with observed taxes, these exposures determine municipal yields in equilibrium, leaving the relative municipal yield curve as a test of the mechanism.[12]

5.3 Model and data yield curves

Figure 7 sets the calibrated model curve (top) against the data (bottom). To cover the full data sample, I keep the estimated coefficients fixed and add a seventh economy for 2018–2020 after the TCJA, built from that period's taxes, rates and quantities. Both rows show a municipal yield near the household benchmark at the short end and a spread over it that widens with maturity. In nominal semiannual yields, the model's implied tax $1 - y_T^M / y_T^C$ is 41.0% at two years, 34.7% at ten and 24.8% at thirty, a decline of 16.2 percentage points. The model's thirty-year municipal and taxable yields are 4.651% and 6.185%, against 4.555% and 5.794% in the adjusted data.

5.4 Accounting for the puzzle

The accounting asks how much of the municipal puzzle the model explains, alongside the measured capital-gains, liquidity and default differentials. To compare these contributions, I express them on the same observed taxable yield curve. Otherwise, a higher taxable yield in the model would mechanically enlarge its gap in basis points, even with the same implied tax.

At each maturity, I keep the model's municipal-to-taxable yield ratio and multiply it by the observed taxable yield. I then subtract the household after-tax benchmark. The resulting gap, denoted $\widetilde{\mathcal{G}}_T$, measures the model's contribution on the data's taxable curve:

$$\begin{aligned} \text{Normalized municipal yield} &= \frac{\text{Model municipal yield}}{\text{Model taxable yield}} \times \text{Observed taxable yield}, \\ \widetilde{\mathcal{G}}_T &= \text{Normalized municipal yield} - \text{Household benchmark}. \end{aligned} \tag{25}$$

I call this the corporate normalization. It preserves the model's implied tax and changes only the scale of this comparison. The solved equilibrium and the curves in Figure 7 are unchanged. Appendix D.1.7 gives the averaging and normalization details.

[12] Appendix C describes the solution method, and Appendix C.3 reports accuracy.

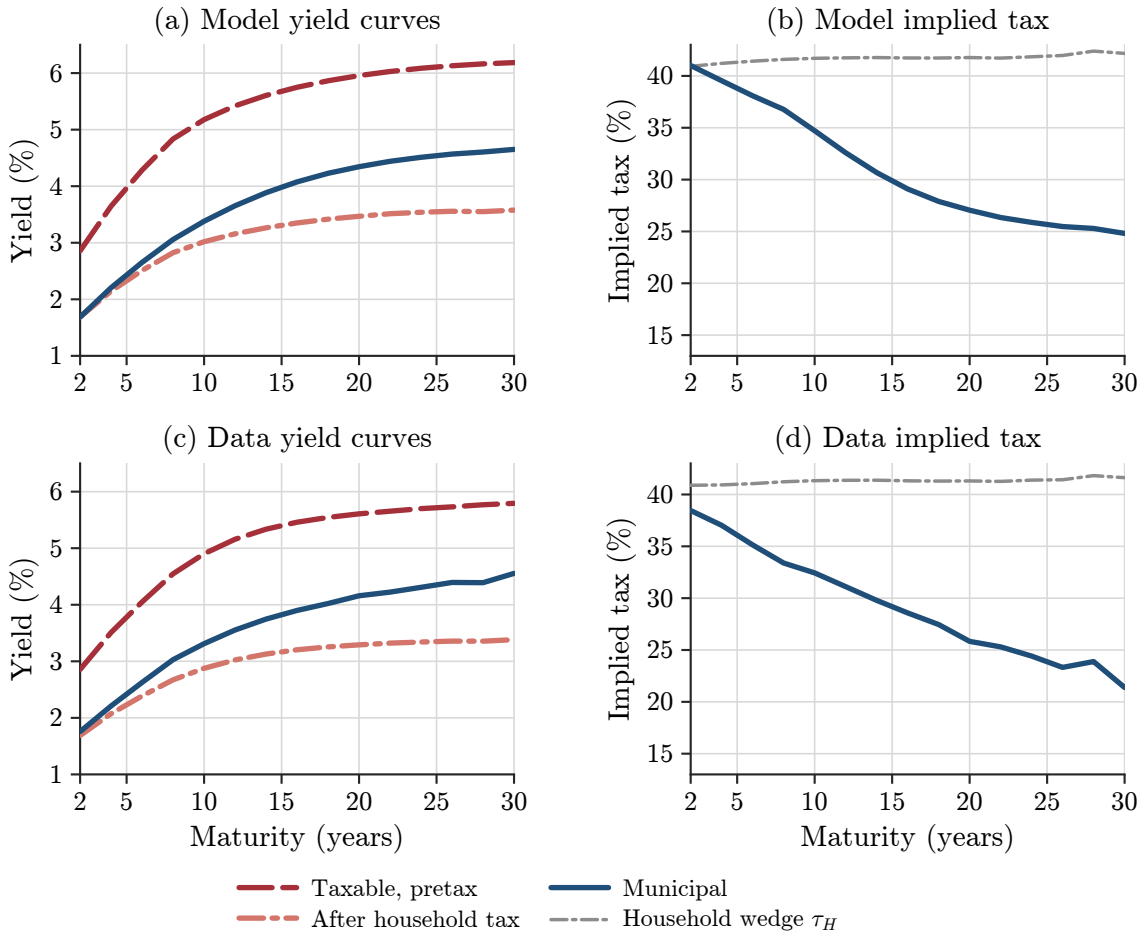


Figure 7: Model (top) and adjusted data (bottom), 2000–2020. Panels (a) and (c) plot the pretax taxable yield, the taxable yield after household tax (the household benchmark) and the municipal yield. Panels (b) and (d) plot the implied tax $1 - y_T^M / y_T^C$ and the household wedge. Corresponding panels share axis scales. The taxable curve is the model's pooled taxable market (top) and the observed corporate benchmark (bottom). The municipal data curve removes callability (Appendix D.1.2) and the issuance premium, and panel (d) uses this same adjusted curve. The model evaluates the estimated coefficients at observed issue-quarter rate states in the six calibration blocks and a 2018–2020 economy. Both rows use the empirical bond, rating and quarter weights and nominal semiannual yields. The household benchmark multiplies each observation's taxable yield by the high-tax household's relative after-tax share for that bond's state and date before averaging. Implied taxes and the household wedge are ratios of averaged yields.

The accounting has three steps at each maturity T:

$$\begin{aligned} \text{Spread}(T) - \underbrace{\big[\text{Call}(T) + \text{Prem}(T)\big]}_{\text{comparability adjustments}} &= \underbrace{\text{Puzzle}(T)}_{\text{municipal puzzle}}, \\ \text{Puzzle}(T) - \underbrace{\big[\text{CG}(T) + \text{Liq}(T) + \text{Def}(T)\big]}_{\text{bond-side adjustments}} &= \underbrace{\text{Puzzle}^{\text{adj}}(T)}_{\text{residual before the model}}, \\ \text{Puzzle}^{\text{adj}}(T) - \underbrace{\widetilde{\mathcal{G}}_T}_{\text{model}} &= \underbrace{\text{Residual}(T)}_{\text{unexplained}}. \end{aligned} \tag{26}$$

The spread, call adjustment and issuance premium follow Section 3.7. The terms CG, Liq and Def denote capital-gains, liquidity and default differentials. Since the model excludes these features, its gap is compared with Puzzle$^{\text{adj}}(T)$. The final residual equals the observed corporate yield times the implied-tax difference (model minus data), with the data reflecting all five adjustments.

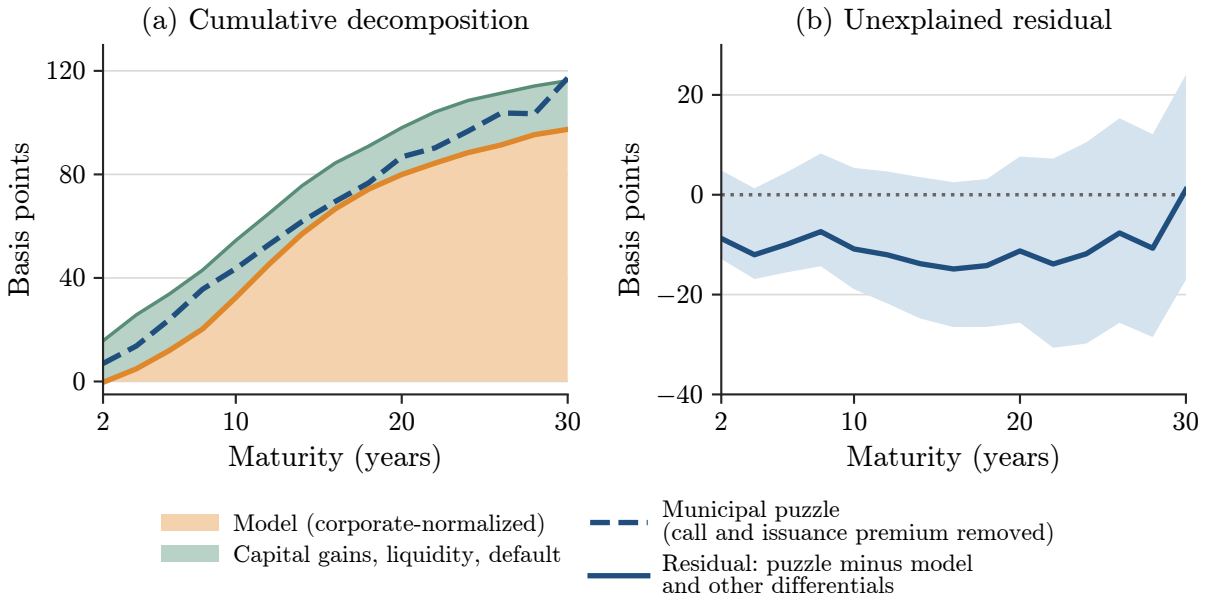


Figure 8: Cumulative decomposition of the municipal puzzle, 2000–2020, as in (26). The puzzle is the observed spread after removing callability and the issuance premium. Panel (a) stacks the model gap $\widetilde{\mathcal{G}}_T$ of (25) and the capital-gains, liquidity and default differentials against that puzzle. Panel (b) shows Residual(T), the puzzle less their sum. Shading spans the ranges of the five empirical adjustments (Appendix Table 27) and excludes model uncertainty. Weights and yield conventions match Figure 7.

Table 4: The model's share of the municipal puzzle. Removing callability and the issuance premium from the raw spread gives the puzzle. The normalized model gap applies the model's implied tax to observed corporate yields, as in equation (25). The unnormalized model gap is shown for comparison.

		comparability adjustments			model				
T	spread	call	premium	puzzle	unnormalized	normalized	model/puzzle	bond-side	unexplained
2	9.9	0.1	2.9	6.9	−0.2	−0.3		16.0	−8.8
6	27.9	0.7	3.3	23.9	14.3	12.0	50%	21.8	−9.9
10	52.2	4.6	4.0	43.6	36.2	32.4	74%	22.0	−10.9
14	87.0	18.0	7.2	61.8	62.1	57.1	92%	18.6	−13.8
18	110.3	24.0	9.6	76.7	81.1	74.3	97%	16.6	−14.2
22	128.8	27.2	11.4	90.2	92.7	84.3	93%	19.8	−13.9
26	145.0	28.7	12.5	103.7	101.1	91.4	88%	20.0	−7.7
30	159.8	29.8	12.8	117.2	107.4	97.4	83%	18.8	1.0

Notes. Basis points, except shares. The accounting is puzzle = normalized model + bond-side + unexplained, as in (26), where the bond-side adjustments sum the capital-gains, liquidity and default differentials. The model share divides the normalized model by the puzzle, using unrounded values, and is omitted where the puzzle is below 20 bp. Model and data both use the 2000–2020 sample, the averaging of Appendix D.1.1 and the nominal semiannual convention. The separately measured components are not jointly estimated. Rows show every second model maturity. Figure 8 shows all fifteen.

At thirty years (Figure 8 and Table 4), the model accounts for 97.4 basis points of the 117.2-basis-point puzzle, or 83%, and the bond-side adjustments account for 18.8, or 16%, leaving 1.0 basis points unexplained. The model's contribution grows with maturity in basis points. Its share of the puzzle is 74% at ten years, 97% at eighteen and 83% at thirty. The unexplained residual is negative at every maturity except thirty years. The model gap exceeds $\text{Puzzle}^{\text{adj}}(T)$ by 11 to 15 basis points from twelve to twenty-four years, and the largest absolute residual is 14.9 basis points at sixteen years. This negative residual is consistent with the convenience premium interpretation in Fleckenstein and Longstaff (2025).

6 Empirical Tests

Table 5 summarizes three tests: household state-tax pass-through (P1), bank qualification and insurer exposure to the 2017 corporate tax cut (P2), and municipal responses to treasury-rate movements (P3). The estimates have the predicted maturity ordering. None is a calibration target or an input to the curve decomposition.

Table 5: Empirical predictions and sources of variation

	Source of variation	Model prediction	Representative-tax benchmark
P1	Residents' state income tax rates	Pass-through near one at the short end, declining with maturity	Pass-through of one at every maturity
P2	Bank qualification, 2017 insurer tax cut	Little effect at the short end, larger farther out	No effect at any maturity
P3	Treasury-rate changes	Long-end response above the coupon benchmark	Response equal to the coupon benchmark

The curve accounting in Sections 3 and 5 uses the A-or-above municipal sample (agency score seven or better), matching the corporate benchmark's credit band. The P1, TCJA and P3 tests use the narrower AA-and-above sample (agency score three or better) to limit credit heterogeneity that could affect both yields and trading patterns. Appendix D.13.2 reports each on the wider sample and shows that the point-estimate maturity ordering is not driven by the rating cutoff. The bank-qualified design uses all new issues with rating-by-year fixed effects. Below, m_b denotes remaining maturity in years, distinct from the investor masses m_i in the model.

6.1 P1. State-tax pass-through

P1 tests whether resident households' state-tax benefit is reflected less in municipal yields at longer maturities. Most states exempt residents' in-state municipal interest while taxing out-of-state interest. A household in state s thus keeps all in-state municipal interest and $1-\tau_s^{\text{st}}$ of out-of-state interest, with the same state tax applied to taxable bonds.

State rates move resident households' exemption benefits, a wedge institutions largely do not face. Pensions and tax-deferred accounts owe no current state interest tax. Insurer premium

taxes generally do not distinguish municipal from taxable interest. Funds pass through the state benefit according to shareholder residence and portfolio eligibility: single-state funds preserve the home-state exemption, while national funds pass through only the eligible home-state share (Appendix E.3).

Under the conditions in Section 4.6, Proposition 6 gives household pass-through of one at the riskless boundary and below one in the long-end limit. Appendix Proposition 14 gives the state-market responses. The representative-tax benchmark predicts one at every maturity.

The ratio y^M/y^C contains both tax layers. Under the pre-2018 federal deduction convention used for taxable interest, the taxable after-tax share is $\xi_s^{\text{st}}\xi_t^{\text{fed}}$. Here $\xi_s^{\text{st}} = 1 - \tau_s^{\text{st}}$ is the state after-tax share and ξ_t^{fed} the federal share. Home-state municipal interest is exempt at both levels. Under the representative-tax benchmark, household indifference gives $y_{s,t}^M = \xi_t^{\text{fed}}\xi_s^{\text{st}}\, y_t^C$. Dividing out the federal after-tax share defines

$$\mathit{ImpliedStateTax}_{s,t}(T) \;\equiv\; 1 - \frac{y_{s,t}^M(T)/y_t^C(T)}{\xi_t^{\text{fed}}}. \tag{27}$$

Under the representative-tax benchmark, this outcome equals the statutory state rate τ_s^{st}, giving unit pass-through. Here ξ_t^{fed} is one minus the dated top federal ordinary-income rate: 0.604 in 2000, 0.609 in 2001, 0.614 in 2002, 0.65 in 2003–2012 and 0.604 in 2013–2017, excluding the net investment income tax. The model uses its own high-tax-household federal after-tax share in each block. The sample ends before the Tax Cuts and Jobs Act capped state and local tax deductions at \$10,000 from 2018. Appendix D.7 reports the later period under its tax mapping.

I partition bonds into five maturity bins by years to maturity ($[0,5)$, $[5,10)$, $[10,15)$, $[15,20)$, $[20,30]$ years) and estimate

$$\begin{aligned}\mathit{ImpliedStateTax}_{b,t} = &\sum_{g=1}^{5} \beta_g\big(\mathit{StateTax}_{s(b),t} \times \mathbf{1}\{m_b \in g\}\big) + \sum_{g=1}^{5} \delta_g'\big(Z_{b,t} \times \mathbf{1}\{m_b \in g\}\big) \\ &+ \lambda' V_b + \alpha_{t\times m\times \mathit{callable}} + \varepsilon_{b,t},\end{aligned} \tag{28}$$

where $\mathit{StateTax}_{s,t}$ is the top statutory rate and (27) uses call-adjusted zeros. The coefficient β_g measures how much the implied state tax changes per percentage point of the statutory rate in maturity bin g. The representative-tax benchmark gives $\beta_g = 1$. Controls $Z_{b,t}$ are the rounded average rating and six lagged state variables: par-weighted outstanding maturity, preceding-year issuance maturity, outstanding par relative to personal income, log income, log preceding-year issuance and the outstanding share in the bond's maturity bin. The stock and issuance variables cover federally exempt bonds, and the stock and income variables are measured at the preceding quarter-end. Bin-specific coefficients allow supply, market size and maturity structure to affect each part of the curve differently. V_b includes log offering par and security-type indicators, and $\alpha_{t\times m\times \mathit{callable}}$ denotes issue-year–month–maturity–callability fixed effects.

The sample covers 2000–2017 and contains 681,839 observations with complete controls. The main estimates use unweighted OLS, so every bond counts once. Appendix D.7 reports equal-state and equal-state-year weighting. Standard errors are clustered by state, allowing dependence among a state's bonds across years and maturities.

For an IV comparison, I instrument each bin interaction of the current statutory rate with its 1990 counterpart, following the policy-history approach of Babina et al. (2021) and using the historical top-rate series of the Institute on Taxation and Economic Policy. Appendix D.7 reports alternative vintages with the same controls and sample.

Figure 9 compares OLS estimates of (28) for 2000–2017 with the full-state model's cross-state slope. The model uses (27), its own high-tax-household federal share ξ^{fed} and nominal-semiannual yields. At each maturity, I take the slope on the state's top rate, with an intercept and equal weights across the 43 income-tax states and the District of Columbia. Within each economy, states are evaluated at a common rate state near the stationary mean. Slopes are averaged across the six pre-2018 blocks.

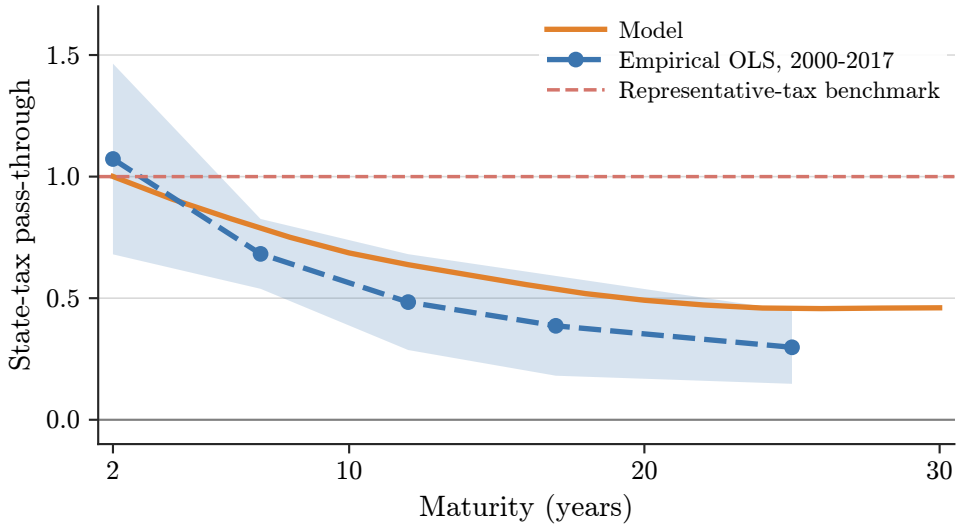


Figure 9: State-tax pass-through by maturity. Blue: controlled, unweighted OLS estimates for 2000–2017 call-adjusted zeros, with state-clustered 95% intervals using $t(46)$. Orange: the model's cross-state slope with its own federal after-tax share and nominal-semiannual yields, using a common rate state across states within each economy and averaging slopes over the six pre-2018 economies. Both use (27). The red dashed line is the representative-tax benchmark.

The OLS profile declines across all five bins, from 1.072 below five years to 0.298 beyond twenty (Appendix Table 30). A ten-percentage-point difference in a state's top rate corresponds to about 10.7 points in the implied tax at the short end and 3.0 points at the long end. The short-to-long decline is 0.774 (s.e. 0.234, $p = 0.002$). Unit pass-through cannot be rejected in the shortest bin and is strongly rejected in the longest. Every bin coefficient is significantly positive at one percent. The IV comparison also declines, from 1.092 to 0.227, with an endpoint-equality p-value below 0.001.

The untargeted model profile declines from 1 at two years to 0.69 at ten and 0.46 at thirty. Appendices D.6 and D.7 retain the main controls and pre-2018 sample while varying yield

construction, credit controls, weights and instruments. They also report state fixed effects, single-state omissions and a separate post-2018 comparison under the changed tax mapping.

6.2 P2. Bank qualification

Bank qualification tests the maturity profile of a bank-only tax benefit. Banks ordinarily cannot deduct interest expense allocated to tax-exempt holdings (Section 265(b) of the Internal Revenue Code). For bank-qualified (BQ) bonds, Sections 265(b)(3) and 291 restore 80% of that deduction. Issuers qualify if reasonably anticipated annual tax-exempt issuance does not exceed $10 million ($30 million in 2009–10 under the American Recovery and Reinvestment Act) and they designate the bonds. Designation changes neither payments nor other investors' taxes. Dagostino (2025) studies how the expanded cap affected issuance and employment.

A bank allocates interest expense across assets in proportion to tax basis, so a dollar of municipal basis carries the same annual funding cost c at every maturity. Designation saves $X = 0.8\,\tau_B c$ a year, where τ_B is the federal bank tax rate. Using Call Report interest expense, X is about 65 basis points in 2005–08 and 9 to 14 from 2011 to 2019 (Appendix D.8). I also scale premia by X_{tm}, the expected annual benefit over the life of a bond issued at t with maturity m (Appendix D.8). Its median is 45 basis points.

Three hypotheses differ in how much of X the designation premium $\varpi_k = y_k^N - y_k^Q$, the non-designated minus the designated yield at maturity k, reflects. Under the representative-tax benchmark, households require $(1-\tau_H)y_k^C$ on both bonds. When their exemption benefit exceeds banks' net benefit, banks prefer taxable bonds at these yields and $\varpi_k = 0$. Under a bank-only benchmark, indifference to taxable bonds gives full capitalization, $\varpi_k = X$, flat in maturity. In the model, the premium is the benefit times the bank's capitalization share θ_k, the average bank share of the total demand response to the bond's yield (Appendix D.8):

$$\varpi_k = \theta_k\, X, \qquad \underbrace{\theta_k = 0}_{\text{representative-tax}}, \quad \underbrace{\theta_k = 1}_{\text{bank-only}}, \qquad \underbrace{\theta_{\text{short}} \simeq 0 < \theta_{\text{long}}}_{\text{model}}. \tag{29}$$

The capitalization share is the bank counterpart of $\mathcal{P}_I^Y(T)$ in Section 4.6. Household after-tax cash returns anchor short yields and banks hold few short bonds, making θ_k near zero. In the calibrated model, banks hold intermediate maturities where their duration benefit offsets their smaller exemption benefit (Section 5.3). Yields there reflect more of the bank benefit. A rising premium in expected-benefit units distinguishes this mechanism from both benchmarks.

The design compares issuers on either side of the cap. The running variable R_{jt} is issuer j's annual tax-exempt issuance minus that year's cap, in millions of dollars. The indicator $E_{jt} = \mathbf{1}\{R_{jt} \le 0\}$ marks below-cap issuer-years, and B_b marks designation. Measured issuance differs from the statutory count, giving a fuzzy discontinuity: below-cap status instruments designation. The Mergent sample covers tax-exempt new issues from 2000 to early 2021 with $0.5 \le |R_{jt}| \le 4$. The half-million donut removes managed bunching just below the cap and issuer-years exactly at

it. With $\mathbf{1}_k$ indicating maturity at issuance (1–2, 2–5, 5–10, 10–20 or more than 20 years), the first and second stages are

$$\begin{aligned} B_b\mathbf{1}_k &= \sum_{\ell} \alpha_{k\ell}\, E_{jt}\mathbf{1}_\ell + V_b'\delta_k + e_{bk}, \qquad k = 1, \ldots, 5, \\ y_b &= \sum_{k} \beta_k\, B_b\mathbf{1}_k + V_b'\delta + u_b, \qquad B_b\mathbf{1}_k \text{ instrumented by } E_{jt}\mathbf{1}_k, \end{aligned} \tag{30}$$

where V_b contains local linear controls in R_{jt} on each side of the cap, log tranche par, and fixed effects for maturity, state, offering month, rating, security type, use of proceeds, coupon and callability, with the interactions listed in Appendix D.8. The outcome is the offering yield in basis points, or that yield divided by X_{tm}, so $-\beta_k$ is the premium in basis points or in expected-benefit units. Standard errors are clustered by issuer. Identification requires that, conditional on these controls, crossing the cap moves offering yields only through designation. Callability and the number of maturities per issue are balanced at the cap. The first-stage designation jump is 0.42 to 0.45 across bins, too similar to explain the much larger long-maturity yield premium. Placebo cutoffs show essentially no designation effects, and no systematic yield effects at \$5 million and \$15 million (Appendix D.8).

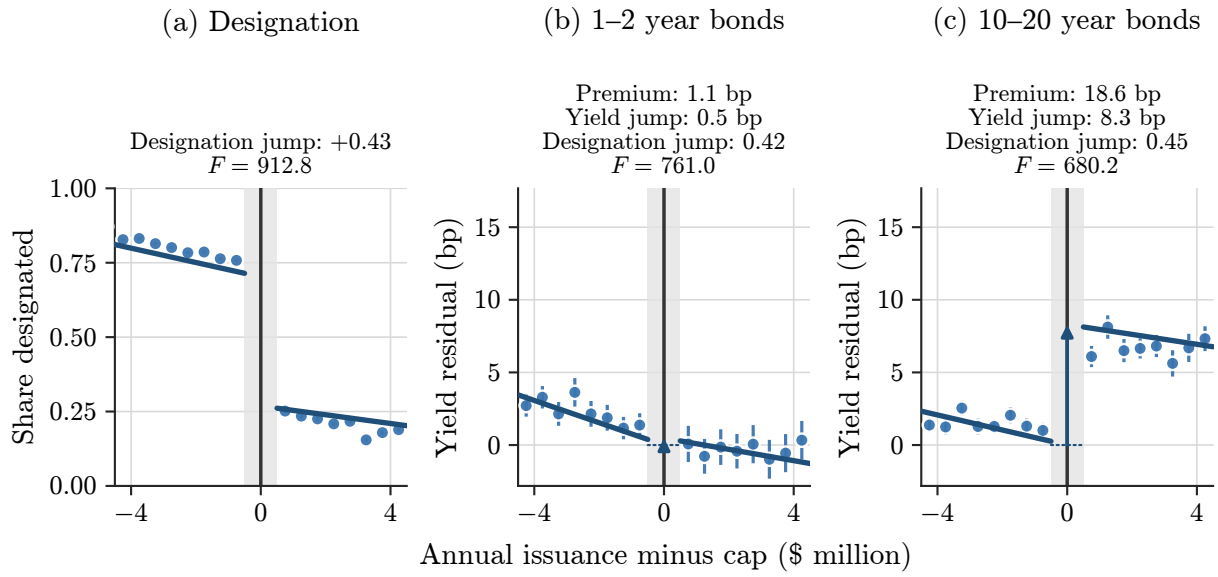


Figure 10: The bank-qualified discontinuity. Panel (a): designation around the cap. Panels (b)–(c): offering-yield discontinuities for 1–2 and 10–20 year bonds. Local linear fits exclude the shaded donut. Bars are 95% intervals. Jumps are differences between the local linear fits at the cap, below-cap minus above-cap for designation and above-cap minus below-cap for yields. First-stage F statistics test the below-cap coefficient with issuer-clustered standard errors. Dividing the bin-specific yield jumps by their first stages gives premia of 1.1 and 18.6 basis points. Appendix Figure 24 shows the other maturities.

Figure 10 shows unconditional designation rates of 0.79 below the cap and 0.21 above. The annotated first stages condition on the regression controls. The yield discontinuity is far larger for 10–20 year bonds than for 1–2 year bonds. Figure 11 reports the joint 2SLS estimates (Appendix

Table 36). The premium is 1.8 basis points at 1–2 years, indistinguishable from zero, 8.7 at 2–5 years, 13.8 at 5–10, 18.4 at 10–20 and 18.2 beyond twenty, and the 10–20 year premium exceeds the 1–2 year premium by 16.5 basis points (s.e. 2.7). In expected-benefit units, the premium rises from 0.17 at 1–2 years to 0.52 at 10–20 years: the bank-only benefit is barely priced in short municipal yields, and about half of it is priced at 10–20 years. The positive premia at 5–20 years reject the representative-tax benchmark, with t statistics of 6 to 8, and the rise in expected-benefit units, 0.34 (s.e. 0.11), rejects the flat premium of full capitalization. The gradient appears in every rate regime and survives alternative donuts and bandwidths, credit-enhancement controls, a general-obligation sample, issuer effects, year clustering and the binned estimator of Dagostino (2025) (Appendix D.8).

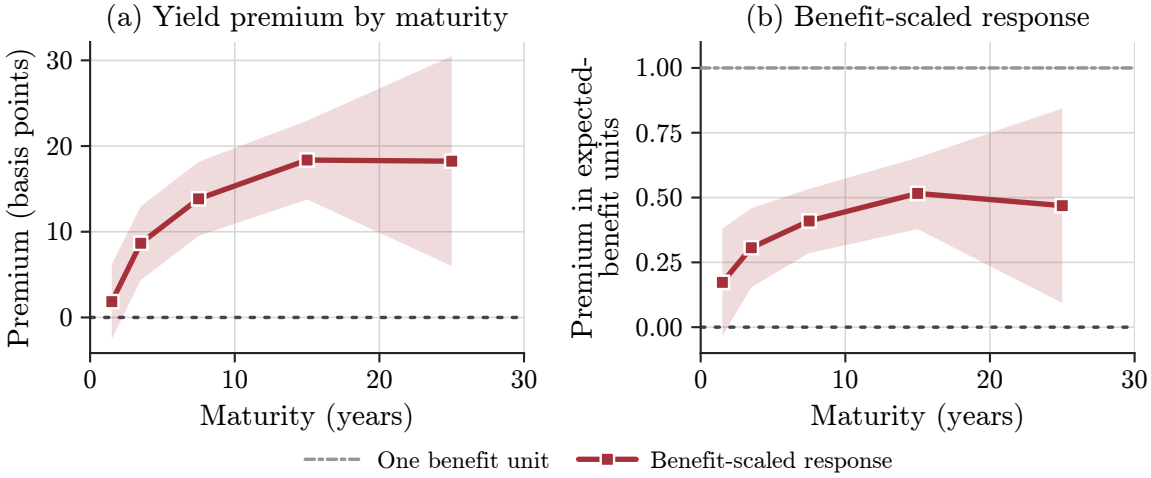


Figure 11: The bank-only benefit and municipal yields. Panel (a): 2SLS yield premium in basis points. Panel (b): the designation response when yield is divided by the expected annual benefit X_{tm}, with the same controls. Funding costs are municipal-holdings-weighted. Expected rates use the Kim–Wright expected-short-rate component, capped at ten years (Appendix D.8). Shading is the issuer-clustered 95% interval. The dashed line marks one benefit unit.

The 2018 cut in the bank tax rate, from 35% to 21%, lowered X by 40% at a given funding cost and lowered the value of the municipal exemption to banks. Relative to 2014–17, the premium fell most at 5–10 years, by 19.2 basis points (s.e. 7.6), while a substantial premium remained at 10–20 years and the first stage did not change (Appendix D.8). The participation margin moves when a clientele's tax treatment changes.

The rising bank profile contrasts with household pass-through, which falls from 1.07 to 0.30 in the controlled pre-2018 OLS estimates (Figure 9). Together, the designs show the opposite maturity gradients predicted for household and institutional benefits. The 2017 corporate tax cut provides a second institutional test: it reduced P&C insurers' exemption benefit, and implied taxes subsequently fell more at longer maturities in states where these insurers held more municipal debt. Appendix D.10 presents the design, estimates and event study.

6.3 P3. Rate responses by maturity

P3 tests whether municipal yields respond more to a given treasury yield change at longer maturities. Each municipal bond is compared with a treasury yield at the same maturity. Under the representative-tax benchmark, municipal zero yields are a constant fraction ξ_H of treasury zero yields, so their relative response is constant across maturities. In the model, long municipal bonds reflect their holders' smaller exemption benefits and have greater rate loadings relative to this benchmark. The response ratio therefore rises from ξ_H toward the holders' tax mix $\xi^* = \omega_H \xi_H + \omega_I \xi_I > \xi_H$ (Proposition 6(iii) and Appendix B.7). I test this maturity pattern.[13]

I compare consecutive trades in the same municipal bond. Let $\Delta y^{\mathrm{M}}_{b,t}$ be its yield change between the two trades, $\Delta y^{\mathrm{Tsy}}_{m_b,t}$ the corresponding change in the maturity-matched treasury yield, and m_b its remaining maturity in years. I estimate

$$\Delta y^{\mathrm{M}}_{b,t} = \beta_0 \, \Delta y^{\mathrm{Tsy}}_{m_b,t} + \beta_1 \big(\Delta y^{\mathrm{Tsy}}_{m_b,t} \times m_b \big) + \beta_2 \, m_b + \varepsilon_{b,t}. \tag{31}$$

The response to a treasury yield change is $\beta_0 + \beta_1 m_b$, so a positive β_1 means that this response is larger for longer bonds. The term $\beta_2 m_b$ allows municipal yield changes unrelated to treasury movements to vary with maturity. Comparing trades within a bond removes its time-invariant characteristics.

The sample covers January 2005 to August 2020 and includes non-callable, federally tax-exempt municipal bonds rated AA and above (agency score three or better). Trades are at most 90 days apart, and the treasury yield must move by at least 10 basis points in either direction to limit transaction noise. Inverse trade-pair weights give each bond about equal weight. Standard errors are clustered by the second trade's date. The treasury comparison follows the bond's remaining maturity at each trade. Appendix D.12 details the construction and also holds treasury maturity fixed across the pair to separate rate movements from the bond's aging.

The estimates give $\hat{\beta}_0 = 0.346$ and $\hat{\beta}_1 = 0.007$ (Appendix Table 42). A 100-basis-point treasury increase is associated with a municipal increase of about 35 basis points at one year and 42 at ten years. Grouping bonds below five years and at five years or more gives the same ordering, with responses of 0.351 and 0.426.

The model's prediction concerns zero-coupon yields, while the data report municipal yields to maturity and treasury par yields. Coupon payments span several horizons, so these reported yields can have a maturity-dependent response even under the representative-tax benchmark. I measure that response by pricing each bond's scheduled payments with municipal zero yields fixed at 0.65 times treasury zero yields, corresponding to a 35% federal tax rate. I recover its yield to maturity at both trade dates and apply the same regression, trade pairs and weights used for the observed yields (Appendix D.12.1).

[13] The observed response level, about 0.35 at one year, also reflects treasury safety and liquidity demand (Krishnamurthy and Vissing-Jorgensen, 2012) and transaction-outcome trimming. The empirical test therefore concerns how the relative response varies with maturity.

On the 918,350 pairs with regular payment schedules and available zero curves, the observed maturity interaction is 0.0080, compared with 0.0024 under this coupon benchmark (Appendix Table 44). The difference is 0.0056 per year of remaining maturity (s.e. 0.0021, $p = 0.007$). Coupon conventions account for 30% of the observed interaction. For a 100-basis-point treasury move, bonds twenty years farther from maturity have about 11.3 basis points more municipal yield response than the benchmark predicts. The remaining maturity pattern supports P3. The binned long-minus-short difference after the same adjustment is 0.0623 (s.e. 0.0123), and after-tax shares from 0.60 to 0.80 also leave a positive residual gradient.

The result survives changes in the minimum treasury move, the time between trades and the credit sample. All thirty combinations of treasury-move cutoffs and trade windows give a positive maturity interaction, from 0.0047 to 0.0104, significant at five percent (Appendix Table 45). After removing the coupon benchmark separately in each case, all thirty continuous and binned estimates remain positive. Twenty-four continuous and all thirty binned estimates remain significant at five percent. Holding treasury maturity fixed and using the monthly corporate curve also leave positive adjusted gradients (Appendix D.12.1). On the wider rating sample, the observed gradient rises from 0.007 to 0.013 in the continuous specification and from 0.075 to 0.113 in the binned specification (Appendix Tables 48 and 49).

7 Policy Counterfactuals

How much does withdrawing a municipal tax benefit raise borrowing costs for each dollar of revenue it generates? I study two reforms. The first removes a state's exemption for residents' interest on its own municipal bonds. The second gives municipal interest the tax treatment of taxable interest for every investor. Each reform changes the relative appeal of bonds to different investors. As they adjust their portfolios, yields change in both municipal and taxable markets. I measure the resulting financing costs, tax receipts and distribution of after-tax bond income.

7.1 Design and fiscal accounting

Each reform starts from the calibrated 2015–2017 economy. I change the relevant tax rules and solve again for investors' holdings and both yield curves. Bond supplies, investor resources, risk preferences and outside exposures remain fixed. The comparison therefore captures how a tax change reallocates the same debt and changes its financing cost.[14]

I measure annual financing costs by multiplying each maturity's yield by the amount of debt outstanding and summing across maturities. The increase after a reform is the replacement-financing cost: the additional annual cost of financing the same debt at the new yields.[15]

[14]Tax changes are applied in each investor's baseline payoff units, preserving preferences and liability hedges. Unless stated otherwise, yield changes average the 36 monthly rate states observed in 2015–2017. Dollar amounts and the California rate experiment in Figure 12(a) use the mean rate state, $z = 0$.

[15]The calculation uses continuously compounded yields to construct coupon-equivalent flows. Existing contractual payments and original-issue-discount schedules remain unchanged. The taxable market pools treasury and corporate

The same calculation applied to investors' holdings divides this interest flow into taxes and after-tax bond income. Receipts use each group's federal and state tax rates, with state taxes assigned by residence. The bank financing-tax cost is counted as federal revenue and deducted from banks' bond income.[16] Since investors hold all outstanding bonds, the additional interest paid by issuers must go either to governments or to investors. Summing across both markets and all governments gives

$$\Delta(\text{Financing costs}) = \Delta(\text{Tax receipts}) + \Delta(\text{After-tax bond income}). \tag{32}$$

Financing costs can thus exceed added receipts because some of the additional interest remains with investors after tax. I report costs per dollar of added receipts, using the reforming state's revenue for home-state repeal and combined federal and state revenue for equal treatment. The state ratio also reflects changes in other governments' receipts, which are outside its denominator.[17]

7.2 The home-state exemption

Most income-tax states exempt residents' interest on their own municipal bonds while taxing interest on other states' bonds. This home-state preference gives residents an additional reason to hold local debt.[18] I first vary the strength of this benefit in California, then remove it in seven separate state experiments.

Panel (a) of Figure 12 first shows how California's tax benefit affects its yield curve. I lower both household groups' state tax rates proportionally, keeping the home-state exemption, federal taxes and household composition fixed, and solve at five rates. A lower state tax reduces the advantage of holding home-state bonds. I measure that advantage in yields by comparing California with the pooled municipal market of states without an income tax. The implied state tax is the percentage by which California's yield is below this comparison yield.[19]

At the riskless two-year maturity, the implied state tax changes one-for-one with the high-tax household rate. At thirty years, each percentage-point reduction in that rate reduces the implied state tax by 0.377 to 0.525 percentage points across the four reductions considered. The household's

debt. Cash income, other liability flows, issuer deductions, refinancing timing and fiscal feedback are outside this bond-income accounting (Appendix E.2).

[16] The absolute municipal rate is $\tau_i^M = 1 - (1 - \tau_i^C)/\xi_i$ by (1). Household groups use resource-weighted federal and state rates by residence, and institutions use their own tax assignments (Section 5.1). Appendix E.2 gives the mapping and fiscal scope.

[17] These ratios compare financing and bond-income flows. Welfare would also depend on changes in risk bearing, investment opportunities and public spending.

[18] The Supreme Court upheld the preference in *Department of Revenue of Kentucky v. Davis* (2008). Illinois, Iowa and Wisconsin generally tax own-state municipal interest, with exemptions for particular issues (Investment Company Institute, 2022). Oklahoma exempts qualifying local obligations issued after July 1, 2001, while Utah exempts its own municipal interest. Since 2018, the federal state-and-local-tax deduction cap has also increased the exemption's value for residents whose deductions it limits.

[19] The ratio is $1 - y^M_{\mathrm{CA},T}/y^0_T$, where y^0_T is the pooled no-income-tax-state municipal yield. Both yields are solved at $z = 0$. This comparison replaces the federally adjusted taxable yield $\xi^{\mathrm{fed}} y^C_T$ in (27), giving a within-state counterpart to P1 (Figure 9).

state tax benefit is therefore fully reflected at the short end and only partly reflected at the long end, as P1 predicts.

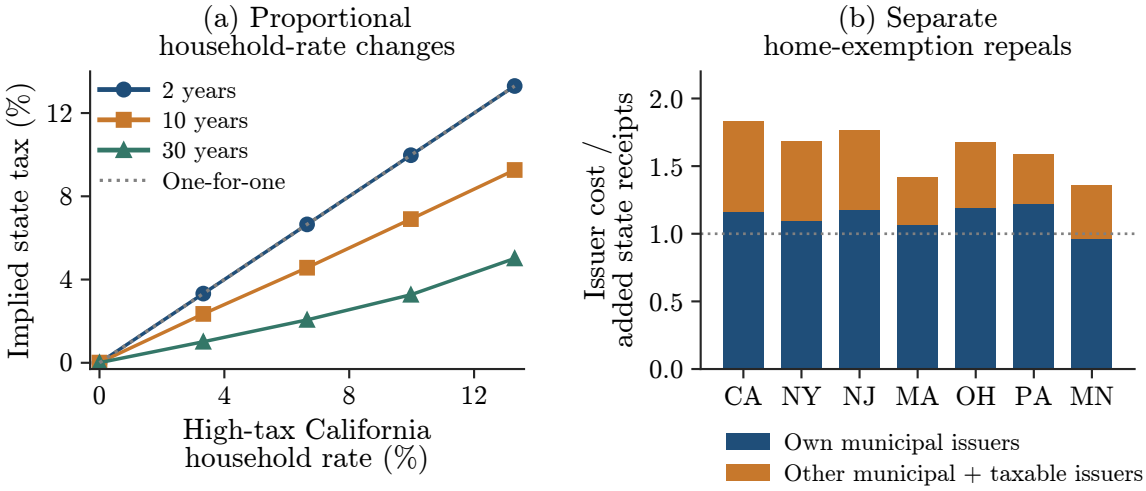


Figure 12: State taxes and the local investor base, with continuous yields at $z = 0$ and group-level absolute tax assignments. Panel (a) scales both California household groups' state rates, keeping the home-state exemption and federal taxes, and plots $1 - y^M_{\mathrm{CA},T}/y^0_T$ at two, ten and thirty years against the high-tax group's rate. Here y^0_T is the municipal yield in the pooled market of the seven states without an income tax, markers are solved equilibria, and the dotted line is one-for-one. Panel (b) removes each listed state's home-state exemption in turn and reports the replacement-financing costs of own-state municipal issuers and of other municipal and pooled taxable issuers per dollar of added state receipts from modeled resident bond portfolios.

Panel (b) turns to repeal of the exemption, holding state tax rates fixed. I remove it separately in California, New York, New Jersey, Massachusetts, Ohio, Pennsylvania and Minnesota, seven states with high supply-weighted tax benefits. In each experiment, residents' home-state municipal interest becomes subject to the same tax as their out-of-state municipal interest. The change applies to all 14 resident household types, the seven balance-sheet bins in each household group. The tax treatment of institutions, nonresidents, cash and taxable bonds stays fixed.

The reforming state's municipal financing cost rises by \$0.97–\$1.22 per dollar of added state receipts, with a ratio below one only in Minnesota. Adding the financing costs borne by other municipal and taxable issuers raises the ratio to \$1.36–\$1.83. With the same state receipts in the denominator, the difference measures the borrowing costs transmitted to other markets.

Portfolio substitution explains this transmission. As home-state bonds lose their tax advantage, residents sell them and buy other municipal bonds, taxable bonds and cash. Nonresident high-tax households absorb short bonds, and banks absorb intermediate bonds. In California at $z = 0$, residents' share of the state's bonds at two to eighteen years falls from over 90% to about 35%, while banks' intermediate-maturity share rises from 10% to 30%. The trades change investors' portfolio rate exposure and hence the returns they require on their other bonds. Only residents

owe the new tax, but all holders receive the new market yield. California's monthly-average yields rise 12.0, 26.4 and 24.7 basis points at two, ten and thirty years, and the thirty-year taxable yield rises 0.9 basis points.

State receipts depend on residents' whole bond portfolios, not just their remaining home-state holdings. In California at $z = 0$, municipal financing costs rise by $1.38 billion and state receipts by $1.19 billion. Of the added receipts, $0.51 billion comes from newly taxed home-state bonds. The rest comes from residents' other municipal and taxable bonds.

The tax base changes as residents reallocate. Moving from home-state to out-of-state bonds preserves taxable bond income, since both are taxed after repeal. Moving into cash reduces the bond-interest tax base measured here, which excludes cash income. Across states, residents' total riskless share rises by 0.03–0.56 percentage points of financial assets (Appendix Table 52). Higher bond yields partly offset the revenue reduction from reallocation. In California, added receipts of $1.19 billion remain below the $1.25 billion obtained by applying the new tax rates to unchanged prices and holdings (Appendix E.2).

7.3 Equal municipal and taxable treatment

The second reform applies each investor's taxable-interest treatment to its municipal interest, removing federal and state exemption benefits as well as banks' financing-tax cost. The change extends across all investor groups.[20] I implement it by moving each municipal after-tax share 99.9% toward its taxable share and removing 99.9% of the bank financing-tax cost.[21] Cash and taxable-interest treatment remain fixed. Bond-specific risk and differences in supply continue to distinguish the two markets.

Table 6: Coupon-equivalent fiscal accounting with 99.9% of the municipal–taxable tax distinction removed.

Coupon-equivalent change	$bn/year	Per receipt dollar
Municipal issuer financing cost	65.21	0.979
Pooled taxable issuer financing cost	10.61	0.159
Total issuer financing cost	75.82	1.139
Federal receipts, including bank financing-tax cost	59.71	0.897
Residence-state receipts	6.88	0.103
Total receipts	66.59	1.000
Net investor bond income	9.22	0.139

Notes. Changes from the same 2015–2017 baseline, with continuous-compounding yields at $z = 0$ and household absolute marginal tax rates assigned within modeled group and residence. Federal receipts include the $1.03 billion decline in the bank financing-tax cost, and net investor bond income is after interest taxes and that cost. Before rounding, the total issuer-cost change equals total receipts plus net investor bond income. The ledger covers bond-income flows and excludes cash income.

Total financing costs rise by $1.14 for each dollar of additional federal and state receipts: $0.98 for municipal issuers and $0.16 for taxable issuers (Table 6 and Figure 13(a)). Taxable issuers bear

[20] The reform is broader than the investor-specific changes in the 2017 Tax Cuts and Jobs Act (Appendix D.10).

[21] Exact equality makes the riskless two-year municipal bond and cash perfect substitutes, leaving their holdings split indeterminate. The 99.9% endpoint avoids this indeterminacy.

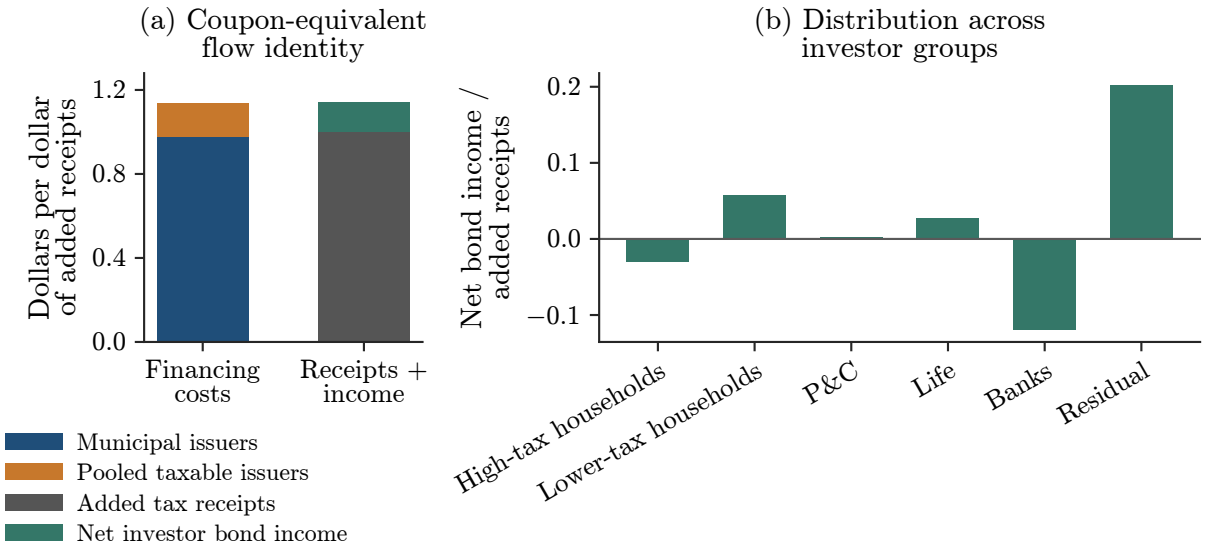


Figure 13: Coupon-equivalent flows after removing 99.9% of each tax distinction in the fixed 2015–2017 economy. Panel (a) reports identity (32) per dollar of added federal and state receipts, including the bank financing-tax cost. Panel (b) allocates net investor bond income across investor groups. Continuous yields and equilibrium holdings at $z = 0$ use group-level absolute tax assignments. Taxable issuers pool treasury and corporate quantities.

14.0% of the financing-cost increase even though their bonds' tax treatment has not changed. Of the added receipts, 89.7% accrue federally, including changes in bank financing taxes, and 10.3% accrue to investors' states of residence.

The spillover to taxable issuers arises because investors choose their holdings in both markets jointly. At the mean rate state, high-tax households and banks sell municipal bonds to the tax-neutral residual group, life insurers and lower-tax households. The tax-neutral group's share of municipal debt rises from zero to 47%, while high-tax households' share falls from 66% to 23% and banks' from 30% to 4%. As tax-neutral investors take on municipal rate exposure, they require higher returns to bear risk on their taxable bonds as well. Their share of taxable debt falls from 69% to 59%, with households and banks absorbing the sales. Appendix Table 52 reports the corresponding changes as shares of each group's own portfolio.

Thirty-year municipal and taxable yields rise by 208.6 and 34.7 basis points, averaged across monthly rate states. After the reform, municipal yields are below taxable yields by only 0.14 basis points at two years and 2.4 at thirty.

The additional \$0.14 of financing costs per receipt dollar becomes additional after-tax investor bond income, by (32). The aggregate gain combines losses for high-tax households and banks with gains for lower-tax households and insurers (Figure 13(b)). The tax-neutral group gains most, benefiting from higher yields and larger municipal holdings without losing a tax exemption.

Under equal treatment, applying new tax rates to old prices and holdings gives $40.3 billion of added receipts and leaves issuer costs unchanged. Re-solving the equilibrium adds $26.3 billion of receipts and produces $75.8 billion of additional financing costs. Both experiments therefore connect fiscal outcomes to who holds the debt after reform and the yields those investors require.

8 Conclusion

Investor taxes shape the yield curve through participation and the allocation of rate risk. The high-tax household's exemption benefit determines relative yields at the riskless short maturity. Lower-tax investors also hold longer tax-exempt bonds when their lower risk costs offset their smaller exemption benefits. Required returns then reflect the taxes and risk costs of different holders. Joint portfolio choice transmits these effects across maturities and markets, including to bonds an investor never holds.

The model accounts for 83% of the 117-basis-point municipal puzzle at thirty years using measured U.S. taxes, ownership and balance sheets, without targeting municipal yields. The empirical tests distinguish its mechanism from the representative-tax benchmark. Household taxes matter less at longer maturities, while bank- and insurer-specific tax benefits matter more. Municipal yields also respond more strongly to treasury-rate changes relative to the household after-tax benchmark at longer maturities. Together, these tax and rate-response patterns support the participation mechanism.

The policy experiments show why tax reforms must be evaluated at the new equilibrium allocation. Removing municipal tax benefits changes the tax base, redistributes bond income and raises borrowing costs even for taxable issuers whose tax treatment is unchanged. The framework applies wherever investor taxes differ across assets sharing rate risk, including treasury and corporate bonds under state taxes, tax-favored accounts and cross-border withholding. By shaping who bears rate risk, investor taxes help determine the cost of long-term public and private finance.

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A Analytical framework and results

This appendix states the general portfolio problem and the formal results for the two-investor affine benchmark. It links portfolio choice to participation, holding returns and yields. Appendix B gives the proofs and extensions.

A.1 The general economy

Assets and investors

Time is discrete and indexed by t. One aggregate rate state z_t follows a stationary AR(1), and the short rate, the pretax log gross return on cash, is bounded below at zero:

$$z_{t+1} = \rho z_t + \sigma_r \varepsilon_{t+1}, \qquad r_t = r(z_t) := \max\{\bar{r} + z_t, 0\}, \qquad \rho \in (0,1), \quad \varepsilon_{t+1} \sim \mathcal{N}(0,1). \tag{33}$$

The mean rate $\bar{r}$ is the cash rate at the stationary mean $z = 0$. The floor affects only the cash payoff, and the state z_{t+1} remains Gaussian. Appendix B.2 explains how the floor makes prices nonlinear.

Let $\mathcal{A}$ be a finite set of bond markets, indexed by a and distinguished by asset class or issuer. In the application, $\mathcal{A}$ contains a taxable market C and a municipal market M, or several state municipal markets in the quantitative model. Each market contains zero-coupon bonds indexed by their remaining maturity $T \in \{1, \dots, T_{\max}\}$, the number of periods until the bond pays one unit. The one-period claim of the taxable market C is the cash account, with return $r(z)$, so separately held taxable bonds begin at $T = 2$. A bond with T periods remaining has rate-state price $P_T^a(z_t)$ and holding-period return

$$1 + R_{T,t+1}^a = \frac{P_{T-1}^a(z_{t+1})}{P_T^a(z_t)} \exp\big(\sigma_{\eta,T}^a \, \eta_{T,t+1}^a\big). \tag{34}$$

The standard normal shocks $\eta_{T,t+1}^a$ are independent of each other and of ε_{t+1}, with $\sigma_{\eta,1}^a = 0$ in every market, so a one-period claim is fully riskless. Bond-specific risk represents residual variation in holding returns, such as an issue-specific change in trading demand, and makes longer bonds imperfect substitutes without adding a persistent price state. Appendix C.8 explains the return convention and shows its small quantitative effect.

Investors belong to finitely many types, indexed by i. Type i has mass $m_i > 0$, risk aversion $\gamma_i > 0$, funding W_i, outside exposure ℓ_i, and an after-tax share $\xi_i^a = 1 - \tau_i^a > 0$ on returns in market a, where τ_i^a is its tax rate on those returns. Because after-tax shares differ across both investors and markets, the same structure covers municipal and taxable bonds, state taxes on treasury and corporate interest, tax-favored accounts and cross-border withholding (Appendix B.14).

Portfolio choice and equilibrium

Investor i holds cash $x_{i,t}^0 = W_i - \sum_{a,T} x_{i,T,t}^a$ and bonds $x_{i,T,t}^a$, in dollars per unit of mass. Aggregate holdings are $m_i x_{i,T,t}^a$, in the same units as supply. Appendix B.1 gives the exact wealth equation and the return approximation. Portfolio choice uses log returns plus a Jensen adjustment, $\mathcal{R}_{T,t+1}^a = \mu_T^a(z_t) + u_{T,t+1}^a$. Here $\mu_T^a(z) := \log \mathbb{E}_t[1 + R_{T,t+1}^a]$, and $u_{T,t+1}^a := \log(1 + R_{T,t+1}^a) - \mathbb{E}_t[\log(1 + R_{T,t+1}^a)]$ is the centered log-return innovation. Substituting $R \approx \mathcal{R}$ and $e^r - 1 \approx r$ gives approximated wealth $\widehat{W}_i'$ and the mean–variance problem:[22]

$$\begin{aligned} \max_{x_i \geq 0,\, x_{i,t}^0 \geq 0} \quad & \mathbb{E}_t[\widehat{W}_i'] - \frac{\gamma_i}{2} \operatorname{Var}_t(\widehat{W}_i'), \\ \text{subject to} \quad & \widehat{W}_i' = W_i + \xi_i^C r(z_t) x_{i,t}^0 \\ & \qquad + \sum_{a,T} \xi_i^a x_{i,T,t}^a \mathcal{R}_{T,t+1}^a + \ell_i \sigma_r \varepsilon_{t+1}. \end{aligned} \tag{35}$$

Positions and cash are nonnegative. Funding and masses are fixed across states. A positive ℓ_i raises net wealth when rates rise by offsetting asset exposure.

[22] Appendices D.1.3 and E.5 give the capital-income tax correction.

Investor i's after-tax excess return in these units is $\widetilde{\mu}^a_{i,T} = \xi^a_i \mu^a_T - \xi^C_i r$, and $\widetilde{\Pi}_i(t+1)$ is the after-tax innovation of its whole portfolio, including the outside exposure. Appendix B.1 derives the reduced problem and explains the return convention. Let $\beta_i(z) \geq 0$ be the multiplier on the funding constraint $\sum_{a,T} x^a_{i,T} \leq W_i$, the shadow value of funding, which is zero whenever the investor holds cash. The portfolio conditions are

$$\begin{aligned} x^a_{i,T}(z) > 0 \quad &\Longrightarrow \quad \widetilde{\mu}^a_{i,T}(z) = \underbrace{\gamma_i \xi^a_i \operatorname{Cov}_z\Big(u^a_{T,t+1}, \widetilde{\Pi}_i(t+1)\Big)}_{\text{marginal portfolio-risk cost}} + \underbrace{\beta_i(z)}_{\text{shadow value of funding}}, \\ x^a_{i,T}(z) = 0 \quad &\Longrightarrow \quad \widetilde{\mu}^a_{i,T}(z) \leq \gamma_i \xi^a_i \operatorname{Cov}_z\Big(u^a_{T,t+1}, \widetilde{\Pi}_i(t+1)\Big) + \beta_i(z). \end{aligned} \tag{36}$$

The funding multiplier also satisfies $\beta_i(z)[W_i - \sum_{a,T} x^a_{i,T}(z)] = 0$.

Every separately held bond market and maturity clears:

$$x^a_{i,T}(z) \geq 0, \qquad \sum_i m_i \, x^a_{i,T}(z) = Q^a_T, \tag{37}$$

An equilibrium consists of prices and holdings satisfying (34), (36) and (37), with Q^a_T the fixed dollar supply. Participation is determined by these conditions.

A.2 From holding returns to yields

A T-period bond becomes a $(T-1)$-period bond next period. Including bond-specific risk, its price satisfies $P^a_T = e^{-\mu^a_T + (\sigma^a_{\eta,T})^2/2} \mathbb{E}_z[P^a_{T-1}(z')]$. Iteration links yields to holding returns along the bond's life.

Write $p^a_T = \log P^a_T$. With affine log prices, $p^a_T = -A^a_T z - C^a_T$, the yield is $y^a_T = (A^a_T z + C^a_T)/T$. Thus A^a_T/T is the yield's response to the rate state and C^a_T/T is its level at the mean state. The response of the one-period return is

$$s^a_T(z) := \frac{\partial \mu^a_T(z)}{\partial z}. \tag{38}$$

The distinction is between the response of one holding-period return, s^a_T, and the response accumulated in the bond's price, A^a_T. Retaining the independent bond-specific return variance $(\sigma^a_{\eta,T})^2 = v^a_T$, substituting affine prices in the resale equation and matching its state and constant terms gives

$$A^a_T = \underbrace{\rho A^a_{T-1}}_{\text{inherited loading}} + \underbrace{s^a_T}_{\text{current response}}, \qquad C^a_T = C^a_{T-1} + \mu^a_T(0) - \tfrac{1}{2}\Big[\sigma^2_r \big(A^a_{T-1}\big)^2 + \big(\sigma^a_{\eta,T}\big)^2\Big]. \tag{39}$$

From $A^a_0 = C^a_0 = 0$, the loading $A^a_T = \sum_{j=1}^T \rho^{T-j} s^a_j$ accumulates the responses, discounted by persistence, and the level term accumulates the mean-state one-period returns less half-variance terms. With a riskless one-period claim, $\sigma^a_{\eta,1} = 0$, the same recursion gives the yield as the average of one-period returns along the bond's life less convexity:

$$y^a_T(z_t) = \underbrace{\frac{1}{T}\sum_{j=0}^{T-1} \mathbb{E}_t\big[\mu^a_{T-j}(z_{t+j})\big]}_{\text{one-period returns as the bond ages}} - \underbrace{\frac{1}{2T}\sum_{k=2}^{T}\Big[\sigma^2_r \big(A^a_{k-1}\big)^2 + \big(\sigma^a_{\eta,k}\big)^2\Big]}_{\text{convexity}}. \tag{40}$$

Risky municipal bonds have positive bond-specific variance in the joint-participation benchmark. The taxable variance may be zero, as in the illustration. Appendix B.2 gives the nonlinear recursion. Portfolio choice determines μ. Its state derivative gives s, and the price recursion accumulates these into C and A.

A.3 The two-investor affine benchmark across maturities

The two-investor benchmark has municipal maturities $T = 1, \ldots, T_{\max}$ and taxable maturities $T = 2, \ldots, T_{\max}$, with cash as the one-period taxable claim and fixed supplies Q^a_T. It retains the

investors, tax normalization and bond-specific risk of Section 4.2, with different risk aversions γ_i. Sections A.4–A.6 analyze this benchmark. I fix a participation branch, solve its portfolio conditions and market clearing, and check that no investor benefits from changing its holdings. Closed forms use three restrictions:

(i) funding constraints are nonbinding;

(ii) the floored cash rate is replaced by the linear rule $r(z) = \bar{r} + z$; and

(iii) prices are computed as if the same participation branch applied at every state, which makes log prices affine, and the branch is verified at the states studied.

On a fixed branch, holdings, one-period returns and log prices are affine in z, and the price coefficients solve a fixed point. Appendix B.4 states the feasibility and uniqueness conditions. The quantitative model lets participation and funding change across states and keeps the rate floor.

Log prices then have constant coefficients,

$$p_T^{a,\mathrm{aff}}(z) = -A_T^{a,\mathrm{aff}} z - C_T^{a,\mathrm{aff}}, \tag{41}$$

and one-period returns are affine as well, $\mu_T^{a,\mathrm{aff}}(z) = \mu_T^{a,\mathrm{aff}}(0) + s_T^{a,\mathrm{aff}} z$, so $s_T^{a,\mathrm{aff}}$ and $A_T^{a,\mathrm{aff}}$ are the state coefficients of the one-period return and of the negative log price. The affine recursions (39) and the yield average (40) of Section A.2 hold exactly.

Investor i's net after-tax exposure to short-rate risk is

$$\widehat{D}_i(z) := \underbrace{\sum_j x_{i,j}^M(z) A_{j-1}^M + \xi_i \sum_j x_{i,j}^C(z) A_{j-1}^C}_{\text{tax-weighted asset exposure}} - \underbrace{\ell_i}_{\text{outside liability exposure}}. \tag{42}$$

Net rate exposure is dollar exposure to the rate factor, before division by wealth. The unit risk cost is $\gamma_i \sigma_r^2 \widehat{D}_i$. Multiplying it by a bond's loading gives its marginal rate-risk cost. Constant loadings imply that $\partial_z \widehat{D}_i$ reflects trading alone. The superscript "aff", where shown, identifies coefficients of this affine benchmark.

Bond-specific risk permits joint holdings over several maturities. Without it, different investors' active municipal conditions agree only where their rate-risk-cost difference equals the tax difference. With municipal after-tax shares normalized to one, the aggregate bond-specific response cancels under fixed supply (Section A.5). Changing its quantitative mean normalization moves the thirty-year gap by 0.20 basis points (Appendix C.8).

At one period, zero risk and nonbinding funding give $\mu_1^M = \xi_H r$ whenever $r > 0$ (Appendix Proposition 8). At longer maturities, participation in the tax-exempt market determines whose taxes and portfolio-risk costs shape equilibrium returns. The next three subsections derive that participation margin, its pricing channels and the yield implications. Appendix B.4 gives the branch construction, and Appendix Proposition 7 and Appendix B.6 give the uniqueness conditions for current-state clearing and for the loading solution.

Figures 4 and 6 illustrate a thirty-maturity branch with joint municipal participation from maturity 17. Appendix B.10 gives its parameters, verifies its KKT conditions at the mean state and establishes a unique loading solution satisfying (S). The household-only comparison has $A_T^M / A_T^C = \xi_H$ at every maturity (Proposition 11). Level panels omit the vertical scale because the illustration is not a calibration.

A.4 The participation margin

Section 4.3 uses the equilibrium participation condition proved in Theorem 1. The comparison below isolates the duration benefit at a household-only allocation, marked by the superscript hh, in which the high-tax household holds all municipal supply and the lower-tax investor holds all risky taxable supply. The lower-tax investor's net rate exposure already reflects its taxable book, which adds exposure, and its liabilities, which offset it.

Proposition 1 isolates the special case $v_T^M = 0$ to show the duration comparison alone. The equilibrium criterion used in the main text retains $v_T^M > 0$. In the zero-variance case, the high-tax household's active KKT equality and the lower-tax investor's condition for a nonnegative first-order purchase incentive are

$$\underbrace{\mu_T^M - \xi_H r}_{\text{offered to the household}} = \underbrace{\gamma_H \sigma_r^2 A_{T-1}^M \widehat{D}_H^{\text{hh}}}_{\text{household marginal risk cost}}, \tag{43}$$

$$\underbrace{\mu_T^M - \xi_I r}_{\text{offered to the lower-tax investor}} \geq \underbrace{\gamma_I \sigma_r^2 A_{T-1}^M \widehat{D}_I^{\text{hh}}}_{\text{lower-tax investor's marginal risk cost}}. \tag{44}$$

If the second inequality fails, the lower-tax investor holds none of the bond. Equality marks indifference, and a strict inequality makes a small position strictly profitable. Substituting the household's equality cancels the common return and leaves one condition:

$$\underbrace{\sigma_r^2 \, A_{T-1}^M \, \Lambda}_{\text{duration benefit (grows with maturity)}} \geq \underbrace{(\xi_I - \xi_H)\, r}_{\text{difference in exemption benefits (flat in maturity)}}, \quad \Lambda := \gamma_H \widehat{D}_H^{\text{hh}} - \gamma_I \widehat{D}_I^{\text{hh}}. \tag{45}$$

The duration benefit is the loading times $\sigma_r^2 \Lambda$, the gap between the investors' costs per unit of rate exposure. Appendix Assumption 1 requires $\Lambda > 0$, which means that risk-weighted net rate exposure is larger for the household at this allocation. The difference in exemption benefits is the same at every maturity. Clearing fixes benchmark holdings, so $s_T^M = \xi_H$ and (39) gives $A_{T-1}^M = \xi_H \sum_{j=0}^{T-2} \rho^j$. Starting from $A_0^M = 0$, the loading rises strictly with maturity, so the duration benefit grows from zero and crosses the difference in exemption benefits at most once.

Proposition 1 (Participation from the long end)**.** *In the two-investor affine benchmark without bond-specific risk, suppose $\Lambda > 0$ at the household-only allocation and $r_t > 0$. At that allocation, the maturities with a nonnegative first-order purchase incentive for the lower-tax investor form the upper interval $\{T : T \geq T^*(z_t)\}$, where*

$$T^*(z_t) \;=\; \min\Big\{T \geq 1 :\; A_{T-1}^M \geq \frac{(\xi_I - \xi_H)\, r_t}{\sigma_r^2 \, \Lambda}\Big\}, \tag{46}$$

with $\min \varnothing = +\infty$. A strict inequality makes participation strictly profitable, and equality need not imply a positive holding.

Proof. See Appendix B (proof of Proposition 1). □

The threshold T^* is set by taxes, balance sheets, and loadings, and no maturity habitat is assigned. At fixed benchmark loadings, a larger difference in exemption benefits moves the threshold out and a larger duration gap Λ moves it in. With positive bond-specific risk, the household's required return also includes compensation for the risk of its whole holding, while a new buyer bears no inventory risk at a zero position. This additional participation incentive appears explicitly in the equilibrium index (78) of Theorem 1.

In equilibrium, prices and both portfolios are re-solved. Appendix Theorem 1 gives conditions under which the lower-tax investor's municipal holdings form an upper maturity interval, using the equilibrium duration gap $\widehat{\Lambda} := \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$. From here on, T^* denotes the start of that interval.

On the branches studied below, the household holds municipal bonds alone below T^* and jointly with the lower-tax investor from T^* on. The next subsection derives how that allocation transmits taxes through both portfolios. Section 4.7 examines other motives that sustain participation in the tax-exempt market.

The illustration has household-only municipal holdings below T^*, joint holdings above it and institutional holdings of all risky taxable bonds. The calibrated economy has several longer-maturity clienteles (Section 5).

A.5 The pricing channels

This subsection derives the three pricing channels described in Section 4.4 and then the rate responses that build the third. Each return response reflects the holders' after-tax cash returns and the change in their portfolio-risk cost.

Fix maturity T, and let $\mathcal{U}_T^M \subseteq \{H, I\}$ denote the investors with positive holdings of that municipal bond. Each active investor satisfies the same KKT equality,

$$\underbrace{\mu_T^M - \xi_i r}_{\text{after-tax excess return offered to } i} = \gamma_i \left[\underbrace{\sigma_r^2 A_{T-1}^M \widehat{D}_i}_{\text{whole-portfolio rate risk}} + \underbrace{(\sigma_{\eta,T}^M)^2 x_{i,T}^M}_{\text{bond-specific risk}} \right]. \tag{47}$$

On the branches studied here the lower-tax investor alone holds the risky taxable bonds. Dividing its taxable condition by its after-tax share gives, for $T \geq 2$,

$$\mu_T^C - r = \gamma_I \left[\sigma_r^2 A_{T-1}^C \widehat{D}_I + \xi_I (\sigma_{\eta,T}^C)^2 x_{I,T}^C \right], \tag{48}$$

and $\mu_1^C = r$ for cash. Subtracting ξ_H times this equality from the equality of a municipal holder separates the gap $g_T = \mu_T^M - \xi_H \mu_T^C$ into after-tax cash returns, duration costs, rate loadings and bond-specific compensation.

Proposition 2 (Three channels of the one-period return gap)**.** *In the two-investor affine benchmark, consider a state at which the lower-tax investor alone holds the risky taxable bonds, and let i be an investor holding the maturity-T municipal bond, the lower-tax investor whenever it does. Then*

$$\begin{aligned} g_T = \; & \underbrace{(\xi_i - \xi_H) r}_{\textit{after-tax cash return}} + \underbrace{\sigma_r^2 A_{T-1}^M \left(\gamma_i \widehat{D}_i - \gamma_I \widehat{D}_I \right)}_{\textit{duration cost}} + \underbrace{\gamma_I \sigma_r^2 \widehat{D}_I \left(A_{T-1}^M - \xi_H A_{T-1}^C \right)}_{\textit{rate loading}} \\ & + \underbrace{\gamma_i (\sigma_{\eta,T}^M)^2 x_{i,T}^M - \xi_H \gamma_I \xi_I (\sigma_{\eta,T}^C)^2 x_{I,T}^C}_{\textit{bond-specific risk}}. \end{aligned} \tag{49}$$

At $T = 1$ the duration cost, the rate loading and the bond-specific term are zero, and so is the after-tax cash return when $r > 0$. Where the household holds the bond alone, the after-tax cash return is zero and the duration cost is at most $(\xi_I - \xi_H) r$. Where the lower-tax investor holds it, the after-tax cash return is $(\xi_I - \xi_H) r$ and the duration cost is zero.

Proof. See Appendix B.6. □

Below the participation threshold, the duration cost is $\sigma_r^2 A_{T-1}^M \widehat{\Lambda}$. Once the lower-tax investor holds the bond, its equality instead assigns $(\xi_I - \xi_H) r$ to the after-tax cash return. At jointly held maturities either investor's equality gives the same total gap. The return responses below determine the third term's loadings.

The rate loading links maturities. Iterating the loading recursion (39) from $A_0^a = 0$,

$$A_{T-1}^M - \xi_H A_{T-1}^C = \sum_{j=1}^{T-1} \rho^{T-1-j} \left(s_j^M - \xi_H s_j^C \right),$$

so the rate loading accumulates the response gaps of the shorter maturities the bond passes through. It is zero at two periods, since the one-period responses are $s_1^M = \xi_H$ and $s_1^C = 1$ when $r > 0$. The final term compensates bond-specific risk, the municipal holder's inventory cost less ξ_H times the taxable holder's.

Figure 5 illustrates the decomposition. The bond-specific term is below 5 percent of the gap. At thirty periods the rate loading accounts for just over half, two thirds of it inherited from the tax mix at jointly held maturities.

A bond's one-period return responds to the rate state for two reasons: each holder's after-tax return on cash rises by its after-tax share per unit of the rate state, and trading changes how much net rate exposure each holder carries. Aggregating the holders' equalities separates the two. Multiply investor i's equality (47) by m_i/γ_i, sum over the active holders, and differentiate with respect to z. The bond-specific terms cancel because fixed supply implies $\sum_i m_i \, \partial x_{i,T}^M(z)/\partial z = 0$. What remains is

$$\left(\sum_{i \in \mathcal{U}_T^M} \frac{m_i}{\gamma_i} \right) s_T^M - \sum_{i \in \mathcal{U}_T^M} \frac{m_i \xi_i}{\gamma_i} = \sigma_r^2 A_{T-1}^M \sum_{i \in \mathcal{U}_T^M} m_i \frac{\partial \widehat{D}_i(z)}{\partial z}. \tag{50}$$

Define the active holders' risk-tolerance shares and aggregate exposure response as

$$\omega_{i,T}^{M} := \frac{m_i/\gamma_i}{\sum_{j\in\mathcal{U}_T^M} m_j/\gamma_j}, \qquad E_T^M := \frac{\sum_{i\in\mathcal{U}_T^M} m_i\,\partial\widehat{D}_i(z)/\partial z}{\sum_{i\in\mathcal{U}_T^M} m_i/\gamma_i}.$$

The weight $\omega_{i,T}^M$ is type i's risk tolerance, m_i/γ_i, as a share of the active holders' total. The term E_T^M is their aggregate net-duration response divided by that total risk tolerance, and E_T^C is defined in the same way from the holders of the taxable bond. Proposition 3 states the resulting decomposition.

Proposition 3 (Holding-return response to the rate state)**.** *Consider a regular state of the two-investor affine benchmark: price moments are differentiable, active risky holdings and inactive KKT slacks are strictly positive, and the stacked active KKT-and-clearing Jacobian is nonsingular, so the active set $\mathcal{U}_T^M \subseteq \{H, I\}$ is locally fixed. Then*

$$s_T^M = \underbrace{\sum_{i\in\mathcal{U}_T^M} \omega_{i,T}^M \xi_i}_{\text{holders' tax mix}} + \underbrace{\sigma_r^2 A_{T-1}^M E_T^M}_{\text{portfolio-risk response}} \tag{51}$$

exactly. The tax-mix term depends on who holds maturity T. The portfolio-risk response depends on how equilibrium trading changes those holders' exposure across their whole portfolios. These are the two contributions to the rate-loading channel.

Proof. See Appendix B.6. □

The tax term is ξ_H under household-only holdings and, under joint holdings,

$$\xi^* := \omega_H \xi_H + \omega_I \xi_I, \qquad \omega_i := \frac{m_i/\gamma_i}{m_H/\gamma_H + m_I/\gamma_I},$$

where ω_i is investor i's share of total risk tolerance. Thus the lower-tax investor's after-tax share enters this term only at maturities it holds.

The sign of the portfolio-risk response follows from reallocation on a joint-participation branch. The lower-tax investor alone holds risky taxable bonds, both investors hold the municipal maturities in a nonempty set $J \subseteq \{2, \ldots, T_{\max}\}$, and the household alone holds every other municipal maturity. Bond-specific risk permits joint holdings over this set (Section A.3). Appendix Proposition 10 shows that at a regular state with $r(z) > 0$, positive loadings and positive bond-specific variance on J,

$$d := \frac{\partial\widehat{D}_H(z)}{\partial z} > 0, \qquad \frac{\partial\widehat{D}_I(z)}{\partial z} = -\frac{m_H}{m_I}\, d < 0.$$

A higher rate raises $(\xi_I - \xi_H)r$, shifting municipal duration to the household. Its risk cost rises even on risky bonds the lower-tax investor never holds. This sign applies to duration-weighted portfolios, not necessarily to each position. At a household-only risky maturity,

$$s_T^M - \xi_H = \gamma_H \sigma_r^2 A_{T-1}^M \frac{\partial\widehat{D}_H(z)}{\partial z} > 0. \tag{52}$$

The portfolio-risk response is zero at the one-period bond, where $A_0^M = 0$, and grows with the bond's loading. The taxable leg moves the other way. Because the lower-tax investor's net rate exposure falls as the rate state rises, the response of its taxable return is

$$s_T^C = 1 - b A_{T-1}^C, \qquad b := \gamma_I \sigma_r^2 \frac{m_H}{m_I}\, d > 0,$$

which lies below one at every risky maturity.

If joint municipal participation begins at T^* and continues to the long end, the municipal responses are

$$s_T^M = \begin{cases} \xi_H + \gamma_H \sigma_r^2 A_{T-1}^M d, & T < T^*, \\ \xi^* = \omega_H \xi_H + \omega_I \xi_I, & T \geq T^*. \end{cases} \tag{53}$$

Below the threshold, the lower-tax investor's after-tax share affects the municipal return response through the household's portfolio adjustment. From the threshold on, both investors hold the bond, clearing cancels their aggregate exposure response, and the tax mix ξ^* remains.

Figure 6(a) plots these municipal and taxable responses.

The taxable leg completes the response gap $s_j^M - \xi_H s_j^C = \partial_z g_j$, whose discounted sum over $j < T$ is the loading difference $A_{T-1}^M - \xi_H A_{T-1}^C$ in the third term of (49). Applying the return decomposition in both markets, with the lower-tax investor the sole holder of risky taxable bonds, gives

$$s_T^M - \xi_H s_T^C = \underbrace{\left(\bar{\xi}_T^M - \xi_H\right)}_{\text{difference in tax mix}} + \underbrace{\sigma_r^2 \left(A_{T-1}^M E_T^M - \xi_H A_{T-1}^C E_T^C\right)}_{\text{difference in portfolio-risk responses}}, \tag{54}$$

where $\bar{\xi}_T^M := \sum_i \omega_{i,T}^M \xi_i$, equal to ξ_H or ξ^*, is the tax mix of the municipal bond's active holders. Both terms are zero at one period, where the bond has no rate exposure and the high-tax household alone holds it. Below the participation threshold, portfolio adjustment raises the household's municipal response and lowers the lower-tax investor's taxable response, so the two legs reinforce each other. From the threshold on, the tax term adds $\xi^* - \xi_H = \omega_I(\xi_I - \xi_H)$. Appendix equation (81) gives the exact counterpart with nonlinear covariances.

Through the resale price, a long bond inherits both terms. On an upper-interval branch, the rate loading of Proposition 2 incorporates portfolio-risk responses from three periods on and the change in tax mix from $T^* + 1$ on.

Figure 6(b) divides municipal loadings by $G_T := \sum_{j=1}^T \rho^{T-j} = A_T^{\text{Tsy}}$, the loading of a reference curve with one-for-one holding-return responses. Since $A_T^M / G_T = \sum_{j=1}^T (\rho^{T-j}/G_T) s_j^M$, this relative yield sensitivity averages the responses with positive weights. It equals ξ_H in the household-only benchmark. Its excess accumulates portfolio-risk responses below T^* and tax-mix differences from T^* onward. Both vanish when the after-tax shares coincide.

A per-dollar benefit or cost that does not vary with the rate state, such as a trading-cost advantage or the calibration's fixed bank financing-tax cost, drops out of these response formulas on the same participation branch. Such a benefit or cost still changes which maturities an investor holds and the levels of one-period returns (Appendix B.6).

A.6 From one-period returns to the yield-implied tax

The channels act on one-period returns, while the puzzle concerns yields to maturity. By (40), a yield averages the one-period returns a bond earns as it ages, less convexity. Applied to both markets, this turns the one-period gap g_T of Proposition 2 into the yield gap:

$$\begin{aligned} y_T^M - \xi_H y_T^C = & \underbrace{\frac{1}{T} \sum_{j=0}^{T-1} \mathbb{E}_t \left[g_{T-j}(z_{t+j}) \right]}_{\text{average gap along the bond's life}} \\ & \underbrace{- \frac{1}{2T} \sum_{k=2}^{T} \left[\sigma_r^2 \left(\left(A_{k-1}^M\right)^2 - \xi_H \left(A_{k-1}^C\right)^2 \right) + \left(\sigma_{\eta,k}^M\right)^2 - \xi_H \left(\sigma_{\eta,k}^C\right)^2 \right]}_{\text{convexity difference}}. \end{aligned} \tag{55}$$

The riskless one-period gap is zero. Longer yield gaps average the one-period gaps of Proposition 2, less the convexity difference. The yield-implied tax is

$$\tau_T^{Y,\text{impl}} := 1 - \frac{y_T^M}{y_T^C}, \qquad \text{the model counterpart of } \textit{ImpliedTax} \text{ in (2).}$$

It equals the household wedge less the yield gap per unit of taxable yield, $\tau_T^{Y,\text{impl}} = (1-\xi_H) - (y_T^M - \xi_H y_T^C)/y_T^C$, so the gap accumulated in (55) pulls the implied tax below the household wedge, by more at longer maturities when the gap grows faster in proportion than the taxable yield (Proposition 4(ii)).

The relative yield response sums the response gaps in (54). On a joint-participation branch the response gap is positive at every risky maturity, through the portfolio-risk response below the participation threshold and through both response terms from that threshold on (Proposition 10), so for $T \geq 2$

$$\underbrace{\frac{\partial y_T^M}{\partial z} - \xi_H \frac{\partial y_T^C}{\partial z}}_{\text{relative yield response}} = \frac{1}{T}\sum_{j=2}^{T} \rho^{T-j} \underbrace{\left(\frac{\partial \mu_j^M}{\partial z} - \xi_H \frac{\partial \mu_j^C}{\partial z} \right)}_{\text{holding-return response gap}} > 0, \tag{56}$$

in model-period units, and both sides are zero for the riskless one-period bond. Long municipal yields therefore move more with the short rate than the household benchmark implies. The lower-tax investor's tax affects these responses even at maturities it does not hold.

Proposition 4 gives the maturity patterns when the lower-tax investor holds municipal bonds from a threshold T^* onward. The restriction on taxes, risk-tolerance shares and rate persistence is

$$\text{(S)} \qquad \omega_H \frac{\xi_I - \xi_H}{\xi_H} \leq \rho. \tag{57}$$

This condition keeps the feedback b of portfolio reallocation into taxable returns, $s_T^C = 1 - bA_{T-1}^C$, below ρ, so taxable loadings rise with maturity. Municipal loadings also rise beyond the participation threshold because the loading there lies below its long-end value, $A_{T^*-1}^M < \xi^*/(1-\rho)$ (Appendix equation (88)). The long-end taxable response $s_\infty^{C,\text{aff}} := \lim_{T\to\infty} s_T^{C,\text{aff}}$ lies in $(0,1)$, because the lower-tax investor's net duration falls as the rate state rises.

To establish the maturity ordering of yield levels, define the forward rate implied by today's prices for the period from $t+T-1$ to $t+T$:

$$f_T^a(z) = \log \frac{P_{T-1}^a(z)}{P_T^a(z)}, \qquad y_T^a(z) = \frac{1}{T}\sum_{j=1}^{T} f_j^a(z). \tag{58}$$

With positive taxable forwards, the yield ratio is a weighted average of municipal-to-taxable forward-rate ratios. Proposition 4(ii) states the condition under which their maturity ordering carries to yields. The forward rate compares adjacent-maturity prices at the same date. The holding return uses next period's resale price. The affine recursions (39) give their relation:

$$f_T^a(z) = \mu_T^a(z) - (1-\rho)A_{T-1}^a z - \tfrac{1}{2}\left[\sigma_r^2 (A_{T-1}^a)^2 + (\sigma_{\eta,T}^a)^2\right]. \tag{59}$$

Mean reversion and convexity distinguish the forward rate from the expected holding return. For taxable bonds, the term premium is the yield less the average expected future short rate. Equation (40) expresses it as average expected bond risk premia less convexity. The municipal–taxable yield gap also reflects differences in holders' tax benefits.

Proposition 4 (The municipal-to-taxable loading ratio rises and the yield-implied tax declines)**.** *In the affine two-investor benchmark, consider the equilibrium of Appendix Proposition 10 in which the high-tax household alone holds municipal bonds shorter than T^*, both investors hold municipal bonds from T^* to $T_{\max}$, and the lower-tax investor holds the taxable bonds. Assume (S).*

(i) Rate sensitivity. *The municipal-to-taxable loading ratio is non-decreasing in maturity. It starts at the household's after-tax share and, when the recursions are extended to infinite maturity with the household's duration response d held at its solved value, converges to the long-end tax mix relative to the taxable response:*

$$\frac{A_1^{M,\text{aff}}}{A_1^{C,\text{aff}}} = \xi_H \qquad \textit{and} \qquad \lim_{T\to\infty} \frac{A_T^{M,\text{aff}}}{A_T^{C,\text{aff}}} = \frac{\xi^*}{s_\infty^{C,\text{aff}}}. \tag{60}$$

(ii) Yield levels. *Suppose also that taxable forward rates are positive and the municipal-to-taxable forward-rate ratio is non-decreasing in maturity at the stationary mean $z = 0$. Then the yield ratio y_T^M/y_T^C is non-decreasing at $z = 0$ (Appendix Proposition 12), so the yield-implied tax $\tau_T^{Y,\text{impl,aff}}$ is non-increasing.*

(iii) Long end. *Extend the recursions to infinite maturity and suppose the loadings, the risk-premium coefficients π_T^a and the inventory terms ι_T^a converge, with limits marked by the subscript ∞ and a positive limiting taxable forward. Then*

$$\lim_{T\to\infty} \tau_T^{Y,\text{impl,aff}} \;=\; 1 - \frac{\xi^*\bar{r} + A_\infty^M\pi_\infty^M - \frac{1}{2}\sigma_r^2(A_\infty^M)^2 + \iota_\infty^M}{\bar{r} + A_\infty^C\pi_\infty^C - \frac{1}{2}\sigma_r^2(A_\infty^C)^2 + \iota_\infty^C}, \tag{61}$$

where $\pi_T^a = \sum_i \omega_{i,T}^a \gamma_i \sigma_r^2 \widehat{D}_i(0)$ is the holders' cost of rate risk per unit of loading, weighted by risk tolerance, and ι_T^a is their compensation for bond-specific risk net of its convexity term (Appendix equation (89)).

Proof. See Appendix B.7. □

Part (i) orders rate sensitivities. Part (ii) gives the separate forward-rate-ratio condition for the ordering of yield levels. Part (iii) adds relative term premia, convexity and inventory costs to the long-end tax mix. With $\widehat{\Lambda} > 0$, small taxable feedback b and negligible convexity and inventory terms, the long-end implied tax is below $1 - \xi^*$ and therefore below $1 - \xi_H$ (equations (91) and (92)).

A.7 Participation from other motives

This subsection states the formal neutrality result behind Section 4.7. Lower trading costs or default diversification can induce long-end participation without an advantage in bearing rate risk.

An advantage that grows with maturity competes with a maturity-invariant tax difference. Proposition 5 separates the resulting gap from the motive for participation.

The proposition sets aside every other source of a gap by making the remaining characteristics neutral. A characteristic is neutral when, for a household that bears it, it adds ξ_H times as much to the required municipal return as to the required taxable return, so it creates no gap when the household holds both bonds. Rate risk is neutral when the household holds both bonds, since municipal loadings are then ξ_H times taxable loadings. A common return shock is neutral when municipal exposure is ξ_H times taxable exposure, and a trading cost is neutral when it reduces after-tax wealth by the same amount in both markets.

Proposition 5 (Participation creates the gap)**.** *In the two-investor affine benchmark without bond-specific risk, suppose every non-tax characteristic other than rate risk is neutral and zero at the one-period claim: a risk common to both markets, independent of rate risk, with municipal exposure ξ_H times the taxable exposure, or a cost that reduces after-tax wealth by the same amount in both markets.*

(i) If the household held every municipal and taxable bond, $g_T = 0$ at every maturity.

(ii) Wherever the lower-tax investor holds the maturity-T municipal bond,

$$g_T \;\geq\; \underbrace{(\xi_I - \xi_H)r}_{\text{after-tax cash return}} + \underbrace{\gamma_I\sigma_r^2\widehat{D}_I\big(A_{T-1}^M - \xi_H A_{T-1}^C\big)}_{\text{rate loading}}.$$

Under neutrality, household holdings of both bonds create no gap. Where the lower-tax investor holds municipals, the gap includes the full tax difference plus the rate-loading term. That term is positive on the branch of Proposition 10 when the lower-tax investor's net rate exposure is positive, but can have either sign under other motives. Appendix B.13.2 proves (i). Proposition 15 proves (ii).

Proposition 17 solves two three-maturity economies with constant cash rates. Lower trading costs give long-end participation and an increasing municipal-to-taxable return ratio. Independent default factors give the same default-net gap at the long end, while the intermediate raw gap reflects diversification. Municipal round-trip costs rise with maturity and fall with trade size (Harris and Piwowar, 2006; Green et al., 2007). Appendix Figure 18 reports the matched-market evidence.

The diversification example rests on the concentration of household credit risk. The state-tax exemption tilts households toward their own state's bonds, which limits their risk sharing (Babina et al., 2021), whereas an institution whose book is mostly taxable bonds holds little municipal default risk and bears it more cheaply. Default exposure rises with maturity, as do the municipal CDS spreads in Appendix Figure 19, so this advantage also grows with maturity and the lower-tax investor again holds only the long municipal bond. With separate municipal and taxable default factors, the same pricing holds once each gap is taken net of the default compensation its holders require (Appendix Proposition 16): the net gaps are zero at the riskless and middle bonds and equal the difference in exemption benefits at the long bond. In both examples, the lower-tax investor holds long tax-exempt bonds in equilibrium because its advantage grows with maturity, and the return on those bonds includes the difference in exemption benefits, as in the rate-risk mechanism. With rate risk added back, numerical examples under both motives show participation only at the long end whenever it occurs. Their long gaps, net of default components in the diversification case, are at least the difference in exemption benefits, or fall short of it by less than 0.1 basis points where the rate-loading term is slightly negative (Appendix B.13.4).

A preference for municipal bonds as an asset class has a different pricing implication. In Appendix Proposition 18, such a preference supports long-end municipal participation while keeping municipal yields at or below the household benchmark. The argument thus requires motives that are neutral across the two markets.

The quantitative model uses the duration motive because rate risk is common to both markets, equilibrium prices determine the bonds' rate loadings, and ownership and the maturity of insurers' portfolios discipline the investors' outside rate exposures. Rate risk plays the central quantitative role. Lowering risk aversion by 20% lowers the thirty-year gap by 13.6 basis points and the taxable yield by 70 basis points, while changing the bond-specific variance by 20% moves it by less than 0.1 basis points (Appendix C.6).

A.8 From mechanism to empirical predictions

The mechanism predicts whose taxes are priced at each maturity. This section turns that prediction into three tests. Two of them concern how tax benefits are priced in yield levels. The theory is sharpest for rate sensitivities, which link maturities through the loadings, so I pair the yield-implied tax of Section A.6 with a loading-implied tax:

$$\tau_T^{Y,\text{impl}} = 1 - \frac{y_T^M}{y_T^C}, \qquad \tau_T^A = 1 - \frac{A_T^M}{A_T^C}.$$

A pass-through varies one investor's after-tax share at a fixed rate state and re-solves the equilibrium, letting holdings, risk costs and both yield curves adjust while non-tax primitives stay fixed. The pass-throughs of investor i's exemption benefit $1-\xi_i$ into the two implied taxes are

$$\mathcal{P}_i^Y(T) := \frac{d\tau_T^{Y,\text{impl}}}{d(1-\xi_i)}, \qquad \mathcal{P}_i^A(T) := \frac{d\tau_T^A}{d(1-\xi_i)}, \qquad i \in \{H, I\},$$

where an argument ∞ denotes the limit as $T \to \infty$. Under the representative-tax benchmark of Section 3.3, $y_T^M = \xi_H y_T^C$ and $A_T^M = \xi_H A_T^C$ at every maturity, so in both measures household pass-through is one and institutional pass-through is zero at every maturity. With heterogeneous taxes, the household alone holds the riskless short bond, whose return is the household's after-tax cash return $\xi_H r$, while both investors hold long bonds, whose returns depend on both after-tax shares through the tax mix $\xi^* = \omega_H \xi_H + \omega_I \xi_I$, with the risk-tolerance shares ω_i of Section A.5. Treasury zero-coupon yields lie outside the two-investor benchmark, and I represent them by the reference curve of Section A.5, whose one-period return moves one-for-one with the rate state, so their loading is $A_T^{\text{Tsy}} = G_T$.

Proposition 6 (Tax pass-through at the short and long ends)**.** *Consider the equilibrium of Proposition 4 (Appendix Proposition 10), with $0 < \rho < 1$, $\sigma_r > 0$, a positive cash rate and slack household funding. Extend its recursions to infinite maturity, and suppose long yields and their tax derivatives converge locally uniformly, with a positive long taxable yield.*

(i) (Short end) Both implied taxes are the household's own, $\tau_1^{Y,\text{impl}} = \tau_1^A = 1 - \xi_H$, so $\mathcal{P}_H^Y(1) = \mathcal{P}_H^A(1) = 1$ and $\mathcal{P}_I^Y(1) = \mathcal{P}_I^A(1) = 0$.

(ii) (Long end, P1 and P2) Suppose the after-tax shares satisfy the tax-difference condition

$$2\omega_H e + \omega_I e^2 \leq 1, \qquad e := \frac{\xi_I - \xi_H}{\xi_H}.$$

Then

$$\mathcal{P}_H^A(\infty) < \omega_H, \qquad \mathcal{P}_I^A(\infty) > \omega_I.$$

The difference $\mathcal{P}_i^Y(\infty) - \mathcal{P}_i^A(\infty)$ shrinks to zero as the mean rate $\bar{r}$ rises, so for every $\bar{r}$ above a threshold set by the other primitives, at every feasible participation threshold,

$$\mathcal{P}_H^Y(\infty) < \omega_H < 1, \qquad \mathcal{P}_I^Y(\infty) > \omega_I > 0.$$

(iii) (Response ratio, P3) The municipal-to-treasury zero-coupon response ratio A_T^M / A_T^{Tsy} is non-decreasing in maturity, equals ξ_H at the short end and converges to $\xi^* > \xi_H$ at the long end.

Proof. See Appendix B.8. Appendix B.7 proves the monotonicity in part (iii). □

Part (i) holds because the household alone holds the riskless one-period bond, so its after-tax share sets both implied taxes. Part (ii) works through the loadings, and the yield-implied tax then stays close to the loading-implied tax. At the long end both investors hold municipal bonds, and Appendix B.8 solves the long-end loading ratio $R_\infty := A_\infty^M / A_\infty^C$ as

$$R_\infty = \xi^* + \varkappa \omega_H (\xi_I - \xi_H),$$

where ξ^* is the tax mix of the two holders and $\varkappa > 0$ measures the feedback from the taxable response.[23]

The first term passes each investor's after-tax share through with its risk-tolerance share, ω_H for the household and ω_I for the lower-tax investor. The second reflects their reallocation of rate risk as the rate state changes. Under the stated tax-difference condition, a larger household exemption benefit strengthens this response, while a larger exemption benefit for the lower-tax investor weakens it. The resulting feedback offsets the household's tax-mix effect and adds to the lower-tax investor's, pushing $\mathcal{P}_H^A(\infty)$ below ω_H and $\mathcal{P}_I^A(\infty)$ above ω_I.

The tax-difference condition ensures this direction. Raising ξ_H narrows the tax difference, weakening the rate-induced transfer of duration to the household. It also raises the tax mix and bond loadings, which strengthens the risk feedback. The first effect dominates when the tax difference is not too large relative to ξ_H. The condition bounds this relative difference, e, and holds whenever $\xi_I < \frac{3}{2}\xi_H$, whatever the risk-tolerance shares or the participation threshold. Lemma 1 proves the derivative bounds with both effects included.

For yield levels, Appendix B.8 shows that $\mathcal{P}_i^Y(\infty)$ equals $\mathcal{P}_i^A(\infty)$ plus two corrections divided by the long taxable yield: the tax response of $y_\infty^M - R_\infty y_\infty^C$, which contains only relative convexity and risk costs and does not depend on $\bar{r}$, and a term for the tax response of the taxable yield itself. The taxable yield rises with $\bar{r}$, so both corrections shrink as $1/\bar{r}$, which gives the yield-level inequalities in part (ii) when the mean rate is high enough.

The tax-difference condition accommodates the tax difference between a top-bracket household and a corporate investor. The mean-rate condition limits how much risk compensation can move relative to interest income. The tax-difference condition and the yield-level inequalities hold with wide margins in the thirty-maturity illustration, where convexity accounts for most of the difference between the lower-tax investor's

[23] The identity $R_\infty = \xi^*[1 + b/(1-\rho)]$ gives $\varkappa = \xi^* b / [(1-\rho)\omega_H(\xi_I - \xi_H)]$. The coefficient and b adjust when an after-tax share changes.

two pass-throughs. In the illustration, the yield-level inequalities also hold for every nonnegative value of the taxable holders' cost per unit of rate loading, π^C_∞. Stronger hedging demand or lower mean rates can enlarge the correction as the long taxable yield in its denominator approaches zero (Appendix B.10).

The proposition establishes the short-end and long-end contrasts that the empirical tests use. At intermediate maturities, the part of $\mathcal{P}^A_i(T)$ that comes from the tax mix alone shifts weight from the household to the lower-tax investor as more of a bond's remaining life falls at jointly held maturities (Appendix Proposition 13). In the thirty-maturity illustration, $\mathcal{P}^Y_H(T)$ falls and $\mathcal{P}^Y_I(T)$ rises strictly with maturity. With more clienteles, each has its own participation threshold and risk-tolerance share (Appendix B.11). These contrasts and the solved profiles give the three predictions tested in Section 6.

Prediction (P1: household state-tax pass-through declines with maturity)**.** A household's home-state tax is part of its exemption benefit on home-state bonds. Its pass-through into the yield-implied state tax of equation (27) is one at the riskless short end and declines with maturity in the solved illustration and calibrated model as other clienteles hold municipal bonds. The empirical test compares pass-through across maturity bins. The representative-tax benchmark predicts one at every maturity.

Prediction (P2: institutional tax benefits affect longer maturities)**.** A change in a lower-tax institution's exemption benefit has no effect at the riskless short end. At longer maturities, the institution's own municipal holdings and the trading it induces make its tax treatment matter for municipal yields. Proposition 6 and the solved profiles predict a larger response at the long end, and the portfolio-risk response also carries the change to shorter risky bonds the institution does not hold. The representative-tax benchmark gives institutional tax treatment no role at any maturity.

Prediction (P3: the muni-to-treasury zero-coupon response rises with maturity)**.** The municipal zero-coupon yield's response to the rate state, relative to the treasury zero-coupon yield's response,

$$\frac{\partial y^M_T(z)/\partial z}{\partial y^{\text{Tsy}}_T(z)/\partial z} = \frac{A^M_T(z)}{A^{\text{Tsy}}_T(z)},$$

rises with maturity, from ξ_H at the short end toward ξ^*. The representative-tax benchmark predicts a constant ratio across maturities.

B Proofs and model extensions

The proofs begin with the general portfolio problem and price recursion, then specialize to participation, the three pricing channels and tax pass-through in the two-investor affine benchmark. Unless stated otherwise, funding is nonbinding ($\beta_i = 0$). These solutions also satisfy the funded problem wherever cash is nonnegative. Appendix C gives the funded system, and Table 7 collects the notation.

B.1 The general economy: the reduced problem and conventions

This appendix derives the reduced portfolio problem of the general economy in Section A.1, compares its return units with simple returns and gives the return on riskless claims when funding binds.

Portfolio choice uses the adjusted return (6), with mean $\mu^a_T = \log \mathbb{E}_t[1 + R^a_{T,t+1}]$ and centered log-return innovation $u^a_{T,t+1}$ as defined in Section A.1. Collecting the cash and bond terms in approximated wealth (35) gives the expected after-tax excess return

$$\widetilde{\mu}^a_{i,T}(z_t) := \underbrace{\xi^a_i \mu^a_T(z_t)}_{\text{after-tax bond return}} - \underbrace{\xi^C_i r(z_t)}_{\text{after-tax cash return}}. \tag{62}$$

The bond and outside innovations have conditional mean zero, so expected approximated wealth separates into the return on the initial funding and the portfolio's excess returns:

$$\mathbb{E}_t[\widehat{W}'_i] = W_i\big(1 + \xi^C_i r(z_t)\big) + \sum_{a,T} x^a_{i,T,t}\widetilde{\mu}^a_{i,T}(z_t).$$

Table 7: Notation for the model

Symbol	Meaning
z, $r(z) = \max\{\bar{r} + z, 0\}$	Latent rate state and the one-period log gross cash return
τ_i^a, ξ_i^a, $\xi_i = \xi_i^C/\xi_i^M$	Investor i's asset-specific tax rate and after-tax share. In the two-market benchmark, ξ_i is its relative after-tax share and $1-\xi_i$ its relative municipal tax benefit
$\xi_H < \xi_I$, ξ^*	Benchmark after-tax shares of the high-tax household and lower-tax investor, and their tax mix $\omega_H\xi_H + \omega_I\xi_I$ with risk-tolerance shares ω_i
$p_T^a(z)$, $A_T^a = -\partial_z p_T^a$, $y_T^a = -p_T^a/T$	Log price of the maturity-T bond in market a, its rate loading (the model's duration), and its yield, with $A_0^a = 0$
$C_T^a(z) = -p_T^a(z) - A_T^a(z)z$	Residual price-level term, the constant intercept in the affine benchmark
$\mu_T^a(z) = \log \mathbb{E}_z[1 + R_T^a]$, $s_T^a = \partial_z \mu_T^a$	Mean adjusted holding return, equal to its holders' required return in the KKT conditions, and its response to the state
$R_{T,t+1}^a$, $\mathcal{R}_{T,t+1}^a = \mu_T^a + u_{T,t+1}^a$	Simple holding-period return and the return used in portfolio choice, with mean μ_T^a and centered log-return innovation $u_{T,t+1}^a$
$\sigma_{\eta,T}^a$	Loading on bond-specific risk, zero for the one-period claim
$\nu_{i,T}^a$	KKT slack: marginal risk cost minus excess return, zero for a positive holding
$x_{i,T}^a$, Q_T^a, m_i, γ_i	Holdings per unit of mass, total supply, investor mass, risk aversion
W_i, W_i', $\widehat{W}_i'$, β_i	Funding per unit of mass, next-period net wealth, its discrete-time approximation, and the multiplier on the funding constraint
D_i, $\widehat{D}_i$, ℓ_i	Tax-weighted asset rate exposure, net rate exposure, and the outside liability loading. Appendix C.7 writes $h_i = -\ell_i$
Λ, $\widehat{\Lambda}$	Duration gap $\gamma_H\widehat{D}_H - \gamma_I\widehat{D}_I$ at the household-only allocation and at the equilibrium allocation
$\omega_{i,T}^a$, $\mathcal{U}_T^a$	Risk-tolerance share of an active holder and the set of active holders of the maturity-T bond in market a
$g_T = \mu_T^M - \xi_H\mu_T^C$	Gap between the municipal one-period return and the high-tax household's after-tax taxable return
$\tau_T^{Y,\text{impl}} = 1 - y_T^M/y_T^C$	Model yield-implied tax, the counterpart of *ImpliedTax* in (2)

The variance is that of the after-tax portfolio innovation, the random part of (35):

$$\widetilde{\Pi}_i(t+1) = \sum_{a\in\mathcal{A}}\sum_T \xi_i^a x_{i,T,t}^a u_{T,t+1}^a + \underbrace{\ell_i\sigma_r\varepsilon_{t+1}}_{\text{liability hedge}}. \tag{63}$$

Dropping the portfolio-independent term $W_i(1+\xi_i^C r(z_t))$ and eliminating cash gives the equivalent reduced problem:

$$\max_{\substack{x_i\geq 0 \\ \sum_{a,T} x_{i,T,t}^a \leq W_i}} \underbrace{\sum_{a\in\mathcal{A}}\sum_T x_{i,T,t}^a \widetilde{\mu}_{i,T}^a(z_t)}_{\text{after-tax excess return}} - \underbrace{\frac{\gamma_i}{2}\operatorname{Var}_t\left(\widetilde{\Pi}_i(t+1)\right)}_{\text{whole-portfolio risk cost}}. \tag{64}$$

A bond is attractive for its after-tax return and costly to the extent that it adds risk to the investor's whole balance sheet.

Rate risk comes from the next-period resale price. With log price $p_T^a := \log P_T^a$ and rate loading $A_T^a := -\partial_z p_T^a$, a bond with T periods remaining has rate innovation $-A_{T-1}^a(\rho z_t)\sigma_r\varepsilon_{t+1}$ to first order, at the conditional mean resale state. Even a riskless one-period bond's price can respond to today's state. A positive ℓ_i offsets the portfolio's rate exposure, and ℓ_i can represent net outside exposure of either sign.

The discrete-time return approximation

With simple holding-period returns $R_{T,t+1}^a$, next-period net wealth is

$$W_i' = W_i + \xi_i^C x_i^0\left(e^{r(z_t)} - 1\right) + \sum_{a,T}\xi_i^a x_{i,T}^a R_{T,t+1}^a + \ell_i\sigma_r\varepsilon_{t+1}.$$

With rate risk alone, the log holding return is the change in log price, $\log(1+R^a_{T,t+1}) = \log P^a_{T-1}(z_{t+1}) - \log P^a_T(z_t)$. Following Greenwood et al. (2024), I replace the simple bond return by the log return plus a Jensen adjustment, and $e^r - 1$ by r for cash. These substitutions give $\widehat{W}'_i$ in (35), with adjusted bond return

$$\mathcal{R}^a_{T,t+1} = \log(1+R^a_{T,t+1}) + \underbrace{\log \mathbb{E}_t[1+R^a_{T,t+1}] - \mathbb{E}_t[\log(1+R^a_{T,t+1})]}_{\text{Jensen adjustment}}.$$

The Jensen adjustment is known at time t. Taking the conditional expectation gives $\mathbb{E}_t[\mathcal{R}^a_{T,t+1}] = \mu^a_T(z_t) = \log \mathbb{E}_t[1+R^a_{T,t+1}]$. Subtracting this mean cancels the adjustment:

$$u^a_{T,t+1} = \mathcal{R}^a_{T,t+1} - \mu^a_T(z_t) = \log(1+R^a_{T,t+1}) - \mathbb{E}_t[\log(1+R^a_{T,t+1})].$$

Thus u is also the innovation of the adjusted return, with $\mathbb{E}_t[u] = 0$. In the affine benchmark, log returns are conditionally Gaussian. The Jensen term then equals $\frac{1}{2}\operatorname{Var}_t[\log(1+R^a_{T,t+1})]$, giving the same expected and unexpected components as equations (7) and (8) of Greenwood et al. (2024). The nonlinear solver evaluates the exponential expectation directly, retaining the full Jensen term. Because the adjustment is conditionally known, it leaves return variances and covariances unchanged. The price recursion is exact for the chosen mean $\mu = \log \mathbb{E}_t[1+R]$. Its use in the linear wealth equation approximates simple returns, whose conditional mean is $e^\mu - 1$. Appendix E.5 compares the two conventions for a deterministic return. I re-solve the one-market economy of Section 5 with simple returns, taxed exactly, re-fitting risk aversion to restore the targeted taxable yields and holding the other coefficients fixed. The thirty-year gap rises from 102.6 to 107.7 basis points.

Funding and riskless claims

The funding constraint in (64) keeps cash nonnegative, so investors cannot borrow. Its multiplier $\beta_i(z)$ in (36) is zero whenever the investor holds cash, since an extra dollar can then stay in cash. A positive multiplier requires the investor to be fully invested, so that buying more of one bond requires selling another, and it raises the required pretax return by β_i/ξ^a_i on each held bond.

For a riskless one-period claim in any market $a \neq C$ with positive supply, the KKT conditions imply

$$\mu^a_1(z) = \min_i \frac{\xi^C_i r(z) + \beta_i(z)}{\xi^a_i}. \tag{65}$$

The minimum runs over investors eligible to hold the claim, so its return is their lowest required pretax return after taxes and funding. For the municipal claim, $\xi^M_H = 1$ and slack household funding give $\mu^M_1 = \xi_H r$ whenever the household attains this minimum.

B.2 The general price recursion

The portfolio conditions determine the equilibrium one-period return μ^a_T, which equals the return each of the bond's holders requires. The price recursion turns these returns into prices and yields in two layers, in which levels accumulate one-period returns and rate sensitivities accumulate their responses to the rate state.

With fixed supplies and funding, I seek an equilibrium log-price function for each market and maturity:

$$p^a_{T,t} = p^a_T(z_t), \qquad p^a_0(z) \equiv 0, \qquad a \in \mathcal{A}. \tag{66}$$

Let $Z' = \rho z + \sigma_r \varepsilon'$. Taking the log of the bond's conditional expected gross return gives

$$p^a_T(z) = \underbrace{\log \mathbb{E}_z\left[\exp\left\{p^a_{T-1}(Z')\right\}\right]}_{\text{log expected resale price}} + \underbrace{\tfrac{1}{2}(\sigma^a_{\eta,T})^2}_{\text{bond-specific payoff term}} - \underbrace{\mu^a_T(z)}_{\text{one-period return}}. \tag{67}$$

Equation (67) discounts the expected resale payoff at the equilibrium holding-period return, so a higher return lowers today's price. The payoff includes bond-specific risk and the price of a $(T-1)$-period bond.

This recursion links adjacent maturities, while the portfolio conditions (36) link markets through investors' holdings. Prices and portfolios solve these equations and clearing (37) jointly at every state. Appendix C gives the numerical implementation. Appendix equations (71)–(72) give the portfolio system with nonbinding funding.

The rate floor, changes in participation and binding funding constraints make prices nonlinear in the state. Resale carries these features across maturities, because today's continuation value averages future prices over states with different participation and funding conditions.

Iterating (67) from $p_0^a = 0$ expresses the price through the one-period returns the bond earns as it ages through maturities $T, T-1, \ldots, 1$:

$$P_T^a(z_t) = e^{p_T^a(z_t)} = \mathbb{E}_t\Big[\exp\Big\{-\sum_{j=0}^{T-1}\Big(\mu_{T-j}^a(z_{t+j}) - \tfrac{1}{2}\big(\sigma_{\eta,T-j}^a\big)^2\Big)\Big\}\Big]. \tag{68}$$

Taking the negative log of (68) and dividing by T gives the yield $y_T^a = -p_T^a/T$. In the affine benchmark, this becomes the explicit average of expected one-period returns less convexity in (40). In the general economy, the yield's dependence on the state is summarized by writing the log price in loading form,

$$p_T^a(z) = -A_T^a(z)\, z - C_T^a(z), \qquad A_T^a(z) := -\frac{\partial p_T^a(z)}{\partial z}, \tag{69}$$

so that $P_T^a = \exp\{-A_T^a z - C_T^a\}$ and, at differentiability points,

$$y_T^a(z) = \frac{A_T^a(z) z + C_T^a(z)}{T}, \qquad \frac{\partial y_T^a(z)}{\partial z} = \frac{A_T^a(z)}{T}.$$

The loading A_T^a measures the fall in log price per unit rise in the state, the model's counterpart of duration, and dividing by maturity gives the yield response. The level term C_T^a carries the price level, including accumulated one-period returns and convexity, and at the mean state the yield is $C_T^a(0)/T$. Both can vary with the state, with $\partial_z C_T^a = -z\,\partial_z A_T^a$ preserving the yield derivative.

The loading recursion links these coefficients to the one-period return response $s_T^a = \partial_z \mu_T^a$ of equation (38), the change in the one-period return required on a T-period bond when the rate state rises. Above the rate floor, cash has response one, and by (65) a riskless one-period claim in another market has response ξ_i^C/ξ_i^a, the active holder's cash after-tax share divided by its asset after-tax share, when funding is slack. At longer maturities the response also includes the change in the holders' risk premia. For locally absolutely continuous continuation prices, an integrable envelope for the Gaussian exponential moment and its state derivative permits differentiation under the expectation in (67):

$$A_T^a(z) = \underbrace{s_T^a(z)}_{\text{current return response}} + \underbrace{\rho\,\frac{\mathbb{E}_z\big[\exp\{p_{T-1}^a(Z')\}\, A_{T-1}^a(Z')\big]}{\mathbb{E}_z\big[\exp\{p_{T-1}^a(Z')\}\big]}}_{\text{continuation price response}}. \tag{70}$$

The fraction averages the continuation loading, giving more weight to states where the resale bond is more valuable.

B.3 Equilibrium conditions and the one-period boundary

The two-investor results assume $0 < \rho < 1$, $\sigma_r > 0$, $m_i, \gamma_i > 0$, $0 < \xi_H < \xi_I$ and positive traded supplies. $C1$ is riskless cash. Affine results use floorless rates and fixed support, whereas the numerical model retains the floor and funding constraints.

One KKT system for all markets. Index risky market–maturity coordinates by $j = (a, T)$ and write $v_j = (\sigma_{\eta,T}^a)^2$, $K_i = \mathrm{diag}(\xi_i^a)$, and the relative after-tax share $\xi_{ij} = \xi_i^C/\xi_i^a$, collected in the vector $\boldsymbol{\xi}_i$. In

the normalized two-market benchmark, $\xi_{ij} = \xi_i$ at municipal coordinates and $\xi_{ij} = 1$ at taxable coordinates. With $p_{j,-} = p^a_{T-1}$, the exact covariance and liability credit are

$$\Sigma_{jk}(z) = \mathrm{Cov}_z(p_{j,-}(Z'), p_{k,-}(Z')) + \mathbf{1}_{\{j=k\}} v_j, \qquad c^L_{ij}(z) = -\mathrm{Cov}_z(p_{j,-}(Z'), \ell_i \sigma_r \varepsilon'). \tag{71}$$

Dividing the first-order condition by the positive asset after-tax share gives

$$\begin{gathered} \nu_{ij} := \xi_{ij} r + \gamma_i [\Sigma K_i x_i - c^L_i]_j - \mu_j \geq 0, \qquad x_{ij} \geq 0, \qquad x_{ij}\nu_{ij} = 0, \\ \sum_i m_i x_{ij} = Q_j. \end{gathered} \tag{72}$$

Concavity of (64) makes these conditions sufficient for investor optimality. Positive bond-specific variances give strict concavity on risky coordinates, but do not establish equilibrium uniqueness.

Proposition 7 (Current-state clearing)**.** *Fix the state and continuation prices, with finite $\Sigma \succeq 0$, finite liability credits, and at least one eligible investor for every positive-supply coordinate.*

(i) The analytical system has a current-state KKT-and-clearing solution.

(ii) If, in addition, $\Sigma = V + \sigma_r^2 aa'$ with V positive diagonal and $a \gg 0$ on the risky coordinates, after-tax shares are positive and supported supplies are positive, the current-state clearing returns and risky holdings are unique, even with endogenous support changes. Deterministic one-period claims are treated separately. Payoff-equivalent holdings in tied boundary claims need not be unique.

Neither part establishes the dynamic fixed point of continuation prices.

Proof. (i) Let $X = \{x \geq 0 : \sum_i m_i x_i = Q\}$, with inner product $\langle x, y\rangle_m = \sum_i m_i x_i' y_i$. Positive masses and finite supplies make X nonempty, compact and convex. For $G_i(x_i) = \boldsymbol{\xi}_i r + \gamma_i(\Sigma K_i x_i - c^L_i)$, Brouwer gives a fixed point of the continuous map $x \mapsto \mathrm{Proj}_X(x - G(x))$. The projection characterization implies

$$\sum_i m_i \langle G_i(x_i), y_i - x_i\rangle \geq 0 \qquad (y \in X).$$

Transfers from positive holders to other eligible investors show that each positive holder minimizes G_{ij}. Thus $\mu_j = \min_i G_{ij}$ and $\nu_{ij} = G_{ij} - \mu_j$ satisfy (72). Conversely, complementarity implies the displayed inequality for every feasible transfer. Predetermined trading exclusions restrict eligibility. (ii) Strict concavity makes individual demand $x_i(\mu)$ continuous, Lipschitz, and piecewise affine. On an active set B_i, its derivative is

$$J_i = \gamma_i^{-1} E_{B_i} K^{-1}_{i,B_i} \Sigma^{-1}_{B_i B_i} E'_{B_i},$$

where E_B embeds active coordinates. Sherman–Morrison gives

$$\Sigma^{-1}_{BB} = V_B^{-1} - \frac{\sigma_r^2 V_B^{-1} a_B a_B' V_B^{-1}}{1 + \sigma_r^2 a_B' V_B^{-1} a_B}, \qquad \Sigma^{-1}_{BB} a_B = \frac{V_B^{-1} a_B}{1 + \sigma_r^2 a_B' V_B^{-1} a_B} \gg 0.$$

Every selection of $J = \sum_i m_i J_i$ therefore has nonpositive off-diagonal entries and $Ja \geq 0$, strictly on held coordinates. If μ^0, μ^1 both cleared, piecewise-affine continuity would give $0 = \overline{J}(\mu^1 - \mu^0)$, with $\overline{J}$ the segment-average selection matrix. Positive supply at μ^0 and demand continuity imply positive holdings on an initial segment for each asset. Finitely many selections then give $(\overline{J}a)_j > 0$ everywhere. Hence $\overline{J}\,\mathrm{diag}(a)$ is strictly diagonally dominant and nonsingular, so returns coincide, and strict concavity gives identical risky holdings. On support-boundary segments, adjacent selections have the same directional derivative. □

Nonlinear continuation covariances need not have this sign structure.

Regularity for derivatives. Derivatives below are taken at regular states, at which the cash rate and price moments are continuously differentiable, complementarity is strict on risky coordinates and the active

KKT-and-clearing Jacobian is nonsingular. The implicit-function theorem then gives locally fixed support and differentiable holdings and returns, with

$$\sum_{i\in\mathcal{U}_j} m_i \frac{\partial x_{ij}}{\partial z} = 0. \tag{73}$$

Individual strict concavity does not imply nonsingularity, because with heterogeneous after-tax shares the normalized operator contains ΣK_i, which need not be symmetric or strongly monotone.

Riskless claims with asset-specific taxes and funding. In any market $a \neq C$, retaining the funding multiplier in (36), $p_0^a = 0$ and $v_1^a = 0$ eliminate one-period covariance with portfolios and liabilities. The KKT conditions become

$$\mu_1^a \leq \frac{\xi_i^C r + \beta_i}{\xi_i^a}, \qquad \text{with equality for a positive holding.}$$

Positive supply makes the return the eligible-holder minimum in (65). The price recursion gives $p_1^a = -\mu_1^a$, hence $y_1^a = \mu_1^a$.

Proposition 8 (Deterministic one-period boundary)**.** *In the normalized municipal–taxable benchmark with nonbinding funding constraints, suppose the one-period municipal claim has $v_1^M = 0$ and positive supply, and $r(z) > 0$. Suppose the high-tax household has the minimal after-tax share, $\xi_H = \min_i \xi_i$. Then only investors with the minimal after-tax share can hold it, and*

$$y_1^M = \mu_1^M = \xi_H r(z), \qquad y_1^C = r(z), \qquad \tau_1^{Y,\text{impl}} = 1 - \xi_H. \tag{74}$$

If the high-tax household is the unique after-tax share minimizer, its tax passes through one-for-one and the tax of an investor with a strictly higher after-tax share has zero effect at this boundary. At a tie the household derivative requires perturbing all tied minimal after-tax shares together. At $r = 0$ the yield ratio is undefined.

Proof. With $\xi_i^M = 1$, $\xi_i^C = \xi_i$ and $\beta_i = 0$, the preceding KKT condition is $\mu_1^M \leq \xi_i r$, with equality for positive holdings. Positive supply gives $\mu_1^M = \min_i \xi_i r = \xi_H r$ and $p_1^M = -\mu_1^M$. Taxable cash has $p_1^C = -r$. Differentiating the yield ratio proves the tax statements. □

Bond-specific risk. Every market has $v_1^a = 0$. Positive risky variances give strict concavity but no uniform bound relative to rate risk, because the cash floor compresses rate variation and inventory premia depend on holdings. Appendices C.3 and C.4 check the funded boundary, grid, barrier and floor knot.

B.4 Affine solutions

Fix a support, set inactive positions to zero, set $r(z) = \bar{r} + z$, and use affine continuation prices on $\mathbb{R}$. Solve the active equalities jointly across maturities, retaining whole-portfolio risk. Then check nonnegative holdings, inactive slacks and, for finite budgets, $W_i - \sum_j x_{ij}(z) \geq 0$.

For affine candidates $p_T^a(z) = -A_T^a z - C_T^a$, Gaussian integration gives exactly

$$\log \mathbb{E}_z e^{p_{T-1}^a(Z')} = -\rho A_{T-1}^a z - C_{T-1}^a + \tfrac{1}{2}\sigma_r^2 (A_{T-1}^a)^2. \tag{75}$$

The current-state covariance $\Sigma = \sigma_r^2 AA' + \text{diag}(v)$, with A collecting predecessor loadings, and liability credit $c_i^L = \ell_i \sigma_r^2 A$ are constant. If the active KKT-and-clearing matrix is nonsingular, holdings and μ are therefore affine in z, and the price recursion returns affine prices with updated loadings $\rho A_{T-1}^a + s_T^a$. At a coefficient fixed point, all active equalities, clearing equations and exact continuation equations hold identically, and, with $A_0^a = 0$,

$$A_T^a = \rho A_{T-1}^a + s_T^a = \sum_{j=1}^{T} \rho^{T-j} s_j^a, \qquad s_T^a = \frac{\partial \mu_T^a}{\partial z}. \tag{76}$$

Affine holdings and slacks make the KKT-valid states an interval, possibly empty, conditional on the global affine continuation. Gaussian transitions leave any proper interval, so interval feasibility is not global equilibrium feasibility. For example, the household-only short-bond slack $(\xi_I - \xi_H)(\bar{r} + z)$ fails at negative rates. The cash floor changes continuation values.

For a fixed support, a continuous coefficient update on a nonempty compact convex invariant set has a fixed point by Brouwer. A contraction gives uniqueness within that set. Nonsingularity, invariance and reinstated KKT feasibility must be checked separately. The numerical diagnostics below verify the reported solution and supply no global existence argument.

B.5 Participation

Assumption 1 (Household duration burden at the benchmark)**.** *At the candidate allocation where the high-tax household holds all munis and the lower-tax investor all taxable bonds,*

$$\widehat{D}_H^{\mathrm{hh}} = \sum_T A_{T-1}^M Q_T^M / m_H - \ell_H, \quad \widehat{D}_I^{\mathrm{hh}} = \xi_I \sum_T A_{T-1}^C Q_T^C / m_I - \ell_I, \quad \Lambda := \gamma_H \widehat{D}_H^{\mathrm{hh}} - \gamma_I \widehat{D}_I^{\mathrm{hh}} > 0.$$

Proof of Proposition 1. The high-tax household's active municipal equality sets

$$\mu_T^{M,\mathrm{hh}} = \xi_H r + \gamma_H \sigma_r^2 A_{T-1}^M \widehat{D}_H^{\mathrm{hh}} + \gamma_H v_T^M Q_T^M / m_H.$$

Substituting it into the lower-tax investor's zero-holding condition gives its slack

$$\nu_{I,T}^{\mathrm{hh}} = (\xi_I - \xi_H) r - \sigma_r^2 A_{T-1}^M \Lambda - \gamma_H v_T^M Q_T^M / m_H. \tag{77}$$

Without bond-specific risk, the slack (77) satisfies $\nu_{I,T}^{\mathrm{hh}} \leq 0$ exactly when (45) holds. Sole-holder quantities are state invariant, giving $s_T^M = \xi_H$ and strictly increasing $A_T^M = \xi_H \sum_{q=0}^{T-1} \rho^q$. If $v_T^M Q_T^M$ is nondecreasing, the slack decreases with maturity and profitable participation forms an upper interval. The inventory term weakly expands it at fixed benchmark loadings and exposures.

Theorem 1 (Upper-interval participation by the lower-tax investor on an affine branch)**.** *At a regular state of a two-investor affine solution, suppose $v_T^M > 0$ on the risky maturities considered. Define the duration gap at the equilibrium allocation, $\widehat{\Lambda} = \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I$, and*

$$\Psi_T = (\xi_I - \xi_H) r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda} - \gamma_H v_T^M Q_T^M / m_H. \tag{78}$$

Then $x_{I,T}^M > 0$ if and only if $\Psi_T < 0$. If $\widehat{\Lambda} > 0$ and the predecessor loadings and $v_T^M Q_T^M$ are nondecreasing, the lower-tax investor's positive support is an upper interval, possibly empty. This identifies the cutoff within a given equilibrium and does not establish uniqueness across equilibria.

Proof. If the lower-tax investor is inactive, $x_{H,T}^M = Q_T^M / m_H$. Substituting the household equality into its inequality gives $\Psi_T = \nu_{I,T} \geq 0$. If both hold, subtract their equalities and use clearing:

$$\Psi_T = -v_T^M \left(\gamma_I + \gamma_H m_I / m_H\right) x_{I,T}^M < 0.$$

If the lower-tax investor alone is active, its equality and the household inequality imply $(\xi_I - \xi_H) r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda} \leq -\gamma_I v_T^M Q_T^M / m_I < 0$, which also gives $\Psi_T < 0$. These cases prove the equivalence. Finally,

$$\Psi_{T+1} - \Psi_T = -\sigma_r^2 (A_T^M - A_{T-1}^M) \widehat{\Lambda} - \frac{\gamma_H}{m_H} (v_{T+1}^M Q_{T+1}^M - v_T^M Q_T^M) \leq 0.$$

The first strictly negative index determines the positive-support cutoff. Equality is a zero-holding boundary. □

Joint holdings. For candidate durations, put $N_T = (\xi_I - \xi_H)r - \sigma_r^2 A_{T-1}^M \widehat{\Lambda}$. For $v_T := v_T^M > 0$, the difference of the two active KKT equalities and clearing give

$$x_{H,T}^M = \frac{\gamma_I Q_T^M + m_I N_T/v_T}{\gamma_H m_I + \gamma_I m_H}, \qquad x_{I,T}^M = \frac{\gamma_H Q_T^M - m_H N_T/v_T}{\gamma_H m_I + \gamma_I m_H}. \tag{79}$$

Both holdings are nonnegative exactly when $-\gamma_I Q_T^M/m_I \le N_T/v_T \le \gamma_H Q_T^M/m_H$. Each maturity's holdings depend on common durations, which must match the resulting portfolios. A nonzero tax difference $(\xi_I - \xi_H)r$ and distinct positive limiting loadings preclude a fixed block of two or more jointly held maturities as every $v_T \downarrow 0$, because bounded holdings require $N_T \to 0$ at each maturity, which assigns different values to the one common duration gap $\widehat{\Lambda}$. The coupled positive-variance formula below therefore describes no feasible joint-block equilibrium in this limit.

B.6 The pricing channels

Proof of Proposition 2. Investor i's active municipal equality (47) gives $\mu_T^M = \xi_i r + \gamma_i[\sigma_r^2 A_{T-1}^M \widehat{D}_i + (\sigma_{\eta,T}^M)^2 x_{i,T}^M]$. The lower-tax investor's taxable equality (48) gives $\mu_T^C = r + \gamma_I[\sigma_r^2 A_{T-1}^C \widehat{D}_I + \xi_I(\sigma_{\eta,T}^C)^2 x_{I,T}^C]$, also covering cash at $T = 1$ because $A_0^C = 0$. Subtract ξ_H times the second equality from the first, then add and subtract $\gamma_I \sigma_r^2 A_{T-1}^M \widehat{D}_I$. This gives the three-channel identity (49).

At $T = 1$, $A_0^a = 0$ and there is no bond-specific risk, so the duration-cost, rate-loading and bond-specific terms vanish. When $r > 0$, the household alone holds the claim (Proposition 8), so $i = H$ and the after-tax cash-return difference also vanishes.

If the household alone holds maturity T, then $i = H$ and $x_{H,T}^M = Q_T^M/m_H$. The lower-tax investor's inactive condition gives $\Psi_T \ge 0$ in (78), with $v_T^M = (\sigma_{\eta,T}^M)^2$, or $\sigma_r^2 A_{T-1}^M \widehat{\Lambda} + \gamma_H(\sigma_{\eta,T}^M)^2 Q_T^M/m_H \le (\xi_I - \xi_H)r$. The duration-cost contribution is the first term on the left and is therefore at most $(\xi_I - \xi_H)r$. If the lower-tax investor holds the bond, use $i = I$ in the identity. The duration-cost contribution then vanishes.

Proposition 9 (Exact active-clientele aggregation)**.** *At a regular state, let* $\Gamma_j = \sum_{i\in\mathcal{U}_j} m_i/\gamma_i$, $\bar{\xi}_j = \Gamma_j^{-1}\sum_{i\in\mathcal{U}_j} m_i\xi_{ij}/\gamma_i$, *and* $\mathcal{C}_j = \Gamma_j^{-1}\sum_{i\in\mathcal{U}_j} m_i[\Sigma K_i x_i - c_i^L]_j$. *Then*

$$\begin{aligned} \mu_j &= \bar{\xi}_j r + \mathcal{C}_j, \\ s_j &= \bar{\xi}_j r' + \frac{1}{\Gamma_j}\sum_{i\in\mathcal{U}_j} m_i\left[(\partial_z\Sigma)K_i x_i + \Sigma K_i \partial_z x_i - \partial_z c_i^L\right]_j. \end{aligned} \tag{80}$$

In the normalized municipal–taxable benchmark, above the cash floor the response difference is

$$s_T^M - \xi_H s_T^C = (\bar{\xi}_T^M - \xi_H) + \partial_z \mathcal{C}_T^M - \xi_H \partial_z \mathcal{C}_T^C. \tag{81}$$

Proof. Weight the active equalities $\mu_j - \xi_{ij} r = \gamma_i[\Sigma K_i x_i - c_i^L]_j$ by m_i/γ_i, sum and divide by Γ_j. Differentiating at locally fixed support gives the second line. In the two-market normalization, $\bar{\xi}_j = \bar{\xi}_T^M$ at municipal and $\bar{\xi}_j = 1$ at taxable coordinates. Subtraction gives the gap. □

With binding funding, each active equality also contains β_i/ξ_i^a by (36), so the return identity in (80) adds $\Gamma_j^{-1}\sum_{i\in\mathcal{U}_j} m_i\beta_i/(\gamma_i\xi_i^a)$ and the response identity adds its state derivative. A per-dollar charge such as the bank financing-tax cost appears in the return identity in the same units and, when it does not vary with the rate state, leaves the response identity unchanged on a given support.

Proof of Proposition 3. With $\widehat{D}_i = A' K_i x_i - \ell_i$, constant affine covariance and liability credits reduce (80) to

$$s_j = \bar{\xi}_j + \frac{\sigma_r^2 A_j}{\Gamma_j}\sum_{i\in\mathcal{U}_j} m_i \frac{\partial \widehat{D}_i}{\partial z} + \frac{v_j}{\Gamma_j}\sum_{i\in\mathcal{U}_j} m_i \xi_i^a \frac{\partial x_{ij}}{\partial z}, \tag{82}$$

where $A_j = A_{T-1}^a$ is the predecessor loading. For normalized munis, (73) cancels the final term, leaving the tax-mix and portfolio-risk terms in the holding-return response.

Taxable counterpart. With $\xi_{ij} = 1$ at taxable coordinates, the taxable version of (82) has tax term one. Its bond-specific term vanishes in three cases. First, with a singleton taxable active set, clearing fixes the sole holder's quantity. Second, equal active taxable after-tax shares factor out of (73). Third, it vanishes when $v_T^C = 0$. Heterogeneous after-tax shares with multiple holders prevent cancellation by unweighted clearing. With the lower-tax investor the sole holder of the taxable bond,

$$s_T^C = 1 + \gamma_I \sigma_r^2 A_{T-1}^C \partial_z \widehat{D}_I. \tag{83}$$

Proposition 10 (Joint participation and propagation across maturities)**.** *At a regular state with $r(z) > 0$ in the two-investor affine benchmark, suppose the lower-tax investor alone holds risky taxable bonds. Let $\varnothing \neq J \subseteq \{2, \dots, T_{\max}\}$ be the fixed set of municipal maturities held by both investors, with the high-tax household alone holding every municipal maturity outside J, including the riskless one-period claim. Assume $A_{T-1}^M, A_{T-1}^C > 0$ for every $T \geq 2$ and $v_T^M > 0$ on J. Define*

$$P_J = \sum_{T \in J} \frac{A_{T-1}^M}{v_T^M}, \quad S_J = \sum_{T \in J} \frac{(A_{T-1}^M)^2}{v_T^M}, \quad d := \frac{\partial \widehat{D}_H}{\partial z}.$$

Then

$$d = \frac{(\xi_I - \xi_H) m_I P_J}{(\gamma_H m_I + \gamma_I m_H)(1 + \sigma_r^2 S_J)} > 0, \qquad \frac{\partial \widehat{D}_I}{\partial z} = -\frac{m_H}{m_I} d. \tag{84}$$

Writing $\omega_i = (m_i/\gamma_i)/(m_H/\gamma_H + m_I/\gamma_I)$, $\xi^* = \omega_H \xi_H + \omega_I \xi_I$, and $b = \gamma_I \sigma_r^2 (m_H/m_I) d$, the slopes are

$$s_T^M = \begin{cases} \xi_H + \gamma_H \sigma_r^2 A_{T-1}^M d, & T \notin J, \\ \xi^*, & T \in J, \end{cases} \qquad s_T^C = 1 - b A_{T-1}^C. \tag{85}$$

Hence $s_T^M - \xi_H s_T^C > 0$ and $A_T^M - \xi_H A_T^C > 0$ at every risky maturity $T \geq 2$, with equality in both comparisons at $T = 1$.

Proof. Clearing fixes sole-held quantities at Q_T^M/m_H for households outside J and Q_T^C/m_I for the lower-tax investor's risky taxable bonds. Only joint muni holdings vary. Differentiate clearing and sum against municipal predecessor loadings. Constant liabilities and the fixed taxable book give

$$m_H \partial_z \widehat{D}_H + m_I \partial_z \widehat{D}_I = 0. \tag{86}$$

At each $T \in J$, subtracting the lower-tax investor's differentiated municipal equality from the household's gives

$$\xi_I - \xi_H = \sigma_r^2 A_{T-1}^M (\gamma_H \partial_z \widehat{D}_H - \gamma_I \partial_z \widehat{D}_I) + v_T^M (\gamma_H \partial_z x_{H,T}^M - \gamma_I \partial_z x_{I,T}^M).$$

With $u_T = \partial_z x_{I,T}^M$ and $k = (\gamma_H m_I + \gamma_I m_H)/m_H$, clearing reduces this to $u_T = -(\xi_I - \xi_H)/(k v_T^M) - \sigma_r^2 A_{T-1}^M \partial_z \widehat{D}_I / v_T^M$. Since $\partial_z \widehat{D}_I = \sum_{T \in J} A_{T-1}^M u_T$, multiplying by A_{T-1}^M and summing over J gives

$$(1 + \sigma_r^2 S_J) \partial_z \widehat{D}_I = -\frac{\xi_I - \xi_H}{k} P_J < 0.$$

Equation (86) gives (84). Household-only quantities have zero derivatives, giving the first muni slope. At joint maturities, risk-tolerance aggregation cancels aggregate exposure and inventory derivatives and leaves ξ^*, and (83) gives the taxable formula.

For propagation, (85) gives the response gap $\partial_z g_T = s_T^M - \xi_H s_T^C$ as

$$\partial_z g_T = \begin{cases} \sigma_r^2 d \left[\gamma_H A_{T-1}^M + \xi_H \gamma_I \dfrac{m_H}{m_I} A_{T-1}^C \right], & T \notin J, \\ \omega_I (\xi_I - \xi_H) + \xi_H b A_{T-1}^C, & T \in J. \end{cases}$$

Thus $\partial_z g_T > 0$ for every $T \geq 2$, while $\partial_z g_1 = 0$ because both predecessor loadings vanish. Iterating (76), with $0 < \rho < 1$, yields

$$A_T^M - \xi_H A_T^C = \sum_{j=2}^{T} \rho^{T-j} \partial_z g_j > 0, \qquad T \geq 2.$$

□

The signed reallocation is the lower-tax investor's duration-weighted portfolio change. Individual jointly held positions can move in either direction.

State-independent holding benefits. Suppose an investor's objective includes $\sum_{a,T} b_{i,T}^a x_{i,T}^a$, with each $b_{i,T}^a$ independent of the rate state. Its active municipal condition becomes

$$\mu_T^M - \xi_i r + b_{i,T}^M = \gamma_i \left[\sigma_r^2 A_{T-1}^M \widehat{D}_i + v_T^M x_{i,T}^M \right],$$

and the taxable condition similarly gains $b_{i,T}^C / \xi_i$. With covariance, financing and constraints fixed, differentiation removes state-independent benefits and liabilities. The preceding proof and loading equations hold on the same feasible joint-holding set. Benefits and liabilities affect return intercepts, gap levels (Appendix B.13) and feasible support. A given loading solution is therefore unchanged on the same branch. Equality across distinct coefficient fixed points additionally requires uniqueness. State-dependent benefits add their derivatives. Fixed participation fees affect global participation without a per-unit KKT term.

Uniqueness of the loading fixed point. For given d, generate municipal loadings from (85) and the loading recursion (76), and let $\mathcal{F}_J(d)$ be the right side of (84) at those loadings. If $\mathcal{F}_J$ maps $[0, \infty)$ into $[0, d_{\max}]$ and $\sup_{[0,d_{\max}]} |\mathcal{F}_J'| < 1$, every nonnegative exposure fixed point lies in $[0, d_{\max}]$, and Banach gives exactly one, with its loading path on this branch. Holdings and inactive inequalities still require verification. Other feasible cutoffs remain possible. The derivative of $\mathcal{F}_J$ is

$$\mathcal{F}_J'(d) = c \frac{P_J'(1 + \sigma_r^2 S_J) - \sigma_r^2 P_J S_J'}{(1 + \sigma_r^2 S_J)^2}, \qquad c = \frac{(\xi_I - \xi_H) m_I}{\gamma_H m_I + \gamma_I m_H},$$

with the derivatives of P_J, S_J obtained by differentiating the same loading recursions.

Proposition 11 (No-participation loading benchmark)**.** *In the two-investor affine benchmark, if the high-tax household alone holds every muni and the lower-tax investor alone holds every taxable bond, then $s_T^M = \xi_H$, $s_T^C = 1$, and $A_T^M = \xi_H A_T^C$ at every maturity.*

Proof. Clearing fixes every holding, so both duration derivatives are zero. Apply (82) and iterate (76). The identity concerns loadings, and different intercept risk premia can still affect the yield ratio. □

B.7 Loading and yield ratios

The sufficient condition and the threshold loading. Specialize Proposition 10 to the upper interval $J = \{T^*, \ldots, T_{\max}\}$. The maturity ordering uses condition (S) and the bound (M) on the loading at the threshold:

$$\textbf{(S)} \quad \frac{\gamma_I m_H (\xi_I - \xi_H)}{\xi_H (\gamma_H m_I + \gamma_I m_H)} \leq \rho, \tag{87}$$

$$\textbf{(M)} \quad A_{T^*-1}^M < \frac{\xi^*}{1 - \rho}. \tag{88}$$

Condition (S) is assumed. The bound (M) holds at every solution of (84) on this branch. Put $n = T^* - 1$ and $\lambda = \gamma_H \sigma_r^2 d$, so that (84) reads $\lambda = \omega_I (\xi_I - \xi_H) \sigma_r^2 P_J / (1 + \sigma_r^2 S_J)$. The branch recursions give $A_n^M = \xi_H \sum_{q=0}^{n-1} (\rho + \lambda)^q$ and $A_{n+l}^M = \rho^l A_n^M + (1 - \rho^l) \xi^* / (1 - \rho)$ for $l \geq 0$. If $A_n^M \geq \xi^* / (1 - \rho)$, every loading on J is at least $\xi^* / (1 - \rho)$, so $S_J / P_J \geq \xi^* / (1 - \rho)$ and $\lambda < (1 - \rho)(\xi^* - \xi_H) / \xi^* < 1 - \rho$, using $\xi^* - \xi_H = \omega_I (\xi_I - \xi_H)$.

Then $A_n^M < \xi_H/(1-\rho-\lambda) < \xi^*/(1-\rho)$, a contradiction. Both statements concern the upper-interval branch of the two-investor affine benchmark and do not prove that this support exists.

Proof of Proposition 4(i). On the household branch, $A_T^M = (\rho+\gamma_H\sigma_r^2 d)A_{T-1}^M+\xi_H$, so A_T^M is increasing from zero. On the joint branch, $A_T^M = \rho A_{T-1}^M + \xi^*$, whose increments have the sign of $\xi^* - (1-\rho)A_{T^*-1}^M$. Condition (M) makes them positive. Since $A_{T-1}^M \geq \xi_H$ on J, $P_J/S_J \leq 1/\xi_H$, and (84) and (S) give $0 \leq b \leq \rho$. Therefore

$$A_T^C = \sum_{q=0}^{T-1}(\rho-b)^q, \qquad s_T^C = 1 - bA_{T-1}^C \downarrow \frac{1-\rho}{1-\rho+b} > 0.$$

The municipal slope rises on the household branch, is constant on the joint branch, and does not jump down when the lower-tax investor begins to participate. Substituting (84) shows that the required inequality $\omega_I(\xi_I - \xi_H) \geq \gamma_H\sigma_r^2 A_{T^*-2}^M d$ is equivalent to $\sigma_r^2 A_{T^*-2}^M P_J \leq 1 + \sigma_r^2 S_J$, which follows from $A_{T-1}^M > A_{T^*-2}^M$ for every $T \in J$, established above using (M).

For any positive e_j and nondecreasing ratios $q_j = a_j/e_j$, the discounted ratio

$$\bar{q}_T = \frac{\sum_{j\leq T}\rho^{T-j}a_j}{\sum_{j\leq T}\rho^{T-j}e_j}$$

is nondecreasing, because it is a positive-weight average of $q_1, \ldots, q_T$, and

$$\bar{q}_{T+1} - \bar{q}_T = \frac{e_{T+1}(q_{T+1} - \bar{q}_T)}{\rho\sum_{j\leq T}\rho^{T-j}e_j + e_{T+1}} \geq 0.$$

Applying this with $a_j = s_j^M$, $e_j = s_j^C$ and (76) proves the loading-ratio ordering, starting from ξ_H. Extending the same branch recursions to arbitrarily long maturities with fixed d and eventual joint participation gives the limits $A_\infty^M = \xi^*/(1-\rho)$ and $A_\infty^C = 1/(1-\rho+b)$, hence the endpoint in (60). This extends the branch and asserts no existence for an infinite-supply economy. Taking $e_j = 1$, for which (M) suffices, proves by the same argument that A_T^M/G_T with $G_T = \sum_{j\leq T}\rho^{T-j}$ is nondecreasing, the treasury-reference ordering of P3 in Section 4.6.

Proposition 12 (Exact forward-ratio bridge)**.** *For any price functions, affine or nonlinear, let* $f_T^a(z) = -p_T^a(z) + p_{T-1}^a(z)$. *If* $f_T^C(z) > 0$ *and* $f_T^M(z)/f_T^C(z)$ *is nondecreasing over the maturities considered, then* $y_T^M(z)/y_T^C(z)$ *is nondecreasing and* $\tau_T^{Y,\text{impl}}(z)$ *is nonincreasing.*

Proof. Telescoping gives $Ty_T^a = -p_T^a = \sum_{j\leq T} f_j^a$. The yield ratio averages forward ratios with positive weights $f_j^C > 0$. Adding a ratio at least as large as its predecessors cannot lower the average. On a two-year grid, T counts steps and annualization cancels. □

The premise concerns full forwards, including inventory and convexity terms, so loading monotonicity alone does not establish it. The nonlinear maturity profiles are checked numerically in Section 5.

Affine forward levels and Proposition 4(iii). For coordinate $j = (a, T)$, put $\Gamma_T^a := \sum_{i\in\mathcal{U}_j} m_i/\gamma_i$ and

$$\bar{\xi}_T^a = \bar{\xi}_j = \frac{\sum_i(m_i/\gamma_i)\xi_{ij}}{\Gamma_T^a}, \quad \pi_T^a = \frac{\sigma_r^2}{\Gamma_T^a}\sum_i m_i\widehat{D}_i(0), \quad \iota_T^a = v_j\left[\frac{\sum_i m_i\xi_i^a x_{i,T}^a(0)}{\Gamma_T^a} - \tfrac{1}{2}\right],$$

where sums run over active holders. Aggregating their level KKT equations at $z = 0$ and subtracting the two Jensen terms in the exact Gaussian mean gives

$$C_T^a - C_{T-1}^a = \bar{\xi}_T^a\bar{r} + A_{T-1}^a\pi_T^a - \tfrac{1}{2}\sigma_r^2(A_{T-1}^a)^2 + \iota_T^a. \tag{89}$$

For munis the weighted holding sum is Q_T^M. For taxable bonds heterogeneous after-tax shares remain inside it. If the displayed coefficients and loadings converge and the limiting taxable forward f_∞^C is positive, Cesàro convergence gives

$$y_T^a(z) \longrightarrow f_\infty^a = \bar{\xi}_\infty^a\bar{r} + A_\infty^a\pi_\infty^a - \tfrac{1}{2}\sigma_r^2(A_\infty^a)^2 + \iota_\infty^a \tag{90}$$

at fixed z, because C_T^a/T converges to the forward limit and $A_T^a z/T \to 0$. On the joint-holding branch, $\bar{\xi}_\infty^M = \xi^*$, and $\bar{\xi}_\infty^C = 1$ because relative after-tax shares on taxable bonds equal one, so the implied-tax limit $1 - f_\infty^M/f_\infty^C$ is (61). Aggregating the joint muni and lower-tax-investor-only taxable branches by risk tolerance gives

$$\pi_\infty^M - \pi_\infty^C = \omega_H \, \sigma_r^2 \, \widehat{\Lambda}, \qquad \widehat{\Lambda} = \gamma_H \widehat{D}_H - \gamma_I \widehat{D}_I, \tag{91}$$

where $\widehat{D}_i = \widehat{D}_i(0)$ is stationary-mean net rate exposure. The premium difference alone cannot bound other level corrections. With small taxable loading feedback ($b \ll 1-\rho$), negligible convexity and inventory terms, and a positive denominator, the first-order endpoint is

$$\tau_\infty^{Y,\text{impl,aff}} \approx \underbrace{(1-\xi^*)}_{\text{clientele-mix tax}} - \underbrace{\frac{\xi^* \, \omega_H \, \sigma_r^2 \, \widehat{\Lambda}}{(1-\rho)\bar{r} + \pi_\infty^C}}_{\text{muni term-premium excess}}. \tag{92}$$

With $\widehat{\Lambda} > 0$, this first-order endpoint therefore lies below the clientele-mix tax $1 - \xi^*$ and hence below its one-period value $1 - \xi_H$.

B.8 Tax pass-through

Proof of Proposition 6.

(i) The riskless short end. The one-period claims have zero variance. With slack household funding, Proposition 8 gives $y_1^M = \xi_H r$ and $y_1^C = r$ at every rate state with $r > 0$, so $y_1^M/y_1^C = A_1^M/A_1^C = \xi_H$. Differentiating the two ratios gives $\mathcal{P}_H^Y(1) = \mathcal{P}_H^A(1) = 1$ and $\mathcal{P}_I^Y(1) = \mathcal{P}_I^A(1) = 0$.

(ii) The long end. Let d be the household's exposure response in (84) and $b = \gamma_I \sigma_r^2 (m_H/m_I) d$, as in (85). The solved long responses and loadings satisfy

$$s_\infty^M = \xi^*, \qquad s_\infty^C = \frac{1-\rho}{1-\rho+b}, \qquad A_\infty^a = \frac{s_\infty^a}{1-\rho}.$$

Thus $R_\infty = A_\infty^M/A_\infty^C = \xi^*/s_\infty^C$. At the mean state, the solved yield formula (90) gives

$$\begin{aligned} y_\infty^M &= \xi^* \bar{r} + A_\infty^M \pi_\infty^M - \tfrac{1}{2}\sigma_r^2 (A_\infty^M)^2 + \iota_\infty^M, \\ y_\infty^C &= \bar{r} + A_\infty^C \pi_\infty^C - \tfrac{1}{2}\sigma_r^2 (A_\infty^C)^2 + \iota_\infty^C. \end{aligned}$$

The terms are interest income, portfolio rate-risk compensation, convexity and bond-specific compensation. Dividing the two yields separates the loading ratio from the remaining yield-level difference:

$$\frac{y_\infty^M}{y_\infty^C} = R_\infty + \frac{y_\infty^M - R_\infty y_\infty^C}{y_\infty^C}.$$

All derivatives below are with respect to ξ_i, with holdings, loadings and both yields adjusting. The assumed convergence permits differentiation of the limiting ratio. The quotient rule gives

$$\begin{aligned} \mathcal{P}_i^Y(\infty) = \mathcal{P}_i^A(\infty) &+ \frac{1}{y_\infty^C} \frac{d(y_\infty^M - R_\infty y_\infty^C)}{d\xi_i} \\ &- \frac{y_\infty^M - R_\infty y_\infty^C}{(y_\infty^C)^2} \frac{dy_\infty^C}{d\xi_i}. \end{aligned} \tag{93}$$

The three terms are the loading-ratio response, the response of relative yield compensation and the response of the taxable denominator. The first retains the endogenous taxable response:

$$\mathcal{P}_i^A(\infty) = \frac{\omega_i}{s_\infty^C} - \frac{\xi^*}{(s_\infty^C)^2} \frac{ds_\infty^C}{d\xi_i}.$$

Lemma 1 proves $\mathcal{P}_H^A(\infty) < \omega_H$ and $\mathcal{P}_I^A(\infty) > \omega_I$ under the tax-difference condition, with uniform positive margins. The calculation below shows that $y_\infty^M - R_\infty y_\infty^C$ and its tax derivatives are independent of $\bar{r}$. Also, $y_\infty^C = A_\infty^C(1-\rho)\bar{r} + O(1)$ and $dy_\infty^C/d\xi_i = O(\bar{r})$. Both correction terms in (93) therefore vanish at rate $1/\bar{r}$. The loading and derivative bounds in the lemma are uniform over all feasible cutoffs, so a threshold depending only on the other primitives makes both yields positive and the corrections smaller than the loading margins. This proves $\mathcal{P}_H^Y(\infty) < \omega_H < 1$ and $\mathcal{P}_I^Y(\infty) > \omega_I > 0$.

(iii) The municipal-to-treasury response ratio. The treasury loading is that of the reference curve, $A_T^{\text{Tsy}} = G_T$. At the short end, $A_1^M = \xi_H$ and $A_1^{\text{Tsy}} = 1$. At the long end, $A_\infty^M = \xi^*/(1-\rho)$ and $A_\infty^{\text{Tsy}} = 1/(1-\rho)$. Their ratio is $\xi^* > \xi_H$. Proposition 4, proved in Appendix B.7, establishes monotonicity between these endpoints. □

Lemma 1 (The loading response)**.** *Under the tax-difference condition in Proposition 6, $\mathcal{P}_H^A(\infty) < \omega_H$ and $\mathcal{P}_I^A(\infty) > \omega_I$. With other primitives fixed, both strict margins have positive lower bounds over all feasible upper-interval cutoffs.*

Proof. For the finite loading calculation, put $n = T^* - 1$ and $\lambda = \gamma_H \sigma_r^2 d$. Equation (84) gives

$$\lambda = \omega_I(\xi_I - \xi_H)\sigma_r^2 f, \qquad f = \frac{P_J}{1+\sigma_r^2 S_J}, \qquad R_\infty = \xi^*\left[1 + \frac{\omega_H \lambda}{\omega_I(1-\rho)}\right].$$

The sums P_J and S_J are those of Proposition 10. With $G_n(p) = \sum_{q=0}^{n-1} p^q$, the loading recursion gives

$$A_n^M = \xi_H G_n(\rho+\lambda), \qquad A_{n+l}^M = \rho^l A_n^M + (1-\rho^l)A_\infty^M, \quad l \geq 0.$$

By (88), $A_n^M < A_\infty^M$. Thus $S_J/P_J \geq A_n^M$ and

$$\lambda G_n(\rho+\lambda) < \omega_I e, \qquad \lambda \frac{G_n'(\rho+\lambda)}{G_n(\rho+\lambda)} < \omega_I e,$$

where the second bound follows from $G_n' \leq G_n^2$. The tax-difference condition implies $e < 1$ and hence $b = (\omega_H/\omega_I)\lambda < \omega_H e < 1$, ensuring convergence of the taxable loadings.

To differentiate the loading sums, let $t_j = \rho^l A_n^M / A_{j-1}^M$ for $j = T^* + l$ and

$$X = \bar{t}_P - \frac{2\sigma_r^2 S_J}{1+\sigma_r^2 S_J}\bar{t}_S.$$

Here $\bar{t}_P$ and $\bar{t}_S$ are the means of t_j with weights proportional to A_{j-1}^M/v_j^M and $(A_{j-1}^M)^2/v_j^M$, respectively. Since $0 \leq t_j \leq 1$, $-2 < X \leq 1$. Let ∂_i denote differentiation with respect to ξ_i holding λ fixed. Differentiating the loadings gives

$$\partial_H \ln f = \left(\frac{1}{\xi_H} - \frac{\omega_H}{\xi^*}\right)X + \frac{\omega_H}{\xi^*}\frac{1-\sigma_r^2 S_J}{1+\sigma_r^2 S_J},$$
$$\partial_I \ln f = \frac{\omega_I}{\xi^*}\left(\frac{1-\sigma_r^2 S_J}{1+\sigma_r^2 S_J} - X\right).$$

Hereafter G_n and G_n' are evaluated at $\rho+\lambda$. Differentiating the fixed point then gives

$$\frac{d\lambda}{d\xi_i} = \frac{\lambda\,\partial_i \ln[(\xi_I - \xi_H)f]}{D}, \qquad D = 1 - \lambda\frac{G_n'}{G_n}X > 1 - \omega_I e > 0.$$

Substitute into $dR_\infty/d\xi_i = \mathcal{P}_i^A(\infty)$. The coefficient of X is positive in the household derivative and negative in the lower-tax derivative. Using $X \le 1$ and $\lambda G_n'/G_n < \omega_I e$ gives

$$\mathcal{P}_H^A(\infty) - \omega_H \le \frac{\omega_H \lambda}{\omega_I(1-\rho)D}\left(\frac{2\omega_H}{1+\sigma_r^2 S_J} - \frac{1}{e} + \omega_I e\right) < 0,$$
$$\mathcal{P}_I^A(\infty) - \omega_I > \frac{\omega_H \lambda}{\omega_I(1-\rho)D}(1/e - \omega_I^2 e) > 0.$$

The tax-difference condition makes the household bracket at most $-2\omega_H\sigma_r^2 S_J/(1+\sigma_r^2 S_J)$, giving a positive margin for each investor. For uniformity, consider $n \in \{1, \dots, T_{\max} - 1\}$ in the finite loading system and trial feedbacks $x \ge 0$ with $xG_n(\rho + x) \le \omega_I e$. This compact primitive domain contains every actual pair. Its positive loading sums give positive minima for f and $\sigma_r^2 S_J/(1+\sigma_r^2 S_J)$, while $D \ge 1 - \omega_I e$ keeps full tax derivatives bounded. At a fixed point, $\lambda \ge \omega_I(\xi_I - \xi_H)\sigma_r^2 \min f > 0$ and $D < 1 + 2\omega_I e$. The displayed margins consequently have uniform positive lower bounds. □

The relative yield compensation. Subtract the two solved long yields, weighting the taxable yield by $R_\infty = A_\infty^M/A_\infty^C$. Equation (91) gives $\pi_\infty^M - \pi_\infty^C = \omega_H\sigma_r^2\widehat{\Lambda}$, while $\xi^* - R_\infty = -bA_\infty^M$. Therefore

$$y_\infty^M - R_\infty y_\infty^C = A_\infty^M(\omega_H\sigma_r^2\widehat{\Lambda} - b\bar{r}) + \tfrac{1}{2}\sigma_r^2 A_\infty^M(A_\infty^C - A_\infty^M) + \iota_\infty^M - R_\infty\iota_\infty^C.$$

The three components are the difference in risk costs on the sole-held books, relative convexity and bond-specific compensation. Subtracting the lower-tax investor's municipal equality from the household's on the shared block and using clearing gives

$$\omega_H\sigma_r^2\widehat{\Lambda} - b\bar{r} = \frac{\omega_H\sigma_r^2}{1+\sigma_r^2 S_J}\left[\gamma_H\left(\sum_{T\notin J}\frac{A_{T-1}^M Q_T^M}{m_H} - \ell_H\right) - \gamma_I\left(\frac{\xi_I}{m_I}\sum_T A_{T-1}^C Q_T^C - \ell_I\right)\right].$$

Every term is independent of $\bar{r}$ on the maintained branch, including its tax derivative. Common duration compensation per unit of loading cancels from $y_\infty^M - R_\infty y_\infty^C$. The joint holders' risk-premium coefficient is also independent of $\bar{r}$. With $\Gamma := m_H/\gamma_H + m_I/\gamma_I$, the value of Γ_T^M on the jointly held block, clearing gives

$$\pi_\infty^M = \frac{\sigma_r^2}{\Gamma}\left(\sum_T A_{T-1}^M Q_T^M + \xi_I\sum_T A_{T-1}^C Q_T^C - m_H\ell_H - m_I\ell_I\right).$$

Since $1/A_\infty^C = 1 - \rho + b$, dividing the taxable yield by its loading cancels $b\bar{r}$ against the mean-rate term in π_∞^C. This establishes $y_\infty^C = A_\infty^C(1-\rho)\bar{r} + O(1)$ used above. The bond-specific terms are $\iota_\infty^M = v_\infty^M(Q_\infty^M/\Gamma - \frac{1}{2})$ and $\iota_\infty^C = v_\infty^C(\gamma_I\xi_I Q_\infty^C/m_I - \frac{1}{2})$.

For the illustration's convexity accounting,

$$\Delta\tau_\infty := \frac{y_\infty^M}{y_\infty^C} - R_\infty = R_\infty(1-R_\infty)\chi_\infty + \psi_\infty,$$
$$\chi_\infty = \frac{\frac{1}{2}\sigma_r^2(A_\infty^C)^2}{y_\infty^C}, \qquad \psi_\infty = \frac{A_\infty^M(\omega_H\sigma_r^2\widehat{\Lambda} - b\bar{r}) + \iota_\infty^M - R_\infty\iota_\infty^C}{y_\infty^C}.$$

The signed convexity contribution is at most $\chi_\infty/4$, since $R_\infty(1-R_\infty) \le 1/4$. The derivatives of the two components sum to $\mathcal{P}_i^Y(\infty) - \mathcal{P}_i^A(\infty)$ in (93).

The tax-mix component across maturities. The following supporting result supplies the intermediate-maturity profile used in Section 4.6 and the state-specific extension.

Proposition 13 (Tax-mix pass-through)**.** *Switch off exposure-response terms in the affine pricing recursions, with support and risk-tolerance weights fixed. Put*

$$G_T = \sum_{j=1}^{T} \rho^{T-j}, \qquad \Theta_S(T) = \frac{\sum_{j=1}^{\min(T,T^*-1)} \rho^{T-j}}{G_T}, \qquad \Theta_L(T) = 1 - \Theta_S(T).$$

For the loading-implied tax τ_T^A of Appendix A.8, the tax-mix pass-throughs $\mathcal{D}_i(T) := -\partial_{\xi_i}\tau_T^A$, the part of $\mathcal{P}_i^A(T)$ that remains with exposure responses switched off, are

$$\mathcal{D}_H(T) = \Theta_S(T) + \omega_H \Theta_L(T), \qquad \mathcal{D}_I(T) = \omega_I \Theta_L(T). \tag{94}$$

They equal $(1,0)$ before participation and converge monotonically to (ω_H, ω_I).

Proof. The channel-off recursions give $A_T^C = G_T$ and

$$A_T^M = \xi_H \sum_{j=1}^{\min(T,T^*-1)} \rho^{T-j} + \xi^* \sum_{j=T^*}^{T} \rho^{T-j}.$$

Differentiating A_T^M/G_T gives (94). For $T \geq T^*$, $\Theta_S(T+1) = \Theta_S(T)\rho G_T/(\rho G_T+1) < \Theta_S(T)$ and $\Theta_S(T) \to 0$. □

Price derivatives in the nonlinear model. On a differentiable nonlinear equilibrium branch, let θ be a tax parameter, such as an after-tax share ξ_i, and put $q_T^a = \partial_\theta p_T^a$ and $\upsilon_T^a = -\partial_\theta \mu_T^a$, including portfolio and covariance responses. Hold the state process, cash rule and bond-specific variances fixed. Under a common integrable envelope for the exponential moments and their derivatives, differentiation of the price recursion gives

$$q_T^a = \mathcal{K}_T^a q_{T-1}^a + \upsilon_T^a, \qquad q_0^a = 0, \qquad (\mathcal{K}_T^a \psi)(z) = \mathbb{E}^{\mathbb{Q}_{T,z}^a} \psi(Z'), \tag{95}$$

where F_z is the conditional law of Z' and $d\mathbb{Q}_{T,z}^a/dF_z = e^{p_{T-1}^a(Z')}/\mathbb{E}_z e^{p_{T-1}^a(Z')}$. The identity follows from $\partial_\theta \log \mathbb{E}_z e^{p_{T-1}^a} = \mathbb{E}^{\mathbb{Q}_{T,z}^a} q_{T-1}^a$. The full implied-tax response is

$$\partial_\theta \tau_T^{Y,\text{impl}} = \frac{q_T^M - (y_T^M/y_T^C) q_T^C}{T y_T^C}, \qquad y_T^C > 0. \tag{96}$$

Both price responses therefore enter the empirical outcome.

B.9 State-specific exemptions and additional clienteles

Multi-state aggregation. For market M_s, put $\xi_{i,s} = \xi_i^C/\xi_{i,s}^M$, $\Gamma_T^{M_s} := \sum_{i \in \mathcal{U}_T^{M_s}} m_i/\gamma_i$, and $\omega_{i,T}^s := (m_i/\gamma_i)/\Gamma_T^{M_s}$. The exact formula is (80) with $\xi_{ij} = \xi_{i,s}$. Applying (82) coordinate by coordinate with the state-specific after-tax shares gives, in the affine construction,

$$\begin{aligned} s_T^{M_s} = &\sum_{i \in \mathcal{U}_T^{M_s}} \omega_{i,T}^s \xi_{i,s} + \frac{\sigma_r^2 A_{T-1}^{M_s}}{\Gamma_T^{M_s}} \sum_{i \in \mathcal{U}_T^{M_s}} m_i \partial_z \widehat{D}_i \\ &+ \frac{\upsilon_{M_s,T}}{\Gamma_T^{M_s}} \sum_{i \in \mathcal{U}_T^{M_s}} m_i \xi_{i,s}^M \partial_z x_{i,M_s}(T). \end{aligned} \tag{97}$$

As for the taxable counterpart before (83), the final term vanishes under common active municipal after-tax shares, a singleton active set, or zero bond-specific variance, but not by unweighted clearing alone.

Proposition 14 (State-tax anchor and tax-mix pass-through)**.** *If the household H_s resident in state s uniquely minimizes $\xi_{i,s}$ and the one-period municipal boundary is deterministic with $r > 0$, then $y_1^{M_s}/y_1^C =$*

$\xi_{H_s,s}$. *Under the multiplicative mapping* $\xi_{H_s,s} = \xi^{\text{fed}}(1-\tau_s^{\text{st}})$, *with federal after-tax share* ξ^{fed} *and state income-tax rate* τ_s^{st}, *define*

$$\tau_{s,T}^{\text{st,impl}} = 1 - \frac{y_T^{M_s}/y_T^C}{\xi^{\text{fed}}}, \qquad \tau_{s,T}^{A,\text{st}} = 1 - \frac{A_T^{M_s}/A_T^C}{\xi^{\text{fed}}}. \tag{98}$$

Then $\tau_{s,1}^{\text{st,impl}} = \tau_s^{\text{st}}$, *with unit pass-through. In the affine channel-off decomposition, if only the home-household effective after-tax share changes and its active weight is one below* T_s^* *and constant at* $\omega_{H_s}^s$ *thereafter, the tax-mix loading-state-tax derivative is* $\Theta_S^s(T) + \omega_{H_s}^s \Theta_L^s(T)$, *where* Θ_S^s, Θ_L^s *are the weights in* (94) *with* T_s^* *in place of* T^*.

Proof. The boundary KKT minimizes $\xi_{i,s} r$ (Proposition 8). Equation (98) gives the anchor. Applying (94) with T_s^* in place of T^* gives the tax-mix derivative, because $\partial \xi_{H_s,s}/\partial \tau_s^{\text{st}} = -\xi^{\text{fed}}$ cancels the factor $1/\xi^{\text{fed}}$ in (98). □

Changes to out-of-state after-tax shares or other clienteles add their derivatives. Other statutory mappings require a corresponding implied-tax transformation. The primitive boundary remains $y_1^{M_s}/y_1^C = \xi_{H_s,s}$. Reported state-tax responses use the full multi-state model.

Many-clientele tax-mix pass-through. Under the nested supports of Assumption 2, fix the support and the risk-tolerance weights. Then $A_T^C = G_T$ and $A_T^M = \sum_{j \le T} \rho^{T-j} \sum_k \omega_{k,j} \xi_k$, so differentiating $\tau_T^A = 1 - A_T^M/G_T$ in ξ_i gives the channel-off affine derivative for type i, $-\partial \tau_T^A/\partial \xi_i$,

$$\mathcal{D}_i(T) = \frac{\sum_{j=T_i^*}^T \rho^{T-j} \omega_{i,j}}{\sum_{j=1}^T \rho^{T-j}}, \tag{99}$$

with an empty numerator equal to zero. It is positive after participation begins and converges to ω_i if the active weight is eventually constant at ω_i, because the finite pre-participation block receives vanishing weight at the long end. Participation ordered by increasing after-tax shares makes the clientele-weighted after-tax share nondecreasing, because adding an after-tax share no smaller than the incumbents' after-tax shares does not lower their risk-tolerance-weighted mean. Such ordered participation does not by itself order total slopes, loading ratios or yields, since other terms in (97) and the taxable leg remain unrestricted.

B.10 The thirty-maturity example

Figures 4 and 6 use $\rho = 0.8$, $\sigma_r = 0.025$, $(\gamma_H, \gamma_I) = (8, 5)$, $(\xi_H, \xi_I) = (0.55, 0.78)$, equal investor masses, and $T^* = 17$ on a thirty-period ladder. For $T \ge 2$, the municipal bond-specific standard deviation is $0.004[1 + 7\sqrt{(T-1)/29}]$. The one-period claim has zero variance. The exposure fixed point in (84) includes only the jointly held block's variances.

For every $d \ge 0$, the predecessor loadings on the joint block are at least $a_{\min} = \xi_H \sum_{q=0}^{15} \rho^q$, so $P_J/S_J \le 1/a_{\min}$ and the continuous map $\mathcal{F}_J$ takes $[0, \infty)$ into $[0, d_{\max}]$ for

$$d_{\max} = \frac{\xi_I - \xi_H}{(\gamma_H + \gamma_I)\sigma_r^2 a_{\min}} < 10.591841.$$

Outward-rounded forty-digit decimal evaluation of $\mathcal{F}_J'$ (Appendix B.6) on 20,000 interval cells gives $\sup |\mathcal{F}_J'| < 0.080131$. Directed endpoint evaluations bracket its unique fixed point:

$$8.310258726885486 < d^* < 8.310258733114512.$$

The resulting A_{16}^M lies between 3.251488563972084 and 3.251488564481132, below $\xi^*/(1-\rho) = 3.457692307692308$, verifying (M). Condition (S) has left side $0.16084 < \rho$. The branch has a unique loading solution.

Figure 6(b) uses $e_j = \gamma_H \sigma_r^2 A_{j-1}^M d^*$ for $j < T^*$ and $c_j = \omega_I(\xi_I - \xi_H)$ for $j \geq T^*$, each zero otherwise, and the recursion gives identically

$$\frac{A_T^M}{G_T} = \xi_H + \frac{\sum_{j \leq T} \rho^{T-j} e_j}{G_T} + \frac{\sum_{j \leq T} \rho^{T-j} c_j}{G_T}.$$

The figure plots their sum.

The level solution uses $\bar{r} = 0.04$, unit masses, $Q_T^M = 0.1$, $Q_T^C = 0.01$ for $T \geq 2$, no taxable bond-specific risk, $\ell_I = 0$ and $\ell_H = 2.985$, chosen so that the duration gap $\widehat{\Lambda} = 4.319$ sits at the midpoint of $(4.285, 4.353]$, the interval that selects $T^* = 17$ under Theorem 1. The threshold is therefore set for the illustration and verified, and the exposure fixed point is solved given it. At $z = 0$, closed-form holdings clear both markets. Active KKT errors are below 10^{-15}, inactive municipal slacks exceed 6.8×10^{-5} and household taxable slacks exceed 0.0026.

Figure 4 reports supply shares. Figure 5 decomposes the one-period return gap by (49), separating the tax-mix and portfolio-risk contributions to its rate-loading term. The identity error is below 10^{-15}. At $T = 16$ the 181-bp return gap comprises 87 bp of duration cost, 89 of rate loading and 5 of bond-specific risk. At $T = 30$ its 205 bp comprise 92 of after-tax cash return, 109 of rate loading (72 from the tax mix and 37 from portfolio-risk responses) and 4 of bond-specific risk.

Full yield-implied-tax derivatives re-solve the exposure fixed point, loadings and holdings with supplies, masses and outside exposures fixed. Over thirty maturities, household pass-through falls from 1 to 0.24 and institutional pass-through rises from 0 to 0.73, including 0.06 at $T = 2$, which the institution does not hold. The smallest adjacent household decrease is 0.0088 and the smallest adjacent institutional increase is 0.0083. Their sum ranges from 0.97 to 1.00. Outward-rounded interval evaluation of the analytic derivatives verifies every adjacent comparison, positive holdings and inactive slacks, with $T^* = 17$ unchanged.

The long-end loadings are $A_\infty^M = 3.458$, $s_\infty^{C,\text{aff}} = 0.885$ and $A_\infty^C = 4.425$. The risk-premium coefficients are $\pi_\infty^M = 0.0118$ and $\pi_\infty^C = 0.0108$, and the long taxable yield is $y_\infty^C = 0.0816$. Loading-implied pass-throughs are $(\mathcal{P}_H^A, \mathcal{P}_I^A) = (-0.012, 1.012)$ and yield-implied pass-throughs $(-0.008, 0.956)$. Proposition 6's tax-difference condition has left side $0.429 < 1$. Convexity absorbs $\chi_\infty = 0.0750$ of the taxable yield, with $R_\infty(1 - R_\infty)\chi_\infty = 0.0128$, $\psi_\infty = -0.0024$ and $\Delta\tau_\infty = 0.0104$. Their derivatives contribute -0.057, 0.002 to institutional pass-through and -0.001, 0.006 to household pass-through, preserving the comparisons with $(\omega_H, \omega_I) = (0.385, 0.615)$.

Raising $\gamma_H \ell_H$ and $\gamma_I \ell_I$ equally lowers both risk-premium coefficients without changing holdings, participation or $\widehat{\Lambda}$. Both endpoint comparisons hold for every nonnegative π_∞^C. At $\pi_\infty^C = 0$, convexity absorbs 0.18 of the taxable yield and contributes -0.168 to institutional pass-through. Yield pass-throughs become $(-0.027, 0.853)$. Institutional pass-through falls below ω_I only when hedging demand lowers the taxable yield to 0.017, against 0.082 at baseline, or to 0.0165 when the mean rate falls with $\pi_\infty^C = 0$.

Equilibrium selection. Enumerated affine configurations are retained only when their linear level KKT-and-clearing system has a unique feasible solution. The uniqueness check applies to each configuration separately. Full-state solutions pass the independent funded-grid checks in Appendix C.3. Their maturity patterns are computed properties.

B.11 Participation and pass-through with many clienteles

Extend the two-investor affine benchmark of Section A.3 to N investor types with slack funding. Order after-tax shares $\xi_1 < \cdots < \xi_N$, so type 1 is the household with the largest exemption benefit. Without bond-specific risk, subtract type 1's active municipal equality from type $j > 1$'s marginal condition (Section A.4):

$$\underbrace{\sigma_r^2 A_{T-1}^M \Lambda_j}_{\text{duration benefit}} \geq \underbrace{(\xi_j - \xi_1) r}_{\text{difference in exemption benefits}}, \qquad \Lambda_j := \gamma_1 \widehat{D}_1 - \gamma_j \widehat{D}_j. \tag{100}$$

At fixed net rate exposures the duration benefit grows with the bond's rate loading while the difference in exemption benefits is flat in maturity. Positive bond-specific risk adds the incumbent's inventory premium to the comparison (equation (72)).

Assumption 2 (Long-end participation, clientele by clientele)**.** *Type 1 is active at every maturity, so* $T_1^* = 1$*. For each type* $j > 1$*,* $\Lambda_j > 0$*, and its active municipal set is an upper maturity interval* $[T_j^*, T_{\max}]$ *with* $T_j^* \geq 2$*. The thresholds need not be ordered by tax advantage.*

Tax-ordered participation makes the tax mix nondecreasing in maturity (Appendix B.9). Clientele-specific pass-through and endpoint comparisons need no tax ordering. Assumption 2 restricts only this affine extension. The calibration does not impose it (Section 5).

Under Assumption 2, the household anchors the riskless boundary. Equation (99) gives each clientele's tax-mix derivative, which is zero before its threshold and positive afterward, converging to its eventual risk-tolerance weight. Full yield responses also include relative compensation and convexity (Section A.6). The portfolio-risk response can transmit a later clientele's tax to shorter risky bonds it never holds. P1 is the always-active household case and P2 the later-participating case.

B.12 Multiple muni states with state and federal taxes

A household resident in state s' pays her home state's income tax on interest from a muni issued by another state $s \neq s'$ but not on a home-state muni. The quantitative institutional types have no modeled state-income-tax wedge on either, so state taxation is a household-specific channel (Appendix E.3). Statutory top state rates vary across states, with a supply-weighted mean of 5.8%. This cross-state variation is both a calibration input and P1's source of identification. The multi-state model replaces the single after-tax share ξ_i with a state-pair-specific ratio $\xi_{i,s} := \xi_i^C / \xi_{i,s}^M$, and the exact KKT aggregation behind (80) applies state by state. Proposition 14 derives the declining tax-mix pass-through in the affine benchmark. The calibrated multi-state implementation supplies P1's full-yield pass-through profile, including level and exposure corrections.

B.13 Participation motives and the municipal gap

For Section 4.7, specialize the economy of Section A.1 to one muni market, the high-tax household H and the lower-tax investor I, with $\xi_H < \xi_I$. The one-period return gap is $g_T = \mu_T^M - \xi_H \mu_T^C$.

B.13.1 One-period return components and neutral characteristics

Add to investor i's objective in (64) a differentiable concave term $\Phi_i(x_i)$ in the same after-tax units. Dividing each KKT condition (36) by its after-tax share gives, under the normalized municipal–taxable benchmark,

$$\mu_T^M \leq \xi_i r + K_{i,T}^M, \qquad \mu_T^C \leq r + K_{i,T}^C, \qquad \text{with equality for a positive holding,} \tag{101}$$

where

$$K_{i,T}^a := \gamma_i \operatorname{Cov}_z\big(u_{T,t+1}^a, \widetilde{\Pi}_i(t+1)\big) + \frac{\beta_i}{\xi_i^a} - \frac{1}{\xi_i^a} \frac{\partial \Phi_i}{\partial x_{i,T}^a}.$$

Additive shocks split $K_{i,T}^a$ into components $k_{ij,T}^a$ for rate risk, bond-specific risk and each additional shock. Additional shocks are independent of the rate and bond-specific shocks.

A common return characteristic is a unit-variance shock q with nonnegative maturity-T exposures e_T^M, e_T^C in both markets, taxed with returns. Its component is $k_{iq,T}^a = c_{iq} e_T^a$, where the cost per unit of exposure, $c_{iq} := \gamma_i \operatorname{Cov}_z(q_{t+1}, \widetilde{\Pi}_i(t+1))$, is the same in both markets. Affine rate risk has $q = -\varepsilon$, $e_T^a = \sigma_r A_{T-1}^a$, $c_{iq} = \gamma_i \sigma_r \widehat{D}_i$ and component $\gamma_i \sigma_r^2 A_{T-1}^a \widehat{D}_i$.

A wealth characteristic changes after-tax wealth by w_T^a per dollar at investor i's intensity $\lambda_i \geq 0$, as with trading costs. For $\Phi_i = -\lambda_i \sum_{a,T} w_T^a x_{i,T}^a$, the component is $k_{iw,T}^a = \lambda_i w_T^a / \xi_i^a$. Funding similarly gives β_i / ξ_i^a.

Definition 1 (Neutral characteristic)**.** *A common return characteristic is neutral if* $e_T^M = \xi_H e_T^C$*, and a wealth characteristic if* $w_T^M = w_T^C$*, at every maturity. For a household that bears it, a neutral characteristic satisfies* $k_{H,T}^M = \xi_H k_{H,T}^C$*, so it creates no gap when the household holds both bonds.*

Neutrality for each investor. A neutral common return characteristic gives $k^M_{iq,T} = c_{iq}e^M_T = c_{iq}\xi_H e^C_T = \xi_H k^C_{iq,T}$ for every investor i. A neutral wealth characteristic gives $k^M_{iw,T} = \lambda_i w_T = \xi_i\,(\lambda_i w_T/\xi_i) = \xi_i k^C_{iw,T}$, so

$$k^M_{Iw,T} - \xi_H k^C_{Iw,T} = (\xi_I - \xi_H)k^C_{Iw,T}. \tag{102}$$

In the affine benchmark in which the household alone holds both bonds, rate risk is neutral. Clearing fixes its exposure, so differentiation gives $s^M_T = \xi_H$ and $s^C_T = 1$, and recursion (39), starting at $A^M_1 = \xi_H$, $A^C_1 = 1$, gives $A^M_{T-1} = \xi_H A^C_{T-1}$ at every maturity by induction, as in Proposition 11.

Neutrality is stated for the household because the puzzle is defined against the household benchmark. A characteristic that weighs more on municipal bonds would add a premium even there.

B.13.2 The gap without and with participation

No gap when the household holds both bonds. Suppose the household alone holds both bonds, there is no bond-specific risk, and every additional characteristic is neutral and zero at the one-period claim. Both household conditions in (101) bind, so $g_T = \sum_j (k^M_{Hj,T} - \xi_H k^C_{Hj,T})$, and its funding terms β_H and β_H/ξ_H cancel. Definition 1 removes the additional components. In the affine benchmark, the neutrality of rate risk makes the rate contribution $\gamma_H\sigma_r^2\widehat{D}_H(A^M_{T-1} - \xi_H A^C_{T-1})$ zero, and with a constant cash rate there is no rate component. At riskless $T = 1$, $\mu^M_1 = \xi_H r$ and $\mu^C_1 = r$. Hence

$$g_T = 0 \qquad \text{at every maturity.} \tag{103}$$

Participation incentive for any motive. At the household-only allocation of Section A.4, with components evaluated there, household-only muni holdings give $\mu^M_T = \xi_H r + K^M_{H,T}$, so the lower-tax investor's first-order gain from buying the maturity-T municipal bond, $\mu^M_T - \xi_I r - K^M_{I,T}$, is nonnegative if and only if

$$\Delta_T := K^M_{H,T} - K^M_{I,T} \;\geq\; (\xi_I - \xi_H)\,r. \tag{104}$$

If $r > 0$, $\Delta_1 = 0$, and Δ_T is nondecreasing in T, the maturities satisfying (104) form an upper interval, possibly empty, that excludes the one-period claim. A nondecreasing Δ_T crosses the maturity-invariant tax difference at most once, and $\Delta_1 = 0 < (\xi_I - \xi_H)r$.

Rate risk gives $\Delta_T = \sigma_r^2 A^M_{T-1}\Lambda$ (Proposition 1). At that allocation, the advantages from several motives add. This comparison holds prices and portfolios fixed. Equilibrium upper-interval participation follows from Proposition 16(i) for the default and trading-cost motives and under the conditions of Theorem 1 for rate risk.

Proposition 15 (The gap when the lower-tax investor participates)**.** *In the economy of Section A.1 with one municipal market, suppose every additional characteristic is a neutral common return or wealth characteristic. If the lower-tax investor holds the maturity-T municipal bond at state z, then*

$$g_T \;\geq\; (\xi_I - \xi_H)\,r + g^{\text{risk}}_T + (\xi_I - \xi_H)\Big(\sum_w k^C_{Iw,T} + \frac{\beta_I}{\xi_I}\Big),$$
$$g^{\text{risk}}_T := \gamma_I\Big[\mathrm{Cov}_z\big(u^M_{T,t+1}, \widetilde{\Pi}_I(t+1)\big) - \xi_H\,\mathrm{Cov}_z\big(u^C_{T,t+1}, \widetilde{\Pi}_I(t+1)\big)\Big],$$

where the covariances run over the rate and bond-specific shocks, with equality if it also holds the maturity-T taxable bond. In the affine benchmark without bond-specific risk, $g^{\text{risk}}_T = \gamma_I\sigma_r^2\widehat{D}_I(A^M_{T-1} - \xi_H A^C_{T-1})$ and funding is slack. With nonnegative costs, this gives Proposition 5(ii).

Proof. The active municipal condition gives $\mu^M_T = \xi_I r + K^M_{I,T}$. Taxable KKT gives $\mu^C_T \leq r + K^C_{I,T}$, with equality when taxable holdings are positive. Subtract ξ_H times the latter:

$$g_T \;\geq\; (\xi_I - \xi_H)r + \sum_j \big(k^M_{Ij,T} - \xi_H k^C_{Ij,T}\big).$$

Rate and bond-specific components give g_T^{risk}. Neutral common-return components cancel. Neutral wealth costs give $(\xi_I - \xi_H)k^C_{Iw,T}$ by (102). Funding terms β_I, β_I/ξ_I give $(\xi_I - \xi_H)\beta_I/\xi_I$. □

The third term prices wealth costs and scarce funding at the lower-tax investor's after-tax share. Nonnegative g_T^{risk} and nonnegative costs suffice for $g_T \geq (\xi_I - \xi_H)r$.

B.13.3 Three motives that produce the long-end gap

Proposition 16 (Three motives that produce the long-end gap)**.** *Let $r > 0$ and consider one of the three cases below. Cases (b) and (c) are set in the two-investor economy with a constant cash rate, no bond-specific risk, and nonbinding funding, so the stated characteristic is the only non-tax component. In each case the characteristic is zero at the one-period claim. In (b), investors have no outside exposure to the default shocks.*

(a) Liability hedging and risk-bearing capacity: the two-investor affine benchmark with $\Lambda > 0$ at the household-only allocation, without bond-specific risk for parts (i) and (iv).

(b) Default diversification: municipal default risk loads on one unit-variance shock and taxable default risk on another, with correlation $\varrho \in [0,1]$ and exposures $\xi_H e_T$ and e_T, where e_T is strictly increasing for $T \geq 1$. Write $X_i^M := \xi_H \sum_T e_T x^M_{i,T}$ and $X_i^C := \xi_i \sum_T e_T x^C_{i,T}$ for investor i's municipal and taxable default exposures, and $k^M_{id,T} := \gamma_i \xi_H e_T (X_i^M + \varrho X_i^C)$ and $k^C_{id,T} := \gamma_i e_T (X_i^C + \varrho X_i^M)$ for its default components. Assume $\gamma_H X_H^{M,\text{hh}} > \varrho\, \gamma_I X_I^{C,\text{hh}}$.

(c) Trading costs: the same round-trip cost $w_T \geq 0$ in both markets, strictly increasing for $T \geq 1$, and trading intensities $\lambda_I < \lambda_H$.

Then:

(i) the participation advantage Δ_T in (104) is zero at $T = 1$ and nondecreasing in T. In (b) and (c), the municipal maturities the lower-tax investor holds in any equilibrium form an upper interval;

(ii) $g_1 = 0$: the household anchors the one-period claim;

(iii) at every maturity at which the lower-tax investor holds the municipal bond, $g_T \geq (\xi_I - \xi_H)r + g_T^{\text{risk}}$ in (a), with g_T^{risk} of Proposition 15, $g_T \geq (\xi_I - \xi_H)r$ in (c), and in (b)

$$g_T = (\xi_I - \xi_H)r + k^M_{Id,T} - \xi_H k^C_{d,T},$$

where $k^C_{d,T}$ is the default component of any investor who holds the taxable bond, so the gap net of these default components is $(\xi_I - \xi_H)r$;

(iv) with the household alone holding both bonds, $g_T = 0$ in (a) and (c), and $g_T = k^M_{Hd,T} - \xi_H k^C_{Hd,T}$ in (b).

Proof. (i) At household-only holdings, (a) gives $\Delta_T = \sigma_r^2 A^M_{T-1}\Lambda$ with $A^M_{T-1} = \xi_H \sum_{j=0}^{T-2} \rho^j$ (Proposition 11); (b) gives $X_H^C = X_I^M = 0$ and $\Delta_T = \xi_H e_T(\gamma_H X_H^{M,\text{hh}} - \varrho\, \gamma_I X_I^{C,\text{hh}})$; (c) gives $\Delta_T = (\lambda_H - \lambda_I)w_T$. All vanish at $T = 1$ and increase weakly. In equilibrium (b), define investor i's unit muni-default cost $c_i := \gamma_i(X_i^M + \varrho X_i^C)$. The lower-tax investor's required municipal return is no greater than the household's exactly when $\xi_H e_T(c_H - c_I) \geq (\xi_I - \xi_H)r$. If the lower-tax investor holds maturity T, this comparison holds there and $c_H > c_I$. Strictly increasing e_T makes it strict beyond T, precluding household holdings and forcing lower-tax-investor holdings. Case (c) uses $(\lambda_H - \lambda_I)w_T$ identically.

(ii) Every non-tax component vanishes at the one-period claim, so the municipal conditions reduce to $\mu_1^M \leq \xi_i r$. Positive supply gives $\mu_1^M = \xi_H r$, while $\mu_1^C = r$, as in Proposition 8.

(iii) Apply Proposition 15 to (a), with no additional characteristic, and (c), with $g_T^{\text{risk}} = 0$ and cost $(\xi_I - \xi_H)\lambda_I w_T/\xi_I \geq 0$. In (b), active conditions give $\mu_T^M = \xi_I r + k^M_{Id,T}$ and $\mu_T^C = r + k^C_{d,T}$. With a constant rate, every taxable holder has the same default component. Subtraction proves the expression.

(iv) In (a) and (c) this is (103). In (b), the household's conditions give $\mu_T^M = \xi_H r + k^M_{Hd,T}$ and $\mu_T^C = r + k^C_{Hd,T}$. □

In (a), Λ is positive from liability hedges or lower duration costs, including lower risk aversion. In (b), households bear all muni default risk at the household-only allocation, while the lower-tax investor's taxable book is exposed to it only through ϱ, so the lower-tax investor can add it more cheaply. Independent

shocks guarantee this advantage. Raw gaps also contain holder-specific default-premium differences driven by relative risk supplies. Section D.1.5 adjusts these with CDS costs. In (c), trading costs reduce after-tax wealth (Section D.1.4). Matched long institutional round-trip costs are 0.53% of price in both markets at 26 to 30 years (Table 21), supporting equal costs. With combined motives, the decomposition of Proposition 15 still applies. Its tax-difference bound requires the remaining risk component to be nonnegative at the resulting equilibrium, and separate default shocks are treated as in (b).

B.13.4 Two solved examples without a duration motive

Consider two investors and three maturities with constant $r > 0$ and no bond-specific risk, eliminating rate-risk advantages. Both investors have risk aversion γ, masses m_H, m_I, after-tax shares $\xi_H < \xi_I$ and slack funding. Supplies are Q_T^M for $T = 1, 2, 3$ and Q_T^C for $T = 2, 3$. Each example adds only its stated non-tax characteristic.

Proposition 17 (Participation for diversification or lower trading costs)**.** *Consider the economy just described.*

(a) Default diversification. Municipal and taxable default risk load on independent unit-variance shocks, with exposures $\xi_H e_T$ and e_T and $0 = e_1 < e_2 < e_3$. Let $X_H^M := \xi_H e_2 Q_2^M / m_H$ and $X_I^M := \xi_H e_3 Q_3^M / m_I$ be the investors' municipal default exposures when the household holds $M1$ and $M2$ and the lower-tax investor holds $M3$, and let

$$X^C := \frac{e_2 Q_2^C + e_3 Q_3^C}{m_H/\xi_H + m_I/\xi_I}.$$

There is an equilibrium with this municipal allocation if and only if

$$\gamma\,\xi_H e_2 (X_H^M - X_I^M) \;\le\; (\xi_I - \xi_H) r \;\le\; \gamma\,\xi_H e_3 (X_H^M - X_I^M).$$

In it both investors hold taxable bonds, each with taxable default exposure X^C, and

$$g_1 = 0, \qquad g_2 = \gamma\,\xi_H e_2 (X_H^M - X^C), \qquad g_3 = (\xi_I - \xi_H) r + \gamma\,\xi_H e_3 (X_I^M - X^C).$$

Net of the default components of the investors who hold each bond, the gaps are 0, 0 and $(\xi_I - \xi_H) r$.

(b) Trading costs. Each investor pays the after-tax cost $\lambda_i w_T$ per period on either bond of maturity T, with $0 = w_1 < w_2 < w_3$ and $0 \le \lambda_I < \lambda_H$. If $(\lambda_H - \lambda_I) w_2 < (\xi_I - \xi_H) r \le (\lambda_H - \lambda_I) w_3$, the household holds $M1$ and $M2$, the lower-tax investor holds $M3$, $C2$ and $C3$, and

$$g_1 = 0, \qquad g_2 = \Big(\lambda_H - \frac{\xi_H}{\xi_I}\lambda_I\Big) w_2, \qquad g_3 = (\xi_I - \xi_H)\Big(r + \frac{\lambda_I w_3}{\xi_I}\Big),$$

with $0 < g_2 < g_3$. The ratio μ_T^M / μ_T^C equals ξ_H at $T = 1$, lies strictly between ξ_H and ξ_I at $T = 2$, and equals ξ_I at $T = 3$.

Proof. (a) Independent default shocks separate market risks. Investor i's KKT conditions are $\mu_T^M \le \xi_i r + \gamma \xi_H e_T X_i^M$ and $\mu_T^C \le r + \gamma e_T X_i^C$, for municipal and taxable exposures X_i^M, X_i^C. Lower taxable exposure implies lower required return and profitable purchases, so both investors hold taxable bonds with equal exposures. Clearing $\sum_i m_i X_i^C / \xi_i = e_2 Q_2^C + e_3 Q_3^C$ gives $X_H^C = X_I^C = X^C$ and $\mu_T^C = r + \gamma e_T X^C$. Each investor's split of its taxable exposure across the two maturities need not be unique. Supply-proportional holdings implement these exposures and clear both maturities. Active muni conditions give $\mu_1^M = \xi_H r$, $\mu_2^M = \xi_H r + \gamma \xi_H e_2 X_H^M$ and $\mu_3^M = \xi_I r + \gamma \xi_H e_3 X_I^M$. Inactive KKT requires $\xi_H r < \xi_I r$ at $M1$, the left threshold inequality at $M2$ and the right at $M3$. Subtracting returns gives the gaps. Remove holder-specific muni component $\gamma \xi_H e_T X_i^M$ and taxable component $\gamma e_T X^C$ to leave $\xi_H r - \xi_H r$ at $T = 2$ and $\xi_I r - \xi_H r$ at $T = 3$.

(b) Investor i requires $\xi_i r + \lambda_i w_T$ for munis and $r + \lambda_i w_T / \xi_i$ for taxable bonds. With slack funding, each bond's return equals the minimum eligible requirement. Since $\lambda_I / \xi_I < \lambda_H / \xi_H$, the lower-tax investor holds

all taxable bonds. It holds muni maturity T when $\xi_I r + \lambda_I w_T \le \xi_H r + \lambda_H w_T$, here only at $T = 3$. Subtract returns to obtain the gaps and

$$g_2 = (\lambda_H - \lambda_I)w_2 + (\xi_I - \xi_H)\frac{\lambda_I w_2}{\xi_I} < (\xi_I - \xi_H)\Big(r + \frac{\lambda_I w_2}{\xi_I}\Big) \le g_3.$$

At $T = 2$ the ratio is below ξ_I because the household's required municipal return is below the lower-tax investor's. At $T = 3$ it is $(\xi_I r + \lambda_I w_3)/(r + \lambda_I w_3/\xi_I) = \xi_I$. □

Case (a) implements default diversification (Proposition 16), requiring $X_H^M > X_I^M$. The intermediate raw gap $\gamma\xi_H e_2(X_H^M - X^C)$ is a default-premium difference. With $\varrho \in (0, 1)$, the same allocation, where it is an equilibrium, has default-net gaps 0, 0, $(\xi_I - \xi_H)r$ (Proposition 16(b)). Case (b)'s intermediate gap reflects higher household trading costs, independent of other holdings.

Examples with rate risk. Adding the affine rate process $z' = \rho z + \sigma_r \varepsilon$, the examples use $\xi_H = 0.6$, $\xi_I = 0.8$, $\rho = 0.8$, $r = 4\%$ and small municipal bond-specific variance, with $\Lambda = 0$ at household-only holdings to remove the initial rate-risk advantage. For each motive, 120 economies with other parameters drawn randomly, including $\varrho \in [0, 0.9]$, are solved and screened for household-only participation incentives. Trading costs give long-only institutional holdings and $0 = g_1 < g_2 \le g_3$, $g_3 \ge (\xi_I - \xi_H)r$, in all 120. Default diversification gives long-only holdings in 72 and no participation in 48. In the 72, the default-net gap is the tax difference plus Proposition 15's rate and bond-specific terms. The latter is below 0.01 bp. The bound holds in 68. Negative rate terms lower it by at most 0.09 bp in four. Rate terms across both motives span −0.09 to 1.7 bp.

B.13.5 A municipal-specific benefit

A muni-only benefit $\zeta_T > 0$, from a mandate, issuer relationship or favorable regulation at matched maturity and credit, violates neutrality, because a household that had it would accept municipal returns below the benchmark. For the lower-tax investor, it adds $-\zeta_T$ to the gap and gives $\Delta_T = \zeta_T$, so a ζ_T that rises with maturity induces long-end participation, and no adjustment removes it.

Proposition 18 (Same participation, opposite pricing)**.** *Consider the two-investor economy without rate risk or bond-specific risk, with a constant cash rate $r > 0$, nonbinding funding, and no additional household characteristic. A fixed per-dollar benefit lowers the lower-tax investor's required municipal return by $\zeta_T \ge 0$, with $\zeta_1 = 0$, ζ_T nondecreasing, and $\zeta_T < \xi_H r$. Compare a neutral benefit, which also lowers the investor's required taxable return by the household-equivalent ζ_T/ξ_H, with a municipal-specific benefit that leaves the required taxable return unchanged.*

(i) In both cases the lower-tax investor holds the maturity-T municipal bond only if $\zeta_T \ge (\xi_I - \xi_H)r$, so participation begins at the same maturity.

(ii) With the neutral benefit, $\mu_T^C = r - \zeta_T/\xi_H$ and

$$g_T = \min\big\{\zeta_T,\ (\xi_I - \xi_H)r\big\} \ge 0,$$

so the gap equals the full difference in exemption benefits wherever the lower-tax investor participates. The ratio μ_T^M/μ_T^C rises with ζ_T.

(iii) With the municipal-specific benefit, $\mu_T^C = r$ and

$$g_T = \min\big\{0,\ (\xi_I - \xi_H)r - \zeta_T\big\} \le 0,$$

and the ratio $\mu_T^M/\mu_T^C = \min\{\xi_H,\ \xi_I - \zeta_T/r\}$ is weakly below ξ_H where the lower-tax investor participates and strictly below it above the participation threshold.

Because the rate is constant, each yield averages the one-period returns $\mu_1^a, \ldots, \mu_T^a$. The yield gap $y_T^M - \xi_H y_T^C$ therefore has the sign of the g_j, and the yield-implied tax is non-increasing in maturity in case (ii) and non-decreasing in case (iii).

Proof. As in Proposition 17(b), minimum required returns determine holdings. The household requires $\xi_H r$ and r. The lower-tax investor requires $\xi_I r - \zeta_T$ and either $r - \zeta_T/\xi_H$ or r. This proves (i). With neutrality, $g_T = \mu_T^M - \xi_H r + \zeta_T$ equals ζ_T when $\xi_H r$ is minimal, otherwise $(\xi_I - \xi_H)r$. The ratios $\xi_H r/(r - \zeta_T/\xi_H)$ below the threshold and $(\xi_I r - \zeta_T)/(r - \zeta_T/\xi_H)$ above increase in ζ_T since $\xi_I > \xi_H$. A muni-only benefit gives $g_T = \mu_T^M - \xi_H r$. Since ζ_T is nondecreasing, μ_j^M/μ_j^C is nondecreasing in j in (ii) and nonincreasing in (iii). Adding a maturity whose ratio is at least (in (iii), at most) the running yield ratio cannot lower (raise) that ratio, which proves the yield statements. □

B.14 Other markets

The mechanism requires common portfolio risk, heterogeneous relative taxes and a shorting friction. For tax-favored and other bonds, $\xi_i = (1-\tau_i^{\text{tax}})/(1-\tau_i^{\text{fav}})$. Symmetric taxation gives $\xi_i = 1$ and removes the relative tax incentive. The slope decomposition (80) applies when after-tax shares differ. Positive taxable forwards and an increasing relative forward ratio imply a declining yield-implied tax (Proposition 12). Condition (S) orders the affine loading ratio (Proposition 4).

Applications include state-exempt treasury versus corporate interest; state-exempt Federal Home Loan Bank and Farm Credit issues versus Fannie Mae and Freddie Mac issues; treaty-dependent withholding on cross-border sovereign debt; and TIPS inflation accrual, taxed before payment in taxable accounts. Taxable accounts, IRAs and pensions also assign different after-tax payoffs to the same security, which retains one market price. In each application, the maturity profile depends on which clienteles hold the bonds. Municipal bonds combine a large federal wedge with tight shorting restrictions.

C Numerical solution and calibration

The quantitative economy, with 643 investor types and 689 bond coordinates, is solved for state-dependent municipal and taxable price functions and funded portfolios.

C.1 State-contingent moments and funded market clearing

Let $\{z_g\}_{g=1}^G$ be the state grid, j a market–maturity coordinate, $p_{j,-}(Z')$ its continuation log price and v_j its bond-specific return variance. Conditional on z, $Z' = \rho z + \sigma_r U$ with $U \sim N(0,1)$. The continuation objects are

$$\mathcal{M}_j(z) = \log \mathbb{E}_z \exp\{p_{j,-}(Z')\} + \tfrac{1}{2} v_j, \tag{105}$$

$$\Sigma_{jk}(z) = \operatorname{Cov}_z(p_{j,-}(Z'), p_{k,-}(Z')) + \mathbf{1}_{\{j=k\}} v_j. \tag{106}$$

The mean of the adjusted return is $\mu_j = \mathcal{M}_j - p_j = \log \mathbb{E}_z[1 + R_j]$, including the full Jensen adjustment of Appendix B.1, and Σ is the covariance matrix of centered log-return innovations. Log prices are piecewise linear in the state and are extrapolated beyond the end nodes along the end segments, so their exponential moments, covariance and rate-innovation covariance have analytic Gaussian integrals. Interpolation gives Σ a low-rank common component, and bond-specific variances add its diagonal component.

Investor i has bond after-tax shares $K_i = \operatorname{diag}(\xi_i^{a(j)})$ ($a(j)$ is coordinate j's market), cash after-tax share ξ_i^C, funding W_i, cash $x_i^0 = W_i - \mathbf{1}' x_i$ and nonnegative-cash multiplier β_i. After-tax shares are after-tax payoffs per dollar of pretax payoff under each block's tax law, normalized so that a household's after-tax share on home-state municipal interest, and an institution's on all municipal interest, is one. Unlike the absolute shares in Section 4.1, the shares here are in these normalized payoff units. Risk aversion and outside exposure are transformed as in Section 4.2. The taxable and cash after-tax share ξ_i^C is then the investor's relative after-tax share, which includes proration for insurers, and a household's after-tax share on another state's municipal bonds is one minus its home-state rate. The liability credit is $c_{ij}^L(z) = -\operatorname{Cov}_z(p_{j,-}(Z'), \ell_i \sigma_r U)$ as

in (71). The two-year financing-tax vector τ_i^{fin} is zero for nonbanks and taxable bonds. For banks it equals twice the annual cost of Appendix C.7 on every municipal coordinate and at every state. Stationarity is

$$\begin{aligned} g_i &= \gamma_i K_i[\Sigma K_i x_i - c_i^L] - K_i\mu + \xi_i^C r\mathbf{1} + \tau_i^{\text{fin}} + \beta_i\mathbf{1}, \\ g_{ij} &= \frac{\epsilon w_j}{x_{ij}}, \qquad \beta_i = \frac{\epsilon}{x_i^0}, \qquad x_{ij} > 0, \quad x_i^0 > 0. \end{aligned} \tag{107}$$

Here $\epsilon > 0$ is the log-barrier weight on holdings and cash (10^{-9} in the baseline), μ_j is the log expected pretax gross return on coordinate j and $r = \max\{\bar{r} + z, 0\}$ is the two-year log gross cash return. Municipal weights w_j equal the issuing state's share of national municipal supply, the same at every maturity, so splitting supply across states does not multiply the barrier. Longer taxable bonds have unit weight. Clearing requires

$$\sum_i m_i x_i(z_g) = Q. \tag{108}$$

Appendix C.2 describes how these equations are solved.

C.2 The solution algorithm

Given candidate log price functions, continuation moments (105)–(106) define a static funded-clearing problem at each state. Its required returns imply the price update

$$\widehat{p}_j(z_g) = \mathcal{M}_j(z_g) - \mu_j(z_g), \tag{109}$$

and the algorithm iterates on prices until $\widehat{p} = p$. Two boundary conditions close the recursion. A one-period municipal claim pays one next period, so its continuation log price is zero and its return is riskless. Its current price is solved in equilibrium. Cash, the one-period taxable claim, is not one of the 689 coordinates. It earns $r = \max\{\bar{r} + z, 0\}$, so its log price is $-r$, which supplies the continuation price of the two-period taxable bond.

Consistent expectations. Moments integrate all segments and extrapolated tails of the candidate functions. A fixed point uses the same prices in expected resale values and current clearing at every grid state. Covariances and funding couple all maturities, so the entire price array is updated together.

Participation. Every investor may hold every bond. The barrier gives $x_{ij} = \epsilon w_j / g_{ij}$, so positions with positive marginal-cost gaps are small, of order 10^{-12} in the reported equilibria. Portfolios are re-solved at each state in every iteration. Holdings are read directly at the three annual nodes: high-tax households hold more than 99.99% of each municipal maturity through six years and less than 0.01% beyond eighteen years.

Step 1: initial prices and portfolios. The block starts from the one-market equilibrium at the same coefficients (Appendix C.6), which is solved by the same algorithm with one municipal market and household after-tax shares averaged across states by supply shares. Its municipal prices are copied to every state market. Each household's municipal holdings are placed in its home-state market, each institution's are split across markets by supply shares, and taxable holdings are unchanged. Required returns start at $\mathcal{M} - p$. Before the first Newton solve, required returns and funding multipliers are adjusted so that the mapped portfolios satisfy the market-level first-order conditions, and positions that the mapping leaves near zero are reset to their barrier values. This only improves the starting point. Every position stays free, so the start does not restrict which investors hold which bonds in the solution.

Step 2: portfolios and clearing at each state, for given prices. At each grid state, the moments $\mathcal{M}(z_g)$ and $\Sigma(z_g)$ and the liability credits follow in closed form from the current price functions. Newton's method then solves stationarity and clearing, (107)–(108), jointly for holdings x_i, cash x_i^0 and required returns μ. Each investor's Hessian is diagonal plus low rank, so it is inverted with the Woodbury identity. Eliminating holdings leaves a 689×689 linear system in μ, the market Schur complement, which is

solved by LU factorization and refined when needed to a relative residual of 10^{-12}. A fraction-to-boundary rule and backtracking keep bond and cash holdings positive. In the first price iteration, the barrier weight falls from 10^{-5} to 10^{-9} in factor-of-ten stages. Later iterations start from the state's previous solution at $\epsilon = 10^{-9}$, and a failed solve restarts from proportional holdings with a larger barrier. Newton iteration stops when the stationarity and clearing residuals are below 10^{-11} and the largest product of a holding and its stationarity residual is below 2.5×10^{-12}. The grid states are solved independently and in parallel.

Step 3: price update. All prices are updated together from (109), across markets, maturities and states. A damped step moves prices a fraction $\min\{0.65, 0.025/\|\widehat{p} - p\|_\infty\}$ of the way to $\widehat{p}$, so no log price changes by more than 0.025. Anderson acceleration combines the last seven price iterates with least-squares weights and mixes at the damped step size. It restarts when the price error more than doubles, is used only when every weight is below 50 in absolute value, and is scaled to the same 0.025 cap.

Step 4: stopping and acceptance. Iteration stops when $\|\widehat{p} - p\|_\infty < 10^{-10}$. All moments are then recomputed without the trimming of singular values below 10^{-12} of the largest used during the solve, and the equilibrium is accepted only if the price, stationarity, clearing and budget residuals of Appendix C.3 meet their thresholds. Yields at the 36 monthly states of the block come from the solved log prices by linear interpolation in z, and the national municipal yield averages the state yields with supply weights. Holdings are read at the annual nodes.

Convergence. Holdings, cash, funding multipliers and required returns are re-solved within each state at every iteration, starting from the previous iteration's solution, so the price fixed point is the only outer loop. In the six calibration blocks, the price error $\|\widehat{p} - p\|_\infty$ starts near 10^{-2} and falls steadily, slowly for the first fifteen to twenty iterations and then by about a factor of ten every five iterations. It reaches 10^{-6} within forty iterations and 10^{-10} within seventy. The first iteration, which carries the barrier stages, is the most expensive. Later iterations need about two Newton steps per state.

The grid combines fifteen equally spaced nodes on $[-6\sigma_z, 6\sigma_z]$, $\sigma_z = \sigma_r/\sqrt{1-\rho^2}$, which include $z = 0$, with the block's three observed annual states, for eighteen nodes in all. An observed state is $z_t = r_t - \bar{r}$, where r_t is twice the continuously compounded two-year corporate zero yield in month t. The annual nodes use December of each block year, and yields are read at all 36 months of the block. The grid has no node at the floor kink $z = -\bar{r}$, so the two-period taxable bond's continuation evaluates the cash price through the same piecewise-linear interpolant as other prices. No state is selected to match long corporate yields.

Policy counterfactuals use the same algorithm, barrier and acceptance checks. The equal-treatment counterfactual (Section 7.3) uses price-update, inner, holding-weighted and linear-solve tolerances of 2×10^{-7}, 2×10^{-8}, 5×10^{-9} and 2×10^{-9}, respectively, and its 99.9% endpoint starts from the solution that removes 99% of each tax distinction. Near binding funding, the cash Newton direction is computed from the low-rank inverse to avoid cancellation. Other policy solves use the baseline tolerances or tolerances between these and the baseline.

Algorithm 1 Full-state equilibrium of one block

Require: after-tax shares, funding, masses, supplies, exposures, rate dynamics, state grid $\{z_g\}$

Initialize p and holdings from the one-market equilibrium; $\mu \leftarrow \mathcal{M} - p$ ▷ Step 1
repeat ▷ outer loop over price functions
  for all grid states z_g, in parallel **do** ▷ Step 2
    Compute $\mathcal{M}(z_g)$, $\Sigma(z_g)$ and liability credits from the current p
    Solve stationarity and clearing (107)–(108) for x, x^0 and μ by barrier Newton
  end for
  $\widehat{p} \leftarrow \mathcal{M} - \mu$ at every grid state ▷ Step 3
  Move p toward $\widehat{p}$ by a damped or Anderson step, with log-price changes capped at 0.025
until $\|\widehat{p} - p\|_\infty < 10^{-10}$ ▷ Step 4
Recompute the moments; accept only if the price, stationarity, clearing and budget checks pass
Read yields at the block's monthly states and holdings at its annual nodes

Local loadings satisfy the nonlinear recursion (70) under the differentiation conditions of Appendix B.2.

C.3 Equilibrium residuals

Table 8 reports four independently reconstructed equilibrium diagnostics for each of the six blocks. With g_i from (107),

$$\begin{aligned} R_p &= \|p - \mathcal{M} + \mu\|_\infty, \qquad R_c = \|Q - \textstyle\sum_i m_i x_i\|_\infty, \\ R_s &= \|g - \epsilon w/x\|_\infty, \qquad R_{\text{comp}} = \max\{\|x \odot g\|_\infty, \|\beta \odot x^0\|_\infty\}. \end{aligned} \tag{110}$$

All six equilibria have zero budget and negative-holding violations, and their riskless claims have zero return variance. Acceptance requires $R_p < 3 \times 10^{-7}$, $R_s < 10^{-7}$, $R_c < 10^{-8}$ and a budget violation $\max_i(\mathbf{1}' x_i - W_i)$ below 10^{-10}, which all six blocks meet. The complementarity measure R_{comp} is reported only, since it equals ϵ at an exact barrier solution. These diagnostics measure equilibrium accuracy on the reported finite grid at the stated barrier weight.

Table 8: Full-state equilibrium diagnostics

Block	R_p	R_s	R_c	R_{comp}
2000–2002	7.511×10^{-11}	5.884×10^{-12}	5.551×10^{-17}	1.002×10^{-9}
2003–2005	4.456×10^{-11}	8.340×10^{-12}	1.110×10^{-16}	1.002×10^{-9}
2006–2008	8.156×10^{-11}	8.547×10^{-12}	1.110×10^{-16}	1.002×10^{-9}
2009–2011	5.498×10^{-11}	8.687×10^{-12}	2.220×10^{-16}	1.002×10^{-9}
2012–2014	9.289×10^{-11}	9.864×10^{-12}	2.220×10^{-16}	1.002×10^{-9}
2015–2017	8.495×10^{-11}	8.352×10^{-12}	2.220×10^{-16}	1.002×10^{-9}

Notes. Maxima over each block's eighteen state nodes and the applicable coordinates, with $\epsilon = 10^{-9}$. Complementarity is the largest unpenalized product of a bond or cash holding and its multiplier, which equals ϵ at an exact barrier solution. The diagnostics assess only the discretized funded equations.

The fixed-coefficient 2018–2020 extension also satisfies these checks: $R_p = 4.519 \times 10^{-11}$, $R_s = 7.777 \times 10^{-12}$, $R_c = 2.220 \times 10^{-16}$ and $R_{\text{comp}} = 1.001 \times 10^{-9}$, with zero budget and negative-holding violations and zero variance on riskless claims.

Riskless boundary and the household benchmark. The 45 one-period municipal claims are priced in funded clearing like other bonds. The lowest return at which an eligible household holds the claim of state s is

$$\min_{i \in \mathcal{H} \text{ eligible}} \frac{\xi_i^C r(z) + \beta_i(z)}{\xi_i^{M_s}},$$

which includes the funding multiplier and so can exceed $r(z)$ times the smallest after-tax-share ratio. In every equilibrium, each state's one-period municipal return is within 0.01 basis points of this frontier at every grid state. The reported household benchmark uses the resident household's after-tax share, which can differ from the frontier household's. At each observed month, the national one-period municipal yield differs from the resident benchmark by the supply-weighted tax-frontier difference plus less than 0.01 basis points.

Analytic integration of the price interpolant. For a standard normal U and an affine segment $p = a + bU$ on $l < U < u$,

$$\mathbb{E}[e^{a+bU}\mathbf{1}_{\{l<U<u\}}] = e^{a+b^2/2}\{\Phi(u-b) - \Phi(l-b)\}.$$

Summing over segments and specified tails eliminates quadrature truncation, leaving interpolation and finite-precision error. The residuals above hold at the grid nodes and do not bound errors uniformly between them. Grid refinement assesses accuracy between nodes (Appendix C.4).

Weighted barriers and funding. Exact stationarity gives $x_{ij}g_{ij} = \epsilon w_j$ and $x_i^0\beta_i = \epsilon$. For concave preferences at fixed prices, the funded utility gap is bounded by $\epsilon(\sum_j w_j + 1)$, provided the stationarity error is controlled.

C.4 Grid, barrier and floor-knot comparisons

Three comparisons fix the seventeen coefficients and measured inputs in 2015–2017. The first increases the regular nodes from fifteen to twenty-nine, retaining the original nodes and endpoints. The second lowers ϵ from 10^{-9} to 10^{-10} with unchanged inner tolerances, and the third adds the exact floor knot. I check each re-solve independently, and Table 9 reports the results.

Table 9: Numerical sensitivity at fixed coefficients, 2015–2017

Comparison	Actual nodes	ϵ	Δ 30-year gap (bp)	Δ HH lower muni book (%)	Δ bank taxable book (%)
Nested grid	32	1.000×10^{-9}	+0.199365	-4.020	-2.585
Smaller barrier	18	1.000×10^{-10}	+0.000138	-0.004	+0.001
Exact floor knot	19	1.000×10^{-9}	-0.228439	-1.689	-2.797

Notes. Each row compares one change with the original eighteen-node, 10^{-9}-barrier economy. Rows are not a sequence. The yield gap is the national supply-weighted municipal yield minus the original resident-household after-tax taxable benchmark, in continuously compounded yields over 36 actual months. Its baseline is 104.270250 bp. Holdings use the three annual measurement nodes, and percentage changes use each group's original book as denominator. The rows are neither confidence intervals nor the seven-block empirical-weight-matched gap. The floor is economically active in every row.

Gap changes are small relative to holding changes. Refinement reduces the lower-tax household municipal book by 4.02%, and adding the floor knot reduces it by 1.69% and bank taxable holdings by 2.80%. Changes in municipal and taxable yields partly offset each other.

Lowering the barrier changes individual riskless positions by at most 77.06% of budgets across grid nodes and 24.84% at annual holdings nodes, compared with maxima of 0.1179% and 0.0709% for risky positions. The largest riskless shift is at a positive rate. Complementarity falls from 1.0016×10^{-9} to 1.0134×10^{-10}, and the tiny residual-group municipal and high-tax household taxable books fall about 90%. Riskless and near-zero holdings remain sensitive to the barrier, although risky holdings and gaps are stable.

C.5 Risk specification and sensitivity

The bond-specific variance profile is $v_n = v_{30}(n/30)^{p_v}$ for maturities of $n = 4, 6, \ldots, 30$ years, with $v_2 = 0$ and the same variances for municipal and taxable bonds at each maturity. The coefficients v_{30} and p_v are fitted jointly with preferences and effective exposures (Table 10) and held fixed in the comparisons above.

Table 10: Common estimated coefficients, held fixed in the full-state evaluation

Symbol	Economic role	Value
Risk. Main information: taxable term spreads and the allocation of long bonds		
γ	Common risk-aversion curvature, per unit of balance-sheet mass	9.6969
v_{30}	Bond-specific variance of a thirty-year bond's two-year return	0.000722
p_v	Maturity exponent of the bond-specific variance	1.2597
Households. Main information: household municipal and taxable stocks		
a_H	Exposure intercept	1.3045
b_A	Loading on the outside-asset ratio	2.704
b_D	Loading on the debt ratio	0.8379
P&C insurers. Main information: P&C stocks, maturities, composition and long fractions		
a_P	Low-type exposure	−1.961
d_P	High-minus-low exposure	3.0311
w_P	Low-type resource fraction	0.0802
Life insurers. Main information: life-insurer stocks, maturities, composition and long fractions		
a_L	Low-type exposure	−5.3927
d_L	High-minus-low exposure	6.3007
w_L	Low-type resource fraction	0.2616
t_L	Middle type's location within the exposure span	0.8625
u_L	Middle type's fraction of non-low resources	0.0162
Banks and residual group. Main information: bank stocks and taxable term spreads		
h_B	Bank exposure	1.2045
μ_R	Residual-group mean exposure	1.0683
s_R	Residual-group exposure dispersion	1.7168

Notes. All seventeen coefficients are estimated jointly to the twenty-two averaged conditions of Table 11. Every coefficient affects every condition through equilibrium prices and clearing, and the headings name the conditions most informative about each block. Exposures are measured like net duration, in the model's two-year units. Resource fractions split each insurer sector's measured resources across its types. Appendix Table 12 reports the search intervals, starting values and step scales.

Effective exposures summarize rate risk without matching every asset position to liability cash flows. The averaged ownership and maturity conditions assess them, and Section 5.4 assesses the fit to the yield curve.

C.6 Calibration conditions, parameter search and full-state evaluation

I estimate the seventeen common coefficients of Table 10 with the criterion (111) in six dated economies with one national municipal market. Table 11 lists the targeted conditions and untargeted tests. Exposure parameters use $h_i = -\ell_i$: positive values add rate risk, while negative values hedge it (Appendix C.7).

Measured inputs. The 1952–2024 annual rate estimates are a persistence of 0.853903 and an innovation standard deviation of 0.016266, and the mean 0.030206 is the 2000–2017 average of the two-year corporate-like yield. Two-year units give $(\rho, \sigma_r, \bar{r}) = (0.729151, 0.042778, 0.060412)$ with floor $r = \max\{\bar{r} + z, 0\}$. Monthly two-year yields set the observed states (Appendix C.2). The pooled treasury and corporate asset uses corporate yields and household taxes, despite the different statutory treatment of treasuries. After-tax shares and bank financing-tax costs are distinct fixed inputs (Table 15), and after-tax shares alone do not identify tax receipts.

Condition construction and normalization. Table 16 lists the twenty-two conditions in evaluation order. Each condition j in block k is a signed model-minus-observed difference g_{kj} with a positive observed normalizer o_{kj}. For the nine ownership conditions, g_{kj} is the model minus the observed stock in national dollars and o_{kj} is the observed stock.

Table 11: Targeted conditions and untargeted tests

A. Targeted: twenty-two conditions, each averaged over the six blocks of 2000–2017	
Ownership stocks (9)	Municipal holdings of high-tax households, lower-tax households, P&C insurers, life insurers and banks. Taxable holdings of households, P&C insurers, life insurers and banks
Insurer maturity (4)	Holdings-weighted mean maturity of the P&C and life-insurer municipal and taxable books
Taxable term spreads (3)	Ten-, twenty- and thirty-year taxable yields over the two-year yield
Insurer composition (2)	Book-weighted mean of squared municipal portfolio shares, P&C and life insurers
Insurer long fractions (4)	Share of each insurer sector's municipal and taxable book maturing beyond twenty years
B. Untargeted tests of the mechanism	
Municipal yield curve	Evaluated on the observed corporate curve, the model accounts for 83% of the 117.2-basis-point thirty-year puzzle after callability and the issuance premium (Figure 8)
State-tax pass-through (P1)	With the model's own federal after-tax share, its cross-state slope falls from 1 at two years to 0.46 at thirty and lies inside the controlled pre-2018 OLS 95% band in the first four maturity bins and 0.01 above its upper bound in the longest (Figure 9)

Notes. Each averaged condition is scaled by one tenth of its observed six-block mean magnitude. Appendix Table 16 reports every scaled deviation at the estimated coefficients.

For each insurer sector and asset, the maturity condition normalizes $B(\overline{T} - \overline{T}^{\text{obs}})$ by $B^{\text{obs}}\overline{T}^{\text{obs}}$, with B its model book and $\overline{T}$ its holdings-weighted maturity. The three taxable-yield conditions average the errors in annual log yields at ten, twenty and thirty years over 36 monthly states. Because observed two-year yields set the states, these are term-spread errors, normalized by the observed block-mean spreads over the two-year yield.

The remaining six conditions use maturities above two through thirty years, excluding the riskless two-year municipal stock included in the ownership and maturity conditions. For investor i in each insurer sector, let M_i and C_i be municipal and taxable holdings in this window and $B_i = M_i + C_i$. The municipal-share second moment $\sum_i (M_i^2/B_i)/\sum_i B_i$ is the book-weighted average of squared municipal portfolio shares. It captures how unevenly insurers allocate their bond portfolios to municipals and is compared with its observed counterpart. For each sector–asset book B with holdings L beyond twenty years and observed long fraction f^{obs}, the long-maturity condition is

$$g^{\text{long}} = \frac{L - f^{\text{obs}}B}{B^{\text{obs}}_{\text{window}}}.$$

Observed window books equal national sector stocks times measured window coverage. These six conditions are normalized by observed second moments and long fractions.

Holdings conditions are evaluated at each block's three observed end-year states and then averaged, because averaging books first would change products and ratios. Let $g_{kj}(\theta)$ be signed condition j in block k after its equilibrium solve at the common coefficient vector θ, and $o_{kj} > 0$ its observed normalizer. The criterion is

$$f_j(\theta) = \frac{6^{-1}\sum_{k=1}^{6} g_{kj}(\theta)}{0.1\,(6^{-1}\sum_{k=1}^{6} o_{kj})}, \qquad \mathcal{L}(\theta) = \frac{1}{2}\sum_{j=1}^{22} f_j(\theta)^2. \tag{111}$$

The book-weighted residuals above enter $\mathcal{L}$ with fixed inverse-squared-normalizer weights, which supply no covariance estimate, parameter standard errors or overidentification test.

Search coordinates. Table 12 gives bounds, initial vector θ^0 and step scales c_k, where a scaled step u_k changes coefficient k by $c_k u_k$. Derivatives and trust radii use these units. All economies start at θ^0. An unoptimized $(\gamma, a_H) = (8.5, 1.1)$ alternative supplies warm starts for the numerical checks without changing targets or portfolios.

Table 12: Parameter-search intervals, initialization and coordinate scales

Coefficient	Search interval	Initial value θ_k^0	Step scale c_k
γ	$[1, 10]$	9	1
a_H	$[-3, 3]$	1	0.3
b_A	$[0, 12]$	2	0.4
b_D	$[0, 20]$	0.5	0.5
a_P	$[-10, 2]$	-1.5	0.5
d_P	$[0.02, 12]$	2.4	0.6
w_P	$[0.01, 0.99]$	0.22	0.08
a_L	$[-12, 2]$	-4	0.6
d_L	$[0.02, 15]$	3.2	0.8
w_L	$[0.01, 0.99]$	0.1	0.08
h_B	$[-3, 3]$	0.6	0.3
μ_R	$[-4, 4]$	1	0.4
s_R	$[0.01, 5]$	2.2	0.4
v_{30}	$[0.000020, 0.006000]$	0.0015	0.0004
p_v	$[0, 3]$	1.15	0.3
t_L	$[0.01, 0.99]$	0.75	0.08
u_L	$[0.01, 0.99]$	0.2	0.08

Joint search. For the scaled conditions f and their Jacobian J in u, central differences refresh J at accepted centers, using feasible one-sided differences at bounds. A bounded trust-region search compares damped Gauss–Newton, backtracked and projected-gradient candidates against the quadratic model

$$\mathcal{L} + (J'f)'u + \tfrac{1}{2}u'(J'J + S)u.$$

Here a structured secant update estimates the residual curvature S.

The initial search uses a finite-difference step of 0.002 and ends after 107 iterations at $\mathcal{L} = 36.357521$. A restart from that vector uses a step of 0.001 after the larger step fails the initial half-step consistency check, and it retains the objective, bounds and equilibrium tolerances. The restart ends after 41 iterations at $\mathcal{L} = 36.286978$, the lowest retained criterion across both fits, and this vector is the quantitative baseline. Subsequent diagnostic probes do not replace it.

Search checks and interpretation. Outer-search convergence is separate from equilibrium accuracy. At the selected vector, steps of 0.002, 0.001 and 0.000125 give projected-gradient infinity norms of 14.57, 11.01 and 0.376, above the tolerance of 0.01. Changes in participation make the derivatives sensitive to the step size. The feasible probes tested improve $\mathcal{L}$ by at most 2.79×10^{-4}, which bounds only those directions. The selected calibration has no local-stationarity certificate.

Full-state evaluation. The full-state evaluation restores 45 issuer markets, separate risks and measured home-state taxes, with the same seventeen coefficients, quantities and condition definitions. These equilibria supply municipal yields, predictions and policies. Solves take 698–2,006 seconds on six workers, compared with about one second for one market. Across the six blocks, the full-state thirty-year gap minus the one-market gap is $(-0.85, -0.72, 2.48, -0.22, -1.15, -0.93)$ bp, and taxable yields differ by at most 0.52 bp.

Coefficient sensitivity. Table 13 changes one coefficient group per one-market re-solve. Every change raises the criterion. Bond-specific variance moves the gap and taxable yield by less than 0.1 basis points. Lower risk aversion produces the largest change. It lowers the thirty-year taxable yield by 70 basis points and puts term spreads about 20% below targets.

Table 13: Sensitivity of the one-market calibration to the estimated coefficients

Coefficient change	30-year gap (bp)	Change (bp)	Criterion $\mathcal{L}$
Estimated coefficients	102.6		36.29
Risk aversion $\gamma \times 0.8$	89.0	−13.6	55.02
Risk aversion $\gamma = 10$ (upper bound)	104.7	+2.1	36.74
Bond-specific variance $v_{30} \times 0.8$	102.7	0.0	36.61
Bond-specific variance $v_{30} \times 1.2$	102.7	0.0	36.59
Maturity power $p_v - 0.2$	102.6	0.0	36.55
Maturity power $p_v + 0.2$	102.7	0.0	36.47
Household exposure intercept $a_H - 0.25$	99.0	−3.7	78.82
Household exposure intercept $a_H + 0.25$	106.7	+4.1	160.84
Household asset loading $b_A \times 0.8$	100.9	−1.7	62.45
Household asset loading $b_A \times 1.2$	105.2	+2.6	400.24
Household debt loading $b_D \times 0.8$	106.2	+3.5	6,720
Household debt loading $b_D \times 1.2$	103.9	+1.3	82.17
P&C exposures ×0.8	102.2	−0.4	56.35
P&C exposures ×1.2	103.3	+0.6	62.51
Life-insurer exposures ×0.8	101.9	−0.7	59.68
Life-insurer exposures ×1.2	103.4	+0.8	46.24
Bank exposure ×0.8	100.5	−2.1	119.11
Bank exposure ×1.2	104.5	+1.8	241.78
Residual exposure mean ×0.8	102.6	−0.1	538.79
Residual exposure mean ×1.2	101.7	−0.9	65.73
Residual exposure s.d. ×0.8	99.3	−3.3	92.74
Residual exposure s.d. ×1.2	105.8	+3.1	1,687

Notes. The one-market economy is averaged over the six blocks of 2000–2017, with all other coefficients at their estimates. Exposure changes scale every type's exposure in the group. The gap is the municipal yield minus the household after-tax taxable yield at thirty years. The criterion $\mathcal{L}$ is the estimation objective, half the sum of squared scaled deviations of the twenty-two conditions.

Elastic bond supply. I allow borrowing to respond to its cost in both markets, holding the estimated coefficients fixed. For market $a \in \{M, C\}$ and maturity T, supply follows

$$Q_T^a(z) = \overline{Q}_T^a \left(\frac{y_T^a(z)}{\overline{y}_T^a} \right)^{-\chi_{a,T}},$$

where y_T^a is the annual continuously compounded yield. Each schedule passes through the original quantity and fixed-supply yield at the mean rate state. These anchors remain fixed when both yield curves are re-solved. Risky maturities from four to thirty years have elastic supply. The two-year riskless claims retain their baseline treatment.

St. Clair (2024) estimates that a one-percent increase in borrowing costs reduces municipal issuance by approximately 0.3–0.6 percent. I use 0.45 as the midpoint of this rounded range and impose the same elasticity in both markets. The estimate concerns annual borrowing near the bank-qualified threshold. Holding financing terms fixed maps proportional borrowing-cost changes locally to proportional yield changes. Applying that response to outstanding model supply defines the comparative-static exercise.

Table 14 reports the one-market economy over the six calibration blocks. At equal elasticities of 0.3, 0.45, and 0.6, the average thirty-year gap lies between 101.0 and 102.0 basis points, compared with 102.6 under fixed supply. At 0.45, municipal yields fall by 14.1 basis points and the household after-tax taxable benchmark by 13.4, so the gap changes by only 0.7 basis points.

I also consider a larger taxable response. The November 2022 estimates of Siani (2022) imply approximately 1.5% less corporate issuance per ten-basis-point increase in credit spreads, holding the benchmark rate fixed. I match that local slope by setting $\chi_{C,T} = 15\overline{y}_T^C$, while keeping $\chi_{M,T} = 0.45$. The thirty-year taxable elasticity is 0.86–0.97 across blocks. Applying this corporate issuance response to the pooled taxable supply is an additional sensitivity assumption. The average gap is 99.2 basis points, retaining 96.7% of its fixed-supply value. Both yields adjust: municipal yields fall by 19.5 basis points and the after-tax taxable benchmark by 16.1. A larger taxable supply elasticity therefore need not widen the gap.

Table 14: Yield gaps with elastic supply in both markets

Municipal elasticity	Taxable elasticity	10 years	20 years	30 years
0	0	35.7	84.9	102.6
0.30	0.30	36.4	84.1	101.9
0.45	0.45	36.7	84.0	102.0
0.60	0.60	36.8	83.4	101.0
0.45	$15\overline{y}_T^C$	37.2	83.2	99.2

Notes. Gaps are in basis points, averaged over the original rate-state observations within each block and equally across the six blocks of 2000–2017. All other coefficients and inputs remain fixed. Elasticities are reported as positive magnitudes: higher borrowing costs reduce supply. The last row converts a corporate spread semi-elasticity to a yield elasticity at each fixed anchor, with yields in annual decimal units.

C.7 Measured inputs and investor construction

One model period is two years, and maturities in this section are in years. Each of the 43 states with a personal income tax and the District of Columbia forms its own municipal market. The seven states without one (Alaska, Florida, Nevada, South Dakota, Texas, Washington and Wyoming) share a single market because every investor treats their bonds alike: their residents pay no state tax on any municipal bond, and all other investors treat them as out-of-state bonds. Each of the 45 municipal markets has fifteen maturities from two to thirty years, and the taxable market fourteen from four to thirty years, for 689 market–maturity pairs. Cash, the one-period taxable claim, pays the short rate, and the two-year municipal bond is also riskless. All bond yields, including the two-year municipal yield, are determined in equilibrium. The taxable market pools treasury and corporate supply. Its yields are compared with the corporate curve, and its interest is taxed like corporate interest.

The inputs set directly from data come in six three-year blocks, 2000–2002 through 2015–2017, centered on the 2001, 2004, 2007, 2010, 2013 and 2016 Survey of Consumer Finances (SCF) waves, and each block uses its own values. The rate state z of Appendix A.1 follows a first-order autoregression whose annual persistence, 0.854, and innovation standard deviation, 1.63 percentage points, are estimated on 1952–2024 short rates. The mean rate is $\bar{r} = 0.0604$ per two-year period, twice the 2000–2017 average annual two-year corporate yield of 3.02%, and actual monthly two-year yields set the observed rate states. Financial Accounts stocks fix sector resources, funding and bond quantities. Mergent outstanding amounts fix the municipal maturity profile and state shares, and Mergent FISD outstanding amounts fix the corporate maturity profile. Tax rates follow statutes and each SCF tax unit's filing status and income, and households' balance-sheet ratios come from the SCF, as described below. Appendix C.6 gives the remaining details, and Appendix Table 15 lists all measured inputs and their sources.

Table 3 lists the six groups. Type i has relative after-tax share ξ_i, resources m_i, funding W_i and outside rate exposure $h_i := -\ell_i$, the negative of the liability loading in Appendix equation (63). This sign convention is the reverse of ℓ_i: positive h_i adds rate risk. Outside assets lose value when rates rise, so long bonds add risk. Negative h_i represents a gain when rates rise, such as a fall in liability value, which long bonds hedge. Total resources m_i, the model's mass of type i, measured in dollars, are debt securities and cash in the Financial Accounts. Funding W_i, the share of these resources available for the modeled bond-and-cash portfolio, is municipal bonds, corporate bonds excluding asset-backed securities, treasury notes and bonds, and cash excluding bank cash, divided by those resources. A type thus invests $m_i W_i$ dollars in bonds and

Table 15: Measured inputs, estimated coefficients and maintained measurement choices

Input	Source, construction and role
National resources and ownership	Financial Accounts vintage of September 8, 2023, with registered household fund look-through, nonprofit separation and sector exempt-stock adjustments. SCF direct and fund holdings split household municipal ownership. These stocks supply conditions 1–9, and tax-exempt interest flows are not used.
Household heterogeneity	Each SCF wave supplies seven resource-weighted balance-sheet bins per household tax group. Outside financial assets exclude retirement accounts and the modeled resource base of deposits, direct bonds and funds. Debt and outside assets are divided by that resource base before bin averaging. Coefficients a_H, b_A, b_D jointly translate the ratios into exposure.
Household tax products	Dated SCF tax-unit supplements and federal calculations, with annual state-rate snapshots for high-tax households and record-level bracket assignments for lower-tax households. The products are computed before weighting. For lower-tax households, the archived bracket tables lack some filing-status cutoffs or related rules in 209 of 792 state-years. These are imputed from the state's nearest-year thresholds or simple proxies, leaving published rates and top rates unchanged. The repeated national geography is a maintained input.
Insurer portfolios	Annual P&C and life bond-position files, matched to security tax status and remaining maturity. Positive-value positions determine four maturity-dollar and six composition conditions, 10–13 and 17–22. Three annual cross-sectional moments are equally averaged within a block. Portfolio moments inform the estimated exposure mixtures.
Institutional tax treatment	The model applies separate P&C proration, life policyholder-share proration, and bank financing-tax rules, and the residual group is tax-neutral. The measured annual bank funding-cost ratio uses interest expense relative to assets, weighted by municipal holdings. Bank-qualification and coverage assumptions are maintained inputs.
Municipal supply and geography	Dated exempt outstanding Mergent amounts, a within-bin maturity projection and state shares. The seven states without an income tax form one pooled market. National household mass in each bin is allocated by state municipal supply share.
Taxable supply	Treasury and corporate national totals from Financial Accounts, corporate maturity shares from FISD outstanding amounts. Treasury maturity shape is separately tied to the maturities measured in the Monthly Statement of the Public Debt. No national quantity is re-estimated against yields.
Rate process and yield conditions	Annual persistence and innovations from the 1952–2024 TB3MS sample, temporally aggregated to two years. The common mean is the mean of the two-year corporate-like yield over 2000–2017. The monthly corporate-like spot curve supplies conditions 14–16, converted to annual log yields. Broad pooled quantities retain the corporate-like yield and tax approximation.
Joint estimated coefficients	The seventeen coefficients in Table 10 enter the same twenty-two averaged conditions. Ownership informs household and bank exposures. Insurer books, maturities and composition inform exposure mixtures, while taxable yields and joint clearing inform risk curvature, residual exposure and the bond-specific variance profile. Equilibrium prices and allocations link all coefficients to all conditions.

Notes. Six three-year blocks cover 2000–2017. Each averaged condition is scaled by one tenth of its observed six-block mean magnitude, and the estimated coefficients are held fixed in the full-state evaluation and the policy endpoints. Licensed inputs remain subject to their original access restrictions.

Table 16: The twenty-two targeted conditions at the estimated coefficients: signed average deviations in units of 10 percent of the observed magnitude

	Condition	Scaled deviation
1	High-tax households: municipal stock	2.518
2	Lower-tax households: municipal stock	−2.672
3	P&C: municipal stock	−0.199
4	Life: municipal stock	0.140
5	Banks: municipal stock	0.591
6	Households: taxable stock	−0.561
7	P&C: taxable stock	−2.711
8	Life: taxable stock	0.218
9	Banks: taxable stock	−0.277
10	P&C: municipal maturity dollars	1.227
11	P&C: taxable maturity dollars	−1.904
12	Life: municipal maturity dollars	2.269
13	Life: taxable maturity dollars	−0.572
14	Taxable term spread, ten years	0.276
15	Taxable term spread, twenty years	0.165
16	Taxable term spread, thirty years	0.145
17	P&C: municipal-share second moment	2.833
18	P&C: long municipal fraction	−0.341
19	P&C: long taxable fraction	−5.630
20	Life: municipal-share second moment	−0.379
21	Life: long municipal fraction	−0.197
22	Life: long taxable fraction	0.386
	Half sum of squared deviations	36.2870

Notes. Each entry averages six signed dated conditions in original units and divides by one tenth of the observed six-block mean magnitude of the compared quantity, so an entry of one is a residual equal to 10 percent of that condition's specified observed normalizer. Entries are evaluated at the estimated coefficients in the economy with one national municipal market and are not t-statistics. Rows 1–9 are stock differences, scaled by the observed stocks. Rows 10–13 multiply the model stock by its mean-maturity difference and are scaled by the observed book times its mean maturity. Rows 14–16 are annual log-yield differences. Since the model's two-year yield equals the observed one in every month, they are term-spread errors, scaled by the observed mean term spread over the two-year yield. Rows 17 and 20 compare book-weighted second moments. The four long-fraction conditions are $(L - f^{\text{obs}}B)/B^{\text{obs}}_{\text{window}}$, with B and $B^{\text{obs}}_{\text{window}}$ the model and observed books in the greater-than-two-to-thirty-year window and L its greater-than-twenty-year holdings. Rows 17–22 are scaled by their observed values. The two-year municipal stock is part of the ownership and maturity-dollar conditions but not of that distributional window. Conditions are evaluated before averaging. Products of averaged books and ratios generally give a different object.

cash. Holdings clear as $\sum_i m_i x_i = Q$. With common risk aversion, the tax-mix weights at a municipal maturity are proportional to its active holders' resources (Appendix Proposition 9).

Households are grouped by each SCF tax unit's federal rate under that year's law. High-tax households face a federal rate of at least 35%. In 2015–2017 they are about 5% of families and hold 46% of household resources. Lower-tax households, banks and insurers hold over half of measured municipal bonds. Within each household group, tax units are sorted into seven equal-resource bins by outside financial assets less debt, relative to the deposits, bonds and bond funds that fund the portfolio. Bins share group tax rates but have exposure $h_k = a_H + (b_A Out_k - b_D Debt_k)/2$, in which measured outside-asset and debt ratios enter with opposite signs. The three common coefficients are estimated, and the factor of one half only normalizes b_A and b_D. High-tax households pay their home state's top rate. Lower-tax households pay each tax unit's bracket rate. The baseline multiplies federal and state after-tax shares for taxable interest tax unit by tax unit. It gives no federal deduction credit for state taxes on out-of-state municipal interest. Appendix E.4 varies that convention.

Institutions face the pre-2018 corporate rate of 35% and no modeled state tax. P&C proration taxes 15% of municipal interest at 35%, so $\xi_P = 0.65/0.9475 = 0.686$. Life insurers' policyholders' share, one half before 2018, is taxed at 35%, so $\xi_L = 0.65/0.825 = 0.788$. Banks keep municipal interest and 65% of taxable interest, giving $\xi_B = 0.65$, but cannot deduct municipal funding costs. Their financing-tax cost, 35% of the ratio of interest expense to assets weighted by municipal holdings, is 1.02% a year in 2000–2002 and 0.11% in 2015–2017. Including it as τ_i^{fin} in the stationarity condition (107) raises the bank's required return by the same amount on every municipal bond. The calibration applies non-bank-qualified treatment (Appendix E.1). The tax-neutral residual group, $\xi = 1$, holds other cash and taxable bonds, mainly through foreign investors, pensions, governments, money market funds and nonprofits. Estimated institutional exposures comprise two P&C and three life-insurer types with separate resource shares, one bank type and a seven-point Gauss–Hermite approximation to a normal distribution of exposures for the residual group.

Each household type is replicated in each of the 45 markets, with its national resources allocated by the state's share of municipal supply and its relative after-tax share set at that state's rate. The quantitative assignment applies a home-state exemption in every income-tax jurisdiction and taxes out-of-state municipal interest at the resident state's rate. Institutions stay national and choose across all markets, whose independent bond-specific risks offer geographic diversification. The full economy thus has $45 \times 14 + 13 = 643$ investor types: fourteen household types in each market and thirteen institutional types.

The federal calculation includes the tax induced when exempt interest makes more Social Security benefits taxable. A local increase in ordinary income identifies both the additional taxable benefits and the associated federal liability. Removing the tax on that ordinary-income increment isolates the induced tax from exempt interest. This is the Social Security taxation proxy used in the household after-tax shares, and explains why their absolute home-municipal shares can be slightly below one.

For the 2018–2020 curve comparison, I hold all seventeen coefficients fixed and update the measured inputs. The extension uses the 2019 SCF composition with annually inflation-adjusted incomes, mean 2018–2020 end-year stocks and dated federal tax rules. Institutions face the 21% corporate rate, P&C proration of 25% and a life-insurer policyholders' share of 30%. The annual bank financing-tax cost is 0.117%, and insurer portfolio fractions are averaged over the available 2018–2019 holdings. This seventh economy supplies the post-2017 curve observations without adding calibration targets. Appendix C.3 reports its equilibrium residuals.

Seventeen coefficients, common to all six blocks, govern what the data do not measure directly and are estimated jointly (Appendix Table 10). Three set the price and amount of risk. Risk aversion γ prices rate and bond-specific risk for every investor, and a bond with n years to maturity has two-year bond-specific return variance $v_{30}(n/30)^{p_v}$ in every municipal market and the taxable market for $n \geq 4$, with the two-year claim riskless. Three household coefficients, a_H, b_A and b_D, map measured balance sheets into rate exposures. Eight set the exposures and resource shares of the P&C and life-insurer types, one sets the bank's exposure, and two set the mean and dispersion of the residual group's exposures. Exposures are measured like net duration: an exposure h has the rate risk of investing all of a type's resources in a bond whose log price falls by h per unit rise in the rate state.

C.8 Bond-specific return risk and its normalization

The independent shock in (34) is a reduced-form component of a bond's holding-period return after conditioning on aggregate rates. For example, an unexpected change in demand for one issue can affect its realized sale proceeds without moving the common rate state. The variance cost grows with the position, making demand finite and allowing several tax clienteles to hold a maturity. The shock vanishes for the one-period claim.

Placing the same factor inside next-period holding value is an exact algebraic regrouping:

$$\frac{P^a_{T-1}(z')}{P^a_T(z)} e^{\sigma^a_{\eta,T}\eta} = \frac{P^a_{T-1}(z') e^{\sigma^a_{\eta,T}\eta}}{P^a_T(z)}.$$

Both forms give identical realized returns, conditional moments and portfolio conditions. The price function describes the aggregate rate component. The residual is specified in holding returns. A separate persistent

bond-price state would require its own evolution and current-price dependence. The chosen convention keeps that additional state out of the term-structure recursion. Bond-specific risk still affects equilibrium yields through risk compensation and the mean of the multiplicative shock.

These effects are quantitatively small. The one-market calibration's 20 percent variance changes move the thirty-year gap by less than 0.1 basis points in either direction (Table 13). The shock's lognormal mean can also be isolated without changing any holding return. Its conditional mean is $\exp\{(\sigma^a_{\eta,T})^2/2\}$. Replacing it by the mean-one factor $\exp\{\sigma^a_{\eta,T}\eta^a_{T,t+1} - (\sigma^a_{\eta,T})^2/2\}$ gives the corresponding log prices

$$\check{p}^a_T(z) = p^a_T(z) - \frac{1}{2}\sum_{k=1}^{T}(\sigma^a_{\eta,k})^2, \qquad \check{P}^a_T(z) = e^{\check{p}^a_T(z)}.$$

The adjacent-maturity price shifts exactly offset the shock normalization:

$$\frac{\check{P}^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta-(\sigma^a_{\eta,T})^2/2}}{\check{P}^a_T(z)} = \frac{P^a_{T-1}(z')e^{\sigma^a_{\eta,T}\eta}}{P^a_T(z)}.$$

Every realized holding-period return, its conditional mean and covariance, and investor optimality therefore remain unchanged. Fixed dollar supplies imply unchanged portfolios and market clearing. The one-period boundary is preserved because $\sigma^a_{\eta,1} = 0$.

For T model periods of $\Delta = 2$ years each, the annualized yield shift is

$$\check{y}^a_T - y^a_T = \frac{1}{2T\Delta}\sum_{k=1}^{T}(\sigma^a_{\eta,k})^2.$$

The fitted variance profile is common to municipal and taxable bonds, so both thirty-year continuously compounded yields rise by 0.855 basis points and their difference is unchanged. Applying the same corporate normalization as in Section 5.4 raises the average thirty-year model gap from 97.39 to 97.59 basis points, a change of 0.20 basis points. The calculation isolates the shock's mean with every realized return and portfolio condition held fixed. Regrouping the factor inside holding value gives exactly the same result.

D Bond-side adjustments and empirical tests

D.1 Comparability adjustments and bond-side differentials

The headline ledger is Table 26 in Appendix D.1.7.

D.1.1 Object and conventions

Common unit and sample. Each adjustment is a basis-point contribution to

$$\text{Spread}(T) = y^{M,\text{raw}}_T - \bar{\xi}_H(T)\, y^C_T. \tag{112}$$

The raw municipal zero curve $y^{M,\text{raw}}_T$ is the curve before either comparability adjustment and includes callable and noncallable issues, and $\bar{\xi}_H(T)y^C_T$ averages each bond's household after-tax corporate yield at its federal and state household tax rates. The household benchmark applies federal and state taxes, the net investment income tax (NIIT) and, from 2018, the cap on state and local tax (SALT) deductions, and its mean wedge is 41.3%. Both curves and the benchmark average equally within maturity × issue-quarter × half-notch rating cells, then across ratings within quarters, then across quarters. The ledger expresses all components in municipal-yield basis points. Their tax treatments follow the constructions below.

The master ledger divides adjustments by the full-sample all-bond spread. The CDS-matched sample used for default starts in 2004 and contains 1,559,971 curve observations. Table 23 reports its own spread under the same dated household tax convention, and the estimated default differential enters the full-sample ledger. Each component retains its eligible sample and estimator.

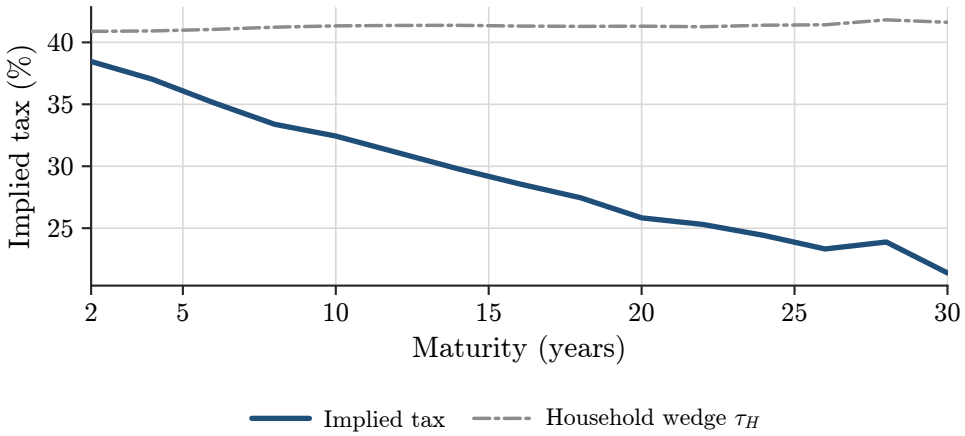


Figure 14: The municipal puzzle in implied-tax units, A-or-above sample, 2000–2020. The implied tax $1 - y_T^M / y_T^C$ uses the call- and issuance-premium-adjusted municipal yields in Figure 2. The household wedge $\tau_H = 1 - \xi_H$ is one minus the ratio of averaged after-tax to pretax corporate yields.

Bands and additive accounting. Each component's band is defined with its estimator below. The first-order ledger estimates each margin separately, sets interactions to zero and adds the contributions in common spread units, so the order in which components are removed leaves the sum unchanged.

Appendices D.1.2–D.1.6 give the five estimators. Table 27 collects their contributions.

D.1.2 Adjustment I: embedded call options

More than 95% of bonds beyond twenty years are callable, so dropping callable bonds would select a small long-end subsample. The bond-level adjustment retains that part of the curve.

The call option. Long municipal bonds conventionally allow par calls on coupon dates from year ten. Their yields compensate investors for the issuer's refunding option. Callable bonds comprise 51.1% of curve observations and 51.8% of those at two to thirty years.

Investment-grade corporates are usually bullet or have make-whole strikes that discourage exercise as rates fall. With few long-end municipal noncallables, I remove each bond's call value,

$$CV_b \;=\; V_b^{\text{bul}} - V_b^{\text{call}} \;\geq\; 0. \tag{113}$$

The callable and bullet valuations share one fitted option-adjusted spread (OAS), which absorbs unpriced credit, liquidity and other factors. The observed callable price identifies a model-conditional counterfactual bullet price.

Step 1: contractual cash flows. Day counts use 30/360 US (NASD), without an end-of-February adjustment or business-day calendar: $\text{DC}(u, v) = [360(Y_v - Y_u) + 30(M_v - M_u) + D_v^* - D_u^*]/360$, where $D_u^* = \min(D_u, 30)$ and $D_v^* = 30$ if $D_v = 31$ and $D_u^* = 30$, otherwise $D_v^* = D_v$. A bond with remaining maturity T_b and $\Delta = 0.5$ has $n = \lceil T_b/\Delta - 10^{-9} \rceil$ coupon dates, a first stub $s = T_b - \Delta(n-1)$ and coupon nodes $t_k = s + \Delta(k-1)$, $k = 1, \ldots, n$.

Let $f = \text{DC}(\text{settlement}, \text{first coupon})$. Then $\text{skip} = \text{round}((f - s)/\Delta)$, and the first paying node is $\underline{k} = \text{skip} + 1$. Each bond must satisfy $|f - s - \Delta \,\text{skip}| \leq 0.02$ years, $0 \leq \text{skip} \leq 2$ and $\underline{k} < n$. Stubs are rounded to 1/360 year to form the groups $(n, s, \underline{k})$, so that t_n remains within half a day of T_b. The accrual a_k is zero before $\underline{k}$, equals $s + \Delta(\underline{k} - 1)$ at $\underline{k}$ and equals Δ afterward, and the cash flow is $100ca_k + 100\mathbf{1}\{k = n\}$ for annual coupon c.

If the dated date precedes settlement, add accrued interest $100c\alpha$ to the clean target price and to the first paying coupon, with $\alpha = \text{clip}(\text{DC}(\text{dated}, \text{first coupon}) - [s + \Delta(\underline{k} - 1)], 0, 1)$. About 29% of observations require this correction.

Step 2: recurring calls. Recorded call times $e = \text{DC}(\text{settlement}, \text{call date})$ in $(0, T_b + 0.15)$ are kept, with the lowest strike taken at ties. Each call maps to grid date $k = \text{round}((e-s)/\Delta)+1$, which must satisfy $\underline{k} \le k \le n$ and $|t_k - e| \le 0.02$ years. Before the first accepted date $K_k = \infty$, and thereafter the last recorded strike is carried forward to every coupon date. Special-call conditions are excluded. A marked callable with no accepted date is excluded from the adjustment. Other bonds without calls have $CV_b = 0$.

Step 3: state-dependent pricing. Issue year selects six calibration economies or the 2018–2020 economy, which uses 2018–2019 insurer fractions and 2018–2020 quantities, rates and taxes. Supply-weighted municipal log prices are interpolated over $\{0, 2, \ldots, 30\}$, including $p_0 = 0$:

$$\log P_T(z) = \text{Interp}_T\left\{\sum_s w_s p_T^{M_s}(z)\right\}, \qquad w_s = \frac{\text{muni supply in state } s}{\text{total muni supply}}. \tag{114}$$

The OAS is based on the mean curve. Spot volatility is measured volatility times a scale m, which is set before the pricing drift is fitted:

$$\log P_T^{\text{scaled}}(z) = \log P_T(0) + m\kappa_T[\log P_T(z) - \log P_T(0)], \qquad \kappa_T = \sigma_T^{\text{data}}/\sigma_T^{\text{model}}. \tag{115}$$

Here $\sigma_T^{\text{model}} = T^{-1}|\partial_z \log P_T|_{z=0}\sigma_y\sqrt{2/(1+\rho_y)}$. Measured volatility is the SD of twelve-month AA-and-above zero-yield changes in the same quarter of consecutive years, after averaging rating cells to maturity–quarter means. Each maturity must have fifty quarters, and the series is smoothed with a count-weighted $\alpha_0 + \alpha_1 e^{-T/\alpha_2}$ fit. The calibrated two-year AR(1) is converted to annual and h-year steps:

$$\rho_y = \sqrt{\rho}, \quad \sigma_y = \frac{\sigma_r}{\sqrt{1+\rho_y^2}}, \quad \rho_h = \rho_y^h, \quad \sigma_h = \sigma_y\sqrt{\frac{1-\rho_y^{2h}}{1-\rho_y^2}}. \tag{116}$$

The pricing drift μ_j differs from the physical drift and is fitted to one-step prices at $T = 1, 1.5, \ldots, 30$, using fifteen-point Gauss–Hermite nodes x_q and weights w_q:

$$\mu_j = \arg\min_{\mu} \sum_T \left[\log P_\Delta(z_j) + \log \sum_q \frac{w_q}{\sqrt{\pi}} P_{T-\Delta}(\mu + \sigma_\Delta\sqrt{2}x_q) - \log P_T(z_j)\right]^2. \tag{117}$$

Prices and drift are interpolated to 201 nodes spanning the model grid. Transitions split quadrature mass between adjacent nodes and clamp tails to endpoints. The first stub uses variance σ_s^2 and drift $\mu^{(s)}(z) = z + (s/\Delta)[\mu(z) - z]$. The median corporate one-year yield in the final month of the issue quarter, in percent, sets $x_0 = \text{clip}(2y_1/100 - \bar{r}, z_1, z_N)$, and contractual dates remain unchanged.

Step 4: callable value and OAS. With $V_n(z) = 100ca_n + 100$, recurse backward at flat continuous spread θ:

$$C_k(z) = P_\Delta(z)e^{-\theta\Delta}\sum_j \Pi_{zj}V_{k+1}(z_j), \qquad V_k(z) = 100ca_k + \min\{C_k(z), K_k\}. \tag{118}$$

Exercise is ex-coupon, and the first stub is discounted using $P_s(z)e^{-\theta s}\sum_j \Pi_{zj}^{(s)}V_1(z_j)$ and interpolated at x_0. The spread θ^* is fitted to the dirty callable price by 42 bisections on $[-0.20, 0.30]$, which leaves an interval of width about 10^{-13} and a repricing error below 10^{-9} per 100 face.

Step 5: bullet value. Removing the call cap in (118), with OAS and cash flows held fixed, gives $CV_b = \max\{V^{\text{bul}}(\theta^*) - V^{\text{call}}(\theta^*), 0\}$ and $P_b^{\text{dec}} = P_b^{\text{dirty}} + CV_b$.

Step 6: zero-coupon yields. On the maturity-interpolated mean discount curve D_0 at $z = 0$, solve for a flat continuous spread $\hat{s}$ by Newton's method, separately for the raw dirty price treated as a bullet and for the adjusted price:

$$100c\sum_k a_k D_0(t_k)e^{-\hat{s}t_k} + 100D_0(T_b)e^{-\hat{s}T_b} = P_b^{\text{target}}, \tag{119}$$

$$y_b^{\text{target}} = 200\left[\exp\left\{\frac{y_0(T_b)+\hat{s}}{2}\right\} - 1\right], \qquad y_0(T_b) = -\frac{\log D_0(T_b)}{T_b}. \tag{120}$$

These nominal semiannual yields give a nonnegative call spread $y_b^{\text{nodecall}} - y_b^{\text{decall}}$, because the adjusted price is no lower.

From bonds to the curve. The 1,828,379 A-or-above pricing bonds are drawn from 1,990,791 issues (Table 1). Bonds must have $T_b \in (1, 30]$, at least two coupons, a complete offering price, coupon and yield, fixed or OID coupons, a price above the deep-discount floor and consistent state-tax flags. Calendar and price screens exclude 16,295 bonds for irregular first coupons, unsupported conventions, missing accrual dates or unusable call schedules. Eligible bonds retain 97.6%–99.8% of maturity-level three-stage weights. Their mean call spread $\Delta_{\text{call}}(T)$ is subtracted from the observed curve.[24]

Identification of the volatility level. The scale m is fitted on AA-and-above callable and noncallable bonds matched within maturity × issue-quarter × rating × half-point-coupon cells. Their mean yield difference, averaged across cells then quarters, is $\text{gap}_T^{\text{mkt}}$, with across-quarter standard error se_T. Model call spreads in the same cells give $\text{gap}_T(m)$, and if callability is the only difference, the correct m equates the two gaps. The fit is applied to A-or-above bonds without re-estimation. Runs at $m = 1$ and 1.15 give

$$e_T \;=\; \frac{\ln \text{gap}_T(1.15) - \ln \text{gap}_T(1)}{\ln 1.15}, \tag{121}$$

giving $\text{gap}_T(m) \approx \text{gap}_T(1)\, m^{e_T}$ locally. The all-bond adjustment elasticity e_T^{all} is a diagnostic (Table 17). The multiplier m is fitted by weighted least squares at integer maturities six through twenty with twelve or more quarters and positive market and model gaps:

$$\ln \text{gap}_T^{\text{mkt}} \;=\; \ln \text{gap}_T(1) \;+\; e_T \ln m \;+\; \varepsilon_T, \qquad \text{weight } w_T = \left(\text{gap}_T^{\text{mkt}}/\text{se}_T\right)^2, \tag{122}$$

whose solution is

$$\ln \hat{m} \;=\; \frac{\sum_T w_T\, e_T\, \ln\left(\text{gap}_T^{\text{mkt}}/\text{gap}_T(1)\right)}{\sum_T w_T\, e_T^2}. \tag{123}$$

The estimate $\hat{m} = 1.089$ has conditional 95% band $[0.972, 1.204]$ from 2,000 independent-quarter bootstrap draws, preserving within-quarter maturity dependence and fixing $\text{gap}_T(1)$, e_T and w_T. The band contains measured volatility ($m = 1$).

Result. At $\hat{m}$, the adjustment for screened A-or-above bonds is small through ten years (4.6 bp) and 29.8 bp at thirty, with directly repriced range [23.0, 36.7] bp. Measured volatility alone gives 24.6 bp.

Direct-repricing validation of the elasticity. At all three multipliers and all fifteen reporting maturities, the all-bond elasticity approximation is within 0.239 bp of direct repricing. This holds although both range endpoints lie outside the estimation pair, below 1.00 and above 1.15.

The call table uses the screened A-or-above pricing sample, while the main puzzle retains the full observed curve and applies the same call component. Screening changes three-stage municipal yields by at most 1.1

[24] Premium bonds with early par calls can require negative OAS and extreme call spreads, and they remain in the sample. Call spreads above 100 basis points contribute at most 0.34 basis points to a three-stage maturity mean, including 0.25 at thirty years, under full three-stage weights.

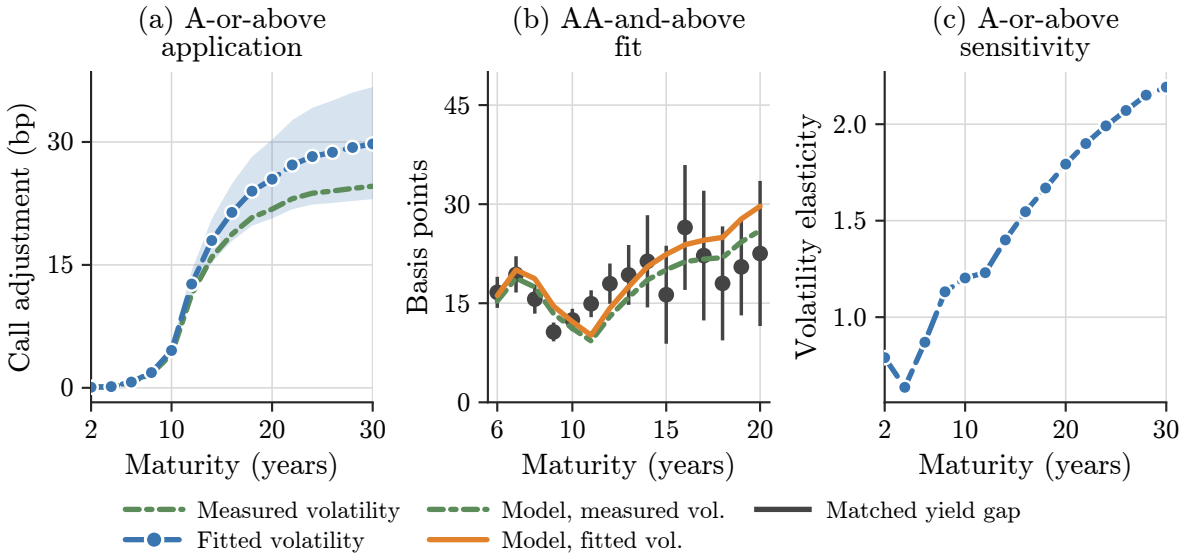


Figure 15: Call adjustment and volatility calibration. (a) A-or-above call adjustments at measured and fitted volatility, with direct-repricing bootstrap endpoints. (b) AA-and-above coupon-matched market gap (±1.96 standard errors) and model gaps, over fitting maturities 6–20 years. (c) All-bond elasticity e_T^{all} of the A-or-above call adjustment, distinct from the matched-cell elasticity e_T used in the volatility fit.

Table 17: Call adjustment by maturity (bp). The central estimate and range are direct repricings at $\hat{m}$ and at the 2.5th and 97.5th percentiles of its bootstrap distribution. Measured volatility and the all-bond elasticity e_T^{all} are diagnostics.

T	measured ($m = 1$)	e_T^{all}	central ($\hat{m}$)	range
2	0.1	0.79	0.1	[0.1, 0.1]
4	0.1	0.64	0.1	[0.1, 0.2]
6	0.7	0.87	0.7	[0.6, 0.8]
8	1.7	1.13	1.9	[1.6, 2.1]
10	4.1	1.20	4.6	[4.0, 5.1]
12	11.4	1.23	12.7	[11.0, 14.3]
14	15.9	1.40	18.0	[15.3, 20.6]
16	18.7	1.55	21.4	[17.9, 24.9]
18	20.8	1.67	24.0	[19.8, 28.2]
20	21.8	1.79	25.5	[20.7, 30.3]
22	23.1	1.90	27.2	[21.8, 32.7]
24	23.7	1.99	28.2	[22.4, 34.2]
26	24.0	2.07	28.7	[22.6, 35.1]
28	24.3	2.15	29.3	[22.8, 36.0]
30	24.6	2.19	29.8	[23.0, 36.7]

bp and the household-tax corporate benchmark by less than 0.1 bp. Tables 27 and 17 report the resulting call adjustments.

Validation against the noncallable subsample. The call-adjusted all-bond and noncallable curves in Figure 16 differ by less than three basis points through ten years. On the displayed two-year maturity grid, the largest absolute difference is eight basis points, at eighteen years, and at twenty years the difference between the adjusted all-bond and noncallable yields is −3.6 basis points.

Robustness to option-pricing dynamics. I repeat the valuation with a normal one-factor Hull–White model (Hull and White, 1994). Its time-varying drift matches each dated input discount curve exactly, while mean reversion and volatility can vary separately. With annual mean reversion fixed at 0.10, fitting volatility to the same AA-and-above coupon-matched gaps at six to twenty years gives diffusion volatility 0.0127, with rates in decimals and time in years. Contractual cash flows, recurring call schedules and the bond-specific OAS procedure are unchanged. Probability-weighted paired repricings across the seven dated economies give A-or-above adjustments of 4.7, 23.8 and 28.9 basis points at ten, twenty and thirty years, compared with 4.6, 25.5 and 29.8 in the baseline.[25] The matched-gap root-mean-square error is 3.8 basis points. On the same noncallable-validation cells, the adjusted curve lies 2.1 basis points below the observed noncallable curve at twenty years, compared with 3.6 in the baseline. The thirty-year call adjustment remains close to thirty basis points under this alternative specification.

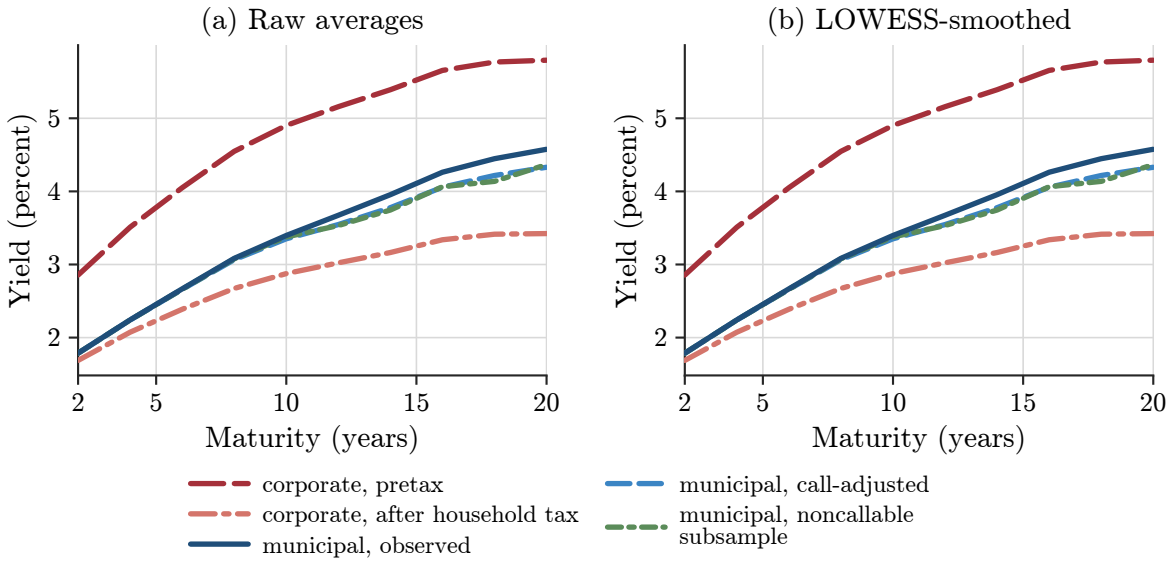


Figure 16: Noncallable validation of Table 17: raw means (a), LOWESS-smoothed (b). Lines show pretax and household after-tax corporate yields alongside observed, call-adjusted and noncallable municipal yields. The latter two share A-or-above maturity–quarter–rating cells containing noncallables, with fifty or more quarters per maturity and no extrapolation past twenty years. Remaining coupon, credit and other differences prevent identification of individual option values.

D.1.3 Adjustment II: capital-gains taxation

Equation (35) applies the interest-income tax treatment to holding-period returns. This adjustment measures the effect of taxing resale gains and losses separately, weighting the correction by empirical first-sale horizons.

[25] The application uses 3,136 probability draws, covering 2,843 distinct callable contracts. Weighted mean differences from baseline pricing are added to the baseline population adjustments, retaining the same three-stage weights and zero call values for noncallables. The conditional numerical-integration standard error of the paired thirty-year difference is 0.056 basis points.

Taxed amounts and rates. Appendix E.5 gives accrual and basis rules. Municipal interest and OID are exempt. Market discount and gains are taxable. OID raises basis, original-issue premium (OIP) lowers it without a municipal deduction, and acquisition premium reduces OID. Taxable interest and OID are ordinary income, and premium amortization assumes the Section 171 election. Original buyers use basis A_h, and secondary buyers start from the purchase price.

Market discount below a quarter point per full remaining year is treated as capital gain. Larger discount is ordinary income at disposition. With n remaining years and face 100,

$$\bar{D}_n = 0.0025 \times 100 \times n. \tag{124}$$

All seven economies use $\tau_{oi} = 0.37$ and $\tau_{cg} = 0.20$, excluding NIIT and state tax. Residual gains and losses are treated symmetrically,

$$\mathcal{T}(G) = \tau_{cg} G, \tag{125}$$

so losses are fully deductible in this calculation.

Benchmark and actual tax rules. For each bond pair, mapping B exempts municipal payments and taxes all corporate income, including discount, as ordinary income. Mapping A separates interest and capital income under the rules above. Both use the fixed rates specified here. Equation (112) uses dated household rates. Under each mapping, Step 5 solves for the municipal issuance-yield shift δ at which the municipal seller earns the corporate seller's return. The difference between these shifts isolates the resale-tax correction.

Step 1: the instruments. Each variant is averaged equally over seven 2000–2020 economies before its range is formed. Discount factors use supply-weighted municipal mean log prices and corporate prices at $z = 0$, interpolated annually from two-year knots. At even $T = 4, \ldots, 30$ and $h = 2, 4, \ldots, T-2$, the two instruments are

$$\text{PAR:} \quad c = \frac{1 - d_T}{\sum_{n=1}^{T} d_n}, \quad \text{offer} = 100, \quad A_j \equiv 100; \qquad \text{ZERO:} \quad \text{offer} = 100\, d_T, \quad A_j = \text{offer} \cdot e^{y_0 j}, \tag{126}$$

with $y_0 = -\log(d_T)/T$.

For a zero purchased at issue, the interest accruing during year $j+1$ is

$$A_{j+1} - A_j = A_j \left(e^{y_0} - 1\right), \qquad A_T = 100.$$

At a sale in year h the original buyer's residual capital gain is $P_h - A_h$, so accrued interest is not taxed again as a gain.[26]

Step 2: rate states at the resale date. Composite Gauss–Legendre nodes z_i and weights ω_i split at both markets' tax boundaries and use the grid-truncated AR(1) density from $z = 0$:

$$w_i \;\propto\; \omega_i \, \frac{1}{\sigma_h} \varphi\!\left(\frac{z_i}{\sigma_h}\right). \tag{127}$$

Here φ is standard normal and σ_h is given by (116). Results with twenty-four and with twelve nodes per regime interval differ by less than 0.001 bp across 2,940 cells, and the residuals of (135) are at most 10^{-12}.

[26] IRS Publication 1212, *Guide to Original Issue Discount (OID) Instruments*, and Publication 550, *Investment Income and Expenses*, describe accrual and basis. Annual effective yield $e^{y_0} - 1$ corresponds to continuous issuance yield.

Step 3: the corporate anchor. The seasoned corporate equilibrium price $P_h^{\text{corp}}(z)$ is held fixed. For each mapping $\mathcal{M} \in \{B, A\}$, its annual after-tax cash flows $X_j^{\mathcal{M}} = \text{cf}_j - \text{tax}_j^{\mathcal{M}}$ determine the continuous return

$$\sum_{j=1}^{T-h} X_j^{\mathcal{M}}\, e^{-r^*_{\mathcal{M}}(h,z)\, j} \;=\; P_h^{\text{corp}}(z). \tag{128}$$

Flows are annual, with continuous discounting and no day-count or semiannual convention. The tax vector $\text{tax}_j^{\mathcal{M}}$ implements the rules above. For the corporate buyer holding to maturity, the mappings differ only in their de minimis treatment of market discount D:

$$\text{rate}_{\mathcal{M}}(D) = \begin{cases} 0, & D \le 0, \\ \tau_{cg}, & \mathcal{M} = A \text{ and } D < \bar{D}_n, \\ \tau_{oi}, & \text{otherwise.} \end{cases} \tag{129}$$

Step 4: the municipal resale price under each mapping. A municipal buyer holding to maturity under $\mathcal{M}$ must earn $r^*_{\mathcal{M}}$. Benchmark flows are exempt, giving

$$P^B(h,z) = \sum_{k=1}^{T-h} CF_k\, e^{-r^*_B(h,z)\, k} \;\equiv\; G_B(h,z). \tag{130}$$

The actual price and tax depend jointly on the discount $A_h - P$. Let G_A be the gross present value at the actual anchor and $q = e^{-r^*_A(h,z)(T-h)}$ the terminal discount factor. Taxing the discount once at maturity gives $P = G_A - q\,\tau\,(A_h - P)$, which is linear in P, so

$$P(\tau) = \frac{G_A(h,z) - q\,\tau\, A_h}{1 - q\,\tau}, \tag{131}$$

and the regime is selected in this order, with $n = T - h$ remaining years:

$$P^A(h,z) = \begin{cases} G_A, & G_A \ge A_h \quad \text{(no discount: premium, untaxed)}, \\ P(\tau_{cg}), & A_h - P(\tau_{cg}) < \bar{D}_n \quad \text{(de minimis)}, \\ P(\tau_{oi}), & \text{otherwise.} \end{cases} \tag{132}$$

When both regime-specific prices satisfy their conditions, this order selects the de minimis price, the higher of the two.

Step 5: the two shifts. A scalar δ shifts the municipal log-discount curve in parallel,

$$\hat{p}_n^{\delta}(z) = \hat{p}_n(z) - \delta n \quad \Longleftrightarrow \quad y_n^{\delta} = y_n + \delta. \tag{133}$$

Both sellers are compared on

$$R \;=\; \frac{1}{h}\, \log \frac{\sum_i w_i\, \mathcal{W}(z_i)}{P_0}, \tag{134}$$

where $\mathcal{W}(z_i)$ is terminal wealth at z_i and P_0 the offer price. Both sellers reinvest interim coupons to h on the unshifted muni curve. Corporate wealth carries forward coupons net of τ_{oi} and the annual OID tax, and the residual gain $P_h^{\text{corp}} - A_h^C$ is taxed at τ_{oi} under B and at τ_{cg} under A. Municipal wealth equals the resale price plus exempt coupons, less τ_{cg} on its residual gain under A. Bisection solves

$$R^{\text{muni}}(\delta_B;\, \text{benchmark}) = R_B^{\text{corp}}, \qquad R^{\text{muni}}(\delta_A;\, \text{actual}) = R_A^{\text{corp}}, \tag{135}$$

and the conditional adjustment is their difference,

$$\Delta_{\text{cg}}(h,T) = 10^4\big(\delta_A - \delta_B\big) \text{ basis points.} \tag{136}$$

The adjustment averages across instrument pairs with issuance weights, in continuously compounded annual yields. The difference $\delta_A - \delta_B$ isolates capital-income taxation from curve mismatch. Each shift recomputes only the municipal issuance coupon or zero price and basis, together with its resale price. States, weights, corporate prices, cash flows and targets, anchors and the reinvestment curve stay fixed.

Instrument types and issuance weights. Issues with positive coupons and prices at least 90 are PAR-like, including premiums above 110. Prices below 90 or zero coupons are ZERO-like. Nonpositive FISD prices denote missing values.

Pooled 1998–2021 amount-weighted ZERO shares are 1.3% municipal and 2.2% corporate (Table 18). Baseline shares match original maturities within three years of each node, pooling when offering amounts are insufficient. At thirty years they are 0.97% and 0.31%. The four pairs in $\{\text{PAR}, \text{ZERO}\}^2$ are weighted by $w(c, m) = \text{share}_c \times \text{share}_m$. The PAR/PAR pair receives 98.7% of the weight at thirty years and drives the adjustment.

Table 18: Four instrument pairs at $T = 30$: pooled interacted issuance weights and horizon-specific adjustments, averaged across seven dated economies. Final columns give each horizon profile's maximum and mean. Baseline maturity-matched shares put PAR/PAR at 98.7% at thirty years.

corp / muni	weight (%)	max over h (bp)	mean over h (bp)
PAR / PAR	96.5	6.51	5.49
ZERO / PAR	2.2	20.07	14.90
PAR / ZERO	1.3	7.71	5.74
ZERO / ZERO	0.0	15.75	10.91

First-sale weights. The conditional adjustment $\Delta_{\text{cg}}(h, T)$ is weighted by the original buyer's first-sale probability at age h. Step 4 already prices the next buyer's tax under hold-to-maturity, so turnover would double-count later dispositions.

Let t_a be seasoned traded par over outstanding par at age a. I measure it from the MSRB dealer-to-customer tape (71.6 million records, 2005–2021), pooling original maturities within ±4 years of T. To exclude primary distribution, I omit when-issued trades and the first quarter after the dated date. A trading segment, a share $\phi(T)$ of par, carries all resales. The remainder is held to maturity. With aggregate turnover fixed, first-sale hazard is

$$\lambda_a = \min\big(t_a/\phi,\, 0.99\big), \tag{137}$$

the cap binding only at the first two ages of thirty-year bonds in the central calibration, where t_a/ϕ would otherwise reach 1.003 and 0.991. The first-sale density and held-to-maturity mass are

$$f_\phi(a) = \phi\,\lambda_a \prod_{j<a}\big(1-\lambda_j\big), \qquad S_\phi(T) = 1 - \phi\Big(1 - \prod_{a<T}\big(1-\lambda_a\big)\Big), \qquad \sum_a f_\phi(a) + S_\phi(T) = 1, \tag{138}$$

and the unconditional adjustment is $\Delta_{\text{cg}}(T) = \sum_h \text{inc}_h(T)\,\Delta_{\text{cg}}(h, T)$, where inc_h sums f_ϕ over the ages assigned to h and the held-to-maturity branch carries a zero adjustment. Each h receives ages $h-1$ and h. The first node also receives age 0, and the last node, with two years remaining, also receives the final pre-maturity age $T-1$.

Calibrating the trading segment. Repeated trades in a narrow segment can generate high turnover without nearly universal resale. Cumulative seasoned CUSIP volume, capped at outstanding par, bounds the par ever sold (Table 19). The parameter ϕ is fitted over the observed window and then extended to full life. At thirty years, par weights within four years give ceiling 0.602 (single-bin 0.617), $\phi = 0.602$ and $S_\phi = 0.398$, with first sales concentrated at young ages.

Result. The envelope crosses three tape windows, three segment multipliers (0.5, 0.75 and 1.0) and two weighting schemes (pooled and maturity-matched). Subunit multipliers are re-fitted on the full window

Table 19: The measured ceiling on the fraction of par ever sold, for selected one-year bins of original maturity. Cumulative traded par counts repeat trades of the same dollar, so the ceiling is an upper bound, and the capital-gains band extends below it. Entries are par-weighted over issues dated from 2005.

orig. maturity	ceiling on P(ever sold)	never traded	obs. window (y)	CUSIPs
4	0.466	0.195	3.9	124520
6	0.571	0.155	5.6	121412
8	0.556	0.227	7.0	118265
10	0.625	0.132	8.0	110693
14	0.558	0.171	9.4	73189
18	0.546	0.216	9.7	46382
22	0.586	0.219	10.0	13056
26	0.612	0.193	10.6	6551
30	0.617	0.159	10.3	22151

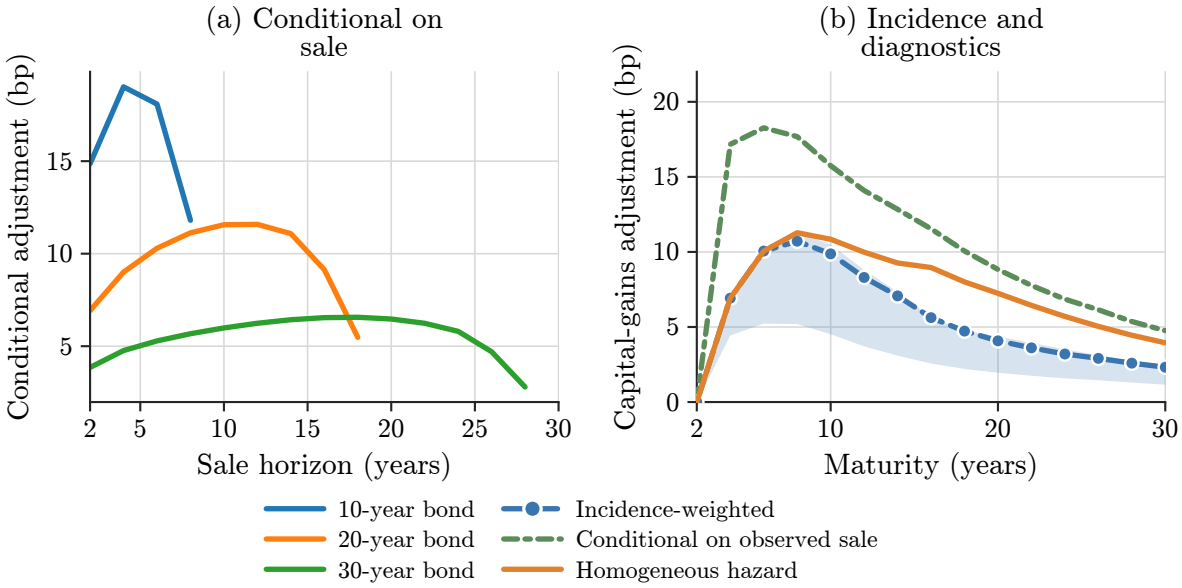


Figure 17: Capital-gains adjustment, averaged over the seven dated economies. (a) Pair-weighted conditional adjustment $\Delta_{\mathrm{cg}}(h, T)$ by sale horizon for ten-, twenty- and thirty-year bonds. (b) Incidence-weighted adjustment with the range of its eighteen variants, and the conditional-on-observed-sale and homogeneous-hazard diagnostics.

before first-sale weights are measured on each half window. The center uses the ceiling, and the effects of the window and the weights have either sign. The homogeneous hazard exceeds the long-bond ceiling and band. At short maturities the fitted segment share is $\phi = 1$, so the reported and homogeneous-hazard estimates coincide.

With (125), the expected tax on the residual gain depends on its signed mean. The thirty-year adjustment is 2.3 bp, with range $[1.2, 2.5]$ bp (Table 20). Conditional-on-observed-sale and homogeneous-hazard ($\phi = 1$) diagnostics are 4.8 bp and 3.9 bp, using the same seven economies, tax rates and trade tape.

D.1.4 Adjustment III: liquidity

Construction. Imputed round-trip cost is $(\max - \min)/\max$ price within CUSIP $\times$ day $\times$ size cells containing customer buys and sells, and the extrema need not belong to one dealer's round trip. The comparison uses 2005–2020 MSRB and TRACE trades of at least \$250,000 made more than three months after issue and more than one year before maturity. Remaining maturity is rounded down to whole years

Table 20: Capital gains by maturity (bp). Conditional and homogeneous-hazard values are diagnostics. The maximum is over sale horizons of the dated-average, pair-weighted conditional curve. The reported series uses first-sale probabilities, and its range is the method envelope. At two years no sale horizon precedes maturity on the two-year grid, so the adjustment is zero.

T	cond. on sale	max over h	ϕ	P(hold)	homogeneous	reported	range
2	0.0	0.0	–	1.00	0.0	0.0	[0.0, 0.0]
4	17.2	17.2	1.00	0.60	6.9	6.9	[4.4, 7.0]
6	18.3	19.2	1.00	0.45	10.1	10.1	[5.2, 10.3]
8	17.7	19.9	0.88	0.40	11.3	10.7	[5.2, 11.3]
10	15.7	19.0	0.82	0.37	10.8	9.9	[4.5, 10.5]
12	14.1	17.8	0.73	0.39	10.0	8.3	[3.7, 8.8]
14	12.8	16.2	0.67	0.40	9.3	7.1	[3.1, 7.3]
16	11.5	14.6	0.58	0.44	9.0	5.6	[2.6, 5.6]
18	10.1	13.0	0.57	0.44	8.0	4.7	[2.2, 4.8]
20	8.8	11.6	0.57	0.43	7.2	4.1	[2.0, 4.4]
22	7.8	10.4	0.58	0.42	6.4	3.6	[1.8, 3.9]
24	6.9	9.3	0.59	0.41	5.7	3.2	[1.6, 3.5]
26	6.1	8.5	0.60	0.40	5.0	2.9	[1.5, 3.1]
28	5.4	7.4	0.60	0.40	4.5	2.6	[1.3, 2.8]
30	4.8	6.6	0.60	0.40	3.9	2.3	[1.2, 2.5]

and assigned to the six five-year bins in Table 21. Costs are averaged to bond-months before matching. The TRACE 1MM+ and 5MM+ caps are retained at their lower bounds, and trade size is not otherwise used.

Cells are matched by month × rating family × maturity, with ten or more bonds per market. Agency letter grades map AAA to notch one, AA to two–four and A to five–seven on the twenty-two-notch scale. Averages weight ratings equally within months, months equally within years, and then years equally.

Let c_m and c_c be the matched municipal and corporate costs in percent of price. One purchase and resale over h years requires annual compensation of c_m/h and c_c/h in the two markets, giving the yield differential in basis points

$$\Delta y^{\text{liq}} = \underbrace{\frac{100\,c_m}{h}}_{\text{municipal cost per year}} - \underbrace{\frac{100\,c_c}{h}}_{\text{corporate cost per year}} = \frac{100(c_m - c_c)}{h}.$$

Both markets use $h = 5$. The construction treats trading costs as reductions in after-tax wealth, with no further multiplication by the interest-income after-tax share. A positive municipal-minus-corporate cost difference raises required municipal yields and is subtracted from the puzzle. Standard errors are the SD of sixteen annual differences divided by $\sqrt{16}$. The band spans $h = 2$, $h = 10$, the regression map and the central estimate ± 1.96 standard errors.

The six bin estimates at $3, 8, 13, 18, 23, 28$ years are interpolated linearly to $2, 4, \ldots, 30$, with end values held flat. Thirty years therefore uses the 26–30 bin, and the interpolation avoids artificial jumps in the ledger.

Result. Institutional muni round trips are slightly cheaper before twenty years and more expensive afterward (Table 21). At 26–30 years, costs of 0.5335% and 0.5310% imply 0.050 bp annually at $h = 5$, which is reported as 0.1 bp, with a band of $[-2.2, 2.3]$ bp. The higher-cost 21–25 bin produces the peak adjustment of 2.9 bp at twenty-four years.

At $h = 5$, standard errors are 0.8–2.3 bp. The 21–25 estimate is about 1.5 standard errors from zero and depends on the holding horizon. Regressing A-or-above offering yields on bond-average round-trip costs, with maturity-bin × rating × offer-year effects, gives 15.4 bp per percentage point of price in round-trip cost, compared with 20 at $h = 5$. Thus the regression map scales $h = 5$ estimates by 0.772. Appendix D.2 gives separate issuer-level regressions on trading frequency and trading costs, which do not enter the band.

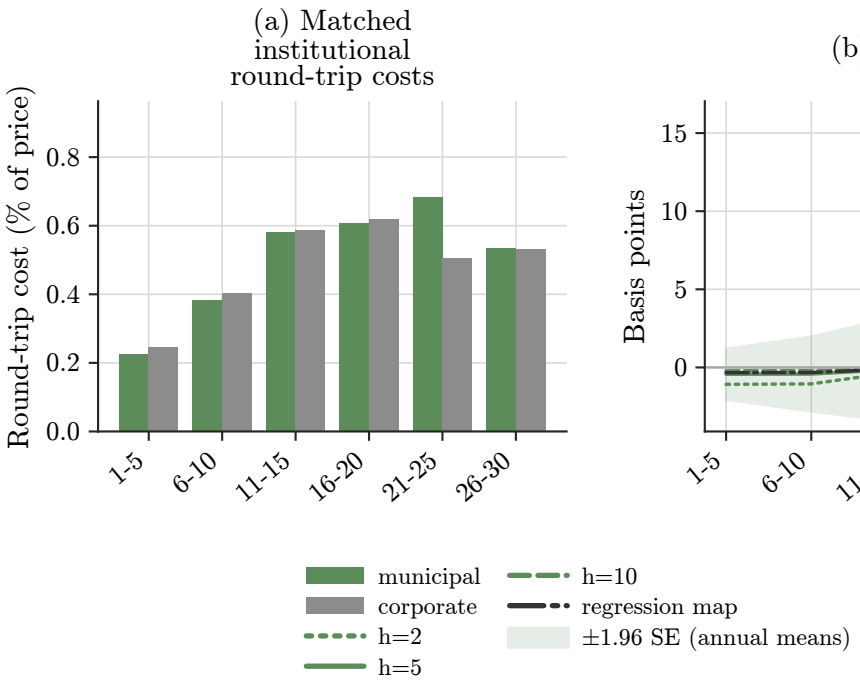


Figure 18: Matched institutional-size customer trades of at least $250,000, 2005–2020. Panel (a): imputed round-trip costs in common month, rating and maturity cells, percent of price. Panel (b): the municipal-minus-corporate cost difference annualized under three holding periods and the regression map. Shading is ±1.96 standard errors calculated across annual means at $h = 5$.

Table 21: Liquidity by maturity bin. Round-trip costs (RTC) in percent of price, adjustments in bp under three holding periods and the regression map.

bin	muni RTC	corp RTC	difference	$h = 2$	$h = 5$	$h = 10$	regr.
1–5	0.225	0.246	-0.022	-1.08	-0.43	-0.22	-0.33
6–10	0.382	0.403	-0.021	-1.05	-0.42	-0.21	-0.33
11–15	0.582	0.588	-0.006	-0.30	-0.12	-0.06	-0.09
16–20	0.609	0.619	-0.010	-0.49	-0.20	-0.10	-0.15
21–25	0.684	0.506	0.178	8.92	3.57	1.78	2.75
26–30	0.533	0.531	0.003	0.13	0.05	0.03	0.04

Sensitivity to the cell rule. Table 22 varies the minimum bond count in each market's cells, using the same underlying trades and size screen. A one-bond minimum admits thin cells and changes the 26–30 year estimate from 0.05 to −6.53 bp. A twenty-bond minimum leaves only 19 common months at 11–15 years and changes the estimate there from −0.12 to −2.99 bp. The ten-bond minimum balances coverage against thin-cell influence.

D.1.5 Adjustment IV: default

Method. I use the simple taxable-equivalent convention in Schwert (2017), which divides municipal interest by the household after-tax share before comparing it with taxable interest. State-government and pooled AA corporate Markit CDS spreads serve as proxies for credit compensation on this common scale. Let s_b^M and s_b^C denote the matched CDS spreads for curve observation b. The default differential in the household's after-tax yield comparison is

$$\mathrm{Def}(T) = \mathrm{Avg}_T[\xi_{H,b} s_b^M] - \mathrm{Avg}_T[\xi_{H,b} s_b^C]. \tag{139}$$

Table 22: Liquidity adjustment at $h = 5$ (bp) under alternative minimum bond counts per market cell.

bin	≥ 1	≥ 5	≥ 10 (used)	≥ 20
1–5	-0.59	-0.50	-0.43	-0.39
6–10	-0.79	-0.49	-0.42	-0.27
11–15	-0.12	0.42	-0.12	-2.99
16–20	-0.09	-0.30	-0.20	-0.31
21–25	2.05	2.20	3.57	1.81
26–30	-6.53	-5.81	0.05	0.30

Both CDS quotes are pretax. Applying the dated household's after-tax share once puts their difference on the scale of $y_T^M - \bar{\xi}_H(T) y_T^C$, adapting Schwert's mapping to these proxies. The after-tax share includes federal and state rates, NIIT from 2013 and the 2018 change in state-tax deductibility. No further corporate-tax or loss-deduction factor is applied.

State-GO CDS are matched to A-or-above bonds by rating, maturity and quarter, and pooled quarterly AA corporate CDS form the corporate leg. After-tax shares are applied before averaging within maturity–quarter–whole-notch-rating cells, then across ratings and quarters, retaining each leg's nonmissing support. AA robustness uses the same construction.

Beyond quoted tenors, cumulative spread $T\,\mathrm{CDS}(T)$ is extrapolated linearly, with negative filled spreads floored at zero and nonnegative quotes left unchanged. The default component is positive when municipal credit compensation exceeds the corporate proxy after the common tax mapping.

Coverage. Exactly observed cells cover 49% at 1–10 years, 47% at 11–20 and 13% at 21–30 in the A-or-above CDS panel. The remaining cells use curve filling (Figure 19, right). The sensitivity range spans the baseline, cells with at least half of their source observations quoted or interpolated, the 2007-onward sample, and CDS–cash basis factors 0.75 and 0.50. Each variant uses the same household tax mapping. The range is their pointwise minimum and maximum.

Result. The thirty-year component is 16.5 bp, with a sensitivity range of $[8.2, 20.8]$ bp. It equals 9.9% of the CDS-matched spread and 10.3% of the full-sample denominator Spread(T), and it remains positive at every maturity (Table 23).

Table 23: Default by maturity. Yields in percent, spread and component in bp. The CDS-matched spread uses each observation's dated federal and state household after-tax share. Its sample differs from the full-sample Spread(T). Both use the same tax convention.

T	muni yield	corp after-tax	spread (bp)	component (bp)	band
2	1.56	1.51	5.7	16.48	[5.69, 17.64]
4	1.99	1.85	14.6	14.35	[6.55, 16.10]
6	2.39	2.15	24.8	12.18	[5.94, 14.20]
8	2.79	2.42	37.0	12.46	[6.23, 14.95]
10	3.12	2.63	49.5	12.46	[6.23, 15.27]
12	3.47	2.79	68.1	11.55	[5.77, 14.50]
14	3.73	2.89	84.0	11.65	[5.83, 14.85]
16	3.93	2.96	97.1	12.26	[6.13, 15.68]
18	4.09	3.01	108.4	12.10	[6.05, 15.54]
20	4.26	3.04	122.2	12.70	[6.35, 16.30]
22	4.36	3.07	129.0	13.34	[6.67, 17.06]
24	4.47	3.08	138.3	14.08	[7.04, 17.94]
26	4.59	3.09	149.3	15.62	[7.81, 19.77]
28	4.58	3.09	149.5	16.16	[8.08, 20.39]
30	4.77	3.11	165.7	16.45	[8.23, 20.75]

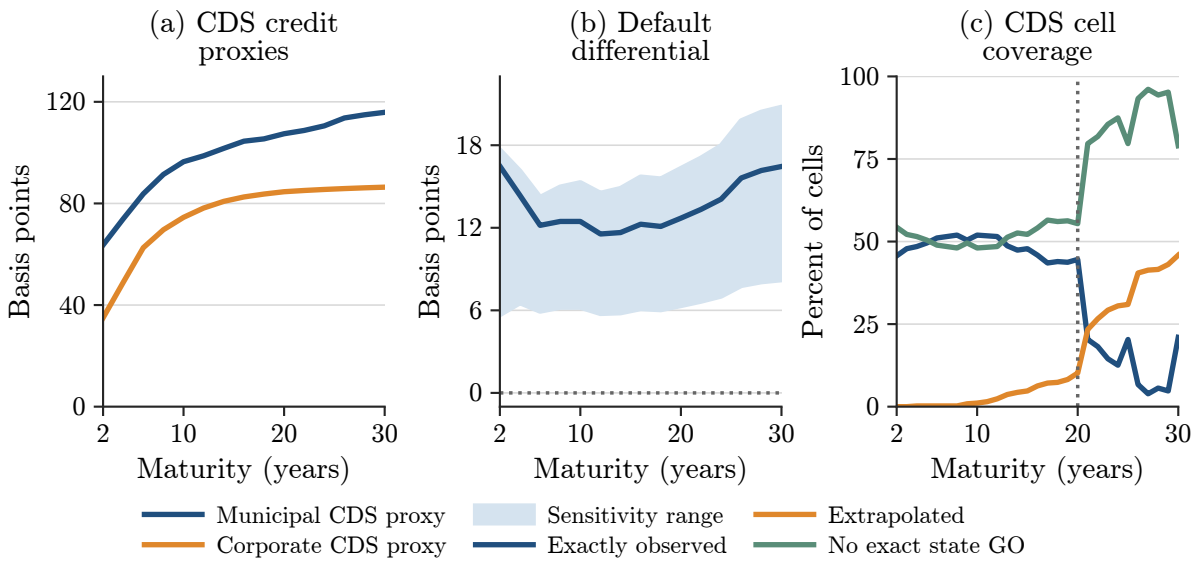


Figure 19: Default adjustment on the A-or-above CDS panel. Left: municipal and corporate CDS proxies in taxable-equivalent units. Middle: their differential after the dated household tax mapping in (139), with the five-variant sensitivity range. Right: exactly observed, extrapolated and no-exact-state-GO shares of rating-matched CDS cells. The last two overlap and the three are not a partition. The vertical rule marks twenty years.

Table 24: Default sensitivity in basis points. Every column uses the dated household tax mapping of (139). The basis columns scale both CDS proxies by the stated factor. Quality-capped cells have at least half of their source observations quoted or interpolated.

T	2007+	basis 0.50	basis 0.75	baseline	quality-capped
2	17.64	8.24	12.36	16.48	5.69
6	14.20	6.09	9.13	12.18	5.94
10	15.27	6.23	9.35	12.46	8.35
14	14.85	5.83	8.74	11.65	8.02
18	15.54	6.05	9.07	12.10	8.76
22	17.06	6.67	10.00	13.34	11.24
26	19.77	7.81	11.71	15.62	17.47
30	20.75	8.23	12.34	16.45	18.66

D.1.6 Adjustment V: the issuance premium

Motivation. The panel has 1,828,379 A-or-above issues, each observed once at issuance. Correlations between zero-coupon and offering yields are 0.998, 0.993 and 0.937, and the conversion from coupon to zero yields raises mean yields by 2.8, 12.7 and 58.0 bp at five, ten and thirty years. In Spread(T), primary municipal yields are compared with a fitted seasoned corporate curve. The adjustment removes the positive placement-related difference between offering and secondary-market yields.

Measurement. Both sides use customer purchases, because an offering is an investor purchase and comparing it with customer sales would add the bid–ask spread and double-count liquidity. Early and seasoned prints must have at least \$250,000 par, and all-size prints give 21.1 bp at thirty years (Table 25). Market drift is controlled with same-week seasoned bonds:

$$\text{Premium}_b = [y_b^{\text{off}} - \bar{y}_k^{\text{seas}}(w_b^{\text{off}})] - [y_{bj}^{\text{tr}} - \bar{y}_k^{\text{seas}}(w_{bj})]. \tag{140}$$

Each bond's eligible prints within thirty offering-date days are averaged, excluding when-issued trades. The benchmark is par-weighted customer purchases in the same week and maturity bucket of bonds older than three years, with at least 25 prints per cell. Differencing removes common rate changes and fixed credit, while changing credit and call exposure can remain.

The sample is restricted to the curve's own filtered population. Using all Mergent offering yields would put only 74% of estimation bonds in that universe. Trade age and benchmark weeks are dated from the offering date. The dated date differs from it by a median of fourteen days, by more than a week for 82.7% and by more than a month for 8.4%, and it precedes the offering for 13.7%.

The three-stage averaging of Appendix D.1.1 is applied within issue-quarter and rating cells and maturity buckets 1–5, 5–8, 8–12, 12–17, 17–23, 23–31. Estimates and bootstrap endpoints are interpolated from midpoints $(3, 6.5, 10, 14.5, 20, 27)$ to the two-year grid, with ends held flat, and subtracted one for one. The longest-bucket mean is 11.9 bp with par weights, which measure dollars, and 12.8 with curve weights, which measure curve observations.

Sensitivity to the trade-age window. In all five windows in Table 25, premia beyond twelve years exceed those below twelve years. From fourteen days, the longest two buckets give 9.3–12.8 bp. Three- and seven-day samples have fewer institutional prints and are noisier. Panel B retains the gradient at higher levels.

Table 25: Issuance premium by maturity bucket (bp) under alternative trade-age windows (column headings, in days, with thirty in the baseline). Panel A, the baseline, requires prints of at least $250,000 par in both legs, the early print and the seasoned benchmark. Panel B uses prints of all sizes.

maturity bucket (yr)	3	7	14	30	60
Panel A: institutional-size prints (baseline)					
1–5	3.7	3.6	1.4	2.9	3.3
5–8	2.8	5.2	3.0	3.3	4.0
8–12	3.4	5.2	3.9	4.0	4.6
12–17	6.2	9.0	7.7	7.5	7.8
17–23	9.1	7.3	9.3	10.8	12.2
23–31	9.5	7.2	11.9	12.8	12.0
Panel B: prints of all sizes					
1–5	4.5	6.7	5.6	3.6	2.2
5–8	5.9	10.1	7.6	6.1	5.0
8–12	7.2	10.5	9.0	7.9	7.4
12–17	15.5	16.2	13.8	12.8	12.4
17–23	15.0	17.3	16.7	16.6	16.7
23–31	18.1	18.7	20.0	21.1	21.0

Notes. Each cell is the drift-controlled difference-in-differences premium, averaged in the same three stages as the curve, for bonds with at least one print inside the window, averaging each bond's prints inside the window.

Timing, conventions and size. Across six buckets, premia average 5.8 bp within three offering-date days, 6.4–7.0 at ages three to thirty, and 8.5 and 11.6 at thirty to sixty and sixty to ninety days, respectively. The changing composition prevents a within-bond interpretation. For par and discount bonds the last two buckets give 10.9 and 13.2 bp, and for premium bonds 9.3 and 10.9, so yield-to-worst differences do not explain the gradient. Long premia are positive in every all-size trade decile and smallest for the largest trades (Figure 20). Small-trade markups also raise the all-size estimates.

Result. The issuance premium rises steeply with maturity, from 2.9 bp at two years to 12.8 bp at thirty (Table 27).

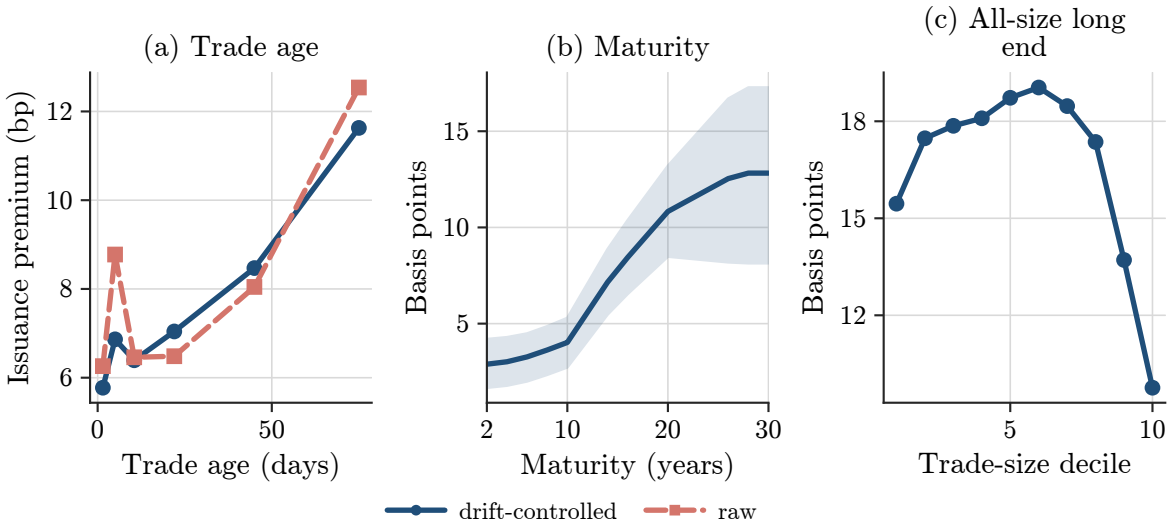


Figure 20: Issuance-premium diagnostics. Left: institutional-size estimates by trade age, with and without drift controls. Middle: institutional-size maturity profile with block-bootstrap band. Right: all-size long-end estimates by trade-size decile.

A 10,000-draw issue-quarter bootstrap samples four-quarter blocks with replacement and endpoint wrapping. Lag-one autocorrelations of quarterly bucket means range from −0.34 (longest bucket) to 0.02. The block and iid-quarter bootstraps give similar short-end intervals, [1.7, 4.2] and [1.8, 4.0] respectively at 1–5 years. The thirty-year interval is [8.1, 17.3] bp, and the intervals exclude zero at every maturity. Intervals condition on the matched sample and seasoned benchmark.

Interpretation. The drift-adjusted post-offering yield change captures the mismatch between primary and seasoned pricing that matters for the curve comparison. Changing dealer markups, ongoing distribution, liquidity seasoning and price discovery may also contribute, so I treat the change as a proxy for the issuer's placement cost.

Sample coverage. Only bonds with an institutional-size print within thirty days of offering contribute, so coverage is 4% of the target curve, and the direction of that selection is not bounded here. The trade tape begins in 2005 and the curve spans 2000–2020, so the estimate is measured on later issues and applied to the whole curve. It also inherits any staleness in recorded offering yields.

D.1.7 The combined effect

Table 26 reports the accounting used in Sections 3.7 and 5.4. Table 27 gives every maturity and component band. The call and issuance-premium adjustments grow with the spread. The three differentials remain near 17–20 bp from twelve years onward. Capital gains peak at intermediate maturities because the resale-tax wedge is amortized over remaining life and the de minimis threshold rises with it. At thirty years, the residual before the model is 98.4 bp, with a component envelope of [80.4, 121.4] bp, and the normalized model leaves an unexplained residual of 1.0 bp (Figure 8). Negative short-end residuals refer to the full curve. Component-specific matched samples can have different signs.

Model and observed curves. Figure 8 and Table 4 report the normalized and unnormalized model comparisons, preserving the empirical household benchmark in (25).

Table 26: What explains the municipal spread over the household after-tax benchmark: contributions in basis points to Spread(T) of equation (112). The two comparability columns remove the call option and the issuance premium from the municipal yield itself, giving the puzzle of Figure 2. The three differential columns are muni-minus-corporate differentials and do not map to a single adjusted curve.

		comparability			differentials					
T	Spread	Call	Issuance premium	Puzzle	Capital gains	Liquidity	Default	Total differential	Share of puzzle	Residual
2	9.9	0.1	2.9	6.9	0.0	−0.4	16.5	16.0		−9.1
6	27.9	0.7	3.3	23.9	10.1	−0.4	12.2	21.8	91%	2.1
10	52.2	4.6	4.0	43.6	9.9	−0.3	12.5	22.0	51%	21.6
14	87.0	18.0	7.2	61.8	7.1	−0.1	11.7	18.6	30%	43.2
18	110.3	24.0	9.6	76.7	4.7	−0.2	12.1	16.6	22%	60.1
22	128.8	27.2	11.4	90.2	3.6	2.8	13.3	19.8	22%	70.5
26	145.0	28.7	12.5	103.7	2.9	1.5	15.6	20.0	19%	83.7
30	159.8	29.8	12.8	117.2	2.3	0.1	16.5	18.8	16%	98.4

Notes. The puzzle is the spread less the call and issuance-premium adjustments. The total differential adds capital gains, liquidity and default, the share divides that sum by the puzzle (omitted where the puzzle is below 20 basis points), and the residual is the puzzle less the sum. Thirty-year component bands, in basis points: call [23.0, 36.7], issuance premium [8.1, 17.3], capital gains [1.2, 2.5], liquidity [−2.2, 2.3], default [8.2, 20.8]. Using unrounded component bounds gives a thirty-year residual before the model of [80.4, 121.4] bp. This component envelope has no joint coverage level. Appendix D.1 gives the construction. The issuance-premium estimate uses the 4% of curve bonds with an institutional-size trade within one month of issuance.

Table 27: Master ledger: five adjustments by maturity, in bp of Spread(T) (equation (112)). Brackets give component bands. The total band is their interval sum and has no joint coverage level. The final column reports the total divided by Spread(T) and is suppressed below 20 bp of observed spread. Table 26 divides the three differentials by the puzzle.

T	call pt band	capital gains pt band	liquidity pt band	default pt band	issuance premium pt band	total pt band	obs. Spread(T)	% expl.
2	0.1 [0.1, 0.1]	0.0 [0.0, 0.0]	−0.4 [−2.1, 1.2]	16.5 [5.7, 17.6]	2.9 [1.7, 4.2]	19.0 [5.4, 23.1]	9.9	
4	0.1 [0.1, 0.2]	6.9 [4.4, 7.0]	−0.4 [−2.2, 1.3]	14.3 [6.6, 16.1]	3.0 [1.8, 4.3]	24.0 [10.7, 28.9]	16.9	
6	0.7 [0.6, 0.8]	10.1 [5.2, 10.3]	−0.4 [−2.5, 1.7]	12.2 [5.9, 14.2]	3.3 [2.0, 4.5]	25.8 [11.3, 31.4]	27.9	92
8	1.9 [1.6, 2.1]	10.7 [5.2, 11.3]	−0.4 [−2.8, 2.0]	12.5 [6.2, 15.0]	3.6 [2.3, 4.9]	28.2 [12.6, 35.2]	41.1	69
10	4.6 [4.0, 5.1]	9.9 [4.5, 10.5]	−0.3 [−3.1, 2.5]	12.5 [6.2, 15.3]	4.0 [2.7, 5.3]	30.6 [14.4, 38.7]	52.2	59
12	12.7 [11.0, 14.3]	8.3 [3.7, 8.8]	−0.2 [−3.3, 3.0]	11.5 [5.8, 14.5]	5.6 [4.1, 7.1]	37.9 [21.2, 47.7]	71.2	53
14	18.0 [15.3, 20.6]	7.1 [3.1, 7.3]	−0.1 [−3.3, 3.0]	11.7 [5.8, 14.8]	7.2 [5.5, 8.9]	43.7 [26.4, 54.7]	87.0	50
16	21.4 [17.9, 24.9]	5.6 [2.6, 5.6]	−0.2 [−3.0, 2.6]	12.3 [6.1, 15.7]	8.4 [6.5, 10.4]	47.6 [30.2, 59.2]	99.4	48
18	24.0 [19.8, 28.2]	4.7 [2.2, 4.8]	−0.2 [−2.6, 2.2]	12.1 [6.0, 15.5]	9.6 [7.5, 11.8]	50.3 [32.9, 62.5]	110.3	46
20	25.5 [20.7, 30.3]	4.1 [2.0, 4.4]	1.3 [−2.0, 4.6]	12.7 [6.4, 16.3]	10.8 [8.5, 13.2]	54.4 [35.5, 68.8]	123.1	44
22	27.2 [21.8, 32.7]	3.6 [1.8, 3.9]	2.8 [−1.4, 7.1]	13.3 [6.7, 17.1]	11.4 [8.4, 14.4]	58.4 [37.2, 75.1]	128.8	45
24	28.2 [22.4, 34.2]	3.2 [1.6, 3.5]	2.9 [−1.3, 7.2]	14.1 [7.0, 17.9]	12.0 [8.3, 15.5]	60.3 [38.0, 78.3]	136.9	44
26	28.7 [22.6, 35.1]	2.9 [1.5, 3.1]	1.5 [−1.7, 4.6]	15.6 [7.8, 19.8]	12.5 [8.2, 16.7]	61.3 [38.3, 79.3]	145.0	42
28	29.3 [22.8, 36.0]	2.6 [1.3, 2.8]	0.1 [−2.2, 2.3]	16.2 [8.1, 20.4]	12.8 [8.1, 17.3]	61.0 [38.1, 78.7]	145.6	42
30	29.8 [23.0, 36.7]	2.3 [1.2, 2.5]	0.1 [−2.2, 2.3]	16.5 [8.2, 20.8]	12.8 [8.1, 17.3]	61.4 [38.4, 79.5]	159.8	38

Bars denote sample averages, and superscripts m and d distinguish model and data. With model municipal and taxable yields $\bar{y}_T^{M,m}$ and $\bar{y}_T^{C,m}$, observed corporate yield $\bar{y}_T^{C,d}$ and observed household benchmark B_T^d, the normalization is

$$\widetilde{y}_T^M = \frac{\bar{y}_T^{M,m}}{\bar{y}_T^{C,m}} \bar{y}_T^{C,d}, \qquad \widetilde{\mathcal{G}}_T = \widetilde{y}_T^M - B_T^d.$$

Here $B_T^d = \bar{\xi}_H(T)\bar{y}_T^{C,d}$, where $\bar{\xi}_H(T)$ is the averaged household benchmark divided by the averaged taxable yield, as in (112). At thirty years the model's taxable yield exceeds the observed corporate yield by 39 basis points, mostly reflecting 2018–2020. Model and data yields are converted to nominal semiannual units before date-by-date after-tax products are formed and averaged with the empirical bond, rating and issue-quarter weights of Appendix D.1.1. Simply rescaling the unnormalized model gap would also carry over the model's own averaged household benchmark. The benchmark here uses observed taxable yields. Date-by-date household after-tax shares covary differently with model and observed yields, so the two calculations differ.

The normalized root-mean-square residual is 11.21 bp (16.97 bp unnormalized), and the maximum absolute residual is 14.87 bp at sixteen years. Five of fifteen absolute residuals are below 10 bp. The signed residuals are −12.03 bp at twelve, −11.26 at twenty and +1.01 at thirty, respectively.

D.2 Issuer-level liquidity and offering-yield regressions

The issuer-level regressions below are separate diagnostics and do not enter the combined adjustment, which uses the matched institutional-size round-trip costs of Section D.1.4.

Table 28 associates offering yields with full-life issuer liquidity proxies. Because the proxies are not predetermined at issuance, the coefficients are neither causal ex-ante premia nor formal bounds. The trading-frequency measure is the issuer mean, across its bonds, of each bond's average daily MSRB trades, with days counted from the bond's dated date to maturity. The spread measure is the issuer mean, across its bonds, of each bond's average imputed round-trip cost, in bond-price units (0.01 is 1% of price, or 100 basis points).

Translation to long-end magnitudes. At 26–30 years, the indicator coefficients in Columns (1), (2), and (4) are about 8–15 basis points in absolute value, and the continuous Column (3) coefficient of 27.262 implies 27.3 basis points for a 0.01 spread contrast. Offering yields thus covary with both liquidity measures.

D.3 Paired issues: Build America Bonds (BABs) vs. tax-exempt

Section 3.8 describes the 2009–2010 BAB program. Scaling BAB offering yields to $0.65 \times y^{BAB}$ puts them on the household after-tax scale and leaves exempt yields unchanged.

Sample and construction. The issue-level (CUSIP) Mergent sample includes non-AMT bonds issued in 2009Q2–2010Q4. Table 29 uses AA-and-above issues (agency score ≤ 3) and A-or-above issues (≤ 7). Figure 3 uses the former and averages yields within integer-maturity $m \in \{1, \ldots, 30\}$ and treatment cells.

Original-issue premium and discount. Exact after-tax IRRs would treat BAB and exempt OID, acquisition premium, interest income and basis separately, so I check the 0.65 yield approximation on bonds issued near par. Restricting the call-matched AA-and-above sample to offering prices of 99–101 leaves a maturity gradient of 2.1 bp per year (s.e. 0.6), against 3.1 in the full sample. The 98–102 restriction gives 2.7 bp (s.e. 0.4). Both remain significant at one percent.

Regression specification. In the notation of equation (4), with BAB as the base category of $EXMP_b$ and m_b the remaining maturity at issuance, I estimate

$$y_b \;=\; \alpha_1\, EXMP_b \;+\; \beta\,(EXMP_b \times m_b) \;+\; \gamma\,(0.65\, y^C_{m_b,t_b}) \;+\; \lambda_{j(b)\times m_b \times q_b} \;+\; \varepsilon_b. \tag{141}$$

Table 28: Issuer liquidity and offering yield, by maturity bin.

	Trading frequency		Transaction-cost spread	
	(1) Above corp median	(2) Top 25% trade	(3) Issuer-avg spread (cont.)	(4) Lowest-spread quartile
Maturity 1–5y	-0.046^{***}	-0.028^{***}	13.446^{***}	-0.057^{***}
	(0.009)	(0.007)	(0.489)	(0.001)
Maturity 6–10y	-0.007^{**}	0.021^{***}	11.624^{***}	-0.048^{***}
	(0.003)	(0.005)	(0.216)	(0.002)
Maturity 11–15y	-0.011^{***}	0.041^{***}	11.860^{***}	-0.040^{***}
	(0.004)	(0.001)	(0.355)	(0.001)
Maturity 16–20y	-0.035^{***}	0.024^{***}	13.447^{***}	-0.043^{***}
	(0.003)	(0.006)	(0.557)	(0.003)
Maturity 21–25y	-0.091^{***}	-0.041^{**}	20.427^{***}	-0.104^{***}
	(0.009)	(0.016)	(1.225)	(0.011)
Maturity 26–30y	-0.093^{***}	-0.082^{***}	27.262^{***}	-0.146^{***}
	(0.030)	(0.018)	(4.815)	(0.037)
Fixed effects	Yes	Yes	Yes	Yes
Maturity cluster	Yes	Yes	Yes	Yes
Observations	785,256	785,256	779,988	785,256

Notes. AA-and-above federally exempt issues, 2000–2021, offering yields in percent. Treatments are issuer averages: (1) daily MSRB trades per bond above the TRACE median for plain-vanilla debentures and medium-term notes; (2) top quartile of muni trade frequency; (3) continuous MSRB spread; (4) bottom spread quartile. Thus -0.046 is -4.6 bp, and column (3)'s 13.4 implies 13.4 bp per 0.01 spread change. Fixed effects interact rounded maturity, rating rounded down, state, issue year–month and call status. Standard errors cluster by rounded maturity, and the sample has no bond-count restriction. $^{*}\,p<0.10$, $^{**}\,p<0.05$, $^{***}\,p<0.01$.

The corporate control $0.65\,y^{C}_{m_b,t_b}$ varies across issue months within matched quarters, identifying γ. Equation (4) uses the unscaled control, with $\gamma_C = 0.65\gamma$. Only the control's coefficient and standard error change, and fitted yields, exempt coefficients and their inference are unchanged.

Results. Table 29 shows a positive maturity gradient under both rating cuts and call matching. All maturity interactions are significant at one percent. Call matching attenuates the long-end gradient by about fifteen percent but leaves $[5, 10)$ unchanged. BAB call flags occur on 49 percent of bonds, versus 39 percent of tax-exempts.

Interpretation. The paired comparison holds issuer and credit fixed, with the BAB yield replacing the corporate benchmark. The widening gap supports the clientele mechanism under the 0.65 approximation and comparable security design and issuance premia. Call-matched columns match call flags only. Exercise terms, coupons, liquidity and OID or premium treatment can differ. The 35% rate is below the AA-and-above curve's 41.1% household average. A lower after-tax share would widen the gap. This short program's comparison agrees most closely with the curve at the endpoints and least at $[15, 20)$.

D.4 The municipal puzzle on unlimited general-obligation bonds

Unlimited-tax general-obligation (GO) bonds pledge full faith, credit and unlimited taxing power. Revenue bonds pledge project revenues. Figure 21 restricts the comparison to AA-and-above unlimited-GO bonds (Mergent code "K") passing the main call-pricing and calendar screens. No default adjustment is applied, because local obligations' credit differs from the state-government CDS proxy (Appendix D.1.5).

Table 29: Paired issues: the tax-exempt-versus-BAB gap widens with maturity, on both rating cuts and with or without call matching.

	AA and above (score ≤ 3)		A− and above (score ≤ 7)	
	(1) base	(2) call-matched	(3) base	(4) call-matched
Panel A. Continuous maturity				
Tax-exempt	-0.102^{***}	-0.091^{**}	-0.062^{**}	-0.043
	(0.034)	(0.036)	(0.027)	(0.030)
Tax-exempt × maturity	0.034^{***}	0.031^{***}	0.033^{***}	0.030^{***}
	(0.003)	(0.003)	(0.002)	(0.003)
After-tax corporate yield	0.379^{***}	0.376^{***}	0.264^{*}	0.261^{*}
	(0.138)	(0.135)	(0.134)	(0.130)
Panel B. Five-year maturity bins				
Tax-exempt	-0.048	-0.050	-0.007	-0.008
	(0.033)	(0.033)	(0.024)	(0.024)
Tax-exempt × $[5, 10)$ y	0.192^{***}	0.191^{***}	0.192^{***}	0.190^{***}
	(0.022)	(0.022)	(0.017)	(0.017)
Tax-exempt × $[10, 15)$ y	0.396^{***}	0.364^{***}	0.368^{***}	0.342^{***}
	(0.033)	(0.033)	(0.026)	(0.030)
Tax-exempt × $[15, 20)$ y	0.450^{***}	0.390^{***}	0.419^{***}	0.403^{***}
	(0.041)	(0.039)	(0.029)	(0.039)
Tax-exempt × $[20, 25)$ y	0.576^{***}	0.540^{***}	0.593^{***}	0.510^{***}
	(0.061)	(0.066)	(0.071)	(0.066)
Tax-exempt × $[25, 30]$ y	0.755^{***}	0.639^{***}	0.996^{***}	0.804^{***}
	(0.079)	(0.039)	(0.074)	(0.060)
After-tax corporate yield	0.380^{***}	0.378^{***}	0.264^{*}	0.262^{**}
	(0.138)	(0.135)	(0.134)	(0.130)
Issuer × maturity × quarter FE	Yes	Yes	Yes	Yes
× call status	No	Yes	No	Yes
State cluster	Yes	Yes	Yes	Yes
Observations	29,839	28,890	61,816	59,886

Notes. The sample covers offering yields in 2009Q2–2010Q4 and excludes AMT bonds, and BAB yields are multiplied by 0.65. The corporate-yield row uses the after-tax control in equation (141). The omitted bin is $[1, 5)$ years. Standard errors are clustered by issuing jurisdiction (51 in columns (1)–(2), 52 in columns (3)–(4)). Call-matched columns add the optional-call flag to the absorbed issuer-by-maturity-by-quarter cell. Columns (1)–(2) use AA-and-above bonds and columns (3)–(4) A-or-above bonds. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

After call and issuance-premium adjustments, GO implied tax falls from 43.1% at two years to 26.0% at thirty, matching the 17.1-point decline of the A-or-above full sample. The screen excludes 4,029 GO bonds (0.86%). Relative to the unscreened GO call-adjusted series, with the issuance premium held fixed, the screen changes the decline by 0.49 points, almost entirely at the short end. The pattern survives both the GO and the AA-and-above restrictions. This agrees with earlier evidence that the long-end residual persists after default and liquidity adjustments and under high-grade or GO restrictions (Longstaff, 2011; Fleckenstein and Longstaff, 2025; Chalmers, 1998).

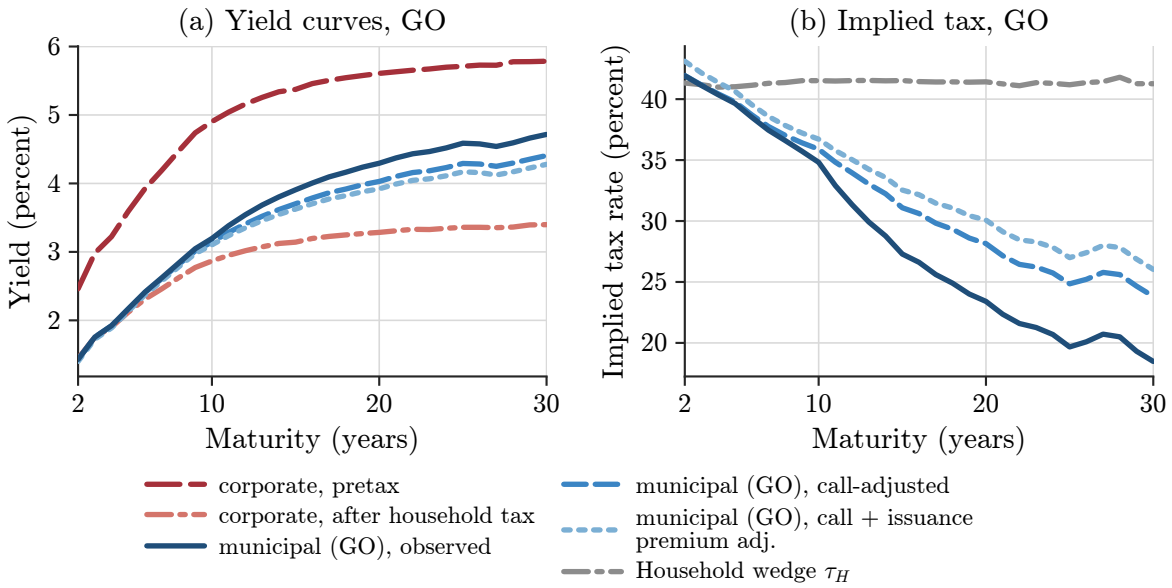


Figure 21: AA-and-above unlimited-tax GO bonds (code "K"): 465,258 bonds pass the main pricing and calendar screens, and curves use the sample's own weights and household benchmark. Calls use the calibrated model. Because GO offering-week matches are scarce, the adjustment uses the all-bond institutional issuance premium (§D.1.6). Adjusted implied tax falls 17.1 points from two to thirty years, matching the A-or-above full sample.

D.5 The single-tax-rate benchmark of Green (1993)

Green (1993) explains rising ratios of municipal to taxable yields through taxable-bond tax timing. Buying two five-percent market-discount bonds and shorting one ten-percent par bond of equal maturity produces no net cash flow before maturity and one dollar of principal then, deferring tax. At rate τ, the package costs P_T and pays $1-\tau(1-P_T)$ after tax at maturity. With d_T the price of a dollar paid at T, it is therefore worth $P_T = (1-\tau)d_T/(1-\tau d_T)$. The limiting friction is the asymmetric taxation of long and short positions, because borrowing and short-sale deductions cannot exceed the investment income they support. The high-tax investor is marginal on municipals and the package, and unrestricted dealers link taxable par prices to it, which lowers long taxable yields below fully taxed returns. Thus one investor can price municipals at every maturity as the yield ratio rises, without the different marginal investors across maturities that Fama (1977) requires.

Relation to participation. Deep market-discount bonds may be scarce. OID accrues annually and shorts consume deduction capacity. Households forgoing the package can accept the conventional benchmark $\xi_H y_T^C$, below Green's relation. With sufficient capacity, these higher-benefit households set municipal prices, leaving strategy investors inframarginal. Raising municipal yields to Green's relation would require shorting municipals, subject to the frictions in Appendix E.6. His relation therefore need not bind. Nor does it account for observed long yields: his own 1987–1990 estimates leave thirty-year municipal yields about twenty taxable-equivalent basis points above his relation, a gap that increases with maturity.

Green rejects earlier clientele explanations because term premia persisted after mid-1980s reforms disrupted bank holdings. Participation driven by tax benefits and duration exposure lets the marginal clientele change. Households anchor short municipal maturities and lower-benefit institutions anchor long ones, and marginal wedges respond to supply and risk capacity. Section 3.4 documents ownership across benefits. I retain $\xi_H y_T^C$, measure tax timing separately and test household state-tax capitalization in P1 and bank and insurer benefits in P2, distinguishing participation from tax timing alone.

D.6 State-tax pass-through: full regression tables

The main P1 regressions use federally exempt AA-and-above issuance (agency score ≤ 3) from 2000–2017, exclude zero maturity and group maturities into $[0, 5)$, $[5, 10)$, $[10, 15)$, $[15, 20)$ and $[20, 30]$ years. The zero-yield columns have 681,839 complete-control observations. Six missing offering yields leave 681,833 in the offering-yield columns. OLS is the main estimator. Inference is state-clustered, using $t(46)$ critical values. For the IV comparison, the 1990 rate interacted with bins gives a Kleibergen–Paap rk Wald F of 10.4 and Sanderson–Windmeijer conditional statistics of 76.1, 115.9, 104.4, 155.7 and 230.4.

The P1 call adjustment prices each bond on the fifteenth of its issue quarter's final month. Fixed effects use actual issue months. The OAS is fitted on the calibrated model's dated supply-weighted curve with a regular semiannual cash-flow grid. Observed call dates map to the nearest grid date within 0.27 years, and exercise recurs from the first mapped date, unlike the main adjustment's contractual calendar. Callable bonds that this method cannot reprice are excluded, because either the first call date lies more than 0.27 years before the first semiannual grid date or the call schedule lacks a usable date and price.

All P1 tables below retain equation (28)'s six lagged state controls and numeric rating interacted with all five maturity bins, log offering par and security-type indicators. Fixed effects are cells defined by issue year, issue month, rounded maturity and callability, and observations with missing controls and singleton cells are excluded. State-clustered tests use $t(G - 1)$, where G is the number of states. Tables specify departures in sample, instrument or weighting.

D.6.1 Comparison across yield constructions

Table 30 compares vendor offering yields (potentially yield-to-worst for premium callable bonds) with zero yields from offering prices, with and without call adjustment. Both OLS and IV reject endpoint equality at one percent for every construction.

D.6.2 Main specification: maturity-interacted state and bond controls

Every P1 specification uses the six lagged state controls and bond controls after equation (28). Stock and issuance controls use all observed federally exempt bonds, regardless of rating. Outstanding par equals the offering amount until the reported outstanding date and the reported outstanding amount afterward. Cells use actual issue months before singletons are removed, and controls are never imputed.

Table 31 restricts the pre-2018 sample to observations with a preceding-month five-year state-government CDS quote and adds that quote interacted with maturity bins. On this common sample of 345,878 bonds from 25 states, the longest-bin OLS estimate rises from 0.323 to 0.401 and IV from 0.309 to 0.353. Both estimators retain a declining profile and reject endpoint equality at five percent in every row.

D.7 State-tax pass-through: instrument vintage and weighting

OLS is the main estimator. The IV comparisons use the state's 1990 top income-tax rate as the baseline instrument. The maintained exclusion restriction is that the historical rate affects current implied taxes only through the current rate after conditioning on the included controls and fixed effects. Table 32 replaces the five current-rate bin interactions in (28) with the current rate and its continuous maturity interaction. The interaction coefficient is the gradient per year. IV uses the historical rate and its maturity interaction. The table compares 1945–2000 vintages with the same pre-2018 sample and state-clustered inference.

All instrument vintages give negative IV gradients under both weightings, as does OLS. Tables 32 and 33 report coefficients, first-stage statistics and state-clustered uncertainty.

Table 30: P1 across yield constructions, 2000–2017.

	Offering yield		Call-adjusted zero		Unadjusted zero	
	OLS	IV	OLS	IV	OLS	IV
β_1: $[0, 5)$ years	0.985^{***}	1.084^{***}	1.072^{***}	1.092^{***}	1.080^{***}	1.102^{***}
	(0.238)	(0.249)	(0.195)	(0.224)	(0.196)	(0.227)
β_2: $[5, 10)$ years	0.635^{***}	0.648^{***}	0.682^{***}	0.705^{***}	0.683^{***}	0.693^{***}
	(0.075)	(0.093)	(0.071)	(0.088)	(0.069)	(0.087)
β_3: $[10, 15)$ years	0.414^{***}	0.410^{***}	0.484^{***}	0.448^{***}	0.515^{***}	0.479^{***}
	(0.053)	(0.075)	(0.098)	(0.121)	(0.100)	(0.117)
β_4: $[15, 20)$ years	0.268^{***}	0.257^{***}	0.386^{***}	0.325^{***}	0.428^{***}	0.360^{***}
	(0.060)	(0.067)	(0.102)	(0.117)	(0.108)	(0.121)
β_5: $[20, 30]$ years	0.193^{***}	0.152^{**}	0.298^{***}	0.227^{***}	0.335^{***}	0.255^{***}
	(0.052)	(0.057)	(0.075)	(0.081)	(0.078)	(0.085)
Decline $\beta_1 - \beta_5$	0.791^{***}	0.933^{***}	0.774^{***}	0.865^{***}	0.745^{***}	0.847^{***}
p-value of decline	0.004	< 0.001	0.002	< 0.001	0.003	0.001
KP F	—	10.4	—	10.4	—	10.4
Observations	681,833	681,833	681,839	681,839	681,839	681,839
States	47	47	47	47	47	47
Full main controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. IV instruments each bin's current rate with its 1990 rate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 31: P1 sample and credit controls, 2000–2017.

Sample	Obs.	States	β_1, OLS	β_5, OLS	Decline p	β_1, IV	β_5, IV	Decline p
Main sample	681,839	47	1.072^{***}	0.298^{***}	0.002	1.092^{***}	0.227^{***}	< 0.001
				(0.075)			(0.081)	
Observed CDS sample	345,878	25	1.210^{***}	0.323^{***}	0.008	1.350^{***}	0.309^{***}	0.008
				(0.082)			(0.098)	
CDS sample + CDS × bin	345,878	25	1.400^{***}	0.401^{***}	0.010	1.392^{***}	0.353^{***}	0.022
				(0.070)			(0.088)	

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. IV uses bin-interacted 1990 rates. Parentheses give standard errors of β_5. CDS is the prior-month five-year state-government quote, without imputation. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 32: Continuous-maturity P1 checks, 2000–2017.

		Unweighted				Equal per state	
Vintage	Corr.	Gradient	s.e.	KP F	SW F	Gradient	KP F
OLS	—	-0.0399^{***}	(0.0131)	—	—	-0.0330^{**}	—
1945	0.40	-0.0548^{**}	(0.0251)	8.0	19.2	-0.0454	2.8
1950	0.31	-0.0622^{**}	(0.0274)	7.8	18.4	-0.0904	1.8
1960	0.38	-0.0636^{**}	(0.0256)	9.0	28.8	-0.1022^{**}	2.5
1970	0.51	-0.0588^{***}	(0.0203)	11.5	45.2	-0.0818^{***}	7.3
1980	0.76	-0.0434^{**}	(0.0164)	10.6	103.4	-0.0396^{**}	26.9
1990	0.82	-0.0447^{***}	(0.0129)	30.1	404.1	-0.0310^{**}	25.9
2000	0.88	-0.0405^{***}	(0.0125)	65.1	998.5	-0.0262^{*}	36.5

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. Gradients are per year. IV uses the indicated historical rate and its maturity interaction. SW is the interaction's conditional statistic. Corr. is the correlation across states between each state's mean current top rate and its top rate in the vintage year, using state means over the main estimation sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Every vintage rejects endpoint equality at five percent. Table 33 compares the profiles with OLS, and Figure 22 plots them.

Table 33: P1 instrument-vintage checks, 2000–2017.

Estimator/vintage	β_1	β_2	β_3	β_4	β_5	Decline p	KP F
OLS (main)	1.072***	0.682***	0.484***	0.386***	0.298***	0.002	—
	(0.195)	(0.071)	(0.098)	(0.102)	(0.075)		
1945	1.267***	0.886***	0.546***	0.359**	0.246**	0.025	3.1
	(0.422)	(0.168)	(0.161)	(0.142)	(0.104)		
1950	1.464***	0.959***	0.613***	0.423***	0.311**	0.022	3.1
	(0.475)	(0.192)	(0.158)	(0.143)	(0.118)		
1960	1.494***	0.954***	0.615***	0.426***	0.327***	0.013	3.6
	(0.441)	(0.176)	(0.152)	(0.141)	(0.115)		
1970	1.404***	0.864***	0.551***	0.392***	0.317***	0.005	4.4
	(0.352)	(0.132)	(0.148)	(0.143)	(0.115)		
1980	1.092***	0.759***	0.520***	0.367***	0.273***	0.009	3.8
	(0.273)	(0.085)	(0.119)	(0.118)	(0.083)		
1990	1.092***	0.705***	0.448***	0.325***	0.227***	< 0.001	10.4
	(0.224)	(0.088)	(0.121)	(0.117)	(0.081)		
2000	1.075***	0.684***	0.482***	0.375***	0.275***	0.001	23.5
	(0.215)	(0.080)	(0.097)	(0.096)	(0.069)		

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. First row: main OLS. IV instruments each bin's current rate with its indicated historical top rate. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

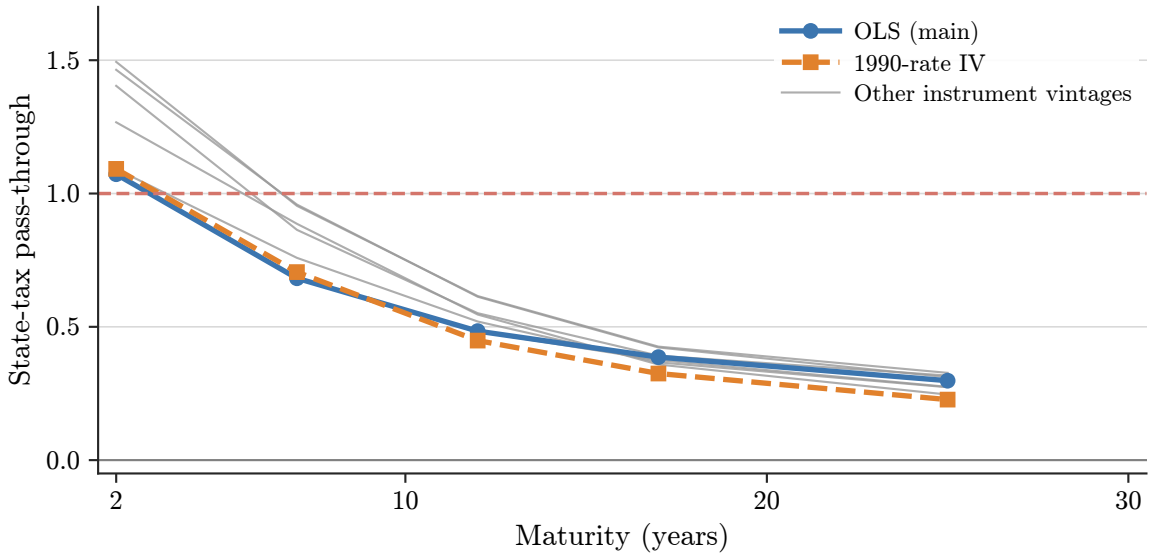


Figure 22: State-tax pass-through using historical tax-rate instruments, 2000–2017. Blue shows the main OLS profile. Orange shows IV using the 1990 rate. Gray lines use the 1945, 1950, 1960, 1970, 1980 and 2000 rates. The dashed red line is the representative-tax benchmark. All estimates use the same sample, controls and fixed effects. Table 33 reports state-clustered uncertainty.

Every weighting in Table 34 retains a declining OLS profile, with endpoint differences significant at one percent.

Table 34: P1 OLS estimates under four weighting schemes, 2000–2017.

Weighting	β_1	β_2	β_3	β_4	β_5	Decline p
Unweighted (main)	1.072^{***}	0.682^{***}	0.484^{***}	0.386^{***}	0.298^{***}	0.002
	(0.195)	(0.071)	(0.098)	(0.102)	(0.075)	
Equal per state	0.869^{***}	0.562^{***}	0.384^{***}	0.314^{***}	0.240^{***}	0.007
	(0.191)	(0.084)	(0.109)	(0.115)	(0.089)	
Equal per state-year	0.915^{***}	0.519^{***}	0.319^{***}	0.261^{***}	0.211^{***}	0.002
	(0.190)	(0.069)	(0.083)	(0.090)	(0.073)	
$1/\sqrt{n_{\text{state}}}$	0.996^{***}	0.630^{***}	0.425^{***}	0.345^{***}	0.281^{***}	0.005
	(0.202)	(0.082)	(0.113)	(0.122)	(0.092)	

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. Weight counts follow complete-case and singleton restrictions. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

State fixed effects. Adding state fixed effects to the pre-2018 OLS specification gives short and long coefficients of 0.658 (s.e. 0.205) and −0.124 (0.211), with a decline of 0.782 (0.220, $p = 0.001$), on 681,839 bonds. State effects absorb fixed tax levels. The five excluded 1990-rate bin interactions sum to the fixed state rate and are collinear with these effects, so this check reports OLS.

Leave-one-state-out. All 47 single-state omissions retain a positive short-to-long decline under both OLS and IV (Figure 23). OLS long-bin estimates span 0.224–0.469 and declines 0.467–0.898. IV long-bin estimates span 0.113–0.309 and declines 0.678–1.007. The largest endpoint-test p-values are 0.054 for OLS and 0.039 for IV.

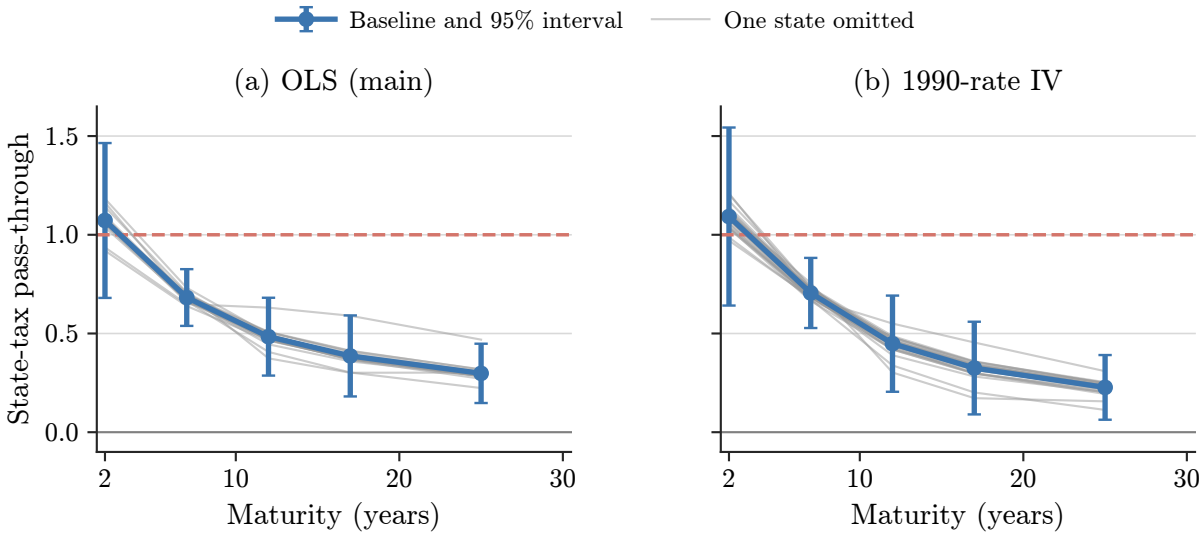


Figure 23: Leave-one-state-out P1 profiles for 2000–2017, retaining all main controls and fixed effects. Gray lines show each of the 47 state omissions, and blue shows the baseline estimates with state-clustered 95% intervals. Panel (a) uses OLS and panel (b) uses the 1990-rate instruments. The dashed line is the representative-tax benchmark.

The two tax regimes. The main P1 sample ends in 2017, matching the compounded federal and state after-tax shares of the six model economies in Figure 9. From 2018, the Tax Cuts and Jobs Act capped state and local tax deductions at $10,000 per return. When the cap binds, an additional dollar of state tax is no

longer federally deductible, and the combined after-tax share on taxable interest is $\xi^{\text{fed}} - \tau_s^{\text{st}}$. The implied state tax for this regime is

$$\mathit{ImpliedStateTax}_{s,t}^{\text{cap}}(T) \;\equiv\; \xi^{\text{fed}} - \frac{y_{s,t}^{M}(T)}{y_t^{C}(T)}. \tag{142}$$

It recovers the statutory state rate under representative-tax indifference for a household with a binding cap.[27] Table 35 keeps the full controls and estimates each regime separately. OLS declines from 1.072 to 0.298 in 2000–2017 and from 0.952 to 0.091 in 2018–2020. IV declines from 1.092 to 0.227 and from 0.912 to 0.082. Both estimators reject endpoint equality at one percent in each period.

Table 35: P1 under the two federal/state tax regimes.

	2000–2017 (main)		2018–2020	
	OLS	IV	OLS	IV
β_1: $[0, 5)$ years	1.072***	1.092***	0.952***	0.912***
	(0.195)	(0.224)	(0.181)	(0.246)
β_2: $[5, 10)$ years	0.682***	0.705***	0.842***	0.900***
	(0.071)	(0.088)	(0.126)	(0.154)
β_3: $[10, 15)$ years	0.484***	0.448***	0.450***	0.466***
	(0.098)	(0.121)	(0.065)	(0.076)
β_4: $[15, 20)$ years	0.386***	0.325***	0.318***	0.326***
	(0.102)	(0.117)	(0.081)	(0.107)
β_5: $[20, 30]$ years	0.298***	0.227***	0.091	0.082
	(0.075)	(0.081)	(0.091)	(0.105)
Decline $\beta_1 - \beta_5$	0.774***	0.865***	0.861***	0.830***
p-value of decline	0.002	< 0.001	< 0.001	0.007
KP F	—	10.4	—	7.9
Observations	681,839	681,839	146,174	146,174
States	47	47	47	47
Full main controls	Yes	Yes	Yes	Yes

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6. IV uses bin-interacted 1990 rates. Outcomes follow equation (27) before 2018 and equation (142) afterward, for a binding SALT cap. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.8 The bank-qualified discontinuity: construction, robustness, and placebos

The two benchmarks. Both benchmarks in §6.2 isolate the bank deduction through matched annual after-tax carry. I suppress dates and let $y_k^C > 0$ denote the taxable yield.

The representative-tax benchmark has $y_k^N = (1-\tau_H)y_k^C$, where τ_H is the top household rate. A designated bond yields the bank $y_k^N - 0.2\,\tau_B c$ after funding-cost disallowance, versus taxable carry $(1-\tau_B)y_k^C$, an advantage of

$$\underbrace{(1-\tau_B)y_k^C}_{\text{taxable alternative}} \;-\; \underbrace{\left[(1-\tau_H)y_k^C - 0.2\,\tau_B c\right]}_{\text{qualified bond at the household yield}} \;= (\tau_H - \tau_B)y_k^C + 0.2\,\tau_B c > 0. \tag{143}$$

This holds for $\tau_H > \tau_B - 0.2\,\tau_B c/y_k^C$, including household rates at least as high as banks' and positive funding costs. Banks prefer taxable bonds even with designation, which leaves household taxes unchanged. Thus $y_k^Q = y_k^N$ and $\varpi_k = 0$ throughout.

[27] See the IRS, 2017 Schedule A instructions, and its summary of the 2018 changes.

Table 36: Bank-qualified designation and yield premia by maturity.

	(1)	(2)	(3)
Dependent variable	BQ designation	Offering yield (bp)	Yield / expected benefit
Estimator	OLS	2SLS	2SLS
Regressor, interacted with each maturity bin	Below-cap indicator	BQ designation	BQ designation
Reported effect	Designation increase	Yield reduction	Scaled yield reduction
1–2 years	0.42***	1.8	0.17
	(0.015)	(2.2)	(0.11)
2–5 years	0.42***	8.7***	0.31***
	(0.015)	(2.2)	(0.08)
5–10 years	0.44***	13.8***	0.41***
	(0.015)	(2.2)	(0.06)
10–20 years	0.45***	18.4***	0.52***
	(0.017)	(2.3)	(0.07)
Over 20 years	0.43***	18.2***	0.47**
	(0.027)	(6.3)	(0.19)
10–20 minus 1–2 years		16.5***	0.34***
		(2.7)	(0.11)
Observations (bonds)	355,372	355,372	355,372
Standard-error clustering	Issuer	Issuer	Issuer
Number of issuers	15,889	15,889	15,889

Notes. Each column is a separate pooled regression containing all five maturity interactions. Column (1) reports the designation increase associated with below-cap status. Columns (2)–(3) instrument designation by below-cap status within each bin, as in (30), and report $-\beta_k$, so positive numbers mean lower yields on designated bonds. Column (3) divides each bond's yield by its expected annual benefit $X_{tm} = 0.8\,\tau_{B,t}\hat{c}(t,m)$ before estimation (Appendix D.8). Contrasts use unrounded estimates. Parentheses give standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Fixed effects in every column. Maturity bucket × year, state × year, state × offering year–month, rating category × year, security type × year, use of proceeds, coupon bin × maturity bucket, and callability × maturity bucket × year. Other controls are log tranche par, maturity-bin indicators, and separate linear issuance slopes on each side of the cap within each bin.

Sample. Tax-exempt new issues in 2000–2021 within \$0.5–\$4 million of the statutory cap (\$30 million in 2009–10, \$10 million otherwise). Maturity is measured at issuance. Appendix D.8 defines the fixed-effect buckets.

In the bank-only benchmark, both municipal bonds satisfy the bank's after-tax indifference conditions:

$$\begin{aligned} y_k^N - \underbrace{\tau_B c}_{\text{full disallowance}} &= (1-\tau_B) y_k^C, \\ y_k^Q - \underbrace{0.2\,\tau_B c}_{\text{qualified disallowance}} &= (1-\tau_B) y_k^C. \end{aligned} \tag{144}$$

Subtracting gives $\varpi_k = 0.8\,\tau_B c = X$, since taxable yields cancel within pairs. Constant funding costs therefore imply a flat basis-point premium. The empirical scale X_{tm} below evaluates this benefit at expected annual funding costs.

Derivation of the designation premium. Compare annual carry on matched bonds of supply $q_k > 0$, fixing outside opportunities and non-tax characteristics. At yield y, household demand is $d_k^H(y)$. Bank demand is $d_k^B(y+u)$ with designation benefit u. Both include baseline funding costs, taxes and risk

compensation. For each $u \in [0, X]$, assume clearing exists, continuous nondecreasing demands with finitely many continuously differentiable regimes, and a strictly positive total-demand slope at clearing. Then

$$d_k^H(y_k(u)) + d_k^B(y_k(u) + u) = q_k, \qquad y_k^N = y_k(0), \quad y_k^Q = y_k(X). \tag{145}$$

For example, a strictly concave quadratic objective, conditional on the remaining portfolio, gives investor demand with reservation return r_{ik} and curvature $a_{ik} > 0$: $x_{ik} = \max\{0, (y + \mathbf{1}\{i = B\}u - r_{ik})/a_{ik}\}$, whose zero-holding region permits changes in participation.

Differentiating (145) on a smooth segment gives

$$-\frac{dy_k(u)}{du} = \frac{\partial d_k^B/\partial y}{\partial d_k^H/\partial y + \partial d_k^B/\partial y} \equiv w_{Bk}(u) \in [0, 1], \tag{146}$$

at clearing net returns. The deduction raises bank demand. Yields fall to offset it. Integrating continuously across participation boundaries gives

$$\varpi_k = y_k(0) - y_k(X) = \int_0^X w_{Bk}(u)\, du = \theta_k X, \qquad \theta_k = \frac{1}{X}\int_0^X w_{Bk}(u)\, du. \tag{147}$$

If banks never participate, $\partial d_k^B/\partial y = 0$ and $\varpi_k = 0$, the representative-tax benchmark. With only banks, $\partial d_k^H/\partial y = 0$ and $\varpi_k = X$, the bank-only benchmark. When both respond, capitalization lies between them. The capitalization share θ_k measures the yield reduction relative to the benefit. It is distinct from banks' share of holdings.

The flat-X prediction concerns annual yield-equivalent carry and makes no claim about a fixed price premium. The ordinary-rule benefit per dollar of basis maps to offering yields through comparable cash flows and basis-to-price ratios.

Scaling by the expected benefit. The scaled specification expresses yield responses in expected-benefit units. The current-year benefit $X_t = 0.8\,\tau_{B,t} c_t$ falls from 104 bp in 2000 to an average of 9 bp in 2013–17, but premia do not track it. Year terciles averaging 8.5, 26.7 and 70.2 bp give 5–20 year premia of 20.8–21.1, 12.3–23.2 and 6.5–13.2, each with a significant long-minus-short contrast. Ratios of period premia to X_t exceed two in 2013–17 and 2020–21 and are at most about 0.3 in the pre-2009 period estimates, making pooled ratios weighting-dependent.

The expected annual benefit scale is $X_{tm} = 0.8\,\tau_{B,t}\,\hat{c}(t, m)$ for issue month t and maturity m. The fitted 2000–2021 annual pass-through from one-year expected rates to muni-weighted funding costs ($R^2 = 0.75$) gives $\hat{c}(t, m) = 0.38 + 0.55\, r_{tm}$. Here r_{tm} is the issue-month Kim–Wright expected-average-short-rate component at the rounded whole-year maturity, capped at ten years. It is computed as the fitted yield less its term premium, because raw zero yields would put a term premium that rises with maturity into the denominator. Bin means of X_{tm} are 33.9, 41.5, 50.5, 56.8, 53.3 bp.

Dividing each offering yield by its own X_{tm} before 2SLS (equation (30)), bond by bond and before any pooling, gives benefit-scaled responses in Table 36 of 0.17, 0.31, 0.41, 0.52 and 0.47, with a 10–20 minus 1–2 contrast of 0.34 (s.e. 0.11). Using the Adrian–Crump–Moench expected-short-rate component subtracts larger term premia, giving 0.17, 0.35, 0.46, 0.55, 0.48 (contrast 0.39).

Within-regime contrasts are 0.57 in low-benefit years, 0.37 in middle years and 0.27 in the high-benefit years 2000–02 and 2005–08, when expected benefits were nearly flat in maturity (62.8–71.2 bp) and curves often inverted. Expected-benefit maturity structure therefore does not drive the gradient. Excluding strong bank-demand years 2020–21 gives 0.18, 0.25, 0.35, 0.39, 0.36 (contrast 0.21).

Yield-level benefit interactions. I also retain offering yields in basis points and interact designation with the expected annual benefit:

$$\begin{aligned} y_b &= \sum_{k=1}^{5} \beta_k B_b X_{t_b m_b} \mathbf{1}_k + \widetilde{V}_b'\delta + u_b, \\ B_b X_{t_b m_b} \mathbf{1}_k &\quad \text{instrumented by } E_{jt} X_{t_b m_b} \mathbf{1}_k, \qquad k = 1, \ldots, 5. \end{aligned} \tag{148}$$

Here $\widetilde{V}_b$ contains equation (30)'s fixed effects and running-variable controls, plus X_{tm} by bin. Table 37 reports yield reductions $-\beta_k$, standard errors and joint first-stage strength. These are different estimators: Table 36 divides the outcome by X_{tm}, whereas this specification multiplies designation and its instruments by X_{tm} and adds benefit controls. Their levels need not coincide. Both give rising profiles. The claim about half capitalization at 10–20 years refers to the outcome-scaled specification. Excluding 2009–2010 leaves the interacted specification's 10–20 minus 1–2 contrast at 0.113 (s.e. 0.032).

Table 37: Bank designation interacted with the expected annual benefit.

Maturity (years)	1–2	2–5	5–10	10–20	> 20
Yield reduction per benefit unit	0.058**	0.096***	0.154***	0.175***	0.233***
	(0.024)	(0.025)	(0.027)	(0.031)	(0.089)
10–20 minus 1–2	0.117***				
	(0.032)				

Notes. Offering yield and expected annual benefit X_{tm} are in basis points. The table reports $-\beta_k$ in equation (148). Each designation–benefit interaction is instrumented by the below-cap indicator times the benefit in that bin. Controls include the benefit separately by bin and the matched fixed effects and running-variable slopes of equation (30). The window is 0.5–4 million dollars from the cap of the year. Observations: 355,372, issuer clusters: 15,889, joint Kleibergen–Paap F: 122.0. Issuer-clustered standard errors are in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Running variable. The running variable (§6.2) sums all tax-exempt par within six-character CUSIP issuer prefixes and offering calendar years, before sample restrictions. Statutory issuance instead aggregates subordinate entities, excludes refundings of designated bonds (deemed designated under Section 265(b)(3)(D)) and private-activity bonds except 501(c)(3), and includes bank loans and leases, mostly absent from Mergent. This measurement noise motivates the fuzzy design. Above the cap, 21% are designated, with single-sale issuer-years above $10.5 million designating 4% and multiple-sale years 45%. The statute tests reasonably anticipated issuance at each sale, so a March designation stands even when an unanticipated October sale takes the year over the cap. Refundings constitute 46% just above the cap. Netting their par reduces over-cap designation to 9%. Designation falls from 24% within $1 million above the cap to 14–16% at $4–8 million and 5% beyond $10 million, consistent with noise.

Controls. In (30), V_b holds a linear function of R_{jt} on each side of the cap, separately by bin, the log par amount of the maturity tranche (floored at $1,000), and fixed effects for maturity bucket × year, state × year, state × offering month, rating category × year, security type × year, use of proceeds, coupon bin × maturity bucket, and callability × maturity bucket × year. The seven maturity buckets, used only in these fixed effects, have upper limits of 2, 5, 10, 15, 20 and 30 years, with a seventh bucket beyond thirty.

Balance at the cap. Baseline-window local linear covariate regressions, with maturity-bucket-by-year and state-by-year effects, find callability and issue maturity counts balanced in every bin. Ratings, observed for 62%, differ only at 1–2 years (0.17 notch, $t = 2.1$). Coupons are 0.09–0.14 points lower below the cap in the four bins beyond two years. Coupons are set at pricing. The baseline includes coupon-bin-by-maturity fixed effects. Log par is lower under the cap at 1–2 years (-0.26) and higher at 10–20 years ($+0.12$), and is always controlled.

The window. The half-million donut excludes managed bunching (2,081 issuer-years within $0.25 million below versus 572 above) and 1,034 issuer-years with 12,076 bonds exactly at the $10 million cap ($30 million in 2009–10). At-cap issuers are eligible (§6.2) and designate 79.6% of their bonds, the below-cap rate, against 27% immediately above the cap. Coding the below-cap indicator with a strict inequality misclassifies them, lowering the no-donut first stage from 0.39 to 0.30 but barely changing 2SLS, because dividing by the first stage offsets the attenuation. The below-cap indicator is always $E = \mathbf{1}\{R \leq 0\}$.

Donuts of zero–$1.5 million and bandwidths of $2–$8 million (Table 38) give 10–20 minus 1–2 contrasts of 13.5–18.4 bp. 1–2 year premia are never significant. In Dagostino (2025)'s $5–$15 million window without exclusion, the contrast is 15.4 (1.8) bp, one of the most precise.

Dropping the $8.5–$13.5 million region that her bunching estimation treats as contaminated compares issuers at least $1.5 million below the cap with issuers at least $3.5 million above it. First stages rise to 0.47–0.54, and the more distant extrapolation gives premia of 7.6 and 10.0 bp at 5–10 and 10–20 years and a contrast of 9.8 (4.2).

Table 38: Bank-qualified discontinuity: sensitivity to the donut, the bandwidth, and the window.

	first stage	1–2y	2–5y	5–10y	10–20y	20y+	10–20 minus 1–2
Donut half-width, bandwidth $4 million							
none	0.39***	1.9 (1.8)	7.5*** (1.8)	13.9*** (1.8)	17.7*** (1.8)	17.7*** (4.5)	15.7*** (2.0)
$0.25 million	0.40***	1.6 (2.0)	7.6*** (2.0)	13.4*** (2.0)	16.7*** (2.2)	16.3*** (5.5)	15.1*** (2.4)
$0.5 million (baseline)	0.42***	1.8 (2.2)	8.7*** (2.2)	13.8*** (2.2)	18.4*** (2.3)	18.2*** (6.3)	16.5*** (2.7)
$1 million	0.44***	1.2 (2.9)	8.2*** (2.7)	13.3*** (2.8)	19.7*** (3.1)	20.6*** (8.0)	18.4*** (3.5)
$1.5 million	0.44***	2.0 (4.4)	7.3* (3.8)	11.8*** (4.0)	15.5*** (4.2)	19.5* (10.2)	13.5*** (5.2)
Bandwidth, donut $0.5 million							
$2 million	0.38***	0.8 (4.7)	7.9 (4.9)	13.3*** (4.8)	16.9*** (5.0)	21.0 (14.1)	16.1*** (4.8)
$3 million	0.40***	2.7 (3.0)	8.9*** (3.0)	15.2*** (2.9)	19.2*** (3.1)	19.9*** (7.6)	16.5*** (3.3)
$5 million	0.42***	−0.9 (1.9)	5.7*** (1.8)	11.4*** (1.8)	14.9*** (1.9)	15.9*** (4.9)	15.8*** (2.3)
$6 million	0.43***	−0.5 (1.7)	5.4*** (1.5)	10.8*** (1.6)	13.2*** (1.6)	11.4*** (4.1)	13.6*** (2.1)
$8 million	0.45***	0.0 (1.5)	6.3*** (1.2)	10.1*** (1.3)	13.5*** (1.3)	12.3*** (3.5)	13.5*** (1.8)
Other windows							
$5–15 million, no donut	0.40***	0.1 (1.6)	5.6*** (1.5)	12.1*** (1.6)	15.5*** (1.5)	16.7*** (3.8)	15.4*** (1.8)
$8.5–13.5 million dropped, $\lvert R\rvert \leq 8$	0.47***	0.2 (3.5)	6.0** (3.0)	7.6** (3.1)	10.0*** (3.4)	−1.8 (9.5)	9.8** (4.2)

Notes. Cells report the 2SLS premium of designation in basis points by maturity bin, pooled over 2000–2021 at each year's cap, with matched fixed effects and issuer-clustered standard errors in parentheses. The first-stage column refers to the 1–2 year bin. The $5–15 million window is the one used in the spread comparison of Dagostino (2025). The last row drops the region her bunching estimation excludes. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Construction, and the relation to Dagostino (2025). Dagostino (2025) uses Ipreo MuniIC issuer-year issuance and documents the same bunching and $30 million first stage in 2009–10. Her regressions on $1 million issuance bins control for size, maturity, insurance, rating and use of proceeds, with year and issuer effects and year clustering. In that study, tax-adjusted treasury spreads jump about 10 bp for GO bonds issued before 2008. In her Call Report data for 2000–2010, designated bonds are about 4% of bank assets, rising to 5% by 2010, about nine times the non-designated share. Table 39 compares these constructions and nets refunding par. All retain a larger long-maturity premium. Issuer effects steepen it, while netting refundings shifts issuer-years and widens intervals.

Using her $1 million issuance-bin estimator on her window, with maturity-bucket, state and rating effects by year and use-of-proceeds controls, gives flat below-cap bins and a step above, matching her Figure 3. Pooled, the step is 4.2 bp in the offering yield and 5.6 in the tax-adjusted spread. It is 1.1 and 1.4 at 1–5 years and 7.8 and 10.2 at 10–20 years. In her pooled-spread setting for GO bonds issued before 2008, the reduced-form jump at the cap is 10.4 bp (s.e. 2.5) with size, maturity, rating, use-of-proceeds, insurance and state fixed effects and issuer clustering, against her approximately 10. With her issuer fixed effects and clustering by year, which leaves eight clusters, the jump is 6.3 (3.8), because within-issuer variation across the cap is thin before 2008.

Placebo cutoffs and the ARRA cap. At $5 million, first stages are 0.00–0.05 and all yield effects are within two bp of zero. At $15 million, first stages are −0.01 to 0.02 and yield effects are zero except in one 20+ cell. With matched effects and a $0.5–$8 million window around $30 million, 2009–10 first stages are 0.19, 0.20, 0.19, 0.16 and 0.26 on 16 thousand bonds from $22–$38 million issuers, and they are jointly significant at five percent. At that cutoff, first stages are 0.01–0.04 in 2004–08 and −0.05 to 0.01 in 2011–15. At $10 million in 2009–10, they are 0.01–0.06 and not jointly significant. The 2009–10 $30 million yield

Table 39: Bank-qualified discontinuity: construction checks.

	1–2y	2–5y	5–10y	10–20y	20y+	10–20 minus 1–2
Baseline (Table 36)	1.8 (2.2)	8.7*** (2.2)	13.8*** (2.2)	18.4*** (2.3)	18.2*** (6.3)	16.5*** (2.7)
Bond insurance and enhancement × year	0.9 (2.2)	8.0*** (2.2)	13.0*** (2.2)	17.7*** (2.3)	17.3*** (6.2)	16.8*** (2.6)
General-obligation bonds only	1.7 (2.3)	7.6*** (2.2)	13.3*** (2.3)	15.2*** (2.5)	15.8** (7.3)	13.5*** (2.9)
Issuer fixed effects	−5.0* (2.6)	1.8 (2.5)	7.5*** (2.6)	17.3*** (2.8)	16.8*** (4.1)	22.3*** (2.4)
Tax-adjusted spread over treasuries	5.3 (3.3)	12.6*** (3.2)	21.4*** (3.2)	26.9*** (3.5)	22.8*** (8.6)	21.6*** (3.8)
Insurance controls, year clusters (22)	0.9 (2.9)	8.0*** (2.6)	13.0*** (3.6)	17.7*** (3.4)	17.3** (7.6)	16.8*** (3.7)
Running variable net of refundings	0.8 (3.1)	4.9 (3.1)	8.1*** (3.1)	15.7*** (3.6)	11.0 (8.2)	14.8*** (3.7)
and refunding bonds excluded	0.9 (3.1)	6.3* (3.2)	8.9*** (3.2)	16.0*** (3.6)	10.9 (7.5)	15.1*** (3.6)

Notes. Cells report the 2SLS premium of designation in basis points, pooled over 2000–2021 at the year's cap in the window \$0.5 to \$4 million, with matched fixed effects and issuer-clustered standard errors unless stated. First stages are 0.40 to 0.51 in every bin of every row. Insurance and enhancement: adds bond insurance × year and credit enhancement × year (47% of window bonds are insured). The general-obligation, issuer-fixed-effect, tax-adjusted-spread and year-clustered rows also include these controls. General obligation: unlimited-tax, limited and double-barreled general-obligation bonds, 70% of the window. Issuer fixed effects: identified from issuers that cross the cap across years. Tax-adjusted spread: $100\,[y/(1-\tau_{B,t})-y^{\text{Tsy}}]$, the outcome of Dagostino (2025). Refunding-net: the running variable excludes refunding par, and the last row also drops refunding bonds from the estimation sample. $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$.

effects are −0.5, −3.2, −1.6, −10.9, −21.5 bp. The average beyond ten minus 1–2 years is −15.7 (6.7), significant at five percent.

By period. Period estimates use 2000–04, 2005–08, 2009–10, 2011–13, 2014–17 and 2018–19, with matched fixed effects and issuer clustering. The ARRA period uses the \$30 million cap and \$0.5–8 million window. Others use \$10 million and \$0.5–4 million. Before 2009, with funding costs between 2.3 and 2.4% and X_t of 65–66 bp, premia are at most about 0.3 of X_t at every maturity: deduction capitalization is weak when non-designated carrying penalties approach a percentage point.

The 2009–10 \$30 million cap has a weaker first stage than \$10 million's 0.42–0.45, reflecting limited use. With costs 1.09% and ordinary-rule X_t 30 bp, premia are 11 (24), 22 (23), 17 (23), 50 (28) and 60 (29) bp. The beyond-ten average minus 1–2 contrast is 43 (20), and the corresponding reduced-form yield contrast, −15.7 (6.7), is significant at five percent.

In 2011–17, with costs of 0.31–0.51% and X_t of 9–14 bp, premia are 2.4–2.5 at 1–2 years and 14–28 at 5–20 years, at or above X_t.

After the 2018 rate cut, premia are 6.9 (4.3), 6.4 (5.0), 7.7 (5.7), 14.9 (7.7), 3.7 (24.2) bp. Difference-in-discontinuities (§6.2), interacting post-2018 with designation in 2014–2019, gives 3.1 (6.5), −9.1 (6.7), −19.2 (7.6), 3.6 (10.0), −1.3 (28.7) bp with unchanged first stages. Annual reduced forms peak at 16–22 bp in 2011–13 and are near zero in 2004–05 and 2017. Premia therefore do not simply track the formula benefit, which was largest when funding costs were highest. The pattern is consistent with changing bank participation. Shareholder-taxed Subchapter S banks, less affected by TCJA, hold 6–14% of bank munis over 2000–2021.

Figure 10 shows the short and long yield discontinuities and Figure 24 all five maturity bins. Table 36 reports the full five-bin estimates.

Call Report items and funding costs. Section 265(b) allocates a bank's interest expense across its assets in proportion to their average adjusted bases, without matching each bond to borrowing of the same maturity. With bank n's annual interest expense I_{nt} and average asset basis $\bar{A}_{nt}$, a dollar of municipal basis held throughout the year is allocated $c_{nt} = I_{nt}/\bar{A}_{nt}$ of interest expense. Designation reduces the nondeductible fraction from one to 0.2, saving $X_{nt} = 0.8\,\tau_{B,t}c_{nt}$ at the bank's federal tax rate $\tau_{B,t}$. With the federal bank rate at 35% before 2018 and 21% thereafter, the municipal-holdings-weighted X_t is about 65 basis points in 2005–08, 14 in 2011–13, 9 in 2014–17 and 11 in 2018–19. December FDIC BankFind reports supply annual interest expense `EINTEXP` (Schedule RI, item 2.e, `RIAD4073`). Year-to-date expense is divided, without multiplying by four, by average assets `ASSET5`. This expense-to-assets ratio is neither a deposit rate nor a maturity-matched funding cost.

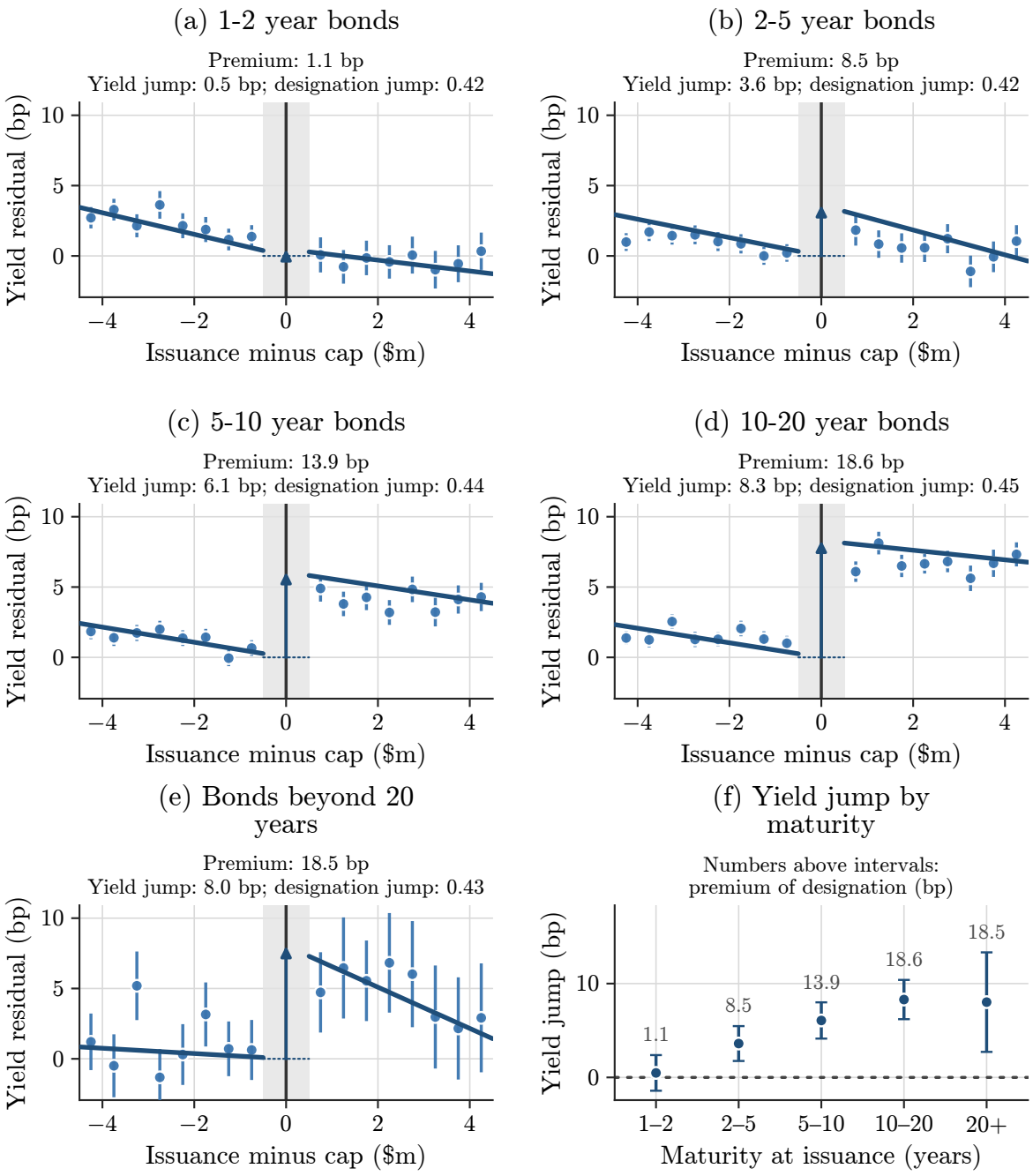


Figure 24: Offering-yield discontinuities by maturity. Panels (a)–(e) show residualized $0.5 million issuance-bin means relative to the below-cap fitted value at the cap, with local linear fits, donut shading and 95 percent bars. Arrows show above-cap-minus-below-cap jumps, and headers divide each yield jump by the designation jump to obtain the premium. Panel (f) shows jumps and premia, the binwise counterparts to Table 36. The common scale covers the central 96 percent of binned means. Sparse outlying cells lie outside it.

Weights use FDIC SCMUNI: state and local securities held to maturity at amortized cost plus available-for-sale at fair value (Schedule RC-B, item 3, columns A/D, 8496/8499 with applicable prefixes). They cover all municipal securities, bank-qualified or not, exclude loans and impose no bank-size cutoff.[28]

With M_{nt} these municipal-security holdings and $\mathcal{B}_t$ the retained banks with positive holdings,

$$c_t = \frac{\sum_{n\in\mathcal{B}_t} M_{nt}(I_{nt}/\bar{A}_{nt})}{\sum_{n\in\mathcal{B}_t} M_{nt}}, \qquad X_t = 0.8\,\tau_{B,t}c_t. \tag{149}$$

The screen retains banks with positive average assets, nonmissing interest expense and an expense-to-assets ratio from zero to below 15%. Mean c_t is 2.34% in 2000–04, 2.31% in 2005–08, 0.51% in 2011–13, 0.31% in 2014–17 and 0.66% in 2018–19, which correspond to federal-rate benefits of approximately 66, 65, 14, 9 and 11 bp.

The Call Report measure approximates the tax allocation using book assets and expense. The statute uses adjusted tax bases and otherwise-deductible interest, which are unavailable taxpayer by taxpayer. The IRS permits specified book-asset averaging conventions, and X_t is an observable ordinary-rule C-corporation scale that does not replicate any bank's tax-return calculation.[29] Weights include Subchapter S municipal holdings. In 2009–10, Section 265(b)(7)'s two-percent-of-assets exception could reduce non-designated disallowance and designation's incremental benefit. Treating 2009–10 through the $30 million cap alone omits this provision. Here X_t remains the ordinary-rule scale, with those years reported separately.

Bank holdings of municipal securities rose from $97 billion in 2000 to $370 billion in 2017 and fell after the 2018 rate cut. Designation, refunding status, security type, bond insurance, and credit enhancement come from the Mergent bond master files.

D.9 Investor holdings by tax status

Portfolio shares vary with exemption benefits (Table 2). In 2000–2017, exempt municipal bonds account for 34–48% of the average P&C insurer's bond portfolio, compared with 6–9% for life insurers (Figure 25). Public pensions' exempt and taxable municipal shares never exceed 0.13% and 0.78%, respectively (Figure 26). After the corporate-rate cut, the P&C exempt share fell from 38.0% in 2017 to 28.6% in 2019, while the other-taxable share rose from 50.2% to 57.5%. This reallocation is the quantity counterpart of P2's repricing (Appendix D.10).

D.10 TCJA insurer-tax shock: static DiD and dynamic event study

The second P2 design uses the 2017 TCJA. Proposition 6 gives zero pass-through of an institution's exemption benefit at the riskless short end, where the household's after-tax cash return anchors the yield, and positive pass-through at the long end. Through the portfolio-risk response, a change in that benefit also reaches risky maturities shorter than those the institution holds. The 2017 Tax Cuts and Jobs Act (TCJA) lowered P&C insurers' exemption benefit, so P2 predicts that after the reform the implied tax fell more at longer maturities in states whose municipal debt P&C insurers held more heavily.

The TCJA cut the corporate rate from 35% to 21% effective 2018Q1, and the proration rule of §832 amplified the resulting decline in P&C insurers' exemption benefit. The disallowed fraction of their tax-exempt interest rose from 15 to 25 percent, so the effective tax on municipal interest stayed at 5.25% ($35\% \times 15\%$ before, $21\% \times 25\%$ after), while the taxable after-tax share rose from 0.65 to 0.79. The P&C relative after-tax share ξ_P therefore rose from 0.686 to 0.834, and the exemption benefit $1 - \xi_P$ narrowed from 0.31 to 0.17.

Other investors' taxes changed too. The top household ordinary rate fell from 39.6% to 37%, and the cap on state and local tax (SALT) deductions changed the value of in-state exemptions. Life insurers' relative after-tax share rose from $0.65/0.825 \simeq 0.788$ to $0.79/0.937 \simeq 0.843$ (Appendix E.1), and banks' taxable

[28] The source definitions are the FDIC's Financial API definitions and the FFIEC December 2017 Call Report, Schedules RI and RC-B. BankFind harmonizes the historical regulatory-report series. The schedule references identify the corresponding commercial-bank items.

[29] See the IRS, Bank Qualified Bonds: Section 265, pp. 13-11–13-12, for the allocation and 20% disallowance. The annual, per-dollar-of-basis benefit is distinct from the present value of deductions over a bond's remaining life.

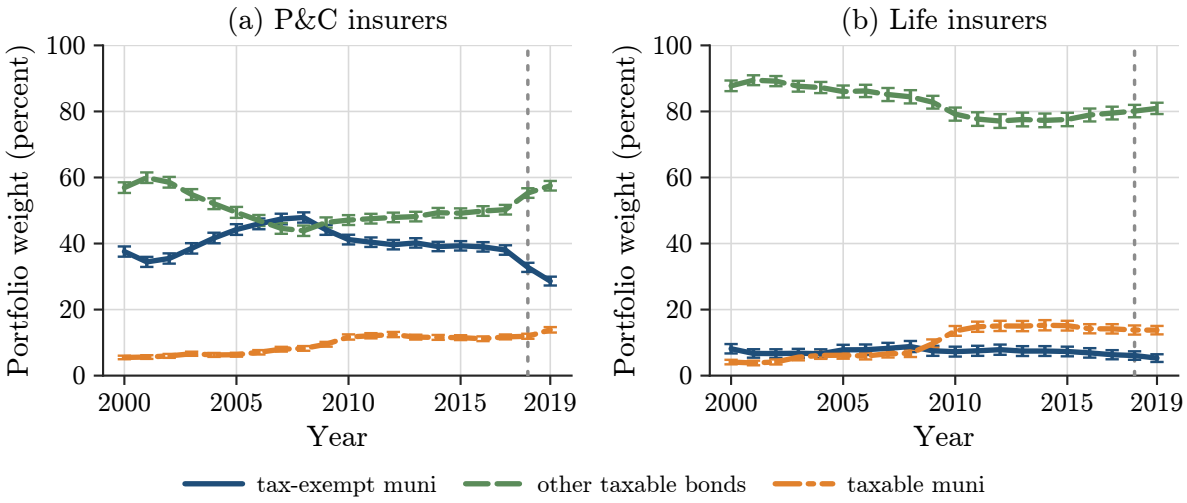


Figure 25: Tax-exempt municipals, taxable municipals and other taxable bonds as shares of insurers' total bond holdings: P&C in panel (a) and life in panel (b), 2000–2019, NAIC Schedule D. The dotted vertical line marks the TCJA corporate-rate cut. Bars are 95 percent intervals for yearly mean insurer portfolio shares, with standard errors clustered by insurer.

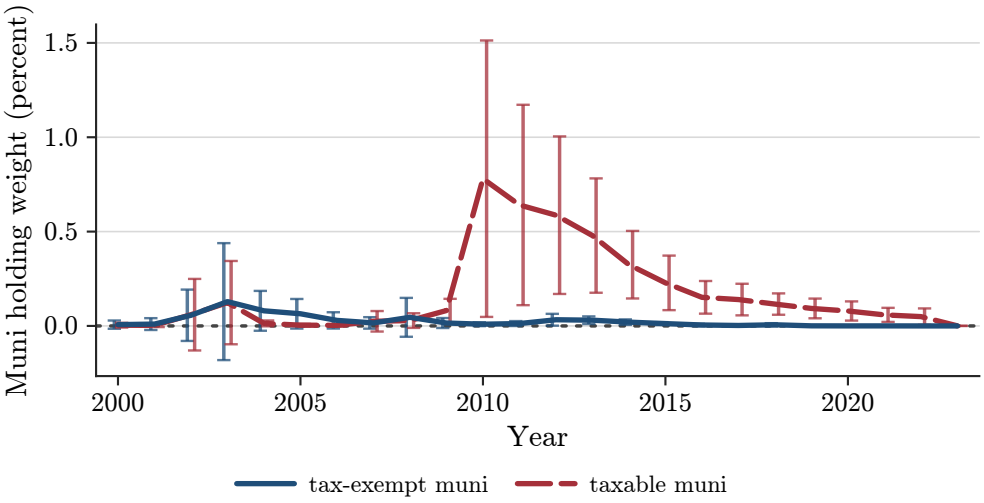


Figure 26: Municipal holdings by federal tax status as shares of total bond portfolios in the public-pension FOIA sample (§3.1), 2000–2023. All thirty plans are covered from 2015. Bars are 95 percent intervals for yearly mean plan portfolio shares, with standard errors clustered by plan. The rest of the plans' bond portfolios is taxable bonds, which the figure omits. Taxable municipals include Build America Bonds.

after-tax share and financing-tax cost changed. The design absorbs these common changes and compares states with different P&C exposure.

The treatment is a state's exposure to P&C insurers, $PCShare_s$: the share of state s's outstanding municipal debt held by P&C insurers at the end of 2015, before the estimation window opens in 2016Q3 and two years before the reform. Across states it averages 0.096 with a standard deviation of 0.039, and each bond b takes the exposure of its issuing state, $PCShare_{s(b)}$. The sample consists of the 1,733,881 customer trades in MSRB data over 2016Q3–2019Q1 in non-callable, AA-and-above, non-AMT municipal bonds. Restricting to non-callable bonds rules out direct repricing of embedded options. The outcome is the implied tax of (2) for each trade, $ImpliedTax_{b,t} = 1 - y^M_{b,t}/y^C_t(m_b)$, computed from the trade yield and the corporate zero yield of the same remaining maturity on the trade date. I estimate

$$\begin{aligned} ImpliedTax_{b,t} = \beta_0\big(PostTCJA_t \times PCShare_{s(b)}\big) + \beta_1\big(PostTCJA_t \times PCShare_{s(b)} \times m_b\big) \\ + \alpha_{\lfloor m_b \rfloor \times Purchase} + \alpha_{issuer \times Purchase} + \alpha_t + \Gamma' Z_{b,t} + \varepsilon_{b,t}, \end{aligned} \tag{150}$$

where $PostTCJA_t = \mathbf{1}\{t \geq 2017Q4\}$ starts the post period at the enactment quarter. The fixed effects compare trades of the same issuer and of the same rounded maturity, purchases with purchases and sales with sales ($Purchase_{b,t}$ is the MSRB dealer-side indicator for a customer purchase), and α_t absorbs each quarter's level. The controls $Z_{b,t}$ contain $PCShare_{s(b)} \times m_b$; the resident state tax rate, pre-period life-insurer holding share and bond rating, each alone and interacted with maturity; and event-time-by-maturity and event-time-by-state-tax-rate interactions. Event-time-by-maturity interactions let the common maturity gradient change each quarter, so β_1 is identified from cross-state differences in $PCShare$ within a quarter. The differential effect at maturity m_b is $\beta_0 + \beta_1 m_b$. The model predicts $\beta_1 < 0$: implied taxes fall more at longer maturities where P&C insurers held more of the debt. The representative-tax benchmark predicts $\beta_0 = \beta_1 = 0$.

Pre-reform P&C ownership may be correlated with state conditions that also move the maturity profile of implied taxes, so the IV specifications instrument $PCShare_s$ with the municipal capacity of the P&C insurers domiciled in the state:

$$IV1_s = \log\Big(\frac{\sum_{i \in D_s} \text{Muni Assets}_i}{\text{Muni Outstanding}_s}\Big), \qquad IV2_s = \log\Big(\frac{\sum_{i \in D_s^{\text{act}}} \text{Bond Assets}_i}{\text{Muni Outstanding}_s}\Big), \tag{151}$$

where D_s is the set of P&C insurers domiciled in state s, D_s^{act} the subset holding any municipal bonds, Muni Assets_i and Bond Assets_i insurer i's par holdings of municipal bonds and of all bonds in NAIC Schedule D, and $\text{Muni Outstanding}_s$ the state's municipal par outstanding. $IV2_s$ uses total bond holdings, so it does not depend on which municipal bonds the domiciled insurers chose to hold. For each instrument, its interactions with m_b, with $PostTCJA_t$ and with their product serve as instruments for the three corresponding exposure terms. Relevance comes from insurer home bias, since locally domiciled insurers disproportionately hold the state's bonds. The Kleibergen–Paap first-stage F statistics are 25.7 for $IV1_s$ and 17.5 for $IV2_s$.

The exclusion restriction is that, conditional on the fixed effects and controls, the domiciled insurers' capacity affects the post-TCJA change in the maturity gradient of implied taxes only through P&C insurers' exposure to the state's bonds. Quarter effects and event-time-by-maturity controls absorb the common component of the federal tax changes. Event-time-by-state-tax interactions absorb level changes associated with state rates. State-tax-by-maturity controls capture a persistent maturity gradient. Under the covariance conditions in Appendix D.11.1, a remaining SALT channel biases OLS toward zero. The appendix states the covariance conditions for this direction. Insurer domicile largely predates the reform.

Appendix Table 40 reports the estimates. The maturity gradient β_1 is -0.017 under OLS and -0.038 and -0.040 under the two IV specifications, with issuer-clustered standard errors of 0.005, 0.010, and 0.015. For two states whose pre-reform P&C shares differ by one standard deviation, 3.9 percentage points, the implied tax in the more exposed state falls after the reform by an additional 0.07 percentage point per year of maturity under OLS and 0.15 under the first IV specification. Between a five- and a twenty-five-year bond, the first IV estimate cumulates to about 3.0 percentage points of implied tax. At a corporate yield of four percent, that is about 12 basis points of municipal yield relative to the taxable yield. All three point estimates have the negative maturity gradient the model predicts, and Appendix D.11 reports additional inference checks.

Table 40: TCJA difference-in-differences: implied-tax response to insurer exposure. The sample is MSRB trades in non-callable, AA-and-above, non-AMT municipal bonds in 2016Q3–2019Q1. The outcome is the implied tax of equation (2), and *PCShare* is the end-2015 P&C ownership share. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

	(1) OLS	(2) IV1	(3) IV2
$PostTCJA \times PCShare$	-0.122^{***}	-0.129	-0.017
	(0.029)	(0.084)	(0.100)
$PostTCJA \times PCShare \times m_b$	-0.017^{***}	-0.038^{***}	-0.040^{***}
	(0.005)	(0.010)	(0.015)
$\lfloor m_b \rfloor \times Purchase$ FE	Yes	Yes	Yes
$Issuer \times Purchase$ FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Issuer cluster	Yes	Yes	Yes
Observations	1,733,881	1,733,881	1,733,881
First-stage F (Kleibergen–Paap)		25.7	17.5

Notes. Columns (2) and (3) instrument *PCShare* with $IV1_s$ and $IV2_s$ of equation (151). All columns include the controls of equation (150), among them the rating and its interaction with m_b. Standard errors are clustered by issuer (six-digit CUSIP, 9,903 issuers) and reported in parentheses.

The event-time estimates, with standard errors clustered by issuer, locate the change at the reform. Across all three estimators, eleven of the twelve pre-TCJA coefficients are individually indistinguishable from zero at the five percent level, while the coefficients become significantly negative at 2018Q1 in every column and remain negative thereafter. Five of the six post-reform quarters exclude zero under IV1 and four under IV2 (Figure 27, Appendix Table 41 reports each quarter).

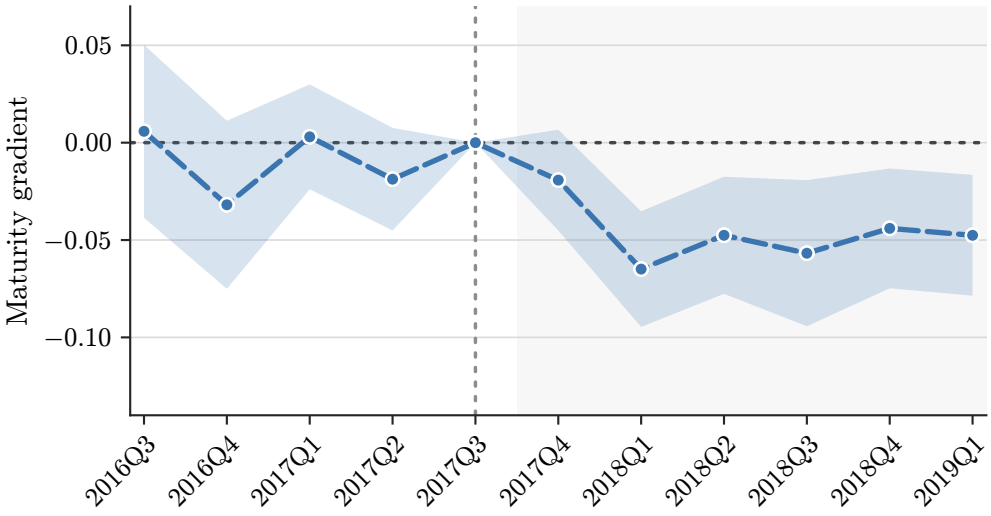


Figure 27: IV1 event-study estimates of the quarter-specific maturity gradient, the coefficient on $\mathbf{1}\{t = q\} \times PCShare \times m_b$, relative to 2017Q3 (dashed line). Shading marks the post-reform quarters from 2017Q4. Bands are 95% intervals clustered by six-digit CUSIP issuer. Appendix D.11 reports further inference checks.

Table 41: TCJA dynamic event study: full triple-interaction coefficients $\hat{\beta}_{1,q}$ on $\mathbf{1}\{t = q\} \times PCShare \times m_b$, by quarter. Standard errors in parentheses are clustered by issuer (six-digit CUSIP) as in Table 40, and 2017Q3 is the omitted base quarter. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Quarter q	(1) OLS	(2) IV1	(3) IV2
2016Q3	0.019**	0.006	−0.020
	(0.010)	(0.023)	(0.036)
2016Q4	−0.014	−0.032	−0.029
	(0.009)	(0.022)	(0.036)
2017Q1	0.001	0.003	0.009
	(0.007)	(0.014)	(0.020)
2017Q2	−0.005	−0.019	−0.021
	(0.006)	(0.013)	(0.020)
2017Q4	−0.001	−0.019	−0.021
	(0.006)	(0.013)	(0.020)
2018Q1	−0.028***	−0.065***	−0.063***
	(0.007)	(0.015)	(0.023)
2018Q2	−0.027***	−0.048***	−0.039*
	(0.007)	(0.015)	(0.023)
2018Q3	−0.031***	−0.057***	−0.063**
	(0.009)	(0.019)	(0.027)
2018Q4	−0.011	−0.044***	−0.051**
	(0.008)	(0.016)	(0.023)
2019Q1	−0.014*	−0.048***	−0.065***
	(0.007)	(0.016)	(0.024)
$\lfloor m_b \rfloor \times$ *Purchase* FE	Yes	Yes	Yes
Issuer × *Purchase* FE	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes
Issuer cluster	Yes	Yes	Yes
Observations	1,733,881	1,733,881	1,733,881

The maturity-gradient estimates persist on the wider sample. With the rating control omitted in both rating samples (Appendix Table 47), moving from the AA-and-above to the A-or-above sample roughly doubles the OLS gradient, from −0.018 to −0.036, leaves the IV1 gradient essentially unchanged, from −0.037 to −0.039, and roughly doubles the IV1 level term.

D.11 TCJA inference and household demand

The insurer regressions cluster standard errors by issuer. As a check for shocks shared within states, I also cluster the same regressions by issuing state (51 clusters). Point estimates are unchanged. State-clustered standard errors are 0.0129, 0.0245 and 0.0342 for the OLS, IV1 and IV2 maturity coefficients, respectively. Their 95% intervals include zero. The post-reform level response remains negative and significant: −0.122 (s.e. 0.061), with $p = 0.044$ from a state-level wild-cluster bootstrap (9,999 replications).

D.11.1 SALT-induced household demand and the insurer-channel regression

A binding TCJA SALT cap can raise high-tax households' exemption benefit and lower municipal yields where they are active. The calculation below gives the conditions under which this household channel attenuates the insurer coefficient.

Write $Y_{b,t} = 1 - y^M_{b,t}/y^C_t(m_b)$ for the implied tax of equation (2), $X_{b,t} = PostTCJA_t PCShare_{s(b)} m_b$ for the insurer interaction, and $W_{b,t} = PostTCJA_t Tax_{s(b)} m_b$ for a household-SALT interaction, where Tax_s proxies SALT exposure. Consider

$$Y_{b,t} = \beta_1 X_{b,t} + \beta_2 PostTCJA_t Tax_{s(b)} + \beta_3 W_{b,t} + \text{controls} + \text{FE} + \varepsilon_{b,t}. \tag{152}$$

The channels under study imply $\beta_1 < 0$ for the insurer mechanism and $\beta_2 > 0$ and $\beta_3 < 0$ for stronger short-end household demand.

Suppose the SALT level term is controlled for, W is omitted, and the remaining disturbance is orthogonal to residualized X. Tildes denote residuals after projecting on all included controls and fixed effects. The omitted-variable formula is

$$\operatorname{plim} \widetilde{\beta}_1 = \beta_1 + \beta_3 \frac{\operatorname{Cov}(\widetilde{X}, \widetilde{W})}{\operatorname{Var}(\widetilde{X})}. \tag{153}$$

Consequently,

$$\beta_3 < 0, \quad \operatorname{Cov}(\widetilde{X}, \widetilde{W}) < 0 \quad \Longrightarrow \quad \operatorname{plim} \widetilde{\beta}_1 > \beta_1. \tag{154}$$

Equation (153) uses residualized variables, accounting for controls, fixed effects and weights. Upward bias attenuates a negative coefficient when its sign is preserved.

IV identifies the insurer coefficient when the instruments are relevant and orthogonal to the remaining household-demand term, under the exclusion restriction stated in Appendix D.10.

D.12 P3: municipal-yield response to treasury yields

Transaction sample. Each MSRB customer purchase or sale records the CUSIP, trade date, par, dollar price, implied yield to maturity and dealer side. January 2005–August 2020 estimation trades are non-callable, federally exempt and AA-and-above (agency score ≤ 3, §6.3). Noncallability removes option contamination. I exclude trades less than 92 days after issue, when offering yields anchor prices, and trades less than 365 days before maturity, where the treasury short-rate substitute is erratic.

Matched treasury changes. At each trade, the daily GSW par-yield curve is interpolated linearly between integer maturities at remaining maturity $m_b(t) = d_{b,t}/365$, where $d_{b,t}$ is days to maturity, and clamped above thirty years. Within each CUSIP and dealer side, consecutive trades form $\Delta y^M_{b,t}$ and $\Delta y^{\text{Tsy}}_{m_b,t} = y^{\text{Tsy}}_{m_b(t),t} - y^{\text{Tsy}}_{m_b(t-1),t-1}$. Here $t-1$ denotes the preceding trade. Comparing purchases with purchases and sales with sales limits bid-ask and intraday contamination. Pairs with gaps of 1–90 days and treasury moves of at least 10 bp are kept, with Table 42's weights and trimming. Over Δt elapsed days, aging contributes $-(\Delta t/365)\partial_m y^{\text{Tsy}}_{m,t} + O((\Delta t/365)^2)$, measured below.

Holding the benchmark maturity fixed. Fixing initial maturity gives $\Delta y^{\text{Tsy,fixed}}_{b,t} = y^{\text{Tsy}}_{m_b(t-1),t} - y^{\text{Tsy}}_{m_b(t-1),t-1}$, removing slide $y^{\text{Tsy}}_{m_b(t),t} - y^{\text{Tsy}}_{m_b(t-1),t}$. On upward-sloping curves slide is negative, averaging -2.6 bp below five years and -1.6 beyond, against absolute changes near 26 bp. Removing it lowers the short response more and increases the gradient (Table 43). Because municipal and treasury curves have different slopes, municipal aging does not offset the treasury slide. Row three also absorbs linear municipal aging through elapsed time and its maturity interaction, which further increases the gradient. I retain the conservative headline specification.

Specifications. Column (1) of Table 42 estimates the continuous interaction in (31), including maturity but no fixed effects. Column (2) replaces maturity by $Long_b = \mathbf{1}\{m_b \geq 5\}$, so its short and long responses are β_0 and $\beta_0 + \beta_1$.

Table 42: P3: observed coupon-yield response to treasury par yield by remaining maturity.

	(1) Continuous $\Delta y^{M}_{b,t}$	(2) Bin $\Delta y^{M}_{b,t}$
$\Delta y^{\text{Tsy}}_{m_b,t}$	0.346^{***}	0.351^{***}
	(0.016) [0.012]	(0.013) [0.006]
$\Delta y^{\text{Tsy}}_{m_b,t} \times m_b$	0.007^{***}	
	(0.002) [0.002]	
$\Delta y^{\text{Tsy}}_{m_b,t} \times Long_b$		0.075^{***}
		(0.012) [0.009]
$Long_b$		0.016^{***}
		(0.002) [0.002]
m_b	0.002^{***}	
	(0.000) [0.000]	
Trade-date cluster	Yes	Yes
Observations	952,874	952,874
Weights	Inverse pair count	Inverse pair count

Notes. The sample consists of pairs of consecutive MSRB trades, within 90 days of each other, in federally tax-exempt, non-callable bonds rated AA and above (agency score ≤ 3) from January 2005 to August 2020, over which the maturity-matched treasury yield moved at least 10 basis points in either direction. The treasury yield is interpolated on the daily GSW curve at the bond's exact remaining maturity on each transaction date. Observations are weighted by the inverse count of their CUSIP's base-sample trade pairs, computed before observations with missing or trimmed outcomes are dropped. The dependent variable is trimmed at its first and ninety-ninth percentiles. $Long_b = \mathbf{1}\{m_b \geq 5\}$. First differencing absorbs time-invariant bond characteristics. Parentheses give trade-date-clustered standard errors (3,864 clusters). Brackets give state-clustered standard errors (54 clusters). Stars are identical under either clustering. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 43: P3 observed coupon-yield responses and treasury-curve ageing: equation (31) on Table 42's 952,874 pairs, inverse-CUSIP-base-pair weighted and trade-date clustered. Row one matches treasury maturity at each trade. Row two fixes first-trade maturity. Row three adds elapsed time and its maturity interaction.

	Bin specification		Interaction	
Treasury benchmark	< 5y	≥ 5y	Bin	Continuous
Benchmark matched at each trade's maturity (Table 42)	0.351^{***}	0.426^{***}	0.0749^{***}	0.0070^{***}
Benchmark held at the first trade's maturity	0.340^{***}	0.423^{***}	0.0825^{***}	0.0084^{***}
plus elapsed time and elapsed time $\times$ maturity	0.334^{***}	0.419^{***}	0.0850^{***}	0.0086^{***}

Notes. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Taxable-market comparison. I also compare municipal changes with the monthly corporate curve. Both reference curves are measured at their monthly averages and the bond's remaining maturity in each trade month. Retaining cross-month pairs with an absolute monthly treasury change of at least 10 bp gives 517,864 pairs, with the original inverse-CUSIP weights. The municipal-to-corporate maturity interaction is 0.0132 (s.e. 0.0026), while the municipal-to-treasury interaction on these pairs is 0.0072 (s.e. 0.0025), both trade-date clustered. These are observed coupon-yield associations. The cash-flow comparison below translates each reference market's zero curve into municipal coupon yields on a common sample.

D.12.1 Coupon cash flows and the zero-yield benchmark

P3 concerns zero-coupon responses. Even when zero yields satisfy $y^{M,0}(u,t) = \xi_H y^{\text{Tsy},0}(u,t)$ at every horizon u, coupon yields need not have a constant response ratio, because their cash flows span horizons with different

yield movements. For this comparison I fix the after-tax share at 0.65 and price each bond's regular coupon and redemption payments at the proportional GSW zero curve,

$$P_{b,t}^{M,0} = \sum_k CF_{b,k} \exp\left[-0.65\, y^{\mathrm{Tsy},0}(u_{b,k,t}, t)\, u_{b,k,t}\right],$$

where $CF_{b,k}$ is payment k, $u_{b,k,t}$ its time to payment, and zero yields are continuously compounded fractions. I then recover the nominal-semiannual yield to maturity, $y_{b,t}^{M,0}$, from this price at each trade date. The simulation retains the bond's coupon and maturity, both transaction dates and the original par-yield regressor. The after-tax share is fixed in this exercise.

Semiannual payments are reconstructed backward from maturity with US 30/360 fractions. The comparison retains fixed semiannual bonds whose first coupon precedes the first trade. Zero-coupon bonds do not face this screen. Unsupported conventions, maturities over thirty years and incomplete curves are excluded. Zero yields are interpolated linearly, with the one-year value held below one year. The 918,350 common pairs retain 94.9% of original weight, with the original trimming and inverse-CUSIP weights. Simulated outcomes are not trimmed.

Table 44 applies the same regression to observed changes, hypothetical changes and their difference. The difference-outcome standard error retains their covariance. The daily benchmark accounts for 30.0% of the observed gradient, leaving 0.0056 (s.e. 0.0021). After-tax shares from 0.60 to 0.80 give shares of 27.6%–37.0%, with a positive residual throughout. The binned residual is 0.0623 (s.e. 0.0123). Fixed-maturity benchmarks, with or without elapsed-time controls, also retain a positive interaction.

Monthly comparisons retain cross-month selection and reference regressors on 500,006 common coupon pairs. Treasury comparisons price at daily proportional GSW zero curves on trade dates. Corporate comparisons use monthly proportional corporate zeros, converted to continuous yields. Both residual interactions remain positive and significant at five percent (Table 44).

Table 44: P3: response gradient after matching coupon cash flows.

	Observed	Coupon benchmark	Difference	Pairs
Daily, continuous	0.0080***	0.0024***	0.0056***	918,350
	(0.0021)	(0.0001)	(0.0021)	
Daily, long minus short	0.0738***	0.0115***	0.0623***	918,350
	(0.0122)	(0.0003)	(0.0123)	
Fixed initial maturity	0.0095***	0.0034***	0.0061***	918,350
	(0.0021)	(0.0001)	(0.0021)	
Fixed maturity + gap controls	0.0098***	0.0036***	0.0062***	918,350
	(0.0021)	(0.0001)	(0.0021)	
Monthly treasury	0.0082***	0.0019**	0.0063**	500,006
	(0.0026)	(0.0009)	(0.0026)	
Monthly corporate	0.0141***	−0.0006***	0.0147***	500,006
	(0.0027)	(0.0001)	(0.0026)	

Notes. The observed and benchmark coupon changes of Appendix D.12.1 use identical pairs, regressors and inverse-CUSIP-base-pair weights, and the difference is estimated jointly. Interactions are per year of remaining maturity, except the five-year-bin contrast. Fixed-maturity rows retain the first trade's treasury maturity. Gap controls are elapsed days and their maturity interaction. Monthly rows use cross-month pairs and monthly regressors. Parentheses give second-trade-date-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Sensitivity to the filter thresholds. Across the cutoffs and trade windows of Table 45, all observed maturity-interaction estimates are positive. Every binned cell is significant at one percent, and every continuous cell is significant at five percent, with twenty-seven of the thirty significant at one percent. Applying the coupon benchmark separately in each cell leaves all thirty continuous interactions positive, from 0.0034 to 0.0104, with twenty-four significant at five percent. All thirty benchmark-adjusted binned differences are

positive and significant at five percent, ranging from 0.0393 to 0.1016. Each comparison retains its original pre-trim CUSIP weights on common coupon pairs.

Table 45: P3 sensitivity: observed maturity interaction across the treasury-move cutoff and trade-gap window (weighted by inverse CUSIP pair counts).

Panel A. Continuous interaction $\Delta y^{\text{Tsy}}_{m_b,t} \times m_b$					
Cutoff \ window	15d	30d	60d	90d	120d
≥ 0.05	0.0053^{***}	0.0057^{***}	0.0051^{**}	0.0071^{***}	0.0090^{***}
≥ 0.075	0.0061^{***}	0.0057^{***}	0.0047^{**}	0.0068^{***}	0.0087^{***}
≥ 0.10	0.0065^{***}	0.0060^{***}	0.0049^{**}	0.0070^{***}	0.0087^{***}
≥ 0.15	0.0080^{***}	0.0071^{***}	0.0059^{***}	0.0077^{***}	0.0091^{***}
≥ 0.20	0.0088^{***}	0.0082^{***}	0.0067^{***}	0.0081^{***}	0.0094^{***}
≥ 0.25	0.0104^{***}	0.0083^{***}	0.0074^{***}	0.0087^{***}	0.0097^{***}
Panel B. Binned interaction $\Delta y^{\text{Tsy}}_{m_b,t} \times Long_b$					
Cutoff \ window	15d	30d	60d	90d	120d
≥ 0.05	0.049^{***}	0.061^{***}	0.070^{***}	0.074^{***}	0.088^{***}
≥ 0.075	0.056^{***}	0.063^{***}	0.068^{***}	0.074^{***}	0.086^{***}
≥ 0.10	0.062^{***}	0.066^{***}	0.069^{***}	0.075^{***}	0.086^{***}
≥ 0.15	0.079^{***}	0.077^{***}	0.079^{***}	0.083^{***}	0.092^{***}
≥ 0.20	0.094^{***}	0.090^{***}	0.085^{***}	0.088^{***}	0.097^{***}
≥ 0.25	0.111^{***}	0.100^{***}	0.096^{***}	0.096^{***}	0.102^{***}

Notes. Each cell re-estimates the headline specification on pairs whose maturity-matched treasury move is at least the cutoff (percentage points) and whose trades are within the window. Weights are inverse CUSIP base-pair counts. Standard errors are clustered by trade date as in Table 42. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

State clustering. Table 42 also reports standard errors clustered across same-state CUSIPs. Date clustering allows common shocks across states. Both headline interactions remain significant at one percent. State clustering leaves all binned grid significance levels unchanged. In the continuous grid of Table 45, the 0.05, 0.075 and 0.10 cutoffs give $p < 0.05$ at thirty days and $p < 0.10$ at sixty days. The 0.15 cutoff gives $p < 0.05$ at sixty days. All other significance levels are unchanged. Thus twenty-seven of thirty continuous cells and all binned cells remain significant at five percent.

D.13 Robustness to the rating cut

Both samples apply their rating cutoffs to the unrounded mean agency score on the twenty-two-notch scale: 1 = AAA, 3 = AA, 7 = A−. Curves in §3 and §5 use seven or better (A− and above), matching the corporate A band (A+, A, A−). The BAB comparison in §3 and P1, TCJA and P3 in §6 use three or better, AA and above. Rating-cell and regression-control transformations follow the specifications of each exercise.

The curve pattern survives the narrower cut, and the BAB, P1, TCJA and P3 tests retain their gradient signs on the wider cut. On both samples, BAB interactions are significant at one percent and the bins rise monotonically. P1's pre-2018 state-tax pass-through declines across all five bins under OLS and IV (Table 46), with endpoint differences significant at five percent under OLS and one percent under IV. TCJA gradients are negative and significant under all three estimators with issuer clustering, and Appendix D.11 reports state-clustered inference separately. P3's observed maturity interactions in the coupon-yield response to par yields are positive and significant at one percent. The wider sample gives larger P3 interactions and a larger OLS TCJA gradient, with similar BAB, P1 and IV TCJA gradients.

D.13.1 Curve construction at AA and above

Figure 28 uses the AA-and-above sample and its weights. Net of calls and issuance premia, implied tax falls from 40.1% at two years to 25.5% at thirty, a decline of 14.6 points versus 17.1 on the A-or-above sample. Each implied tax is one minus the ratio of mean municipal to mean taxable yields, averaged with the sample's bond, rating and month weights. The model comparison repeats the accounting of Section 5.4 on this sample. The normalized model gap exceeds the residual before the model at every maturity, 96.2 against 85.2 basis points at thirty years, with a root-mean-square residual of 11.1 basis points.

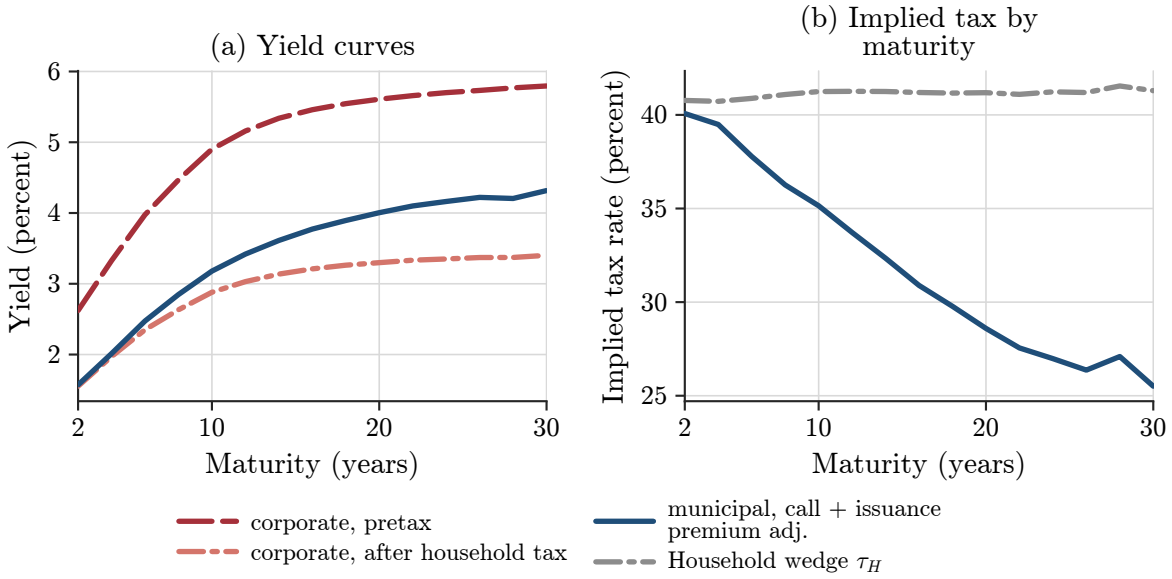


Figure 28: The AA-and-above municipal puzzle, 2000–2020, with this sample's own weights and household benchmark. Panel (a) shows pretax and household after-tax corporate zero-coupon yields and the municipal yield after removing callability and the issuance premium, and panel (b) shows the implied tax $1 - y_T^M / y_T^C$ and the household wedge τ_H. Calls use dated calibrated prices, and the sample's all-size issuance premium is scaled by the full sample's institutional-to-all-size ratio (as described below).

The AA-and-above model comparison uses this sample's bond, rating and month weights and its model ratio of municipal to taxable mean yields applied to observed corporate yields. Call, liquidity and default estimates are sample-specific. The issuance adjustment scales the sample's all-size premium by the full-sample institutional-to-all-size ratio, giving 11.5 bp at thirty years, and capital gains use the common method. All normalized residuals are negative, with 7/15 below ten bp in absolute value, RMS 11.1 bp and thirty-year residual −11.0 bp.

D.13.2 Empirical tests at A− and above

The tables compare scores ≤ 3 and ≤ 7. The insurer comparison omits the headline rating and rating-by-maturity controls in both samples. The P3 tables use state clustering, while the headline estimates use trade-date clustering. The table notes identify these choices. Table 29 compares BAB rating cuts in columns (1) and (3), with call matching in columns (2) and (4).

Table 46: P1 by rating cut, 2000–2017.

	OLS		IV	
	Score ≤ 3	Score ≤ 7	Score ≤ 3	Score ≤ 7
β_1: $[0, 5)$ years	1.072^{***}	0.990^{***}	1.092^{***}	1.084^{***}
	(0.195)	(0.256)	(0.224)	(0.234)
β_2: $[5, 10)$ years	0.682^{***}	0.618^{***}	0.705^{***}	0.680^{***}
	(0.071)	(0.106)	(0.088)	(0.122)
β_3: $[10, 15)$ years	0.484^{***}	0.379^{***}	0.448^{***}	0.382^{***}
	(0.098)	(0.084)	(0.121)	(0.101)
β_4: $[15, 20)$ years	0.386^{***}	0.284^{***}	0.325^{***}	0.262^{***}
	(0.102)	(0.087)	(0.117)	(0.097)
β_5: $[20, 30]$ years	0.298^{***}	0.239^{***}	0.227^{***}	0.216^{***}
	(0.075)	(0.077)	(0.081)	(0.080)
Decline $\beta_1 - \beta_5$	0.774^{***}	0.751^{**}	0.865^{***}	0.868^{***}
p-value of decline	0.002	0.010	< 0.001	< 0.001
KP F	—	—	10.4	12.6
Observations	681,839	1,397,769	681,839	1,397,769
States	47	47	47	47
Full main controls	Yes	Yes	Yes	Yes

Notes. Full controls, fixed effects and state-clustered inference as in Appendix D.6, 2000–2017. IV uses bin-interacted 1990 rates. All columns use call-adjusted zeros. Pricing availability differs across rating bands. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 47: P2: TCJA insurer difference-in-differences, by rating cut

	OLS		IV1		IV2	
	≤ 3	≤ 7	≤ 3	≤ 7	≤ 3	≤ 7
Post × *PCShare*	-0.118^{***}	-0.161^{***}	-0.124	-0.255^{**}	-0.015	-0.175
	(0.029)	(0.035)	(0.085)	(0.110)	(0.102)	(0.131)
Post × *PCShare* × m_b	-0.018^{***}	-0.036^{***}	-0.037^{***}	-0.039^{***}	-0.038^{**}	-0.037^{***}
	(0.005)	(0.005)	(0.011)	(0.009)	(0.015)	(0.012)
$\lfloor m_b \rfloor$ × *Purchase* FE	Yes	Yes	Yes	Yes	Yes	Yes
Issuer × *Purchase* FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Issuer cluster	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1,733,881	3,043,035	1,733,881	3,043,035	1,733,881	3,043,035
First-stage F			25.6	45.5	17.5	34.6

Notes. Standard errors are clustered by issuer (six-digit CUSIP), with 9,903 issuers at score ≤ 3 and 17,634 at score ≤ 7. First-stage F is Kleibergen–Paap. Both samples omit Table 40's rating and rating-by-maturity controls. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 48: P3: observed coupon-yield response to treasury par yields, by rating cut

	(1) continuous		(2) bin <5y vs ≥5y	
	score ≤ 3	score ≤ 7	score ≤ 3	score ≤ 7
$\Delta y^{\text{Tsy}}_{m_b,t}$	0.346***	0.281***		
	(0.012)	(0.011)		
$\Delta y^{\text{Tsy}}_{m_b,t}\times$ maturity	0.007***	0.013***		
	(0.002)	(0.002)		
$\Delta y^{\text{Tsy}}_{m_b,t}\times$ (≥5y)			0.075***	0.113***
			(0.009)	(0.006)
State cluster	Yes	Yes	Yes	Yes
Observations	952,874	2,023,637	952,874	2,023,637
Weights	Inverse pair count			

Notes. The estimates use Table 42's first-differenced specification for coupon yields and par yields, weighted by inverse CUSIP pair counts and clustered by state. Both observed maturity interactions are positive and significant at one percent. The wider-sample continuous interaction roughly doubles and the binned one rises by about half. Appendix D.12.1 maps the zero-coupon benchmark into the observed yield conventions on the main sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 49: P3 observed-response sensitivity grid, binned specification, by rating cut

	AA and above (score ≤ 3)					A− and above (score ≤ 7)				
cutoff	15d	30d	60d	90d	120d	15d	30d	60d	90d	120d
≥ 0.05	0.049***	0.061***	0.070***	0.074***	0.088***	0.078***	0.099***	0.098***	0.114***	0.121***
≥ 0.075	0.056***	0.063***	0.068***	0.074***	0.086***	0.080***	0.098***	0.096***	0.112***	0.118***
≥ 0.1	0.062***	0.066***	0.069***	0.075***	0.086***	0.088***	0.101***	0.097***	0.113***	0.119***
≥ 0.15	0.079***	0.077***	0.079***	0.083***	0.092***	0.095***	0.110***	0.105***	0.117***	0.123***
≥ 0.2	0.094***	0.090***	0.085***	0.088***	0.097***	0.108***	0.118***	0.111***	0.122***	0.128***
≥ 0.25	0.111***	0.100***	0.096***	0.096***	0.102***	0.120***	0.126***	0.119***	0.128***	0.134***

Notes. Cells report the observed interaction Δtreasury × (≥5y) in the regression of coupon-yield changes on par-yield changes, with state-clustered standard errors. Rows give minimum absolute treasury moves in percentage points. Columns give maximum days between consecutive trades, as in Table 45. All thirty cells are significant at one percent on both samples. Every wider-sample coefficient is larger, with the largest proportional gains at the lowest cutoffs. Appendix D.12.1 reports the corresponding coupon-benchmark contrasts on the main sample. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E Institutional detail and data

This appendix gives Section 3's investor-, state- and security-specific tax rules and trading restrictions, separating statutory treatment from quantitative assumptions.

E.1 Investor-by-investor federal tax treatment since 2000

For investor i, pricing after normalization by the municipal payoff uses the ratio $\xi_i = \xi_i^C/\xi_i^M$ of the absolute interest-income after-tax shares $\xi_i^C = 1-\tau_i^C$ and $\xi_i^M = 1-\tau_i^M$. The exemption benefit $1-\xi_i$ differs from the tax rates τ_i^C, τ_i^M, and financing deductions supply a further institutional margin. Table 50 restates Table 2 under the tax treatment applied to the 2018–2020 sample.

Households. Municipal interest is generally federally exempt and taxable bond interest faces ordinary-income tax, so the federal after-tax shares are approximately $(\xi_h^M, \xi_h^C) = (1, 1-\tau_h^{\text{ord}})$. Exempt interest is reported in both high- and lower-income brackets (Figure 1), so the relevant tax benefit varies within the household clientele.

Table 50: Investor heterogeneity in the municipal market after the 2017 Act

Investor type	Federal treatment	State treatment	Net exemption benefit
High-tax household	Interest exempt	Exempt on in-state bonds only	0.35–0.41
Middle-tax household	Interest exempt	Exempt on in-state bonds only	0.22–0.32
Low-tax household	Interest exempt	Exempt on in-state bonds only	0.10–0.12
P&C insurer	Exempt, less proration	Premium taxes, no distinction	0.17
Life insurer	Exempt, less proration	Premium taxes, no distinction	0.16
Bank	Exempt, funding cost disallowed	Varies by state	0.17–0.20
Other (pensions, etc.)	Varies	Varies	< 0.05

Notes. The table shows the tax treatment applied to the 2018–2020 sample. Household values are federal marginal rates, grouped as 35% and above, 22% to 32%, and below 22%, and the top value includes the 3.8% net investment income tax. The $10,000 cap on the state and local tax deduction makes the home-state exemption more valuable to households whose cap binds. Institutions face the 21% corporate rate. P&C proration rises to 25% of exempt interest, $\xi = 0.79/0.9475$, and the life insurer's policyholders' share is set at 30%, $\xi = 0.79/0.937$. The bank value is the break-even implied tax of Table 2 at the 21% rate with the 2015–2017 funding cost and taxable yields. The Act narrows the difference between the P&C and life-insurer exemption benefits.

Table 51 gives top rates. The 2001 phase-down, accelerated in 2003, cut the top ordinary rate to 35% in 2003. The rate rose to 39.6% in 2013 and fell to 37% in 2018. The top long-term capital-gains rate fell to 15% in 2003 and rose in 2013. Since 2013, the 3.8% net investment income tax (NIIT) covers taxable interest and many gains, excluding exempt interest. The calibration computes federal and state rates for each Survey of Consumer Finances tax unit under each year's law (Section 5).

Table 51: Top federal tax rates relevant to municipal pricing, 2000–2020

Period	Top ordinary rate	+ NIIT (on taxable interest)	Top corporate rate
2000	39.6%	39.6%	35%
2001	39.1%	39.1%	35%
2002	38.6%	38.6%	35%
2003–2012	35.0%	35.0%	35%
2013–2017	39.6%	43.4%	35%
2018–2020	37.0%	40.8%	21%

Banks. Interest on debt that finances non-bank-qualified exempt bonds at r_f is generally nondeductible, costing r_f after tax versus $r_f(1-\tau_c)$, a penalty $\tau_c r_f$. Section 265(b)(3) excepts bank-qualified bonds. Section 291 reduces their deduction by 20%, leaving 80% and a cost of $r_f(1-0.8\tau_c)$. Allocation of interest to exempt bonds uses adjusted tax bases.[30]

TEFRA's 15% disallowance became 20% in 1984. The 1986 Act generally denied financial institutions' interest deductions allocable to exempt obligations acquired after August 7, 1986. The ordinary bank-qualified exception covers designated small-issuer obligations under the $10 million annual limit and other conditions. Temporary 2009–2010 rules expanded eligibility. Each dollar of model bank muni holdings bears a financing-tax cost equal to the bank tax rate times the measured funding-cost ratio times $(1-0.8q_{BQ})$, with bank-qualified share q_{BQ}. The aggregate calibration sets $q_{BQ} = 0$. The bank-qualified design applies the deduction to designated bonds. Appendix D.8 derives capitalization.

Property and casualty insurers. P&C insurers pay the corporate rate τ_c on taxable bond interest, and proration reduces their municipal advantage. Its pre-TCJA percentage of 15% gave an effective municipal

[30] See the IRS, Bank Qualified Bonds: Section 265, pp. 13-9 and 13-11. Special rules apply to certain 2009–2010 issues and grandfathered obligations.

rate of $0.15 \times 0.35 = 0.0525$. TCJA replaced it with 5.25% divided by the highest corporate rate, 25% at $\tau_c = 21\%$. The municipal after-tax share thus stays at 0.9475 while the taxable after-tax share rises from 0.65 to 0.79, raising the relative after-tax share from 0.686 to 0.834 and cutting the exemption benefit from 31.4% to 16.6%. Holding proration fixed in a corporate-rate counterfactual would impose a different rule.

The calibration applies the top corporate rate to tax-paying insurers. Before 2018, marginal corporate rates were 15–34% below $100,000, 34–39% from $100,000 to $18,333,333 (including bubble rates recapturing lower-bracket benefits), and flat at 35% above that threshold. Since 2018 the rate is 21% at every taxable-income level.

Life insurers and other institutions. Section 812 divides life-insurer net investment income into company and policyholder shares, with exempt interest reducing reserve deductions. Baseline absolute after-tax shares are 0.825 on municipal interest (a one-half policyholders' share taxed at 35%, an effective municipal rate of 17.5%) and 0.65 on taxable interest, giving a ratio of 0.788. Post-2017 assumptions use a 30% policyholders' share and a 21% corporate rate, which give an effective municipal rate of 6.3%, after-tax shares of 0.937 and 0.79, and a ratio of 0.843. These are maintained mappings and do not estimate individual tax returns. Tax-exempt institutions and tax-deferred accounts gain little incremental benefit from the exemption.

E.2 Relative after-tax shares, absolute liabilities, and fiscal scope

Tax receipts require absolute tax rates, not only relative after-tax shares. For example, the pre-2018 P&C ratio 0.6860 reflects positive municipal tax and a 35% corporate tax. Treating municipal income as untaxed and taxing corporate income at $1 - 0.6860$ would misstate both receipts. Preferences are transformed with the payoff units of Section 4.2 to preserve the same absolute objective. Household assignments retain dated federal responses, the Social Security taxation proxy of Appendix C.7 and state approximations. State receipts follow residence. Before 2018, federal taxes are the residual consistent with the multiplicative federal-state after-tax share, without a new taxpayer-level calculation. Group assignments do not identify individual tax units' bond holdings.

The funded model includes taxable cash. The fiscal ledger covers bond income, excluding cash and other liability flows. Bank financing-tax changes are separate federal receipts and reduce investor bond income without double-counting funding expense or issuer interest. Reform-specific denominators differ (Section 7). The annual $z = 0$ flow $y_T Q_T$ measures coupon-equivalent replacement financing, without resetting existing coupons, OID or redemption promises. Nominal-semiannual reporting is a sensitivity. The equilibria use stated absolute taxes. Fiscal forecasts additionally require cohorts, refinancing, deductions, taxpayer provisions and feedback.

Portfolio substitution and bond-interest receipts. After repeal, both home- and out-of-state municipal interest are state-taxable. Switching issuers does not avoid tax. Residents' cash share rises by 2.42–2.78 percentage points across the seven states, exceeding the riskless-share increase because residents also sell riskless municipal bonds. In California, first applying reform yields to baseline holdings raises annual state receipts by 135.5 million relative to the fixed-price calculation. Reallocation at those yields then reduces receipts by 200.7 million: 177.6 million from a smaller bond portfolio and 23.1 million from its composition. The quantity term uniformly rescales each household type's initial bond basket to its new total holdings. The composition term is the remainder. Together, these effects reduce added receipts by 65.2 million, from $1.253 billion to $1.187 billion. The ledger excludes cash interest. This decomposition concerns bond-interest receipts rather than all state tax receipts.

E.3 State tax exemption and who receives it

Resident households. The baseline normalizes the household's home-state municipal after-tax share to one and assigns an out-of-state share of $1 - \tau_s^{\text{st}}$, without an additional federal deduction credit. Through 2017, the taxable share multiplies the federal relative after-tax share by $1 - \tau_s^{\text{st}}$, tax unit by tax unit. The post-2017 mapping treats the state and local tax deduction cap as binding, as in equation (142). These are the effective tax conventions used in the quantitative model.

Table 52: Portfolio allocation under the two tax reforms

Panel A. Home-state exemption repeal: resident households			
	Home-state municipal	All risky	Riskless
California	16.47 → 6.02	36.01 → 35.45	63.99 → 64.55
New York	16.42 → 5.02	35.06 → 34.69	64.94 → 65.31
New Jersey	15.49 → 2.05	34.95 → 34.69	65.05 → 65.31
Massachusetts	15.60 → 1.81	34.37 → 34.27	65.63 → 65.73
Ohio	14.84 → 1.70	34.23 → 34.13	65.77 → 65.87
Pennsylvania	14.52 → 2.24	33.80 → 33.77	66.20 → 66.23
Minnesota	17.52 → 0.97	35.44 → 35.19	64.56 → 64.81
Panel B. Equal municipal and taxable treatment: investor groups			
	Risky municipal	Risky taxable	Riskless
High-tax households	27.33 → 6.16	0.00 → 27.30	72.67 → 66.54
Lower-tax households	0.79 → 4.36	39.87 → 36.51	59.35 → 59.13
Banks	70.06 → 9.95	29.94 → 90.05	0.00 → 0.00
P&C insurers	8.03 → 7.24	91.97 → 92.76	0.00 → 0.00
Life insurers	0.01 → 20.03	99.99 → 79.97	0.00 → 0.00
Tax-neutral investors	0.00 → 7.65	71.62 → 61.85	28.38 → 30.50

Notes: Entries are baseline → reform percentages of each population's modeled financial assets, including cash and excluding outside assets and liabilities. Panel A aggregates both resident household groups in each state. Each row is a separate repeal. Panel B aggregates each investor group across states under 99.9% equal treatment. Both panels use investor masses, the fixed 2015–2017 economy and $z = 0$. Riskless holdings comprise cash and one-period municipal bonds (two years on the numerical grid). All longer bonds are risky. In Panel A, home-state municipal holdings include both riskless and risky bonds and overlap the other columns. The last two columns sum to 100%. In Panel B, the three columns sum to 100%, subject to rounding. These portfolio shares differ from shares of market supply.

Funds and institutions. Exempt-interest dividends generally pass the federal exemption through municipal funds, while state treatment depends on the shareholder's residence and the fund's in-state share, so funds can attenuate the household state incentive. Institutions face different wedges. Tax-deferred accounts and many pensions owe no current state interest tax. P&C insurers commonly pay state premium taxes, which do not distinguish municipal from taxable interest, and banks' and corporations' state taxes depend on the tax base and business apportionment. None of these is the issuer-state-by-resident-state exemption exploited by P1, so the model assigns that wedge only to households and emphasizes federal proration and financing deductions for insurers and banks.

E.4 Federal deduction credit for out-of-state municipal interest

Before 2018, itemizers could deduct state income taxes on federally exempt interest against ordinary federal income tax. The deduction reduces tax on other taxable income even though the municipal interest itself is federally exempt. Its full benefit is not available to every household: taking the standard deduction, paying the alternative minimum tax, or choosing a larger sales-tax deduction can eliminate the incremental saving.[31]

To test sensitivity to this treatment, I grant full credit at the ordinary-income component of the baseline federal tax response, excluding NIIT and forming tax-unit products before aggregation. This experiment imposes full use of the credit without reducing it for each household's deduction eligibility. I re-solve the six pre-2018 economies, holding all other inputs and calibrated coefficients fixed. The post-2017 binding-cap convention is unchanged.

The credit can reduce state-tax pass-through by making out-of-state bonds more attractive to high-tax-state households. For example, in the 2015–2017 California high-tax group, the effective after-tax shares are 0.498 for taxable interest and 0.919 for out-of-state municipal interest with the credit. At a riskless taxable

[31] IRS Publication 550 (2017), "State income taxes"; 2017 Instructions for Form 6251, line 3; and 2017 Instructions for Schedule A, line 5. 2016 Instructions for Form 8960, line 9b, restrict NIIT deductions to taxes attributable to net investment income, which excludes exempt interest.

yield of 5%, a municipal yield of 2.71% equates the two after-tax returns, compared with 2.87% for the no-state-tax high-tax group. The taxable alternative already incorporates the baseline deduction treatment. The added municipal credit lowers the yield needed to match it. High-tax-state residents can consequently accept lower out-of-state yields than local residents, compressing cross-state yield differences even at the riskless maturity.

The aggregate puzzle changes very little. Using the main sample's 2000–2020 weights and corporate normalization, the thirty-year model gap falls from 97.39 to 97.16 basis points, a reduction of only 0.23 basis points. Six-block mean P1 pass-through, with its original taxable-yield normalization, falls from 1.001 to 0.904 at the two-year riskless maturity and from 0.458 to 0.412 at twenty-five years (interpolated on the two-year grid). Full use of the credit lowers the level of state-tax pass-through while preserving its declining maturity profile. The model's large long-maturity gap and its explanation of declining household tax pass-through therefore remain under this stronger deduction incentive.

E.5 Accrual, price changes, and secondary-market taxation

Equation (35) applies the interest-income after-tax share to the model return $\mathcal{R} = \mu + u$.[32]

Gains and basis. For individuals, federal law separates interest accrual from gains at disposition (Sections 171, 1272–1278 and 1288). For face 100, annual coupon payment c in dollars, original price A_0 and annual effective issue yield y_0, the adjusted issue price satisfies

$$A_t = (1 + y_0)A_{t-1} - c, \qquad t = 1, \ldots, n, \qquad A_n = 100. \tag{155}$$

Discount $A_0 < 100$ is OID. Premium $A_0 > 100$ is OIP. Annual interest is $y_0 A_{t-1} = c + (A_t - A_{t-1})$, including noncash accrual, and is taxed at the ordinary-income rate τ_{oi}:

$$\text{tax}_t = \begin{cases} \tau_{oi} y_0 A_{t-1}, & \text{taxable bond,} \\ 0, & \text{tax-exempt bond.} \end{cases} \tag{156}$$

The basis change $A_t - A_{t-1}$ prevents taxing accrued interest again at redemption. OID raises basis, including for zero coupons, and OIP lowers it. Taxable amortization requires the assumed Section 171 election, and exempt amortization gives no deduction. At maturity, basis equals face, so there is no capital gain. For taxable bonds, de minimis OID below a quarter point per full year can be treated as zero, with the discount recognized as capital gain at maturity. This OID exception generally does not apply to tax-exempt obligations.[33]

A sale at P_h gives $G_h = P_h - \text{basis}_h$, long-term after one year and taxed at the capital-gains rate τ_{cg}. A secondary buyer starts at purchase price P. Above face, premium amortizes at the buyer's yield, mandatorily without deduction for exempt bonds and electively for taxable bonds. On an OID bond bought at date s for $A_s < P \leq 100$, where A_s is the adjusted issue price, the acquisition premium $P - A_s$ reduces remaining OID by fraction $(P - A_s)/(100 - A_s)$, leaving basis 100 at maturity.

Market discount is the shortfall below A_s for OID bonds or below face for par and premium bonds. It accrues ratably, or at constant yield by election. Absent annual inclusion, accrued discount D_h is ordinary income at sale or redemption, limited to positive gain. This also covers exempt bonds bought after April 30, 1993. Below threshold (124), discount is treated as zero and the gain is capital income. Thus

$$\text{tax at sale} = \tau_{oi} \min\{\max(G_h, 0), D_h\} + \tau_{cg}[G_h - \min\{\max(G_h, 0), D_h\}], \qquad G_h = P_h - \text{basis}_h. \tag{157}$$

[32] The after-tax share multiplies the log of the expected gross return $\log \mathbb{E}[1 + R]$ and the centered log-return innovation. Applying the tax to a simple return R gives after-tax gross return $1 + \xi R$, with $\log \log(1 + \xi R)$. For a deterministic two-year simple return of 5.4 percent and $\xi = 0.62$, its difference from $\xi \log(1 + R)$ is 3.2 basis points per two-year period (1.6 a year), and it is zero for a fully exempt leg. Appendix B.1 reports the effect of taxing simple returns exactly in the solved model. The return convention matches the empirical implied tax and the after-tax yield mappings of Appendix D.1.3.

[33] IRS Publication 550 (2017), "Original Issue Discount (OID)," and Publication 1212, "De minimis rule." The market-discount threshold in (124) applies separately.

Individual capital losses offset gains and up to $3,000 of annual ordinary income, with carryforwards (Sections 1211–1212). The adjustment assumes full capital-rate offsets (125). Corporations tax gains at their ordinary rate but offset capital losses only against gains. Banks treat gains and losses on debt as ordinary income (Section 582(c)). Institutional after-tax shares apply one rate to the full return. Appendix D.1.3 implements these accrual and basis mappings.

E.6 Institutional foundations of the no-short-sale constraint

Tax asymmetries, financing and residual risk limit short arbitrage. The nine strategies below explain how these restrictions sustain different marginal duration valuations after hedging, the condition in (167), and motivate nonnegative bond positions.

Household tax rules matter because high-income households hold much of the market. In custodial data, munis are wealthy households' second-largest liquid risky asset and portfolio shares rise with wealth (Gabaix et al., 2026). In tax years 2008, 2016 and 2019, returns with adjusted gross income of $1 million or more, about 0.3 percent of all returns, reported 26–33 percent of tax-exempt interest.[34]

Define the after-tax coupon gap $g = c_M - (1-\tau)c_C$ from matched-maturity par coupons and the investor's marginal tax rate τ. This coupon measure differs from the 117-basis-point thirty-year puzzle in zero-coupon yields (Section 5.4) and from the model's one-period return gap g_T. Carry assumes equal principal before fees, margin and collateral. Duration and price risk need not match.

(i) Financing municipals with debt or a corporate short. Let c_F be the interest on a dollar borrowed to buy a dollar of municipal principal. With a deduction, carry would be $c_M - (1-\tau)c_F$. Section 265(a)(2) denies deductions for interest on debt incurred or continued to purchase or carry tax-exempt obligations, so carry becomes

$$c_M - c_F = \{c_M - (1-\tau)c_F\} - \tau c_F. \tag{158}$$

At parity $c_M = (1-\tau)c_F$, carry loses τc_F. Corporate shorts supply no restored shield, since Section 265(a)(5) treats compensatory coupons c_C as interest under Section 265(a)(2), leaving $c_M - c_C$. This is negative at every maturity because the sample-average municipal zero curve, and hence the implied par yield, lies below its corporate counterpart at every maturity. At thirty years with annual compounding, the municipal and corporate par yields are 4.498% and 5.443%.[35] Fund documents tell investors that debt interest allocable to exempt-interest dividends is nondeductible.[36] Depositories used deductible financing at scale, holding 51.8% of munis in 1972 and $235.1 billion (27.3%) in 1985. After the 1986 denial (Appendix E.1), holdings fell to $120.4 billion in 1990 and $95.4 billion (7.5%) in 1995. Market outstanding grew from $862 billion in 1985 to $1.27 trillion in 1995.[37]

(ii) Pairing corporate positions. If short proceeds buy a corporate long, the investor's own capital finances the muni and the remaining trade is hedge (iii). If proceeds carry the muni, Sections 265(a)(2) and (a)(5) deny deductions for the payments. Revenue Procedure 72-18 tests purpose through tracing, exempt collateral or facts and circumstances. Its business-borrowing presumption lapses when borrowing exceeds business needs. Offsetting straddles also capitalize interest and carrying charges net of taxable position income under Section 263(g), deferring deductions.[38]

[34] IRS Statistics of Income, Individual Income Tax Returns, Table 1.4, all returns by size of adjusted gross income, tax years 2008, 2016 and 2019. Tax-exempt interest includes exempt-interest dividends from funds.

[35] 26 U.S.C. §265(a)(2), (5), and (b), 2018 U.S. Code. Section 265(a)(5)(B) excepts specified short-sale payments when the taxpayer posts cash collateral that earns no material return. That exception does not remove the opportunity cost of the collateral. See also Hines and Horton (2024) on banks' financing-tax restrictions.

[36] BlackRock Municipal Bond Fund, Inc. (2025), Part II, "Dividends and Taxes," p. II-131; iShares Trust (2026), "Taxes," p. 210; Nuveen Municipal Trust (2026), "Tax Matters," p. S-80.

[37] Board of Governors of the Federal Reserve System, Financial Accounts of the United States (Z.1), release of September 8, 2023, Table L.212, year-end levels. Depository institutions comprise U.S.-chartered depository institutions, foreign banking offices in the U.S., and banks in U.S.-affiliated areas.

[38] Rev. Proc. 72-18, 1972-1 C.B. 740, §§3.01–3.04 and 6.01, as summarized in Revenue Ruling 2004-47 and in IRS Field Service Advice 200129009, which is nonprecedential. On straddles, see 26 U.S.C. §263(g)(1)–(2) and

(iii) Hedging duration with a corporate short. An unlevered muni duration hedge shorts long corporates and buys short corporates with proceeds. Let c_ℓ and c_s be the long and short corporate coupons, so that $g = c_M - (1-\tau)c_\ell$, and let $\delta \in [0,1]$ be the currently deductible fraction of the short payment. Equal-principal annual carry relative to the short corporate bond is

$$\Delta = \{c_M + (1-\tau)c_s - (1-\delta\tau)c_\ell\} - (1-\tau)c_s = g - (1-\delta)\tau c_\ell. \tag{159}$$

Denial under Section 265 gives $\delta = 0$ and hence $\Delta = c_M - c_\ell$, which is negative at every maturity and amounts to a loss of 95 basis points a year at thirty years. Section 263(g) capitalization and Section 163(d) carryforwards defer deductions, so current carry approaches $c_M - c_\ell$, with later basis or deduction benefits and a timing and liquidity cost.

Where neither Section 265 nor Section 263(g) applies, individuals deduct investment interest under Section 163(d) only by itemization and up to net investment income, carrying excess forward. Coupons on the short-maturity corporate long position add capacity. Wages and exempt coupons do not. With principal q on both corporate legs, $k = c_\ell - c_s > 0$, unused capacity $C \geq 0$ and no prior carryforwards, current hedge cash flow is

$$\Pi(q) = -(1-\tau)qk - \tau[qk - C]_+, \tag{160}$$

where $[x]_+ = \max\{x, 0\}$, so $\delta < 1$ once qk exceeds C.[39]

Even with $\delta = 1$, the hedge removes only common rate risk. Bond-specific risk (Section A.1) and credit, liquidity and call differences (Appendix D) remain. Shorting requires available bonds and incurs fees, recall risk and margin demands against an illiquid callable long.[40] Corporate borrowing fees are 10–20 bp in Asquith et al. (2013)'s major-lender sample.

(iv) Hedging with swaps. A pay-fixed, receive-floating swap hedges duration without borrowing or shorting a bond, but household access is restricted. A person that is not an eligible contract participant may enter into a swap only on, or subject to the rules of, a designated contract market. An individual qualifies only with more than $10 million invested on a discretionary basis, or more than $5 million when the swap manages the risk of an asset owned or a liability incurred.[41] Ordinary swap payments are not automatically interest on indebtedness, so Section 265(a)(2) alone does not deny them. Household Section 212 expenses attributable to exempt interest fall under Section 265(a)(1).[42] Independently of that classification, let X be a net taxable amount and $d \in [0,1]$ the denied fraction of its loss tax shield:

$$A_d(X) = (1-\tau)X - d\tau X^-, \qquad X^- := \max\{-X, 0\}. \tag{161}$$

Symmetric offsets give $d = 0$. Taxing gains and denying losses gives $d = 1$, lowering expected proceeds by $\tau\mathbb{E}[X^-]$.

Netting gives the swap a tax advantage when the corporate short payment is disallowed. With matched periodic receipts and payments $F, K \geq 0$, the bond trade's cash is $A_C = (1-\tau)F - K$ when the corporate short payment is wholly nondeductible, and the swap's is $A_S = F - K - \tau(F-K)^+$ under the rule that taxes net gains and denies losses. Hence

$$A_S - A_C = \tau \min\{F, K\} \geq 0, \tag{162}$$

§1092(c)(1)–(2). Positions are offsetting when holding one substantially diminishes the risk of loss from holding the other.

[39] 26 U.S.C. §163(d)(1)–(3), which includes otherwise deductible short-sale expenses in investment interest; IRS Publication 550 (2016), Chapter 3, "Interest Expenses" and "Interest and carrying charges on straddles"; IRS Form 4952 and instructions (2016). Net capital gain and qualified dividends count as investment income only through an election that relinquishes their preferential rates to that extent.

[40] FINRA Rule 4210, paragraph (e)(2), sets margin requirements for bond shorts.

[41] Commodity Exchange Act §2(e) and §1a(18)(A)(xi), 7 U.S.C. §§2(e), 1a(18)(A)(xi); see also the CFTC's 2012 final-rule summary.

[42] Treasury Regulations §1.446-3 and §1.265-1(a)(2), (c). IRS Field Service Advice 200037034 treats the ordinary swap payments before it as non-interest in the absence of a loan. The advice concerns a corporate taxpayer and is nonprecedential.

and the two coincide when the corporate payment is deductible against F. Additional deduction capacity favors the corporate short. A pay-fixed swap has the rate exposure of a short fixed-rate bond and a long floating-rate bond, without the corporate short's credit and borrowing exposures, though with its own collateral and counterparty exposure. It weakly dominates a corporate hedge in the mean–variance objective whenever the latter delivers the same exposure with only an additional nonnegative cost and uncompensated risk orthogonal to the existing portfolio and the swap.[43] An outside dealer can also absorb swap risk without trading domestic bonds, whereas physical shorts change the gross long positions required to clear those markets. Because the wealthiest households can clear the eligibility thresholds, the equilibrium test in Appendix E.6.1 grants every household a thirty-year pay-fixed swap from an outside dealer with no fee, margin or notional limit. The test taxes swap gains and denies losses ($d = 1$), keeps physical bond shorts excluded and institutions' hedging unchanged, and re-solves both bond markets. The model's thirty-year gap for 2015–2017 falls only from 104.3 to 94.8 basis points, retaining 90.9% of its baseline value.

(v) Treasury futures. Regulated futures avoid swap eligibility rules but require cash margin and annual marking to market, with gains and losses treated as 60% long-term and 40% short-term. Individuals offset capital gains fully, deduct at most $3,000 against ordinary income, and carry excess forward. Net Section 1256 losses may be carried back three years against net Section 1256 gains. Section 1092 defers losses to the extent of unrecognized gains on offsetting bonds. Gains can therefore be taxed before losses yield tax benefits, creating timing and liquidity costs represented by partial present-value loss denial, $0 \le d < 1$ in (161).[44]

(vi) Shorting municipals. A taxable lender's substitute payment for an exempt coupon loses the coupon's exemption, so a lender requiring the after-tax amount of an exempt coupon c_M must receive L satisfying

$$(1-\tau_L)L = c_M, \qquad L - c_M = \frac{\tau_L}{1-\tau_L} c_M. \tag{163}$$

A short also requires a willing lender of the specific issue and satisfaction of the contract's collateral terms.[45] The reverse trade shorts a municipal bond and buys an equal-principal corporate bond. With τ_B the short seller's tax rate and δ the currently deductible fraction of the substitute payment, as in (iii), its coupon carry under this lender compensation is

$$\pi_{M\to C} = (1-\tau_B)c_C - (1-\delta\tau_B)\frac{c_M}{1-\tau_L}. \tag{164}$$

Even with full deduction, $\delta = 1$, equal borrower and lender rates τ give $\pi_{M\to C} = (1-\tau)c_C - c_M$. Together with (i), both coupon trades have nonpositive carry throughout the band

$$(1-\tau)c_C \le c_M \le c_C, \tag{165}$$

and long-end municipal coupons lie in it, above the household after-tax benchmark and, on the sample-average curves, below the corporate coupon. A municipal curve trade that shorts long-maturity municipals and buys short-maturity municipals pays the gross-up in (163) on its short leg. To the extent its proceeds buy the short-maturity municipal bond, Sections 265(a)(2) and (a)(5) deny a deduction for that payment.

(vii) Tax-exempt floaters. A tender-option-bond partnership divides a long municipal bond into variable-rate equity and an inverse interest receiving the residual. Exempt income passes through, avoiding

[43] In common payoff units, let the corporate hedge be $B = \alpha S - \varsigma + \varepsilon$, where S is the swap payoff, $\alpha > 0$, $\varsigma \ge 0$, $\mathbb{E}\varepsilon = 0$, and $\mathrm{Cov}(\varepsilon, W) = \mathrm{Cov}(\varepsilon, S) = 0$ for the investor's existing risk index W. With $\mathrm{CE}(Y) = \mathbb{E}Y - \gamma\,\mathrm{Var}(Y)/2$, one has $\mathrm{CE}(W + qB) - \mathrm{CE}(W + q\alpha S) = -q\varsigma - \gamma q^2\,\mathrm{Var}(\varepsilon)/2 \le 0$ for $q \ge 0$, provided the replacement is feasible at no greater funding cost.

[44] 26 U.S.C. §1256(a)(1), (a)(3), (b) and (g)(1); §1211(b); §1212(b)–(c); §1092(a)(1)(A). Section 1256(a)(4) removes Sections 1092 and 263(g) only for straddles consisting entirely of Section 1256 contracts.

[45] IRS Publication 550 (2016), "Reporting Substitute Payments"; Treasury Regulation §1.6045-2(a)(4).

an interest deduction. The inverse holder receives the municipal curve's slope and bears its duration, so capturing g still requires one of hedges (iii)–(v). Floater demand limits the scale of the strategy.[46]

(viii) The de minimis presumption. Absent tracing or collateral, Revenue Procedure 72-18 ordinarily presumes no exempt-carrying purpose for insubstantial holdings. A corporation qualifies only up to 2% of average business assets. Direct evidence overrides the presumption. Larger holdings face the facts-and-circumstances test. Sections 163(d), 263(g), 1092 and 1211(b) still constrain deductions and losses.[47]

(ix) Other entities. Financial institutions outside the bank-qualified exception face pro rata interest disallowance under Section 265(b), based on adjusted tax bases, without tracing. Revenue Procedure 72-18 applies a similar ratio to dealers' untraceable business borrowing. Insurer proration reduces the exemption regardless of financing. Section 832(b)(5)(B) reduces P&C loss deductions and Sections 807 and 812 reduce life insurers' reserve deductions by the policyholders' share of exempt interest (30% since 2018). Tax-exempt accounts gain little from the exemption. Partnerships pass its character through under Section 702(b), but their interest remains subject to Section 265(a)(2) under Section 703(a). Over 2000–2020, dealers held a mean 0.85% and maximum 1.94% of munis. Hedge funds, reported from 2012, held at most 0.36%.[48]

The model constraint. With concavity and linear shorting costs, a zero short position is optimal when the marginal risk-adjusted benefit does not exceed the tax, borrowing and financing cost. The quantitative model maintains this corner, funds long purchases and charges measured bank financing taxes. The restrictions exclude unconstrained after-tax arbitrage as a benchmark. Measured costs do not certify every investor's corner, and some hedges remain profitable.

Residual risk and participation after hedging. The mechanism requires only a difference in marginal duration valuations after available hedges and is compatible with continued derivative trading. To isolate the rate-hedging tradeoff in this calculation, suppose bond funding is nonbinding and set bond-specific risk to zero. Let a hedge remove q_i units of duration at an expected charge $\kappa_i q_i$ and a differentiable convex additional cost $\psi_i(q_i)$, all in the investor's normalized payoff units. With residual duration $\widehat{D}_i^{\text{hedge}} = D_i - \ell_i - q_i$, the hedge-dependent objective is $-\kappa_i q_i - \psi_i(q_i) - \gamma_i \sigma_r^2 (\widehat{D}_i^{\text{hedge}})^2/2$, so an interior hedge choice satisfies

$$\lambda_i := \gamma_i \sigma_r^2 \widehat{D}_i^{\text{hedge}} = \kappa_i + \psi_i'(q_i). \tag{166}$$

Here λ_i is the marginal risk cost of another unit of duration. A binding hedge limit replaces the equality by the appropriate one-sided condition. At a household-only municipal allocation, with hedges and other holdings optimized, the lower-tax investor's municipal slack is

$$\nu_{I,T}^{M,\text{hh}} = (\xi_I - \xi_H) r - A_{T-1}^M (\lambda_H^{\text{hh}} - \lambda_I^{\text{hh}}). \tag{167}$$

This is the participation comparison (45) applied to residual, post-hedge risk costs. Liability differences sustain unequal duration valuations through unequal hedge access, binding limits or increasing marginal costs, even when both investors hedge.

If unlimited hedging in either direction spans rate risk, $\psi_i = 0$ and both investors face a common normalized charge κ, then $\lambda_H = \lambda_I = \kappa$. Equal duration costs and a positive tax difference exclude lower-tax municipal holdings. This conclusion requires a common after-tax hedging opportunity. Zero dealer fees alone do not suffice. Within the model, observed insurer holdings (Figure 25) therefore require a departure from this benchmark, which any of the documented frictions can sustain. Calibrated outside exposures are not directly measured net of derivatives.

Hedging is already part of insurers' portfolio management. At year-end 2015, interest-rate swaps accounted for 64% of the notional value of U.S. insurers' interest-rate hedges, and derivative users represented

[46] Revenue Procedure 2003-84, Section 2; 26 U.S.C. §702(b).

[47] Rev. Proc. 72-18, §3.05, as summarized in Revenue Ruling 2004-47 and Field Service Advice 200129009.

[48] Rev. Proc. 72-18, §§5.03 and 7, as summarized in Revenue Ruling 2004-47; 26 U.S.C. §265(b)(1)–(2), §832(b)(5)(B)–(C), §807(a)–(b) and §812. Holdings: Financial Accounts of the United States (Z.1), release of September 8, 2023, Tables L.212 and B.101.f, quarterly levels.

87% of life-insurer assets but 19% of P&C assets.[49] The household hedge-access equilibrium below measures how much of the gap survives when households receive the hedge.

E.6.1 Household swaps: incentives and equilibrium

I first evaluate the incentive to add a pay-fixed swap while holding baseline bond portfolios fixed. The calculation uses the 2015–2017 mean rate state and 630 household cohorts (14 types in 45 markets). I then allow households to choose swaps and bonds jointly and re-solve both bond markets. Historical quotes use 95 dates within 10 bp of the 3.0206% two-year rate, all in 2001–2008, outside the comparison block. Fixed rates at ten, twenty and thirty years are 4.8217%, 5.0976% and 5.3736%, and the twenty-year rate is interpolated.[50]

Floating payments are quarterly, fixed payments semiannual, and liquidation at two years uses an approximate calendar. One discount and forward curve and Gaussian pricing drifts fit quotes and liquidation values. Quarterly physical dynamics preserve the calibrated two-year persistence and variance. Annual net gains and the second-year liquidation payoff are taxed as ordinary income. Losses are denied in the baseline and offset symmetrically in a sensitivity. Municipal funds likewise distribute swap gains as taxable ordinary income or capital gains and allocate net payments pro rata against taxable and exempt income.[51] Segregated swap cash earns taxable interest when positive and incurs nondeductible interest when negative. Existing cash is not optimized.

Objective and tax units. Let $\widetilde{W}_i$ be the calibrated log-return risk index, including outside exposure $-h_i z'$, and M_i^0 its fixed baseline mean. The absolute home-municipal after-tax share $\xi_{i,\mathrm{abs}}^M$ is ξ_i^M of Appendix E.1 on home-state bonds. Multiplying it by the relative after-tax share ξ_i gives the swap's absolute after-tax share. Higher- and lower-tax home-municipal after-tax shares are 0.999905 and 0.990429, and swap tax rates span 21.49%–50.25%. After-tax cash A_i per unit notional changes normalized wealth by $qA_i/\xi_{i,\mathrm{abs}}^M$. The objective and annualized marginal score are

$$\mathcal{W}_i(q) = \xi_{i,\mathrm{abs}}^M \left[M_i^0 + \frac{q\mathbb{E}A_i}{\xi_{i,\mathrm{abs}}^M} - \frac{\gamma_i}{2} \operatorname{Var}\left(\widetilde{W}_i + \frac{qA_i}{\xi_{i,\mathrm{abs}}^M} \right) \right], \tag{168}$$

$$b_i = \frac{10^4}{2} \mathcal{W}_i'(0+) = \frac{10^4}{2} \left[\mathbb{E}A_i - \gamma_i \operatorname{Cov}(\widetilde{W}_i, A_i) \right]. \tag{169}$$

The score uses the model's log-return approximation. Bond-specific shocks remain in baseline risk but, by independence, have zero covariance with a rate-only swap.

Results. Table 53 reports the scores. At every tenor, 360 cohorts have negative and 270 positive scores. Positive cohorts, the top three outside-exposure bins of both groups, hold 42.9% of resources and 9.8% of household munis. Scores up to 3,100 bp are zero-notional slopes, and the quadratic cost of a finite trade is $-\gamma_i q^2 \operatorname{Var}(A_i)/(2\xi_{i,\mathrm{abs}}^M)$. Dispersion gives negative medians but positive weighted means.

Symmetric offsets benefit 585 of 630 cohorts. Loss denial lowers expected cash for all, but covariance changes lower scores for 450 and raise them for 180. Doubling pricing resolution and draws preserves signs (maximum tail change 2.57 bp), as does using terminal dollar payoffs and their covariance. A 25-bp quote window gives 285, 302 and 275 positive scores at ten, twenty and thirty years, some near zero. Asymmetric taxes leave some profitable fixed-portfolio hedges. The next exercise adjusts portfolios and prices.

[49] NAIC, "Update on the Insurance Industry's Use of Derivatives and Exposure Trends," year-end 2015, Tables 1 and 8 and the accompanying discussion. Notional shares measure instrument use, not the fraction of liability risk eliminated.

[50] Federal Reserve H.15 DSWP10 and DSWP30 are mid-market fixed rates against three-month LIBOR and differ from household quotes.

[51] BlackRock Municipal Bond Fund, Inc. (2025), Part II, pp. II-20, II-126 and II-132; iShares Trust (2026), "Taxes," p. 213.

Table 53: Fixed-portfolio swap benefits with gains taxed and losses denied.

Household group	Years	Cash	Risk benefit	Total	Median	Positive
Higher tax	10	−408.6	852.8	444.3	−14.4	135/315
Higher tax	20	−473.5	996.0	522.6	−13.1	135/315
Higher tax	30	−473.9	988.2	514.2	−17.2	135/315
Lower tax	10	−387.9	675.6	287.7	−28.8	135/315
Lower tax	20	−450.0	787.6	337.7	−31.5	135/315
Lower tax	30	−449.5	783.3	333.9	−33.3	135/315

Notes. The table reports annualized marginal scores in basis points per unit notional, as defined in equation (169). Cash, risk benefit and total use calibrated household-mass weights within each group, and identical budgets make resource weights equivalent. Medians and positive counts weight cohorts equally, and components are rounded separately. The segregated financing and tax assumptions are stated above.

Equilibrium with household hedge access. With the 2015–2017 calibration fixed, households jointly choose bonds and nonnegative thirty-year pay-fixed swap notionals. Outside dealers quote without trading modeled bonds. Swaps have zero upfront value at the state's par rate. Two-year liquidation uses the same pricing rule and historical quote family by contract age. Both price surfaces and all portfolios adjust across states, retaining nonnegative bonds and cash, gains taxation, loss denial and segregated interim financing.

Swap access reduces the thirty-year gap by 9.48 bp (9.1%, Table 54), with municipal and taxable yields falling 15.95 and 12.04 bp. At the mean rate state, 270 cohorts hedge, but over 99.99% of their municipal holdings remain riskless two-year bonds, since they chiefly hedge outside exposure. Portfolio responses at other rate states affect continuation prices. The exercise isolates hedge access in the 2015–2017 economy.

Table 54: Household swap access and the equilibrium municipal gap.

Maturity	Baseline	Swap access	Change
2 years	0.00	0.00	0.00
10 years	23.68	22.24	−1.44
20 years	72.63	65.42	−7.20
30 years	104.27	94.79	−9.48

Notes. Entries are the supply-weighted municipal yield minus the household after-tax taxable yield, in basis points, averaged over 36 monthly states of 2015–2017 with continuous yields. After-tax shares, parameters and dollar supplies are fixed. Both yields adjust. Changes are computed before rounding. Refining the simulation fourfold moves either yield curve or the gap by less than 0.003 bp, and price and optimality residuals are below 10^{-10}.

Hedging costs and interpretation. The test includes asymmetric taxes and nondeductible negative-cash interest but excludes dealer fees, bid–ask spreads, margin, collateral opportunity costs and notional limits. The largest mean-state notional is 6.88 times budget, substantially expanding household access.

At fixed bond prices and holdings, let $c_i \geq 0$ be additional two-year certainty-equivalent cost per notional in A_i's absolute units. Equation (168) gives

$$q_i^*(x; c_i) = \frac{\xi_{i,\mathrm{abs}}^M}{\gamma_i \operatorname{Var}(A_i)} \left[\mathbb{E}A_i - \gamma_i \operatorname{Cov}\left(\widetilde{W}_i(x), A_i\right) - c_i \right]_+ . \tag{170}$$

Here $[u]_+ = \max\{u, 0\}$ and $\widetilde{W}_i(x)$ is bond portfolio x's risk index. Costs reduce conditional hedge demand. Margin and position limits restrict feasible hedges. These frictions oppose risk transfer and are expected to attenuate yield responses. Equilibrium gaps depend on both yields. The 9.1% reduction excludes these costs.

E.7 Data sources

Table 55 lists the sources and their coverage.

Table 55: Data sources, variables and coverage.

Source	Variables	Coverage
Municipal issues: Mergent Municipal Bond Database (FTSE Russell)	CUSIP, issue & maturity dates, coupon, callability, tax status (federal/state), issuer state, purpose	1986–2023
Corporate issues: Mergent FISD (FTSE Russell)	CUSIP, issue & maturity dates, coupon, callability, rating, issuer industry, seniority	1986–2023
Insurers: NAIC Schedule D (annual, Q1–Q3 supplements)	CUSIP-level par, book, and market values for U.S. P&C and life insurers	1991–2023
Banks: FDIC BankFind (Call Reports RI, RC-B)	Annual interest expense, average assets and municipal securities holdings by bank	2000–2021
Public pensions: FOIA, CAFR/ACFR investment schedules	CUSIP-level muni/taxable par and market values, 30 state/local plans (coverage broadens from 2015)	2000–2023
Sector holdings: Financial Accounts (Z.1, L.212, B.101.f)	Quarterly munis: households, funds, banks, P&C, life, state/local pensions, other	1972–2022
Households: SCF, Federal Reserve	Tax-unit income and filing status, financial assets and debt	2001–2019 waves
Household taxes: IRS SOI	Filer counts, taxable and tax-exempt interest, effective federal rates by AGI bracket	2000–2022
Municipal trades: MSRB RTRS	Trade-level price, par volume, customer-side flag, trade-condition codes	2005–2023
Corporate trades: FINRA TRACE Enhanced (Academic)	Trade-level price, par volume, customer-side flag, condition codes	2002–2023
Treasury curve: GSW, Federal Reserve	Daily par and zero yields, 1–30y. Par yields enter P3 and zeros price its hypothetical coupon cash flows	2000–2023
Short rates: TB3MS, FRED	Monthly three-month treasury bill rates for the rate process	1952–2024
Treasury supply: Monthly Statement of the Public Debt	Security-level amounts and maturities for the treasury maturity profile	2001–2020
Corporate zeros: HQM (FRED), Mergent FISD robustness	Monthly A-or-above zero curve, 1–30y	2000–2023
Credit risk: Markit/IHS	State-government and AA corporate CDS at 1–30y tenors, coverage of the matched adjustment sample	2004–2020
Expected rates: Kim–Wright, Federal Reserve	Monthly fitted yields and term premia, expected-average-short-rate component capped at 10y	2000–2021
Swaps: Federal Reserve H.15 (FRED)	Par interest-rate swap rates for household swap pricing	2001–2008
State taxes: Tax Foundation	Top marginal rate, bracket thresholds, treatment of out-of-state muni interest	2000–2022
State income: BEA	Quarterly state-level personal income	2000–2022
Historical state rates: Institute on Taxation and Economic Policy (ITEP)	Top statutory rate by state, the P1 instruments	1945–2000 vintages